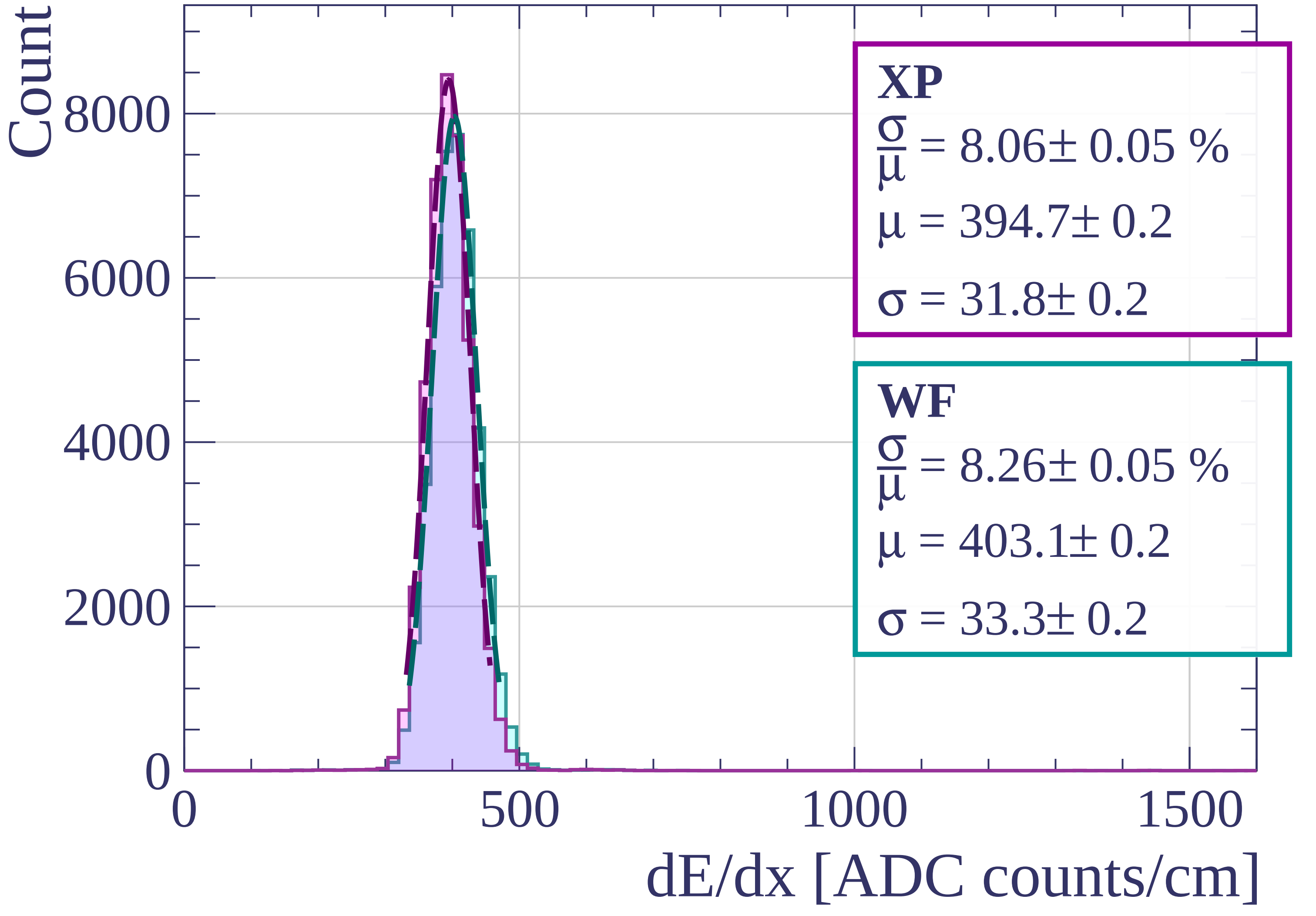
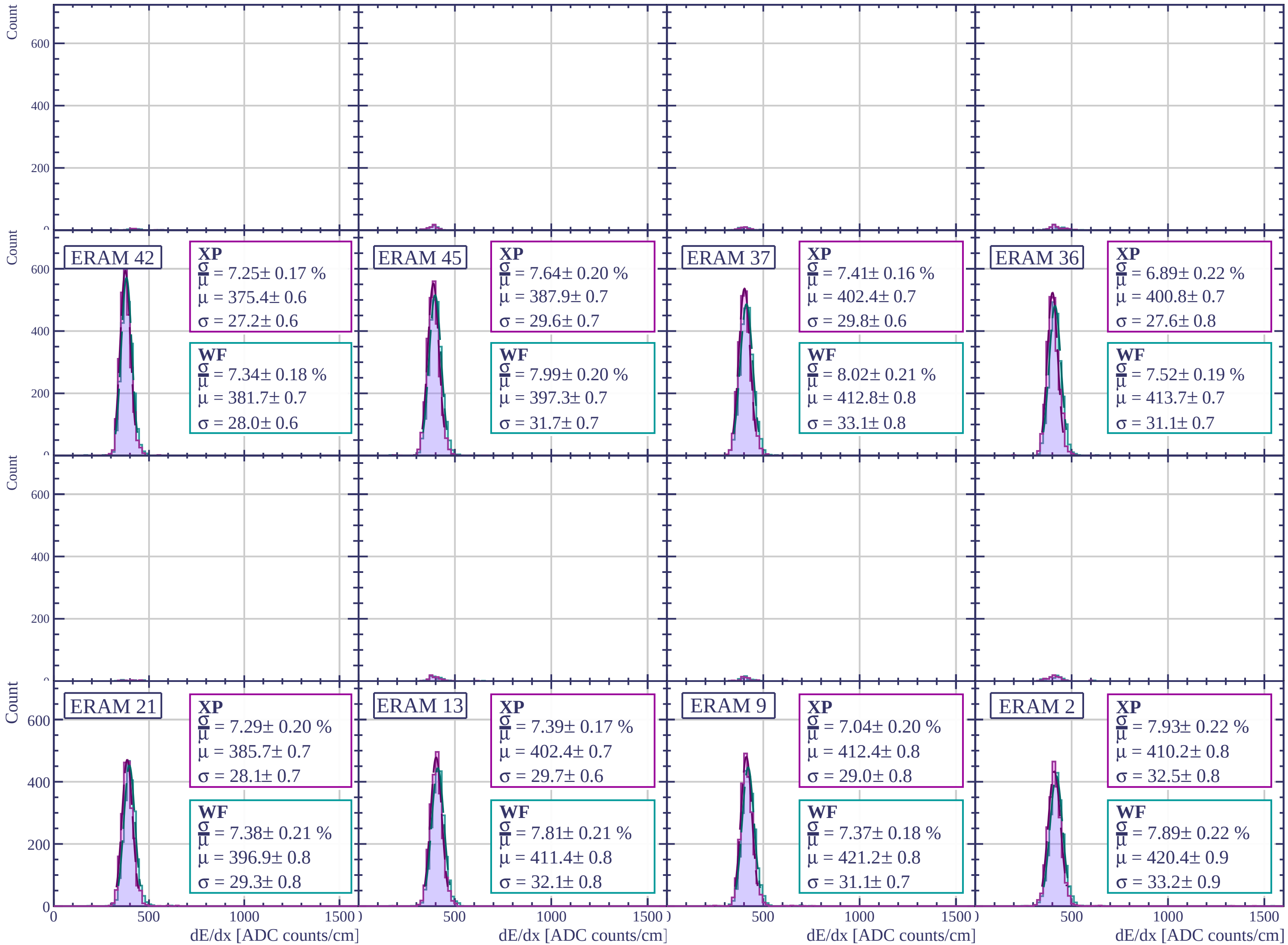
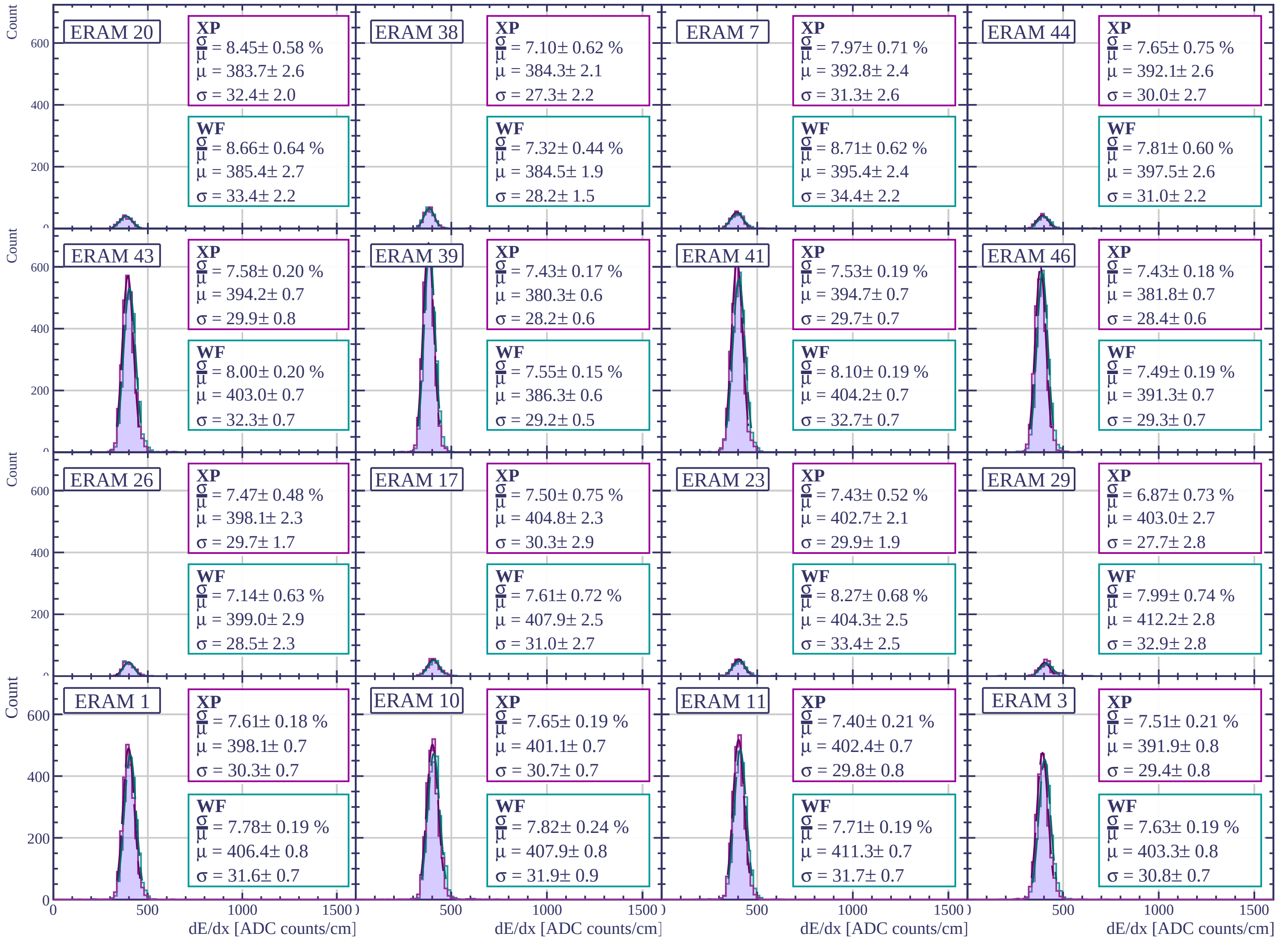
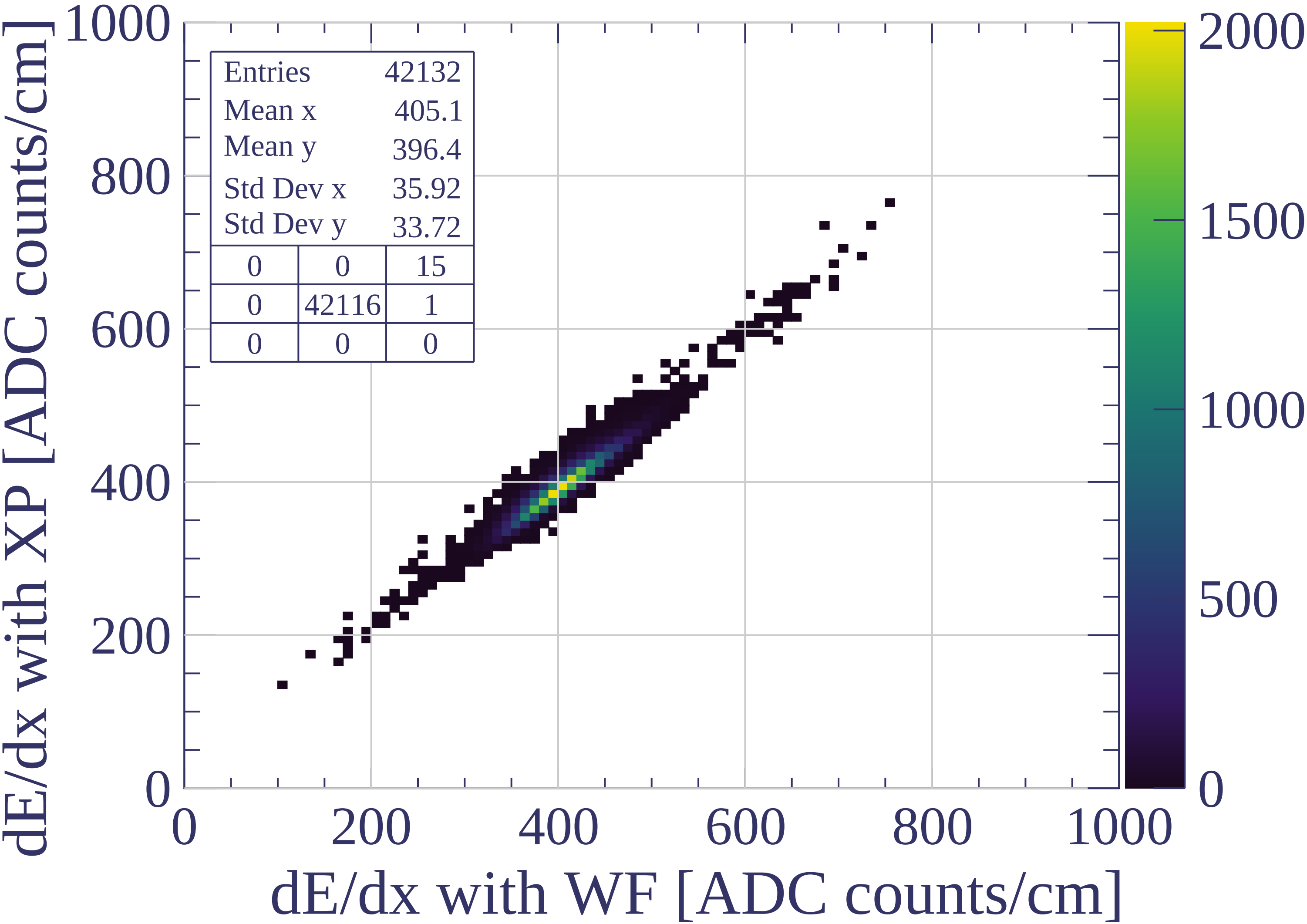
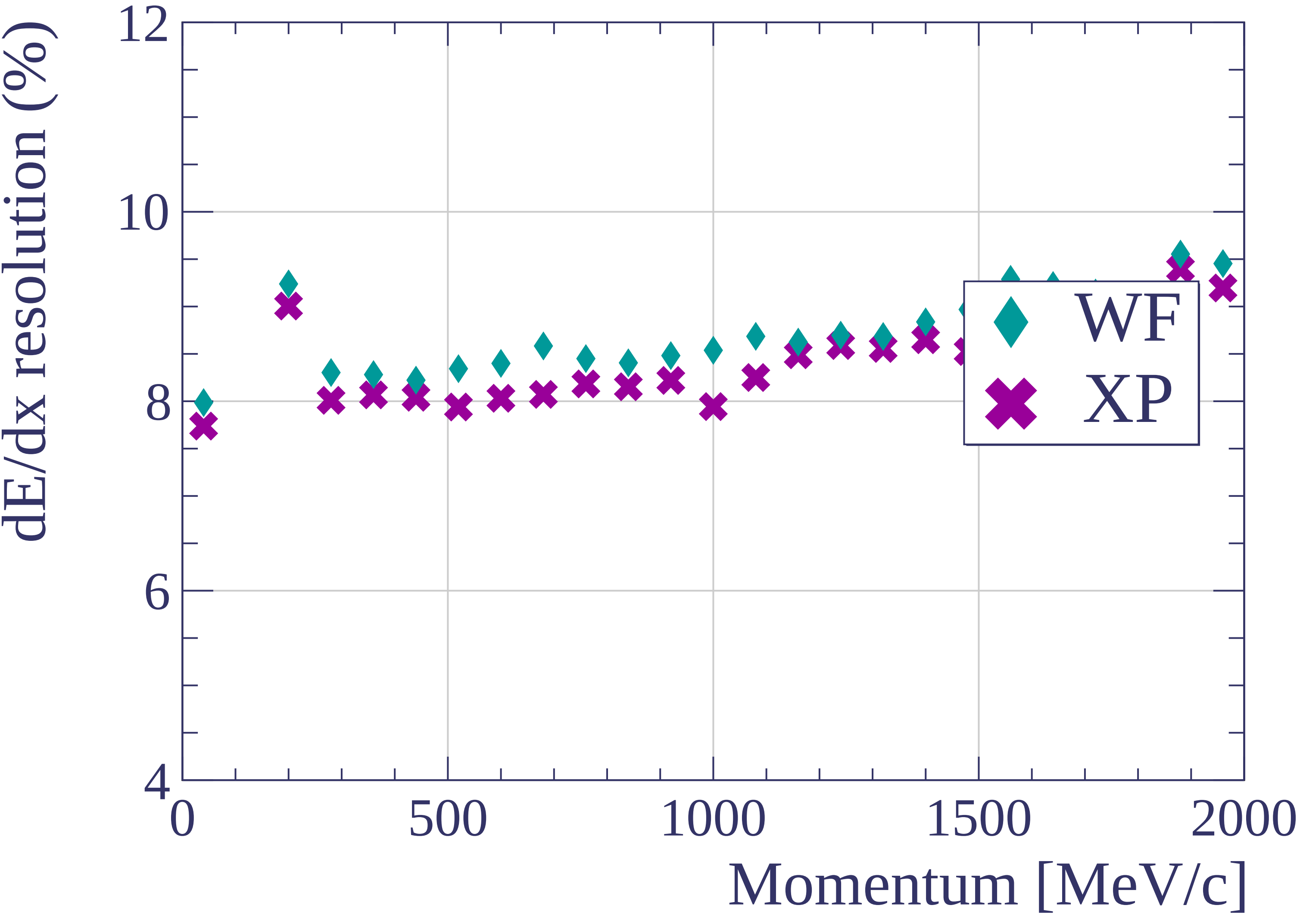


Count
8000
6000
4000
2000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.06± 0.05 %
μ = 394.7± 0.2
σ = 31.8± 0.2
WF
$\frac{\sigma}{\mu}$ = 8.26± 0.05 %
μ = 403.1± 0.2
σ = 33.3± 0.2

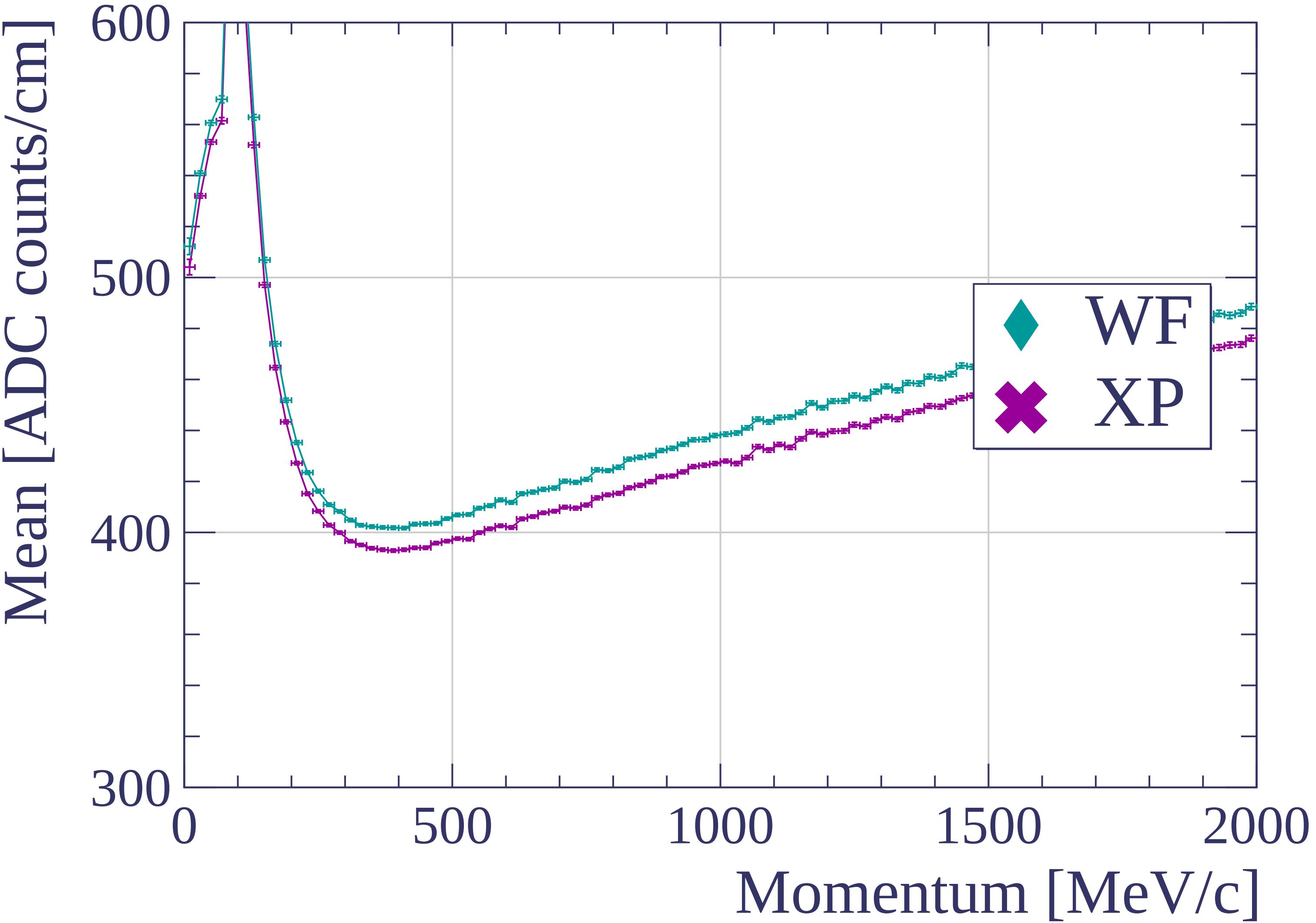

ERAM 42
XP σ/μ = 7.25± 0.17 % μ = 375.4± 0.6 σ = 27.2± 0.6
WF σ/μ = 7.34± 0.18 % μ = 381.7± 0.7 σ = 28.0± 0.6
ERAM 45
XP σ/μ = 7.64± 0.20 % μ = 387.9± 0.7 σ = 29.6± 0.7
WF σ/μ = 7.99± 0.20 % μ = 397.3± 0.7 σ = 31.7± 0.7
ERAM 37
XP σ/μ = 7.41± 0.16 % μ = 402.4± 0.7 σ = 29.8± 0.6
WF σ/μ = 8.02± 0.21 % μ = 412.8± 0.8 σ = 33.1± 0.8
ERAM 36
XP σ/μ = 6.89± 0.22 % μ = 400.8± 0.7 σ = 27.6± 0.8
WF σ/μ = 7.52± 0.19 % μ = 413.7± 0.7 σ = 31.1± 0.7
ERAM 21
XP σ/μ = 7.29± 0.20 % μ = 385.7± 0.7 σ = 28.1± 0.7
WF σ/μ = 7.38± 0.21 % μ = 396.9± 0.8 σ = 29.3± 0.8
ERAM 13
XP σ/μ = 7.39± 0.17 % μ = 402.4± 0.7 σ = 29.7± 0.6
WF σ/μ = 7.81± 0.21 % μ = 411.4± 0.8 σ = 32.1± 0.8
ERAM 9
XP σ/μ = 7.04± 0.20 % μ = 412.4± 0.8 σ = 29.0± 0.8
WF σ/μ = 7.37± 0.18 % μ = 421.2± 0.8 σ = 31.1± 0.7
ERAM 2
XP σ/μ = 7.93± 0.22 % μ = 410.2± 0.8 σ = 32.5± 0.8
WF σ/μ = 7.89± 0.22 % μ = 420.4± 0.9 σ = 33.2± 0.9
Count
dE/dx [ADC counts/cm]

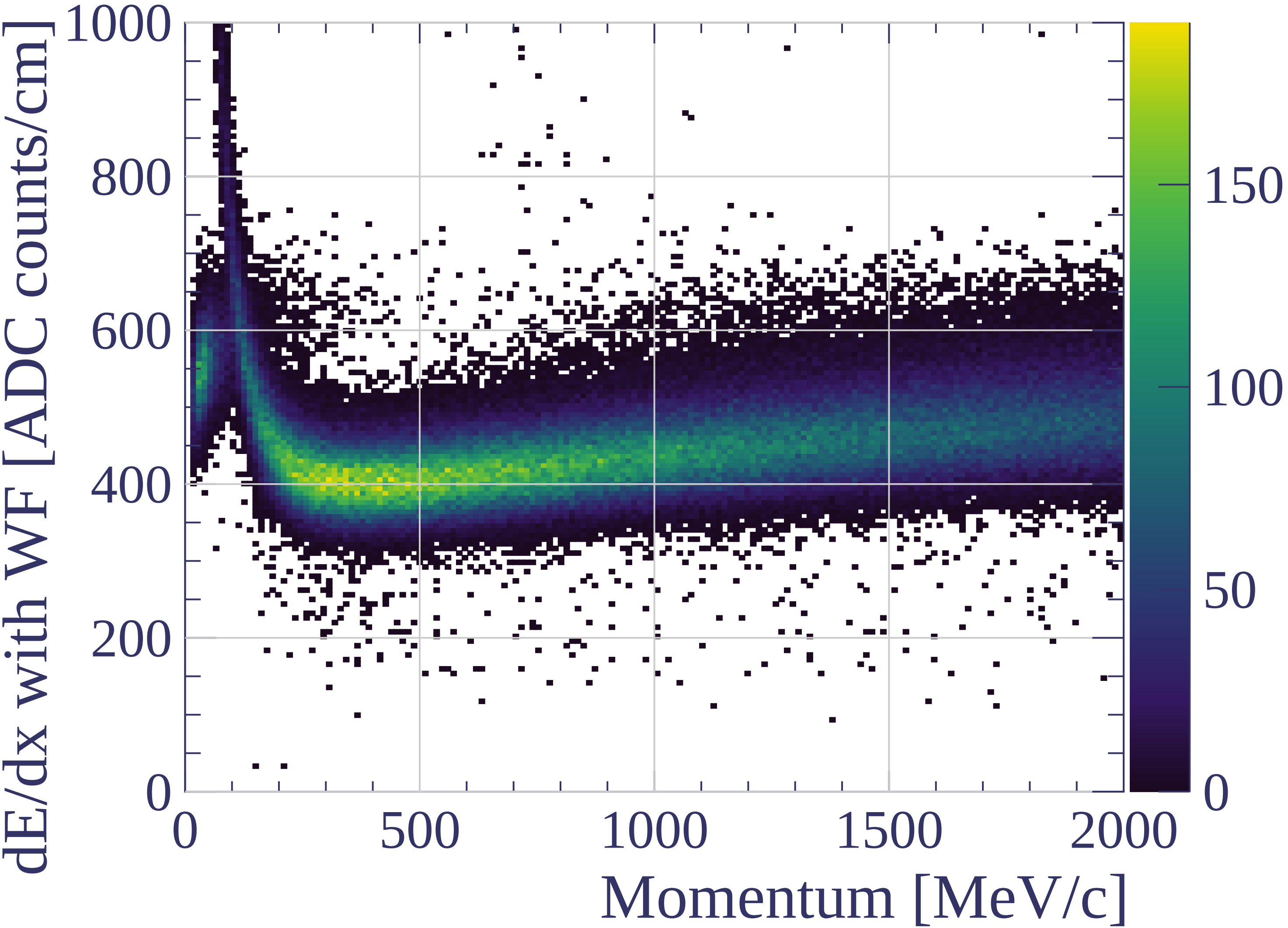

ERAM 20
XP
σ/μ = 8.45± 0.58 %
μ = 383.7± 2.6
σ = 32.4± 2.0
WF
σ/μ = 8.66± 0.64 %
μ = 385.4± 2.7
σ = 33.4± 2.2
ERAM 38
XP
σ/μ = 7.10± 0.62 %
μ = 384.3± 2.1
σ = 27.3± 2.2
WF
σ/μ = 7.32± 0.44 %
μ = 384.5± 1.9
σ = 28.2± 1.5
ERAM 7
XP
σ/μ = 7.97± 0.71 %
μ = 392.8± 2.4
σ = 31.3± 2.6
WF
σ/μ = 8.71± 0.62 %
μ = 395.4± 2.4
σ = 34.4± 2.2
ERAM 44
XP
σ/μ = 7.65± 0.75 %
μ = 392.1± 2.6
σ = 30.0± 2.7
WF
σ/μ = 7.81± 0.60 %
μ = 397.5± 2.6
σ = 31.0± 2.2
ERAM 43
XP
σ/μ = 7.58± 0.20 %
μ = 394.2± 0.7
σ = 29.9± 0.8
WF
σ/μ = 8.00± 0.20 %
μ = 403.0± 0.7
σ = 32.3± 0.7
ERAM 39
XP
σ/μ = 7.43± 0.17 %
μ = 380.3± 0.6
σ = 28.2± 0.6
WF
σ/μ = 7.55± 0.15 %
μ = 386.3± 0.6
σ = 29.2± 0.5
ERAM 41
XP
σ/μ = 7.53± 0.19 %
μ = 394.7± 0.7
σ = 29.7± 0.7
WF
σ/μ = 8.10± 0.19 %
μ = 404.2± 0.7
σ = 32.7± 0.7
ERAM 46
XP
σ/μ = 7.43± 0.18 %
μ = 381.8± 0.7
σ = 28.4± 0.6
WF
σ/μ = 7.49± 0.19 %
μ = 391.3± 0.7
σ = 29.3± 0.7
ERAM 26
XP
σ/μ = 7.47± 0.48 %
μ = 398.1± 2.3
σ = 29.7± 1.7
WF
σ/μ = 7.14± 0.63 %
μ = 399.0± 2.9
σ = 28.5± 2.3
ERAM 17
XP
σ/μ = 7.50± 0.75 %
μ = 404.8± 2.3
σ = 30.3± 2.9
WF
σ/μ = 7.61± 0.72 %
μ = 407.9± 2.5
σ = 31.0± 2.7
ERAM 23
XP
σ/μ = 7.43± 0.52 %
μ = 402.7± 2.1
σ = 29.9± 1.9
WF
σ/μ = 8.27± 0.68 %
μ = 404.3± 2.5
σ = 33.4± 2.5
ERAM 29
XP
σ/μ = 6.87± 0.73 %
μ = 403.0± 2.7
σ = 27.7± 2.8
WF
σ/μ = 7.99± 0.74 %
μ = 412.2± 2.8
σ = 32.9± 2.8
ERAM 1
XP
σ/μ = 7.61± 0.18 %
μ = 398.1± 0.7
σ = 30.3± 0.7
WF
σ/μ = 7.78± 0.19 %
μ = 406.4± 0.8
σ = 31.6± 0.7
ERAM 10
XP
σ/μ = 7.65± 0.19 %
μ = 401.1± 0.7
σ = 30.7± 0.7
WF
σ/μ = 7.82± 0.24 %
μ = 407.9± 0.8
σ = 31.9± 0.9
ERAM 11
XP
σ/μ = 7.40± 0.21 %
μ = 402.4± 0.7
σ = 29.8± 0.8
WF
σ/μ = 7.71± 0.19 %
μ = 411.3± 0.7
σ = 31.7± 0.7
ERAM 3
XP
σ/μ = 7.51± 0.21 %
μ = 391.9± 0.8
σ = 29.4± 0.8
WF
σ/μ = 7.63± 0.19 %
μ = 403.3± 0.8
σ = 30.8± 0.7
Count
0
200
400
600
0
500
1000
1500
dE/dx [ADC counts/cm]


Entries 42132
Mean x 405.1
Mean y 396.4
Std Dev x 35.92
Std Dev y 33.72
0 0 15
0 42116 1
0 0 0
dE/dx with XP [ADC counts/cm]
dE/dx with WF [ADC counts/cm]
0
200
400
600
800
1000
0
500
1000
1500
2000

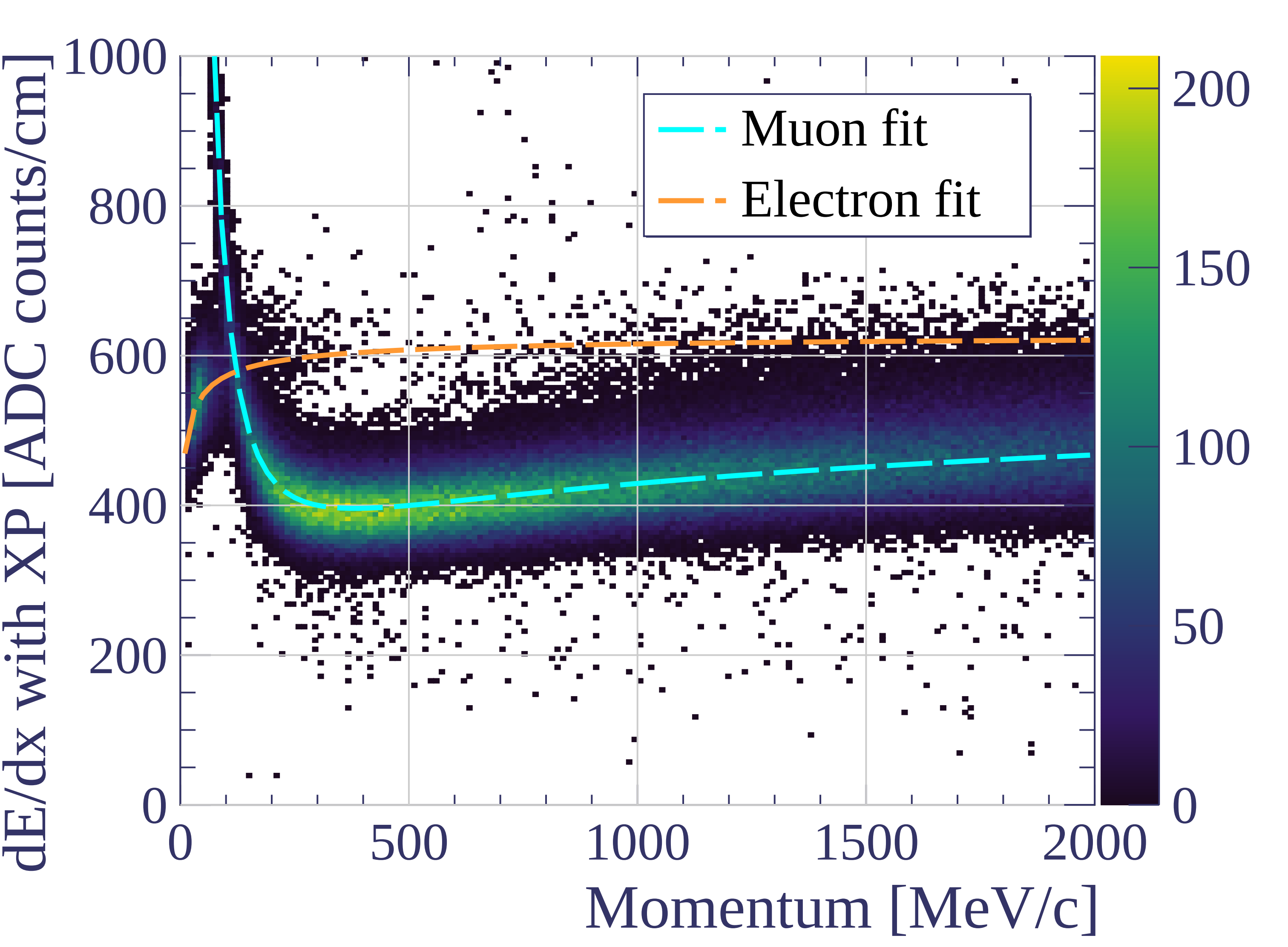


dE/dx resolution (%)
12
10
8
6
4
0
500
1000
1500
2000
Momentum [MeV/c]
WF
XP

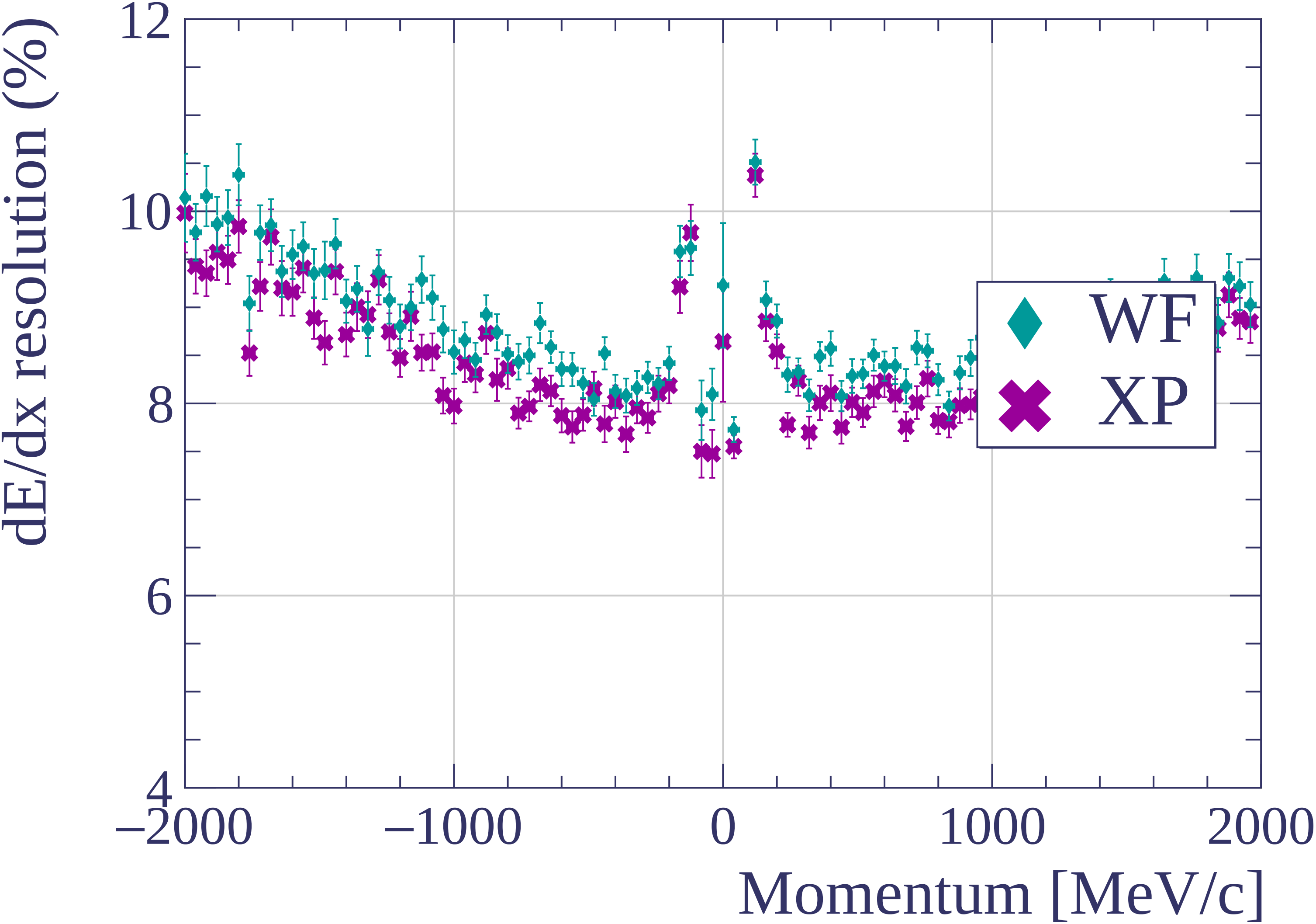

Mean [ADC counts/cm]
600
500
400
300
0
500
1000
1500
2000
Momentum [MeV/c]
WF
XP



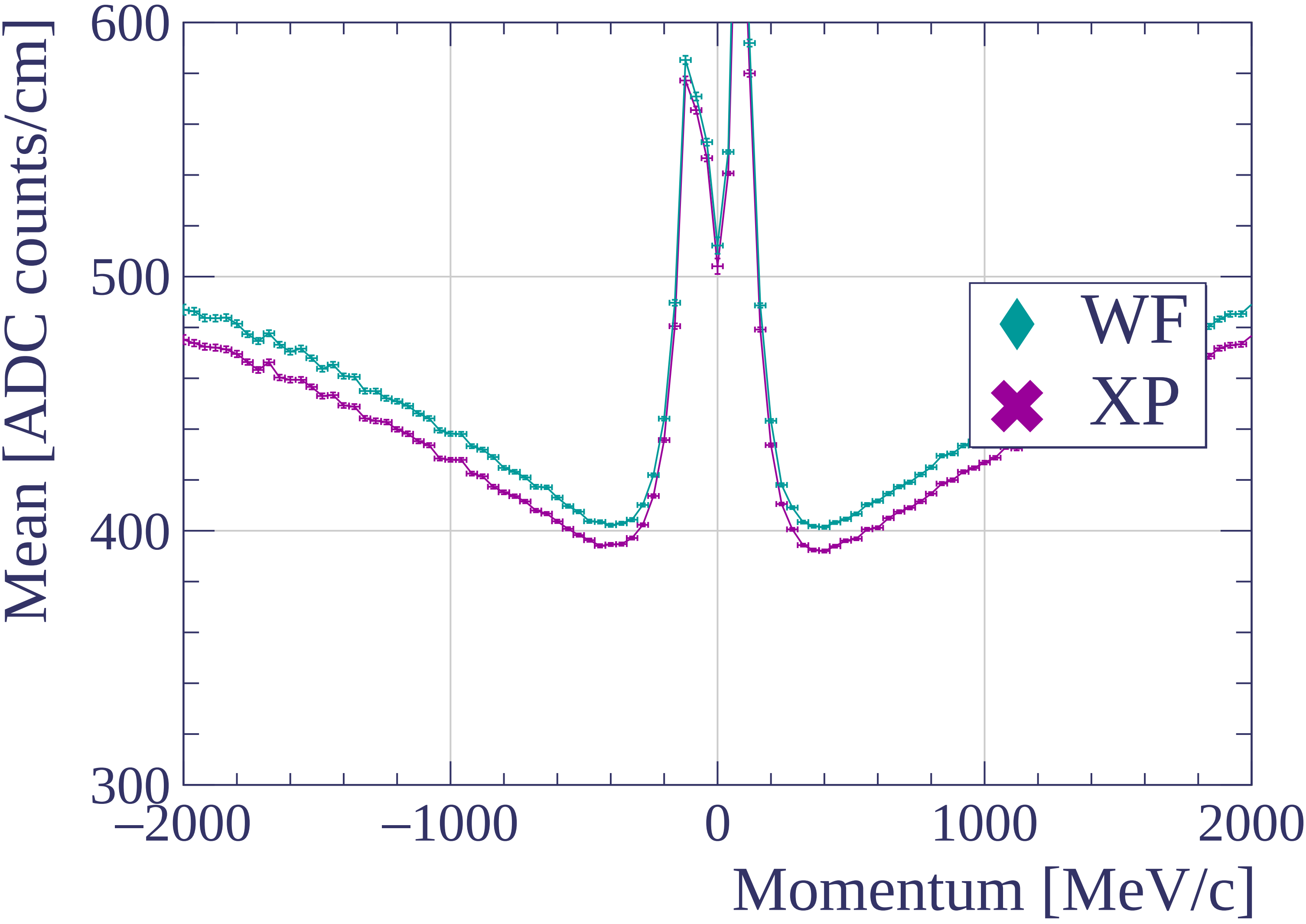
dE/dx with WF [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
0
500
1000
1500
2000
0
50
100
150

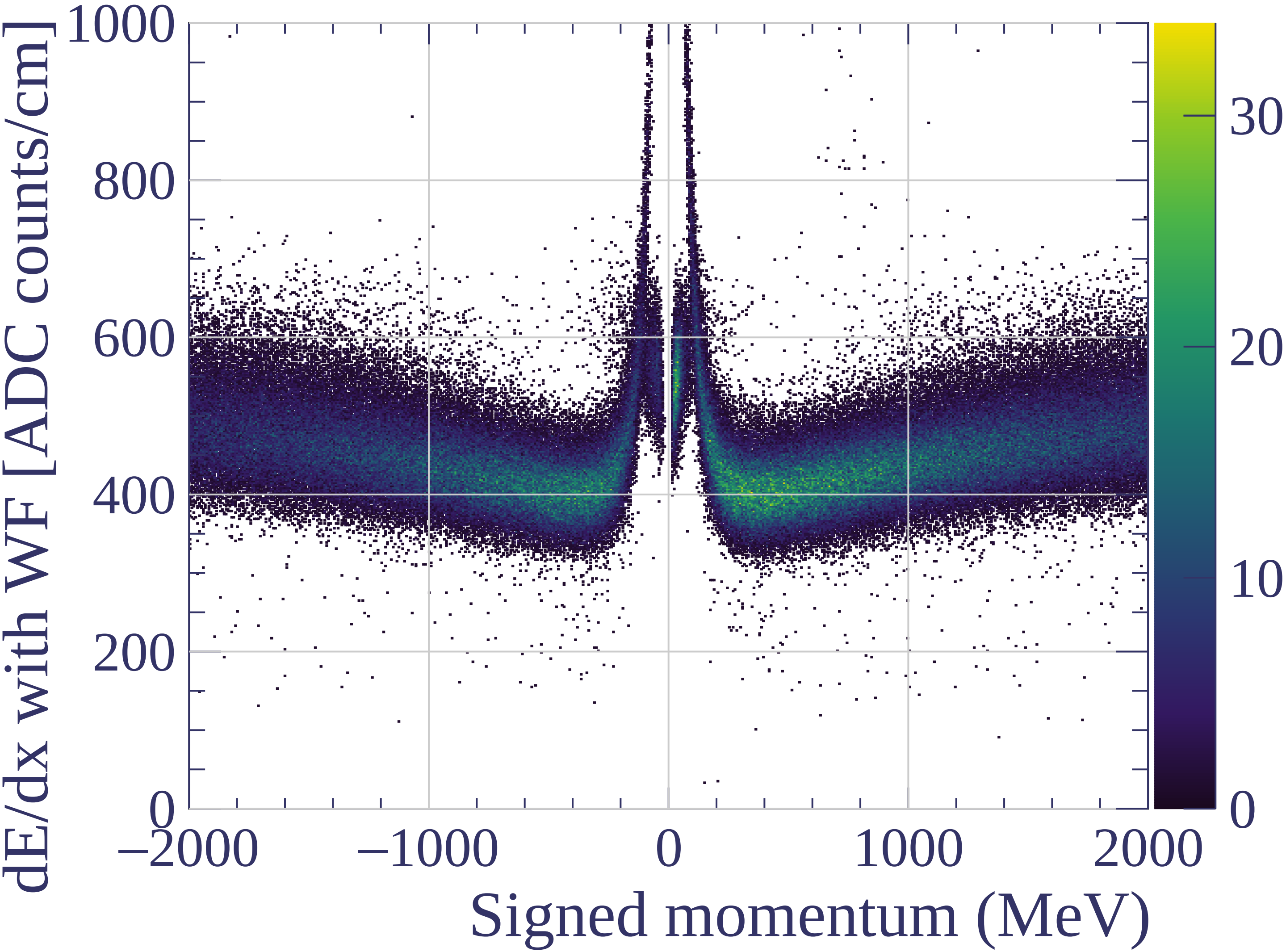


Muon fit
Electron fit
dE/dx with XP [ADC counts/cm]
Momentum [MeV/c]
0
200
400
600
800
1000
500
1000
1500
2000
0
50
100
150
200

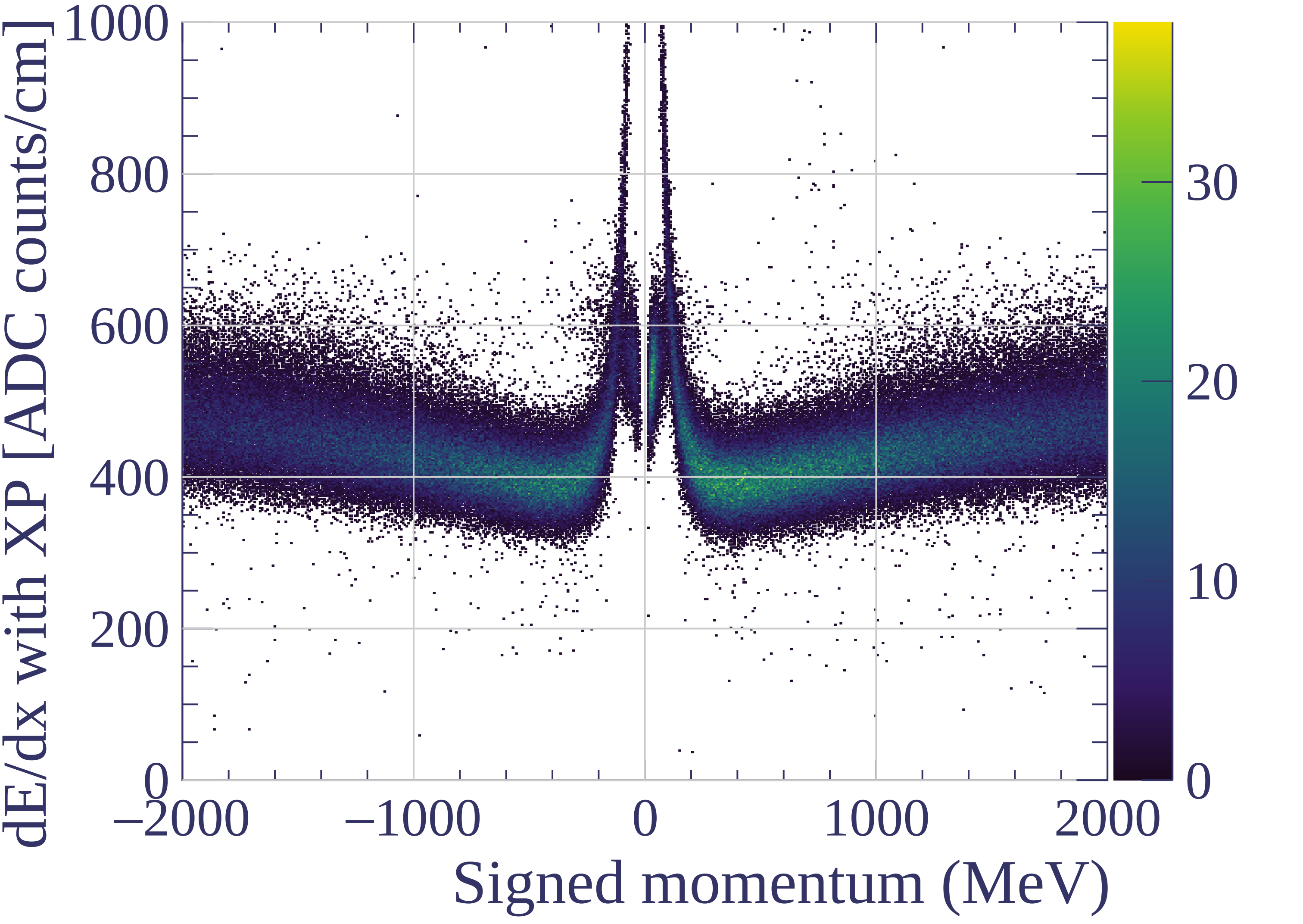


dE/dx resolution (%)
Momentum [MeV/c]
WF
XP
12
10
8
6
4
−2000
−1000
0
1000
2000

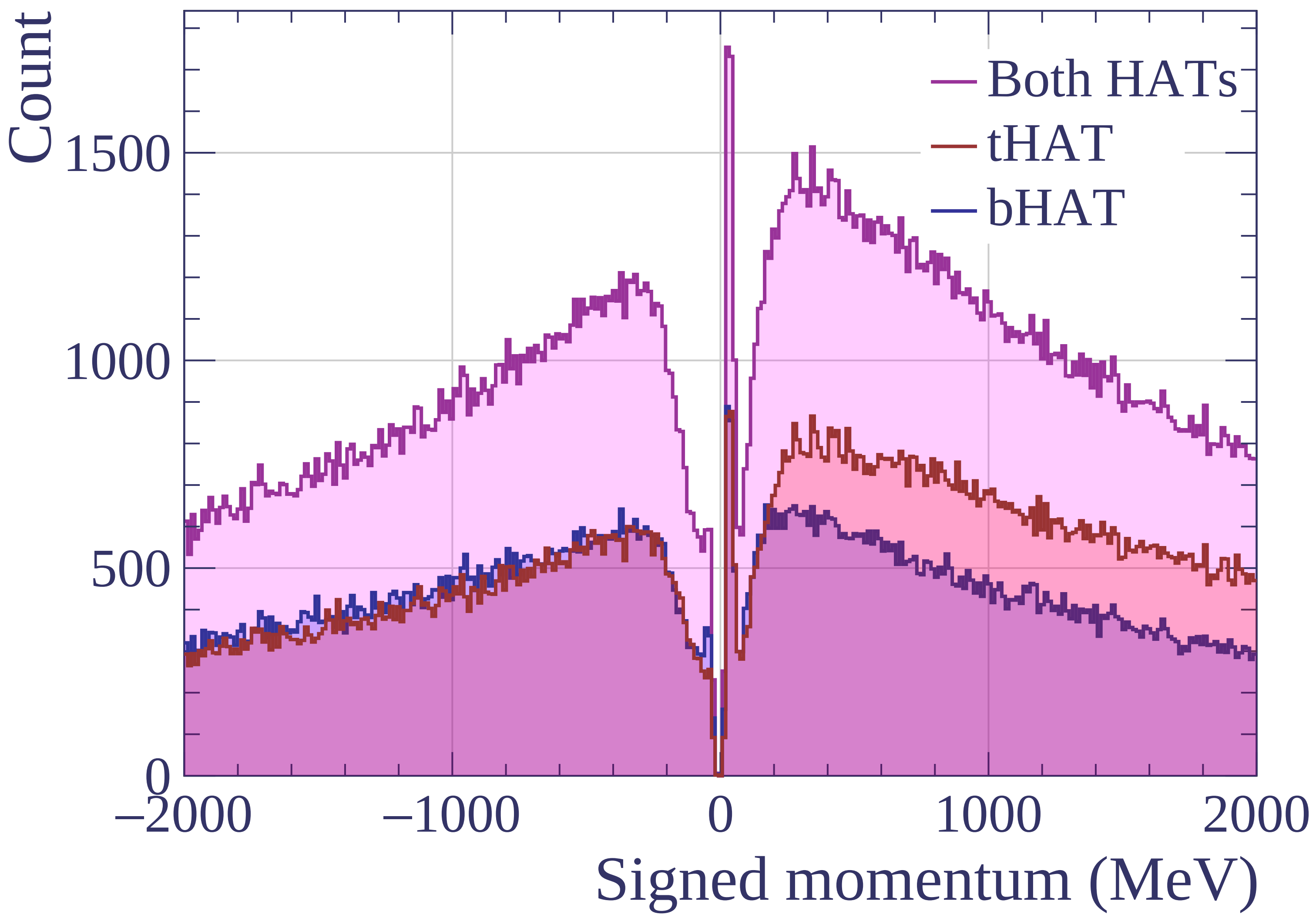

Mean [ADC counts/cm]
Momentum [MeV/c]
WF
XP
600
500
400
300
−2000
−1000
0
1000
2000

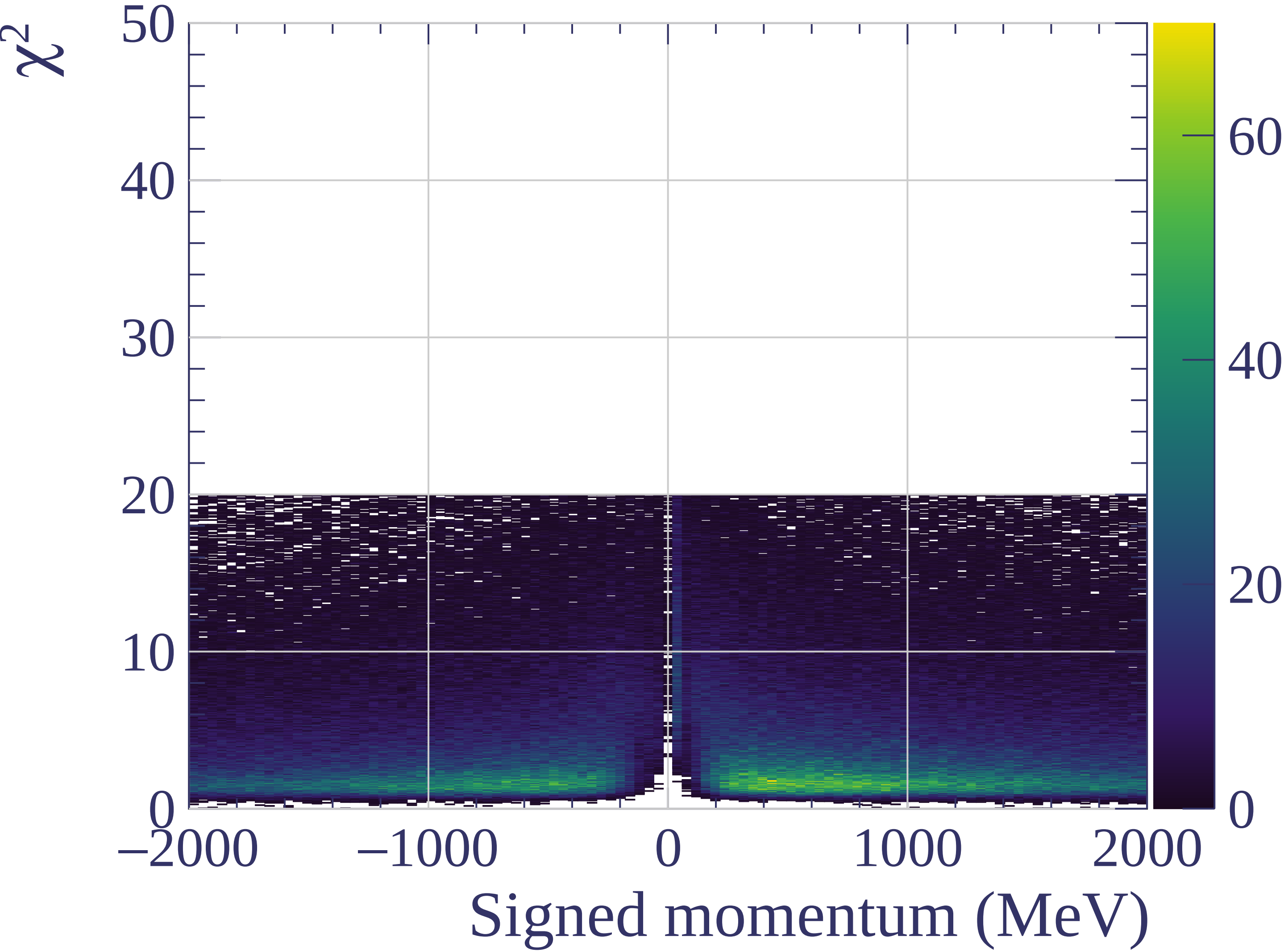


dE/dx with WF [ADC counts/cm]
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
30
20
10
0

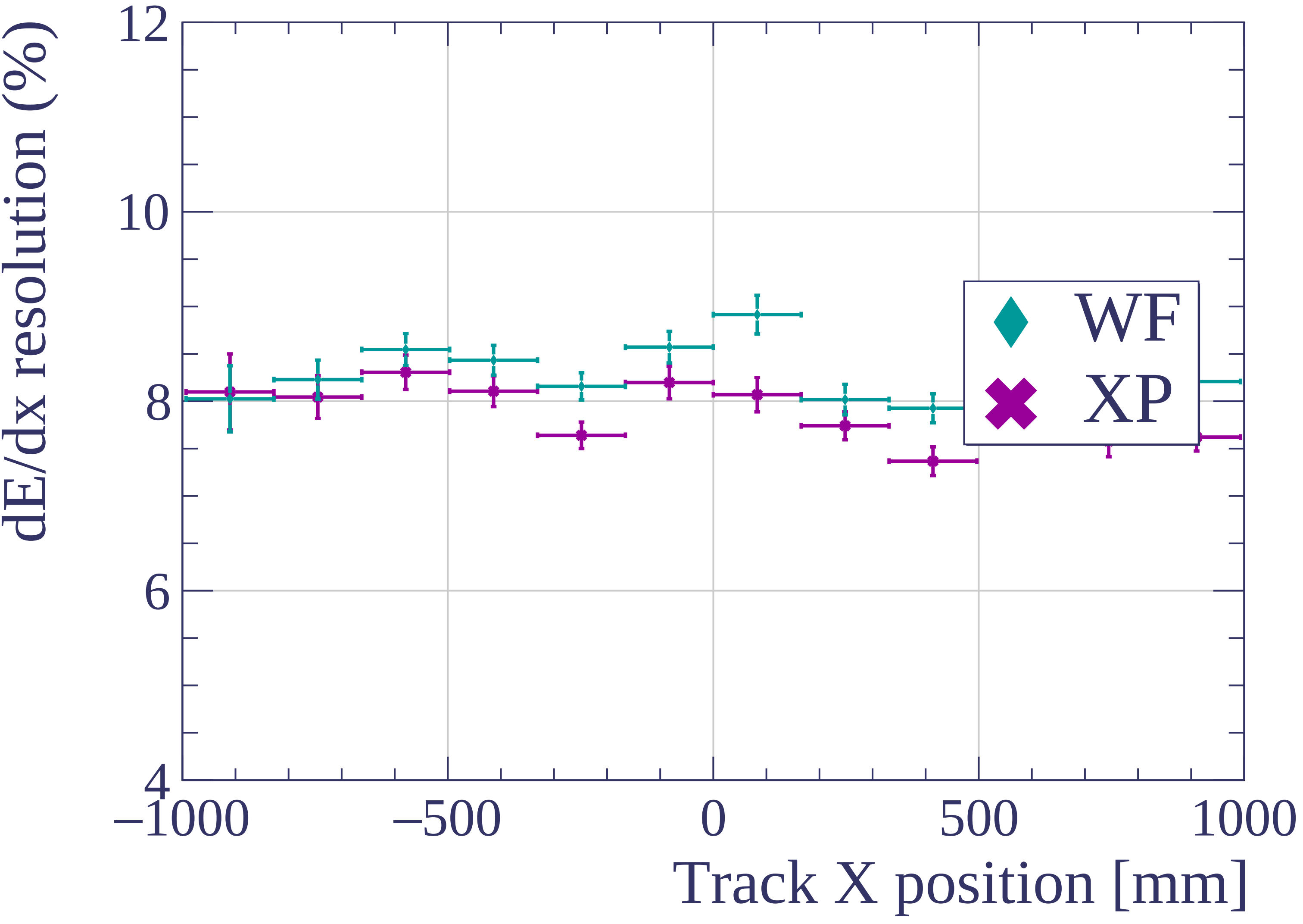


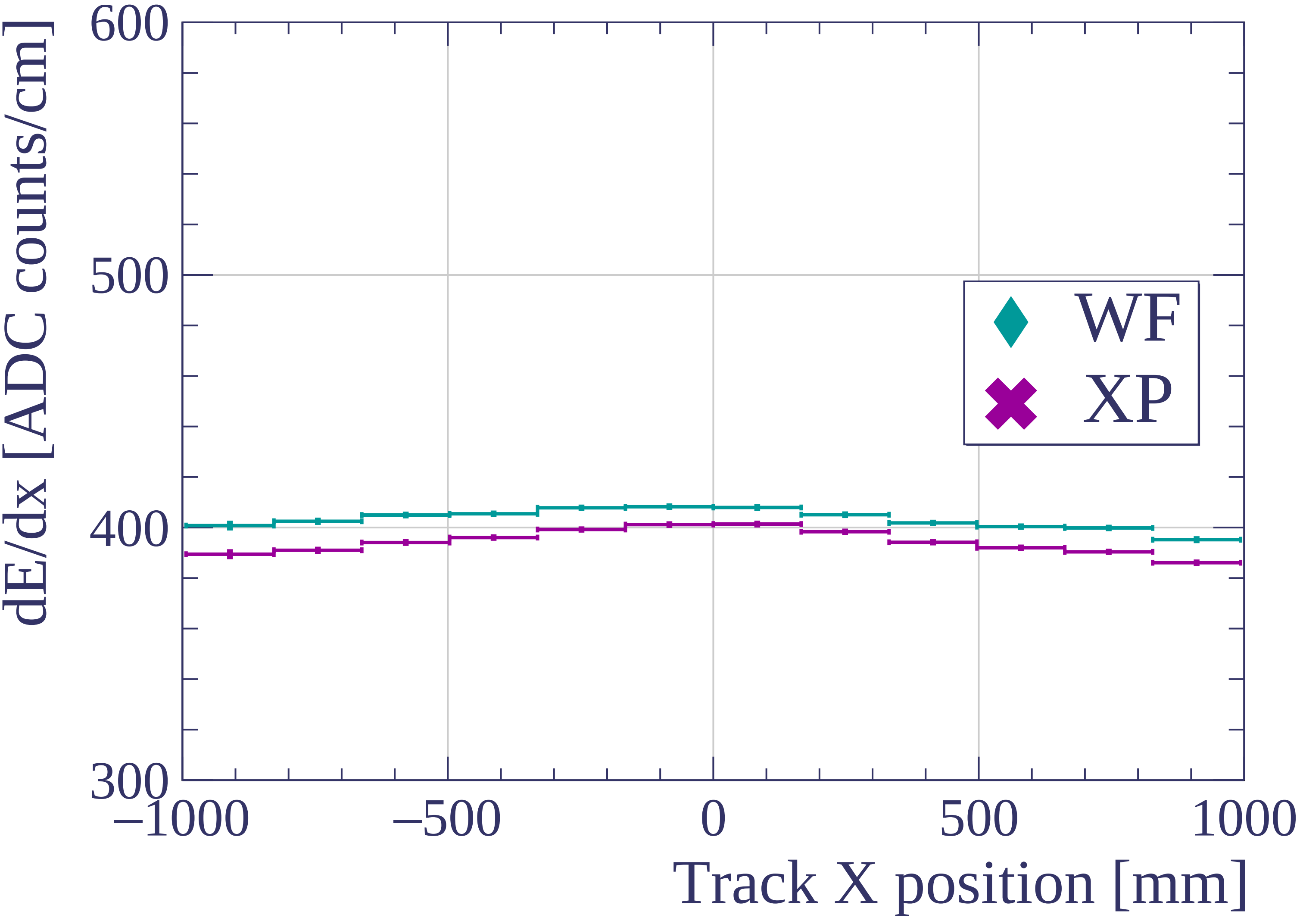
dE/dx with XP [ADC counts/cm]
Signed momentum (MeV)
1000
800
600
400
200
0
−2000
−1000
0
1000
2000
30
20
10
0


Count
1500
1000
500
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
Both HATs
tHAT
bHAT

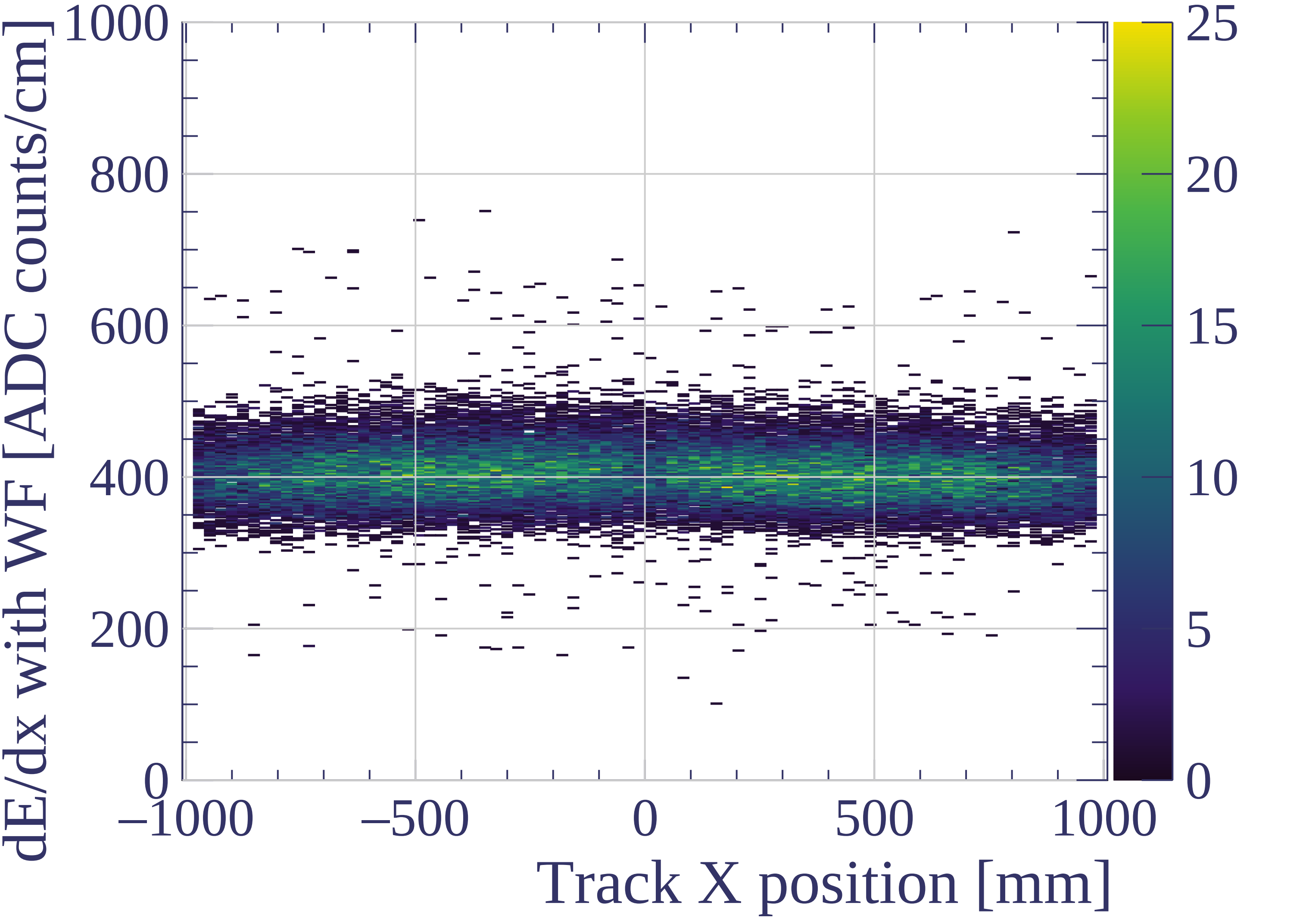


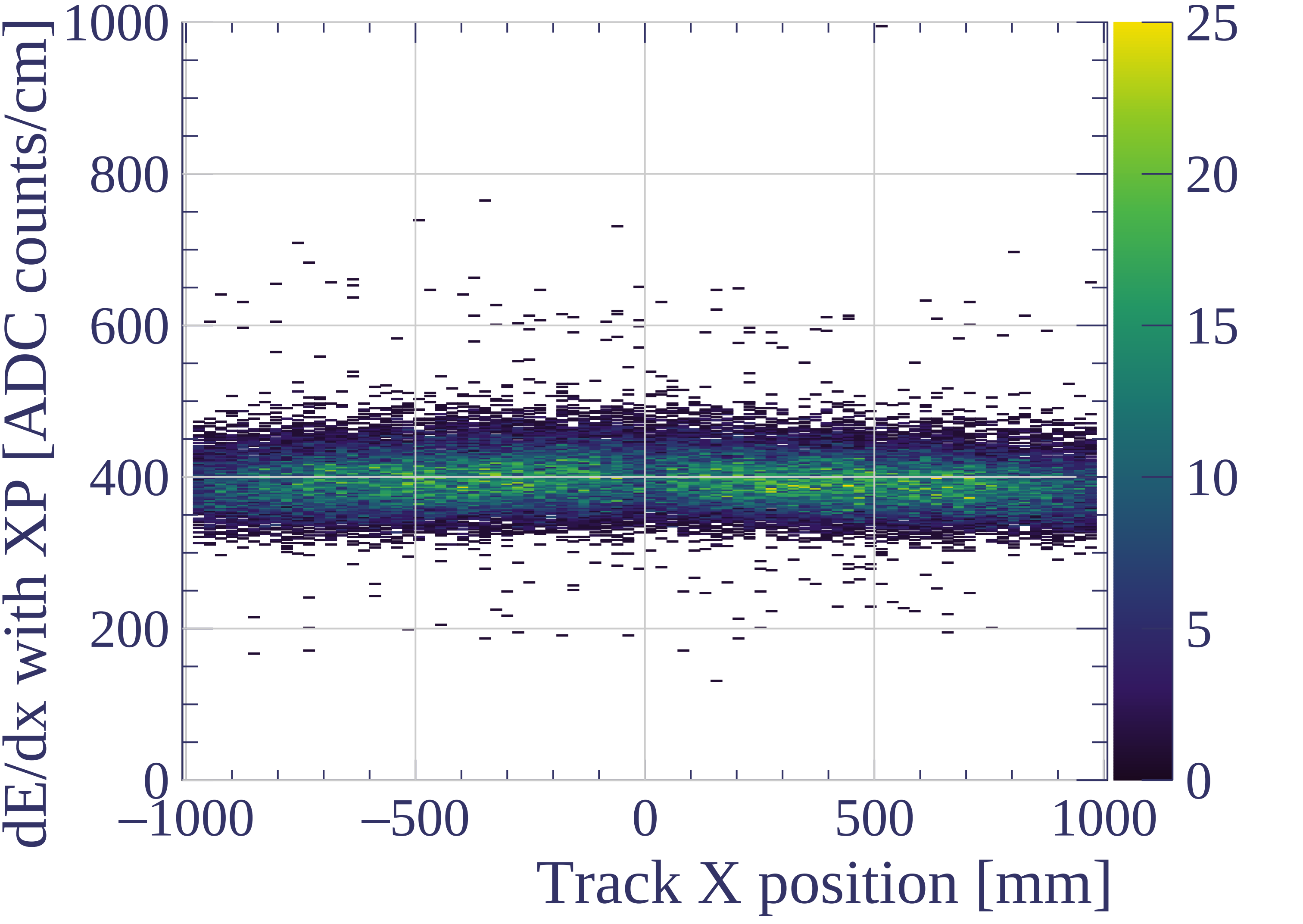
$\chi^2$
50
40
30
20
10
0
−2000
−1000
0
1000
2000
Signed momentum (MeV)
60
40
20
0

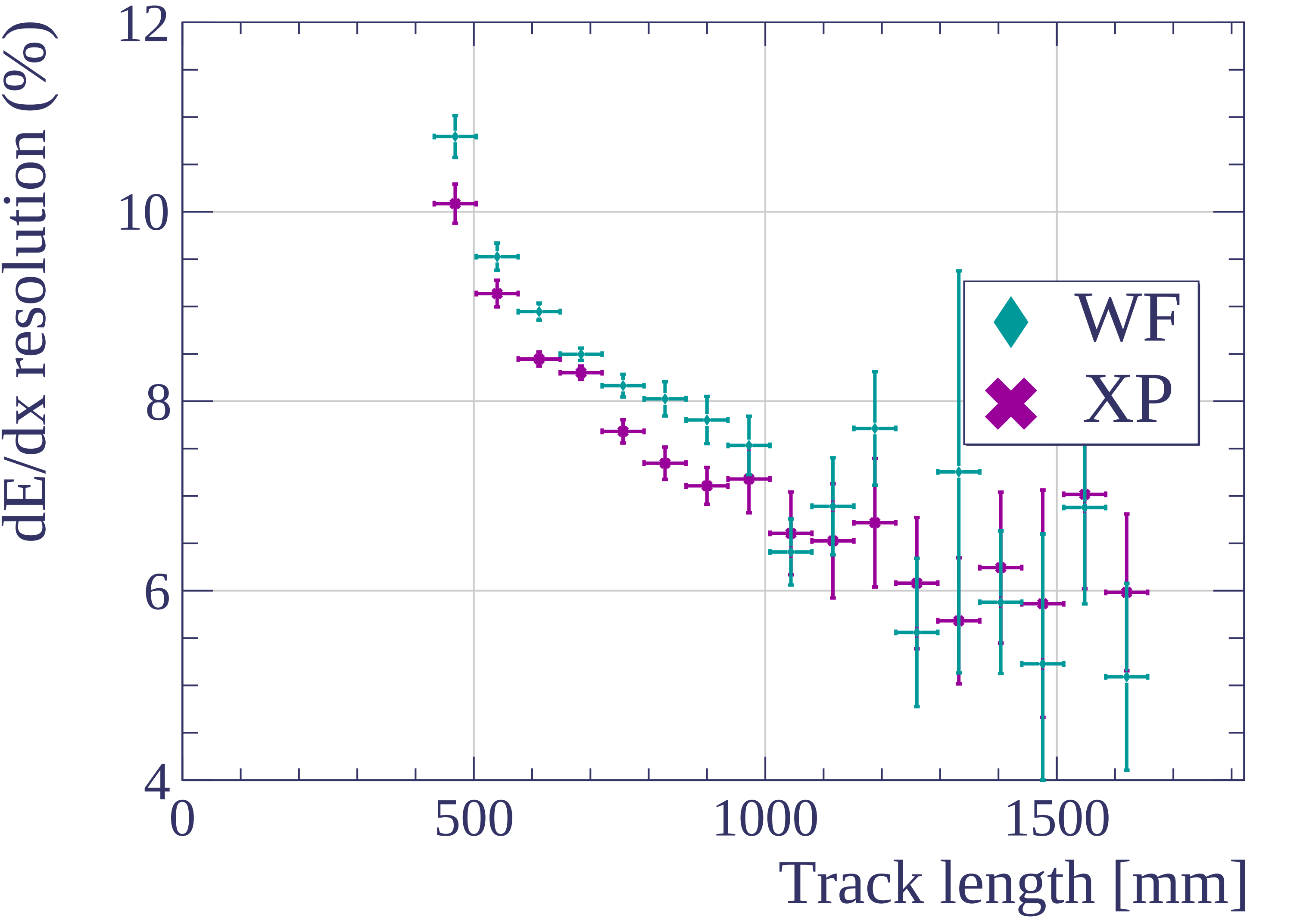

dE/dx resolution (%)
12
10
8
6
4
−1000
−500
0
500
1000
Track X position [mm]
WF
XP



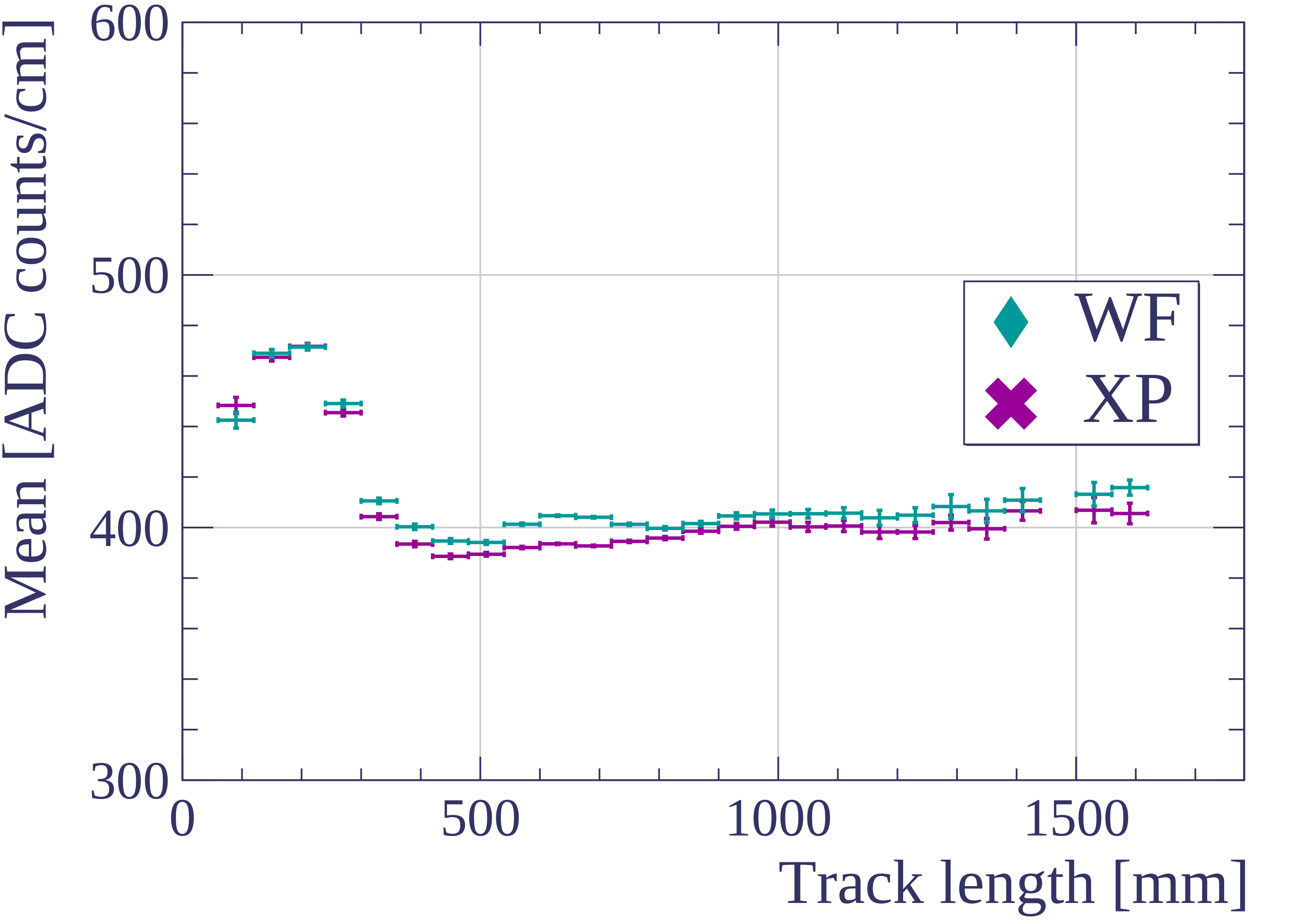
dE/dx [ADC counts/cm]
600
500
400
300
−1000
−500
0
500
1000
Track X position [mm]
WF
XP


dE/dx with WF [ADC counts/cm]
Track X position [mm]
1000
800
600
400
200
0
−1000
−500
0
500
1000
25
20
15
10
5
0

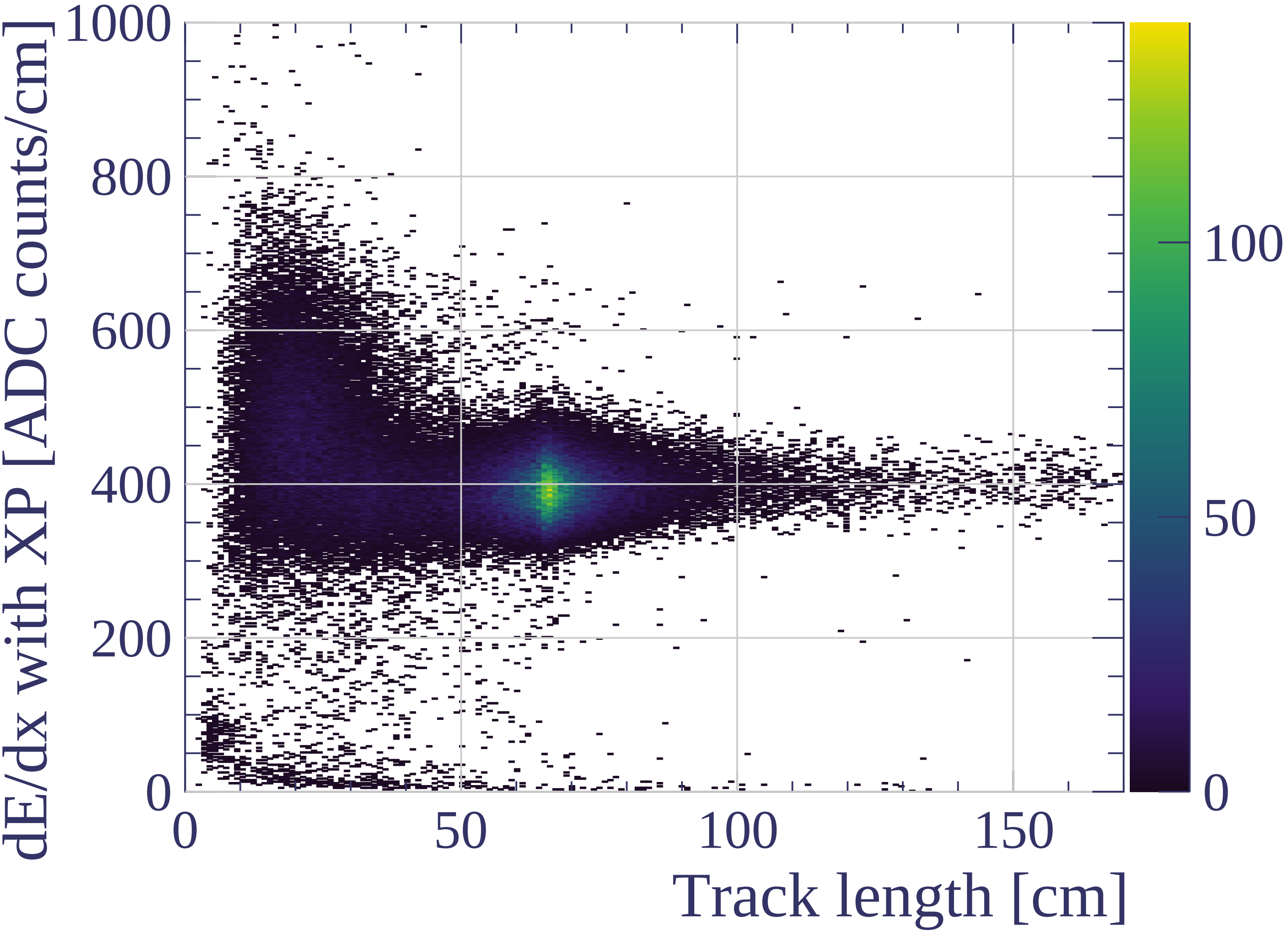

dE/dx with XP [ADC counts/cm]
Track X position [mm]
0
200
400
600
800
1000
−1000
−500
0
500
1000
0
5
10
15
20
25



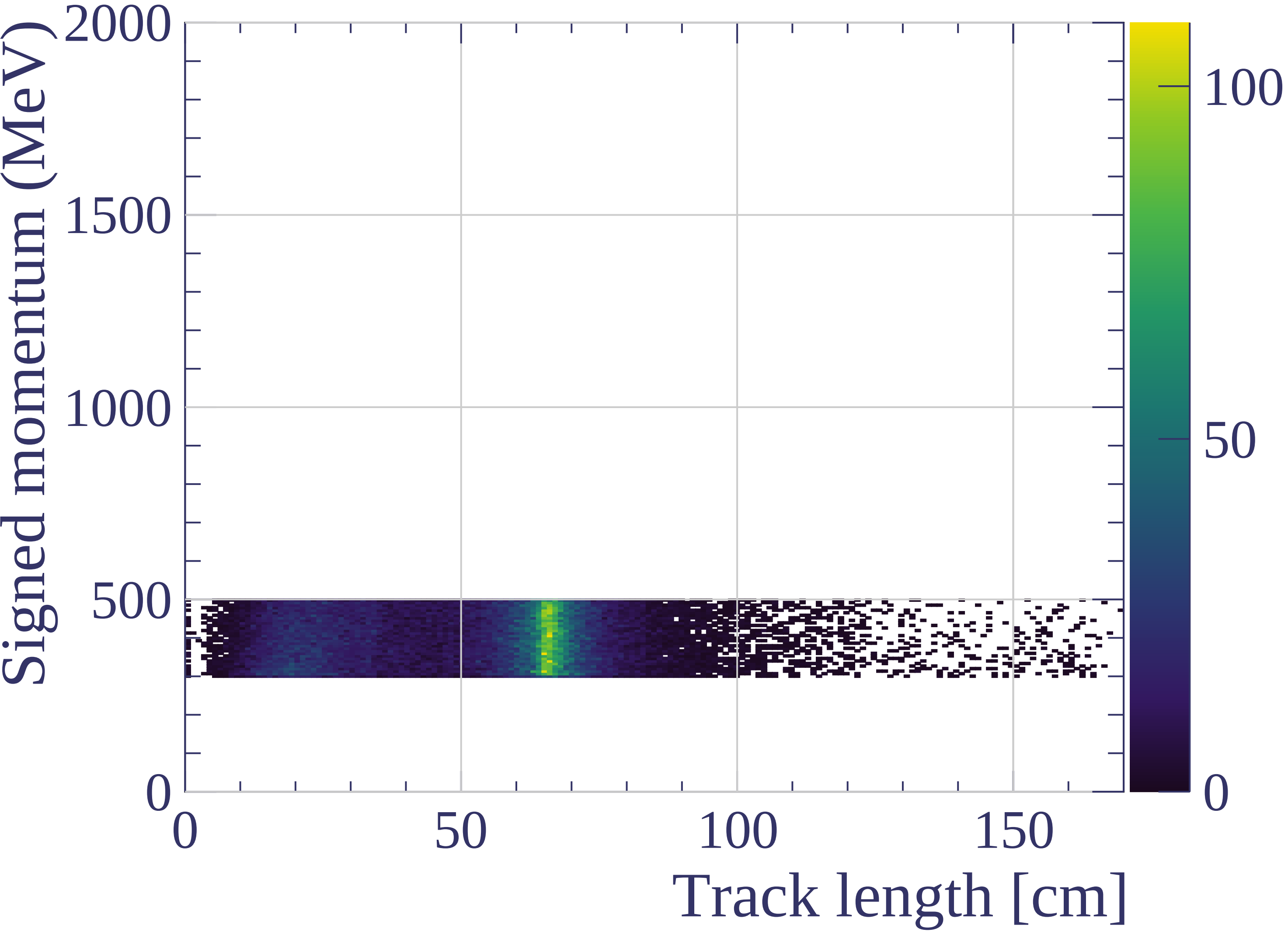
dE/dx resolution (%)
12
10
8
6
4
0
500
1000
1500
Track length [mm]
WF
XP



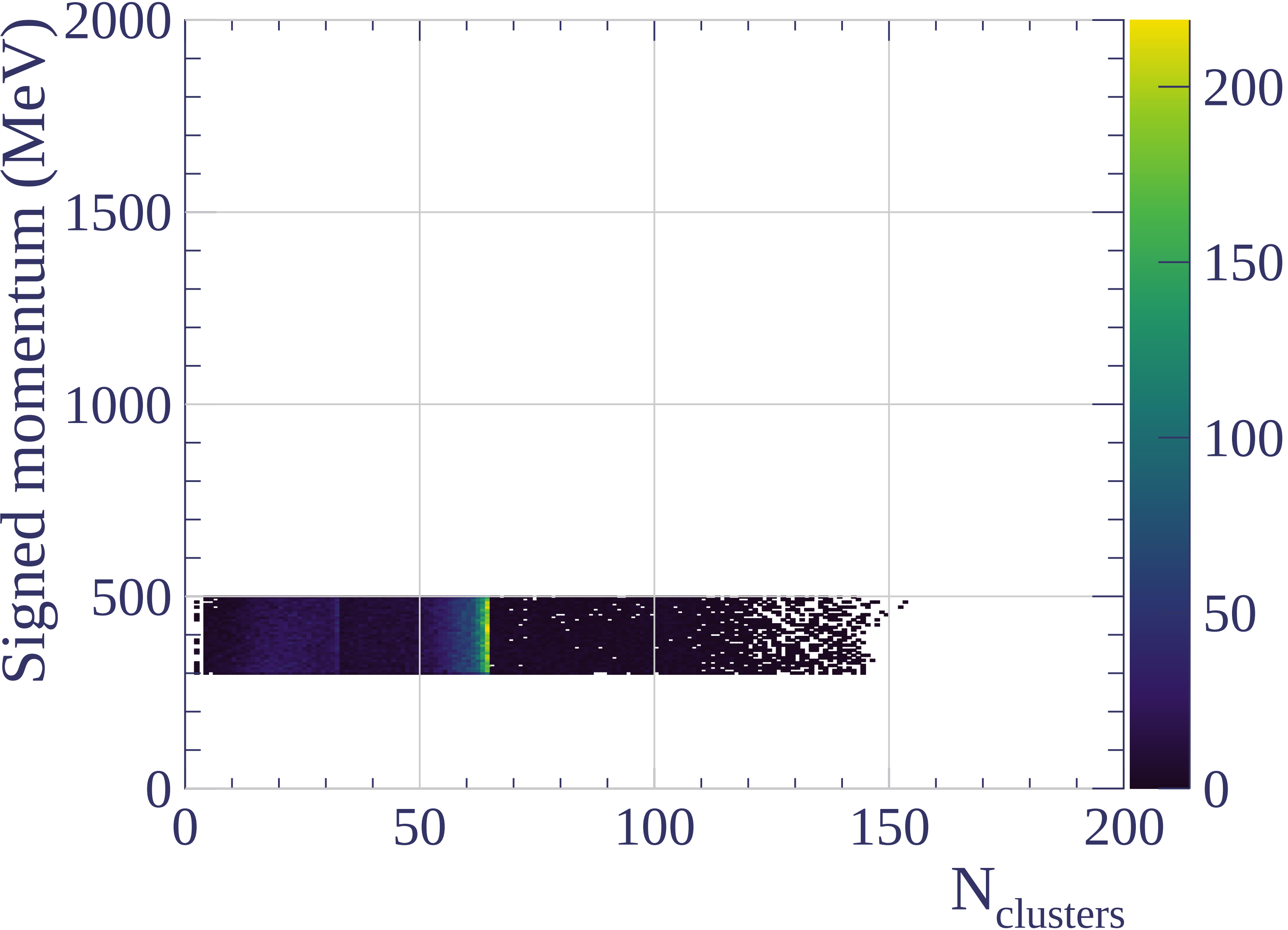
Mean [ADC counts/cm]
600
500
400
300
0
500
1000
1500
Track length [mm]
WF
XP



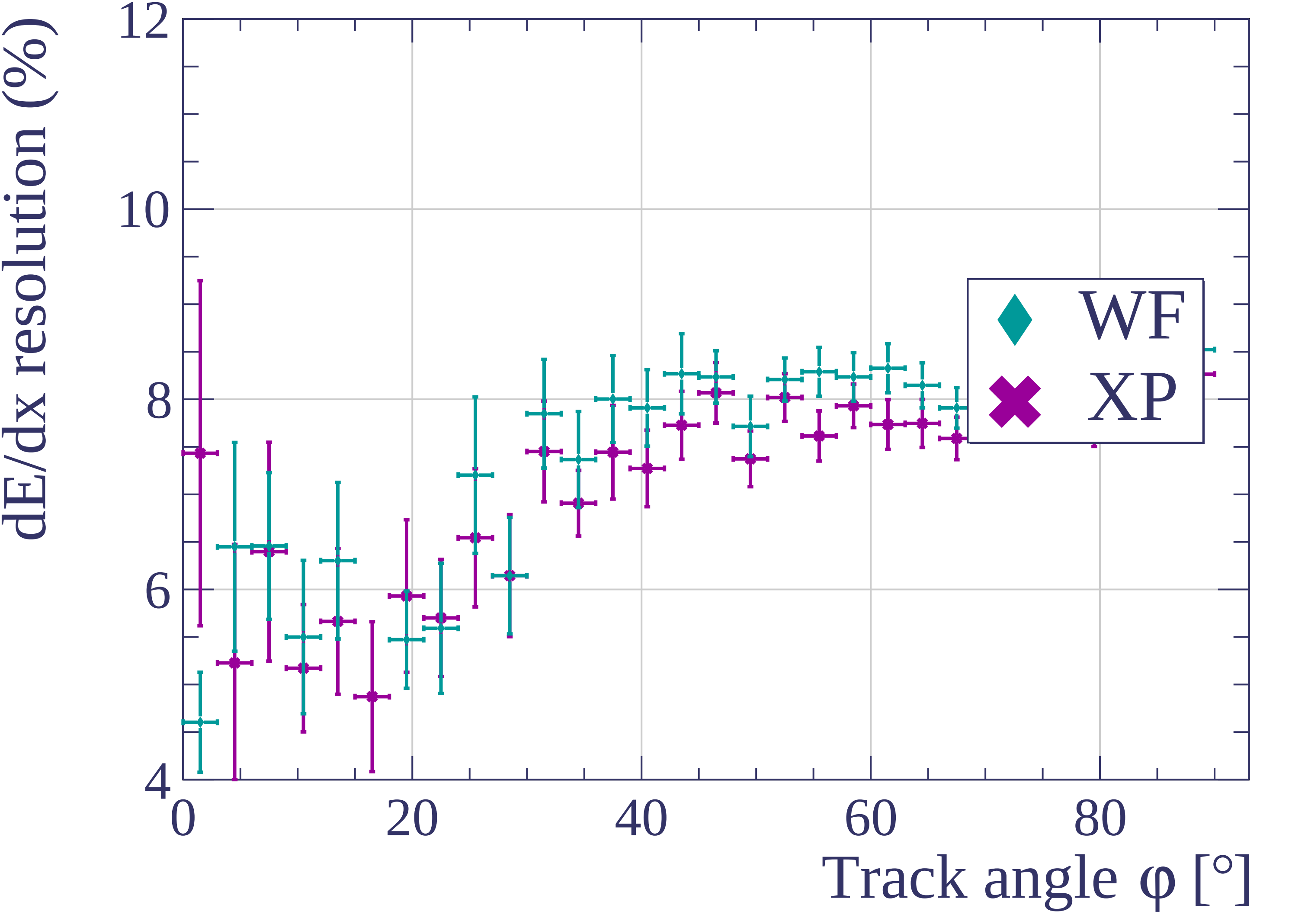
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
0
50
100
150
Track length [cm]
100
50
0



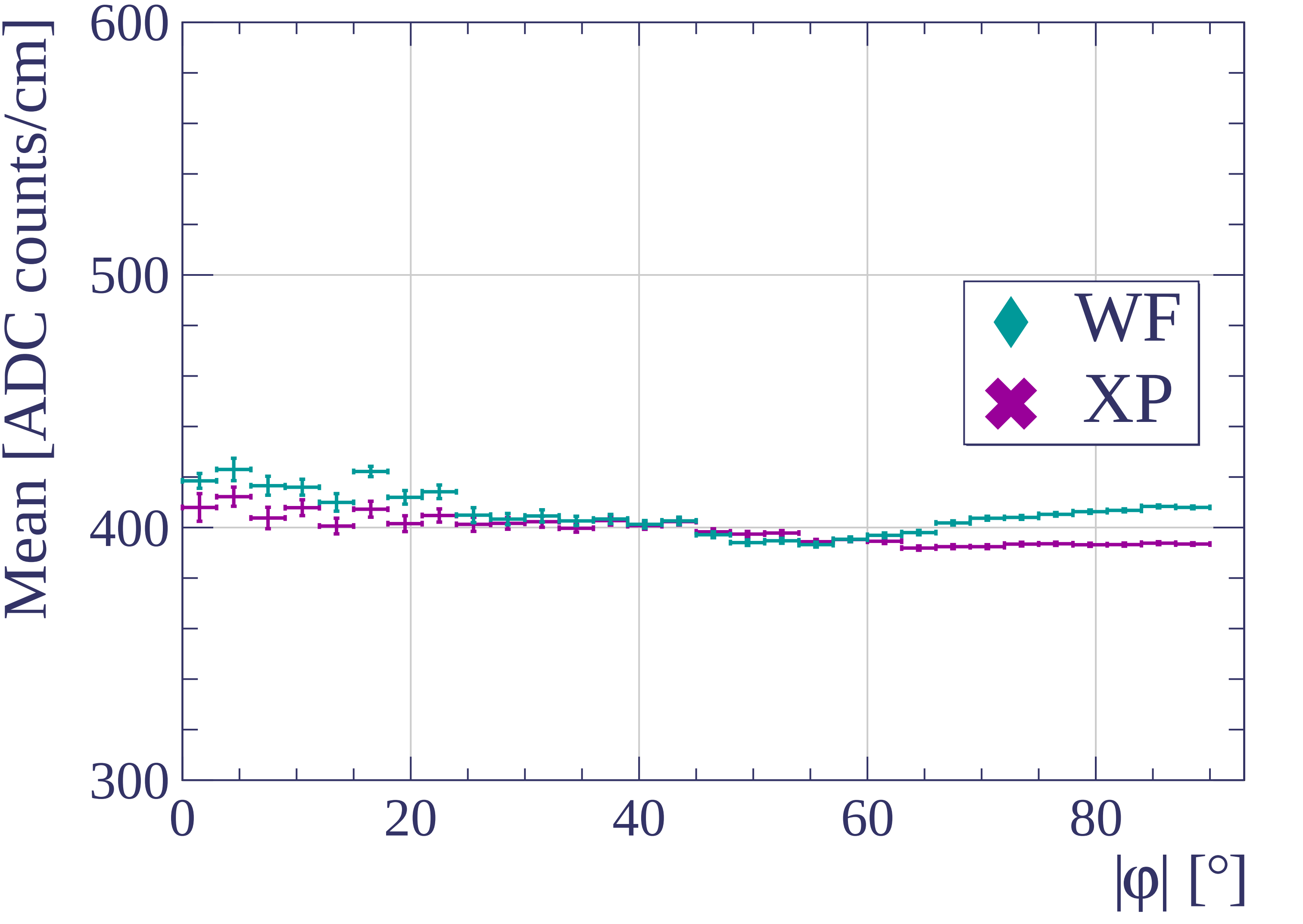
Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
Track length [cm]
100
50
0



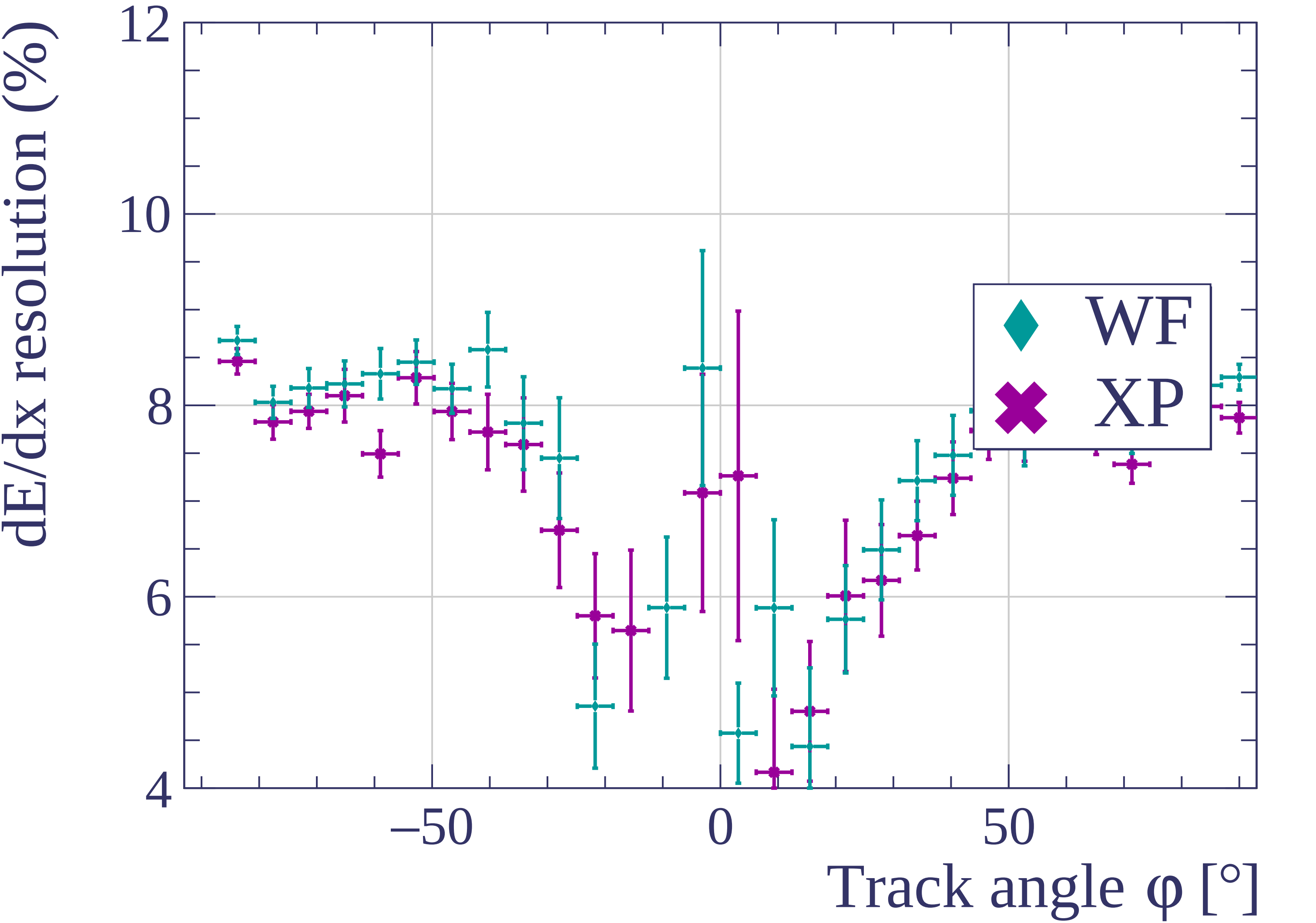
Signed momentum (MeV)
2000
1500
1000
500
0
0
50
100
150
200
N clusters
200
150
100
50
0



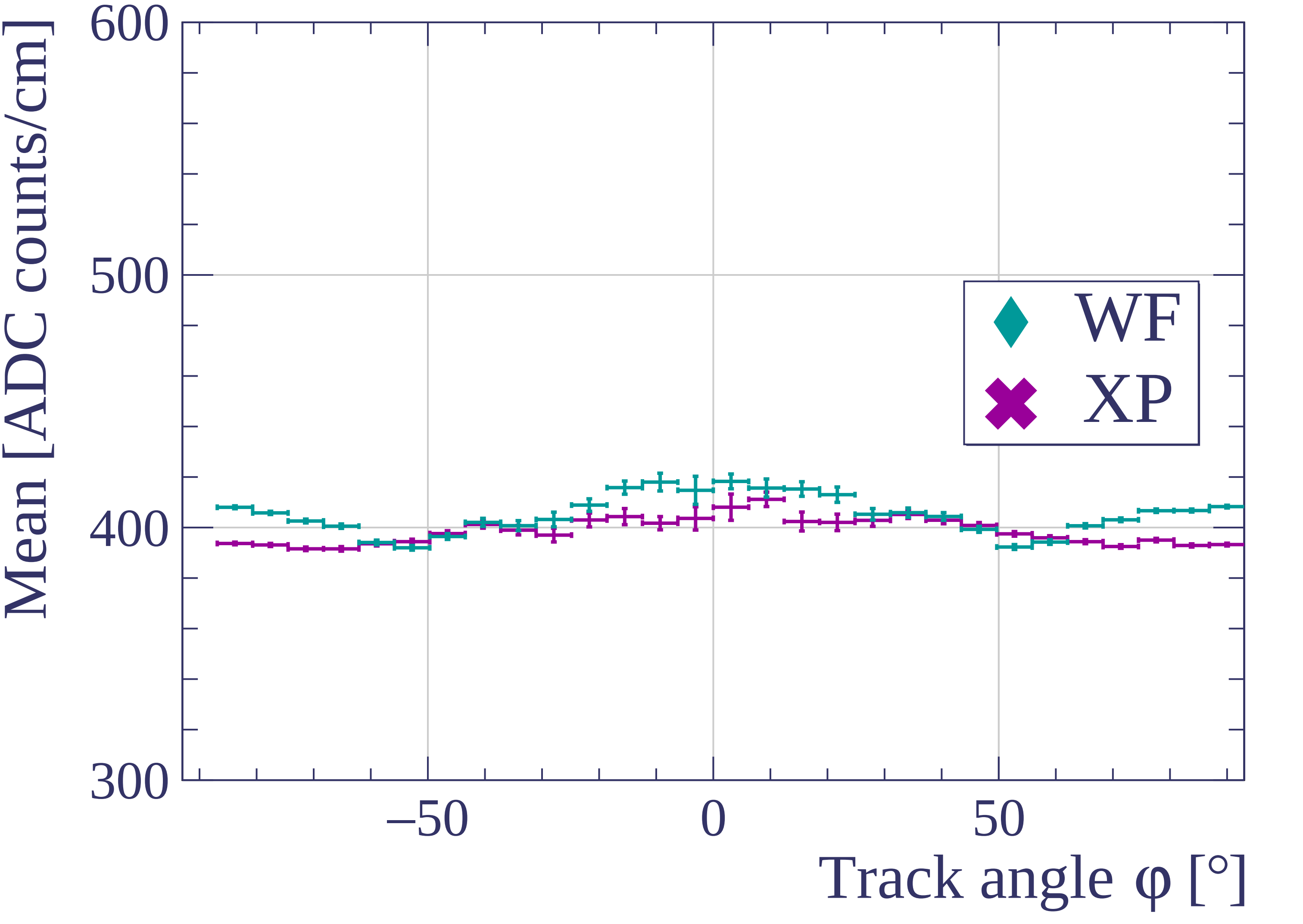
dE/dx resolution (%)
12
10
8
6
4
0
20
40
60
80
Track angle φ [°]
WF
XP



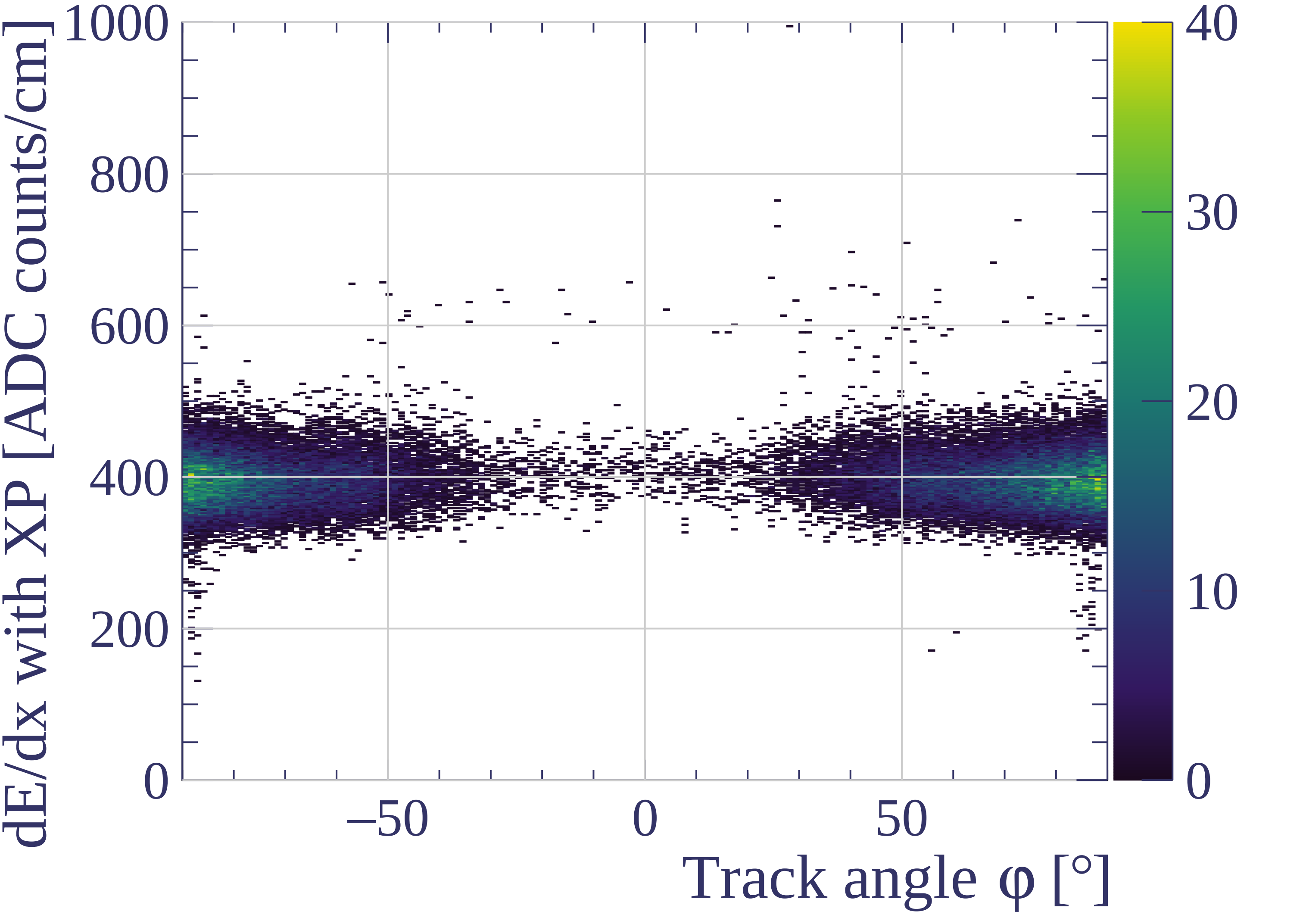
Mean [ADC counts/cm]
600
500
400
300
0
20
40
60
80
|φ| [°]
WF
XP



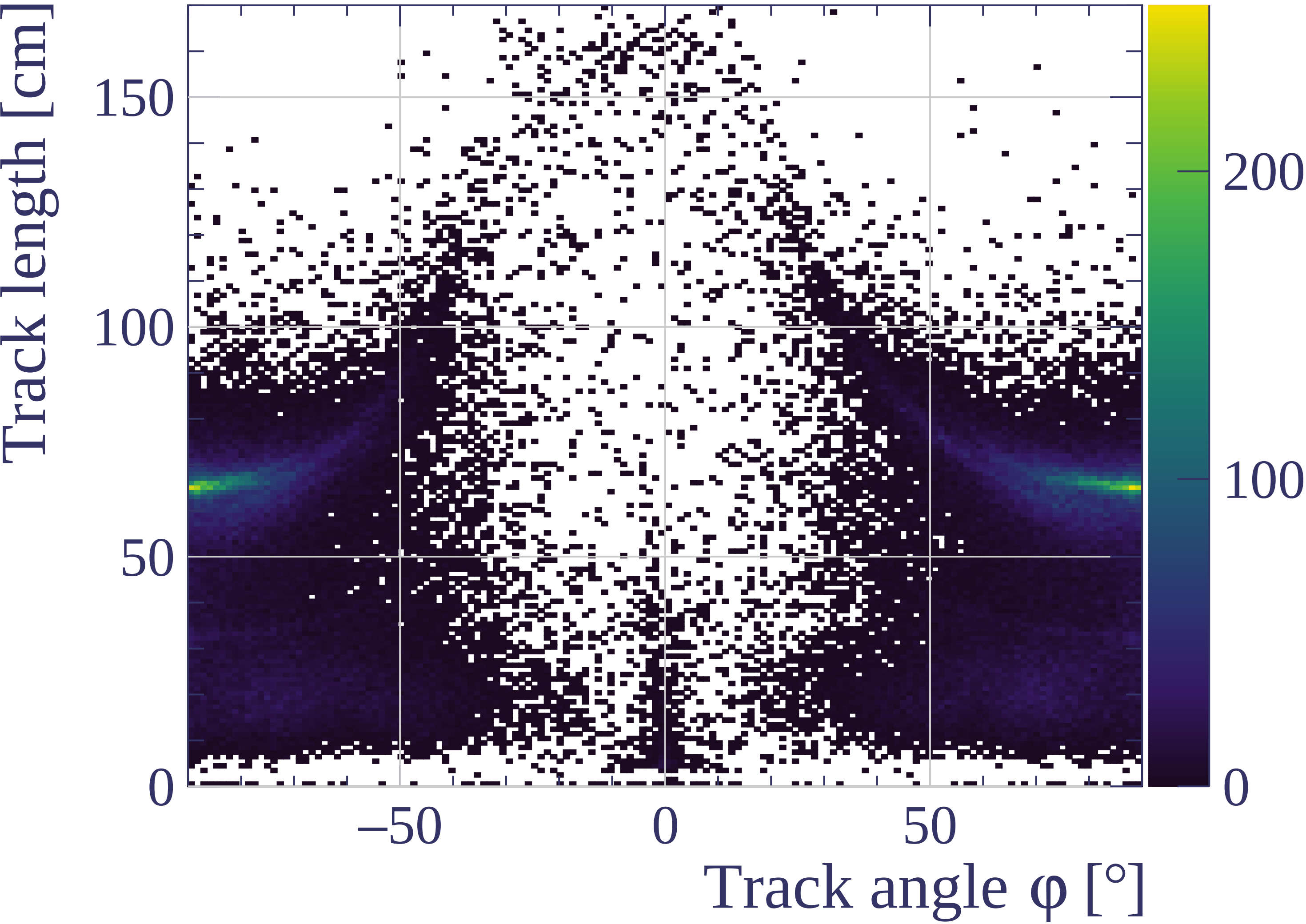
12
10
8
6
4
−50
0
50
dE/dx resolution (%)
Track angle φ [°]
WF
XP



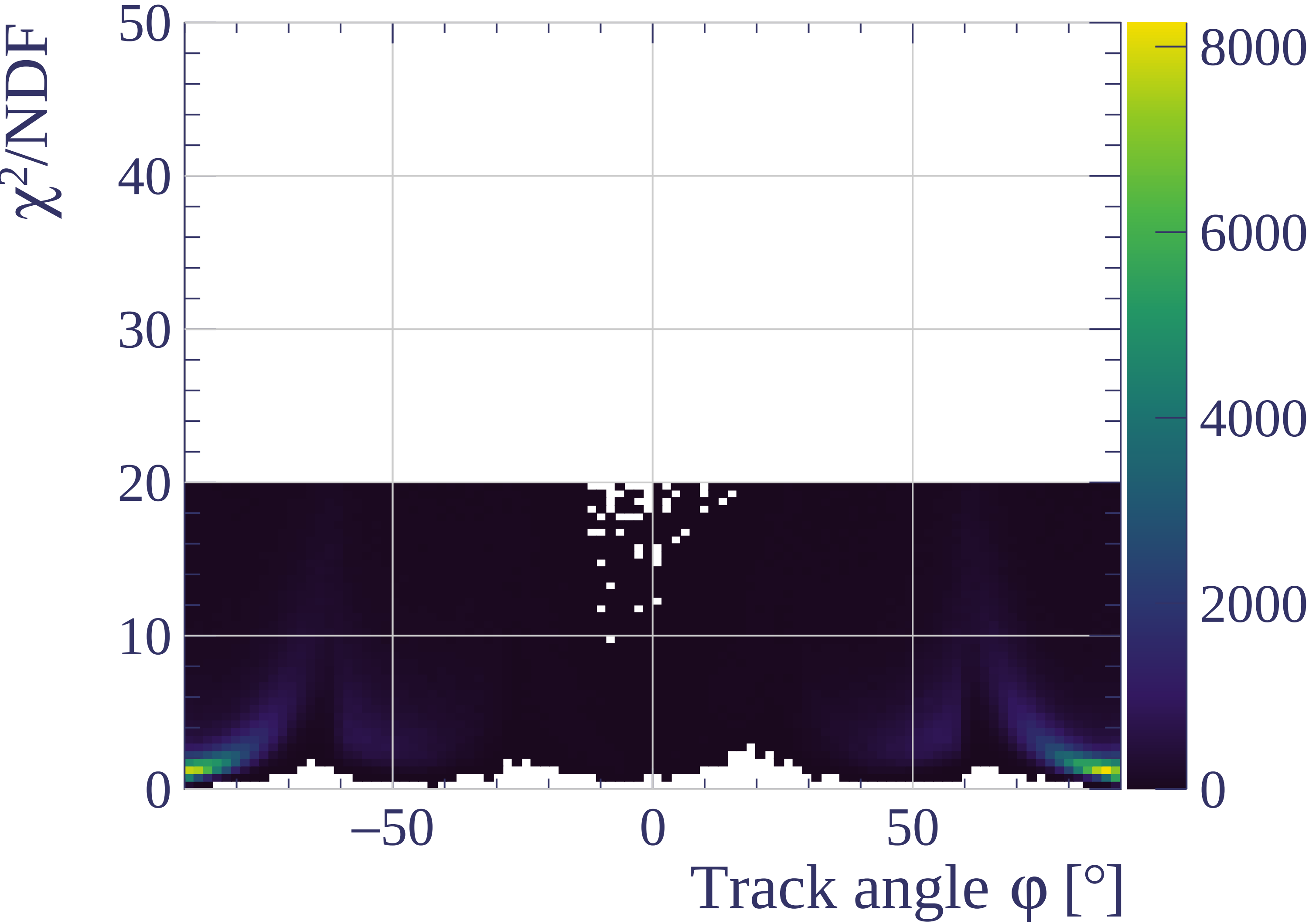
Mean [ADC counts/cm]
600
500
400
300
−50
0
50
Track angle φ [°]
WF
XP



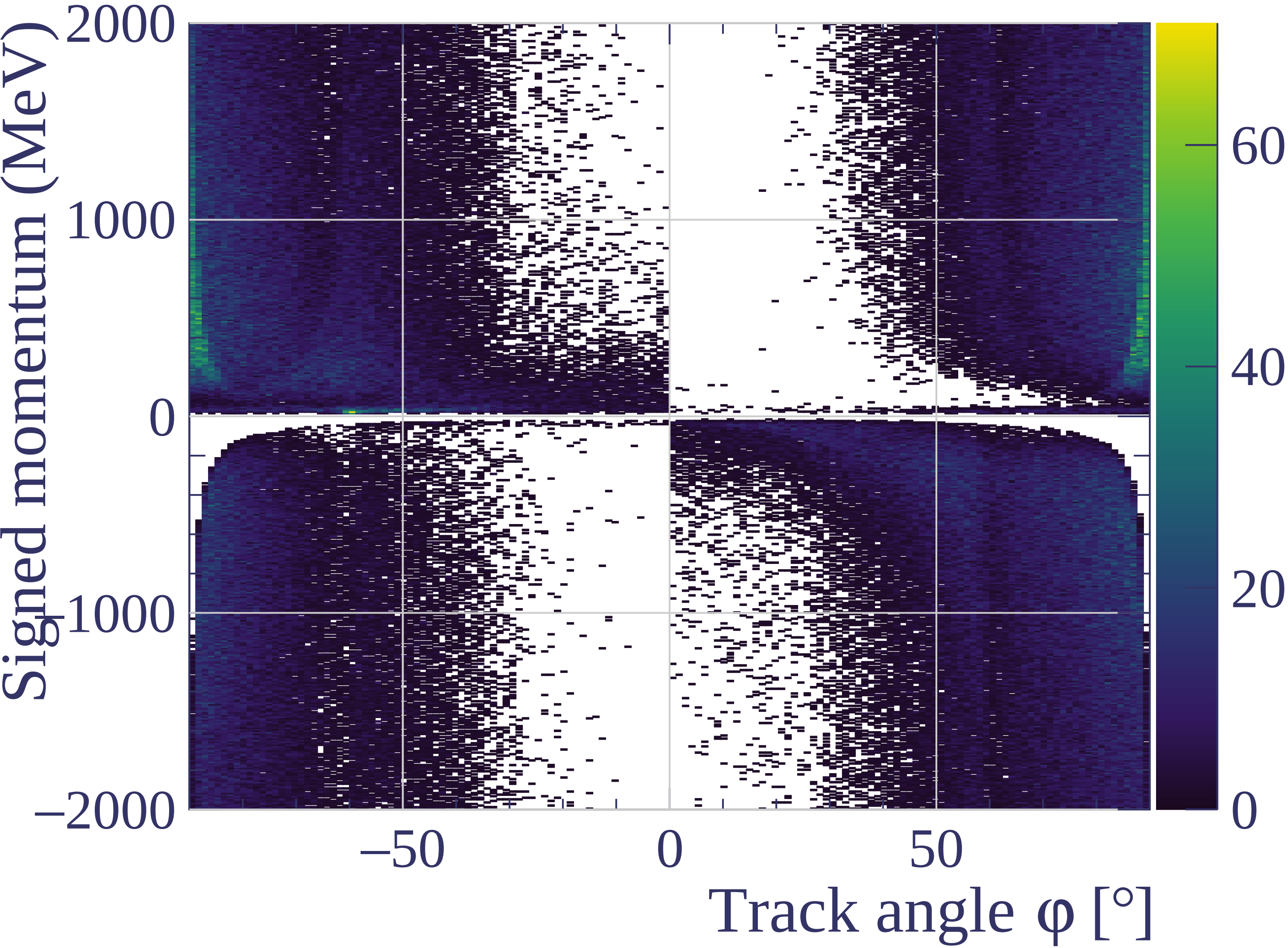
dE/dx with XP [ADC counts/cm]
1000
800
600
400
200
0
−50
0
50
Track angle φ [°]
40
30
20
10
0



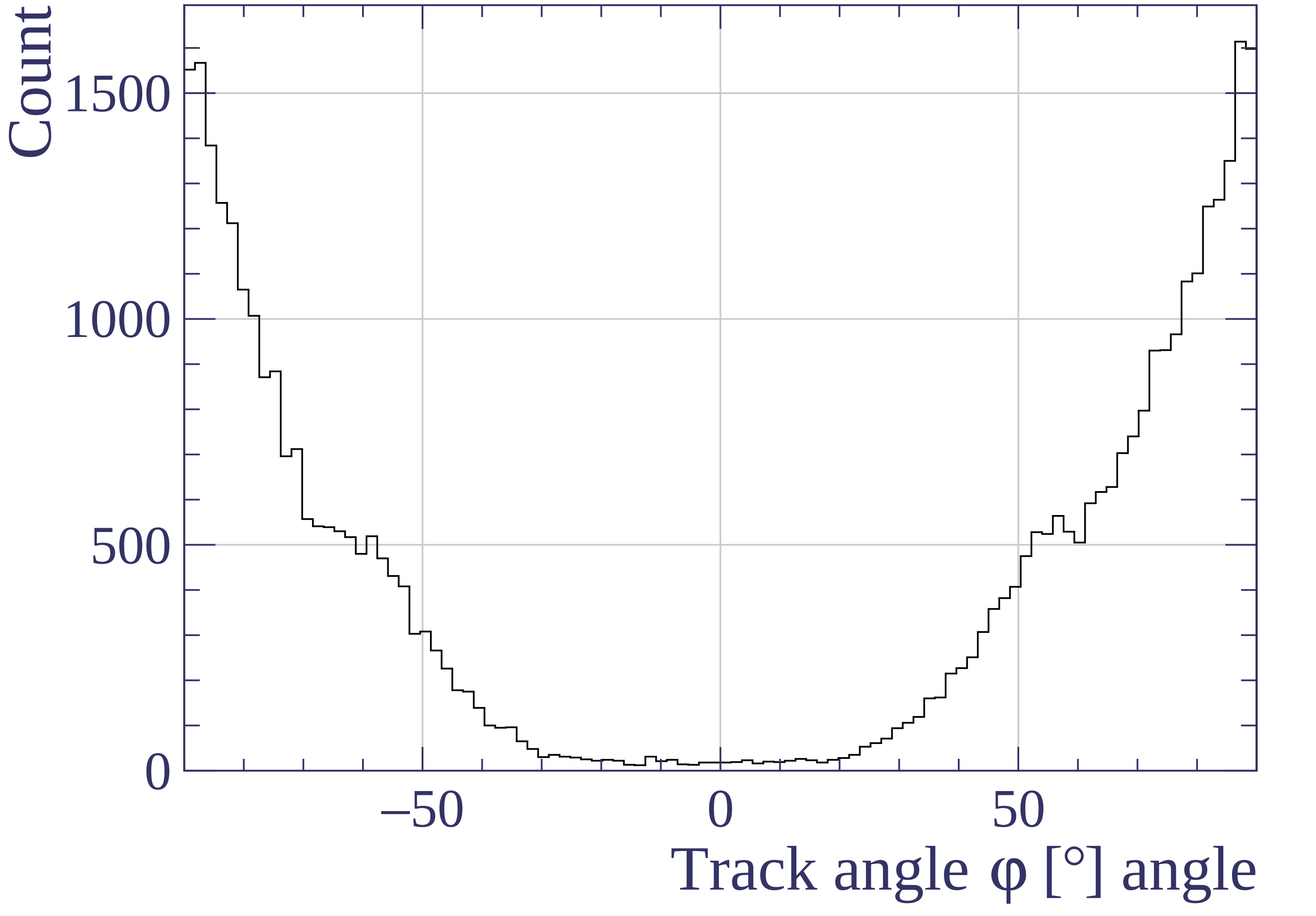
Track length [cm]
0
50
100
150
−50
0
50
Track angle φ [°]
0
100
200

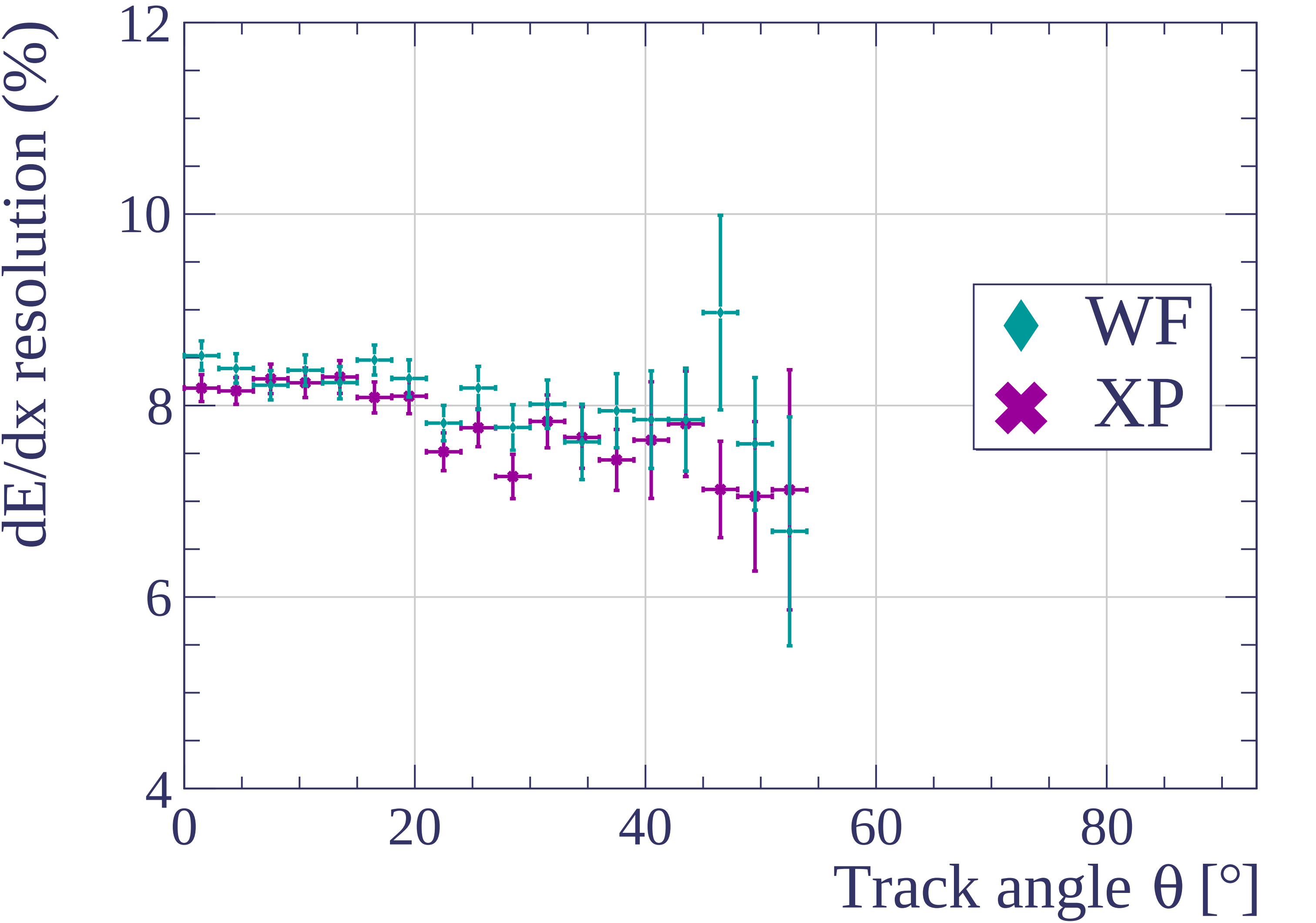


$\chi^2$/NDF
50
40
30
20
10
0
−50
0
50
Track angle φ [°]
8000
6000
4000
2000
0

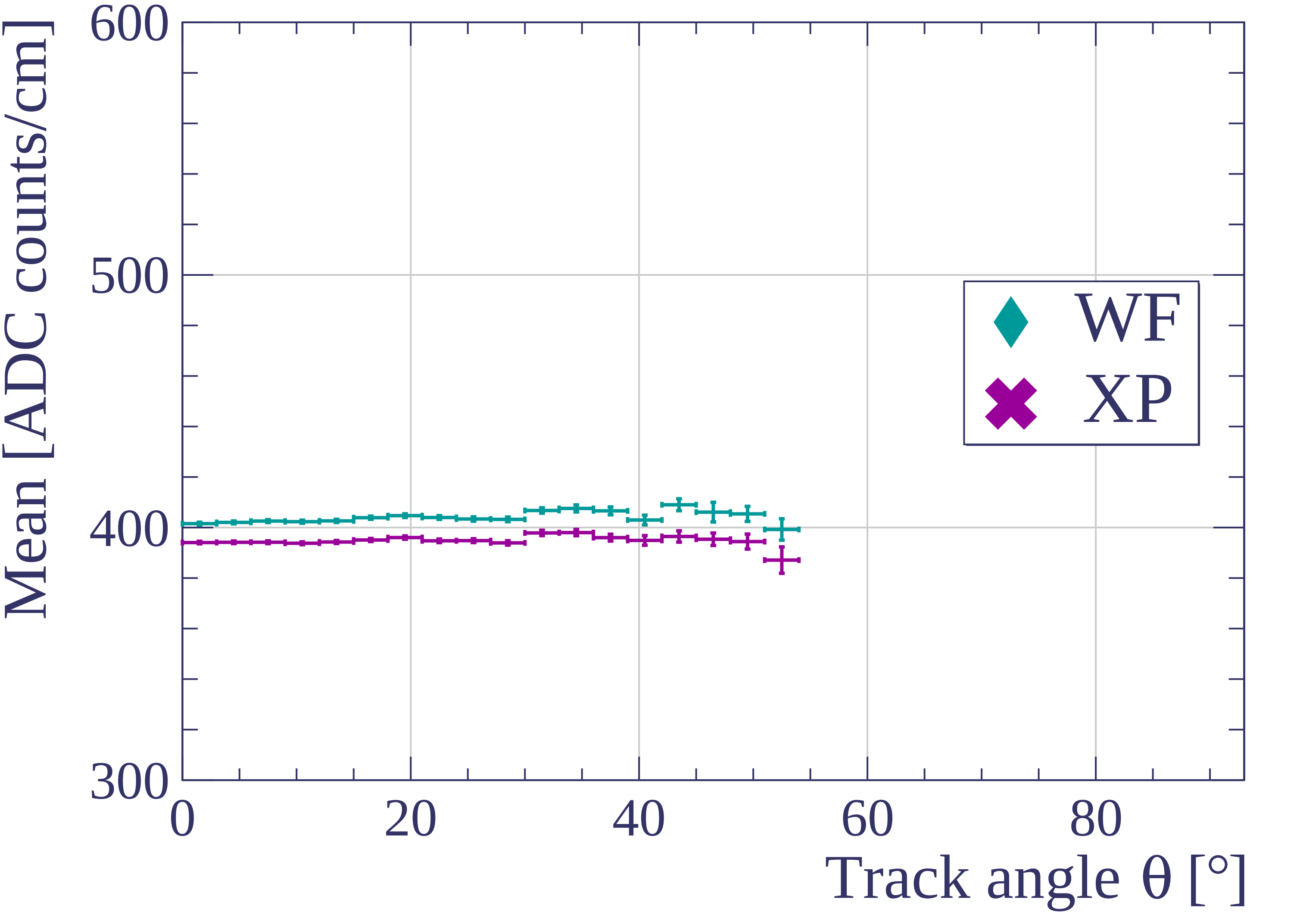


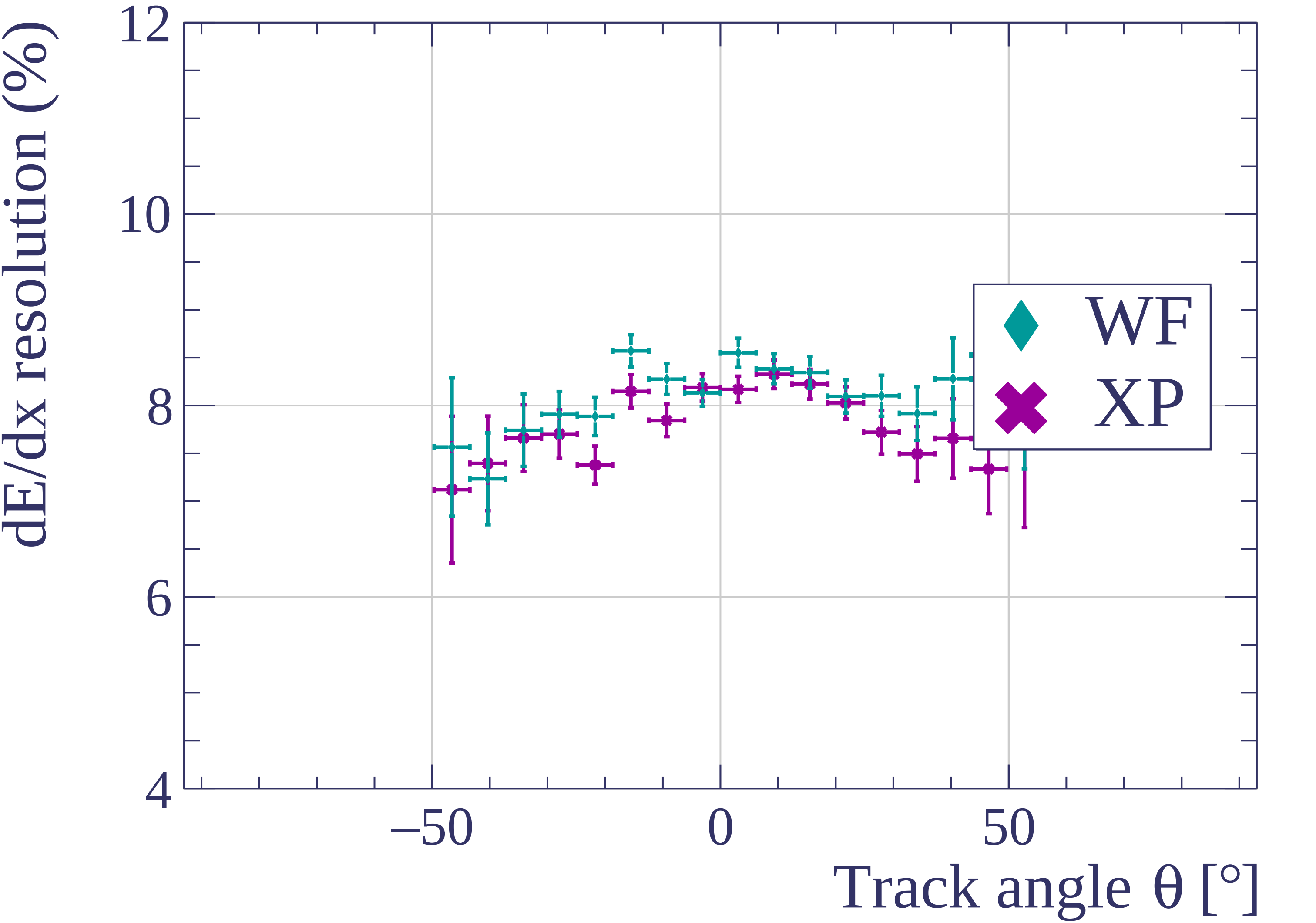
2000
1000
0
−1000
−2000
Signed momentum (MeV)
−50
0
50
Track angle φ [°]
60
40
20
0

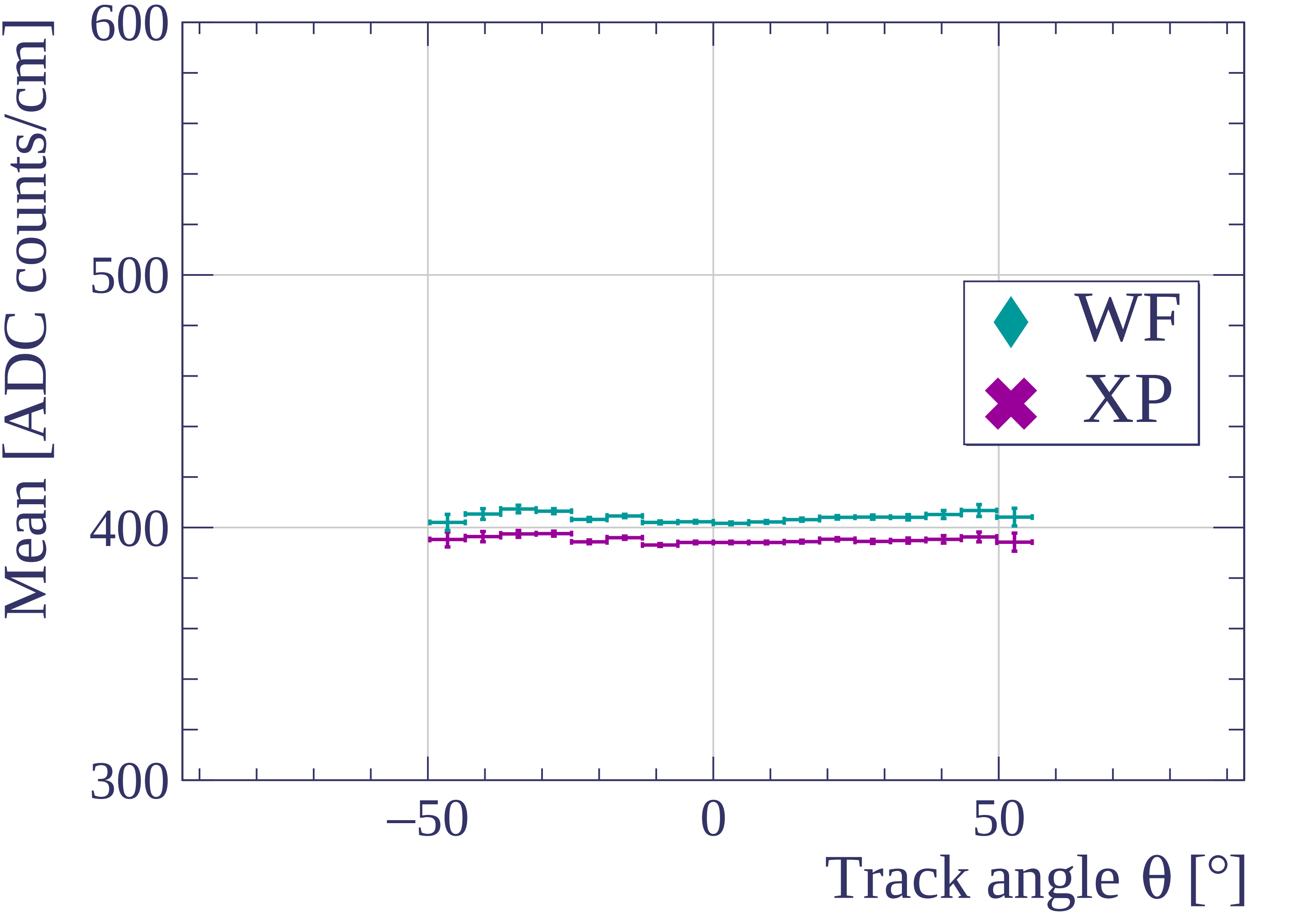

Count
1500
1000
500
0
−50
0
50
Track angle φ [°] angle



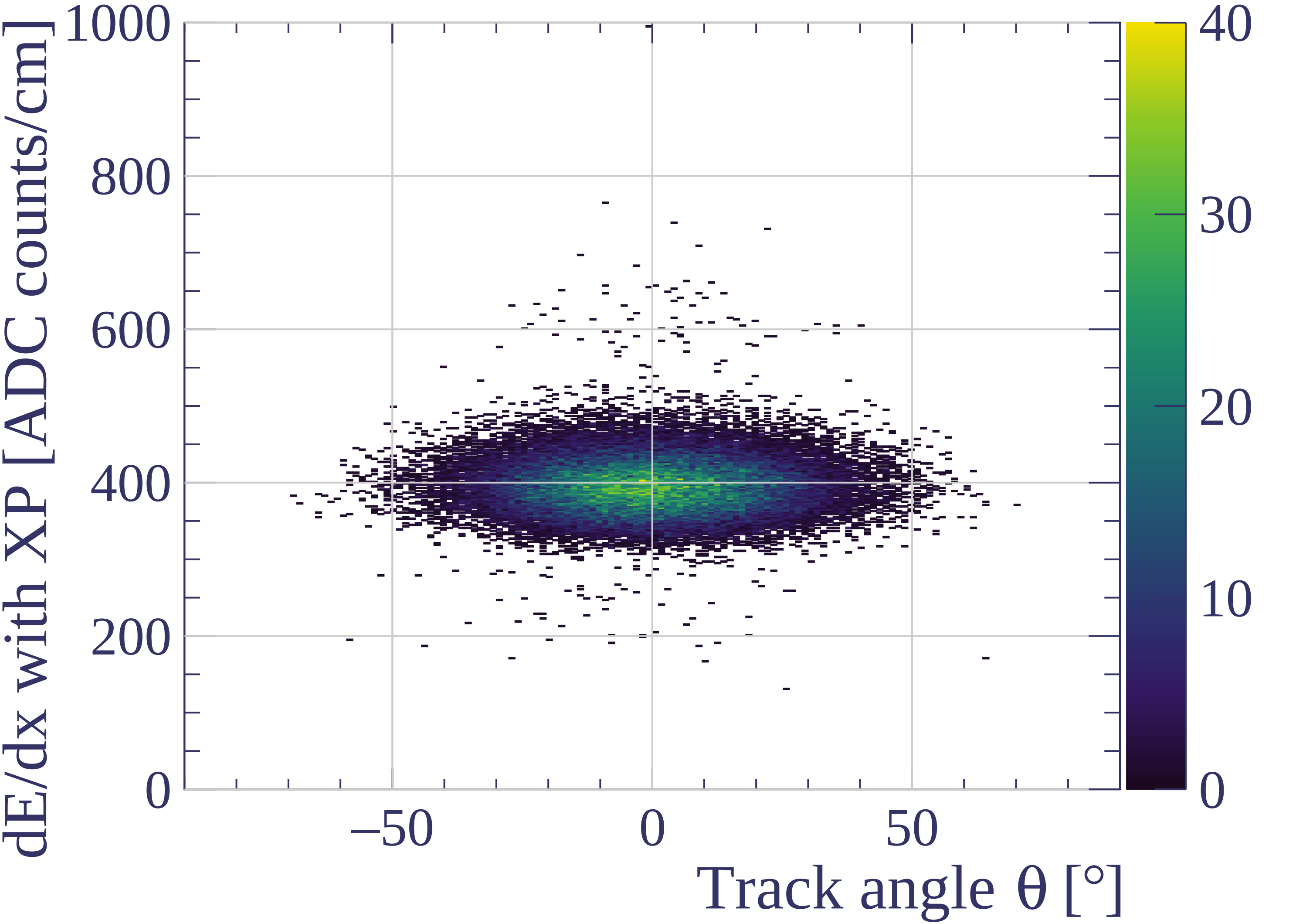
dE/dx resolution (%)
12
10
8
6
4
0
20
40
60
80
Track angle θ [°]
WF
XP


Mean [ADC counts/cm]
600
500
400
300
0
20
40
60
80
Track angle θ [°]
WF
XP

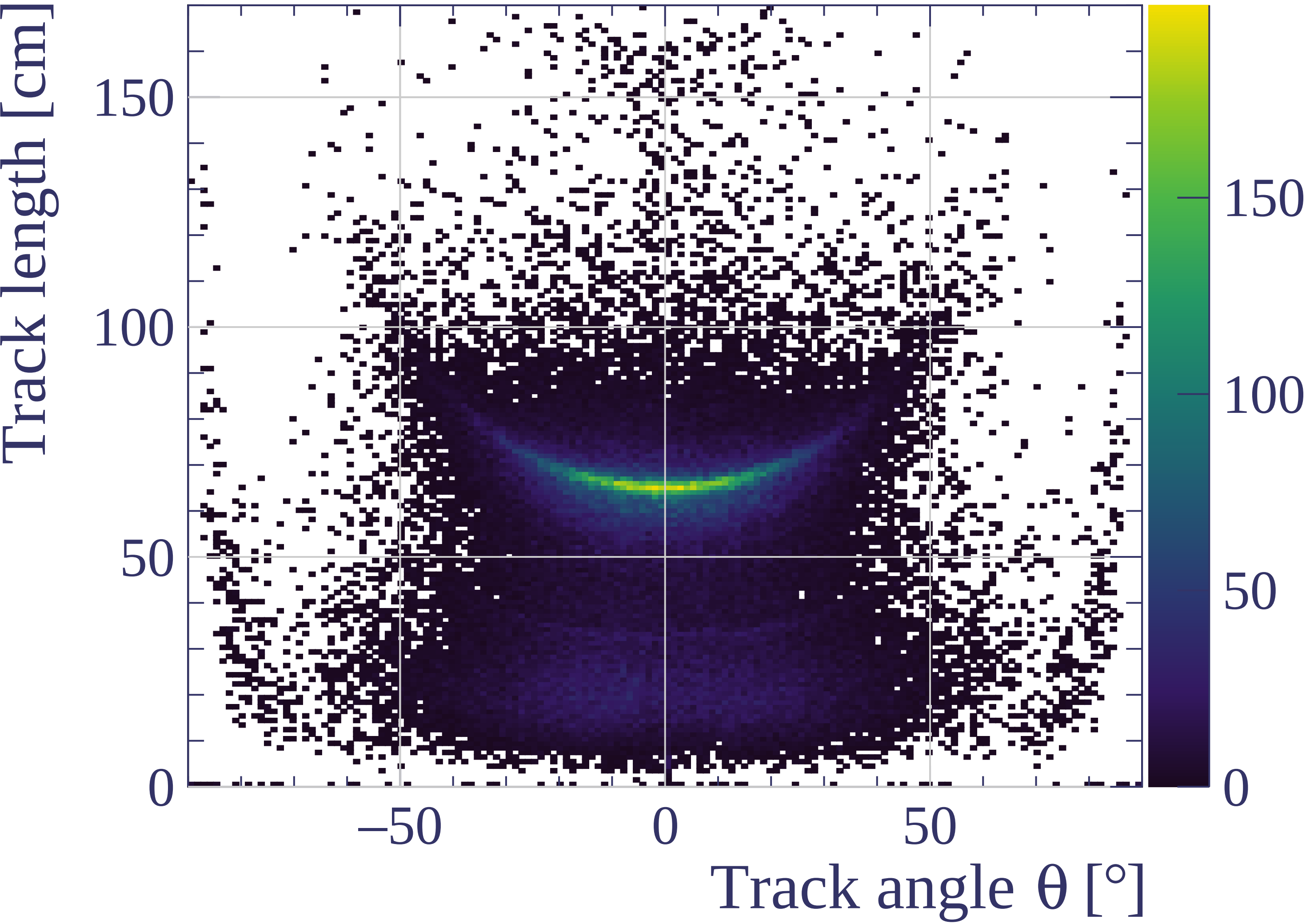

dE/dx resolution (%)
12
10
8
6
4
−50
0
50
Track angle θ [°]
WF
XP



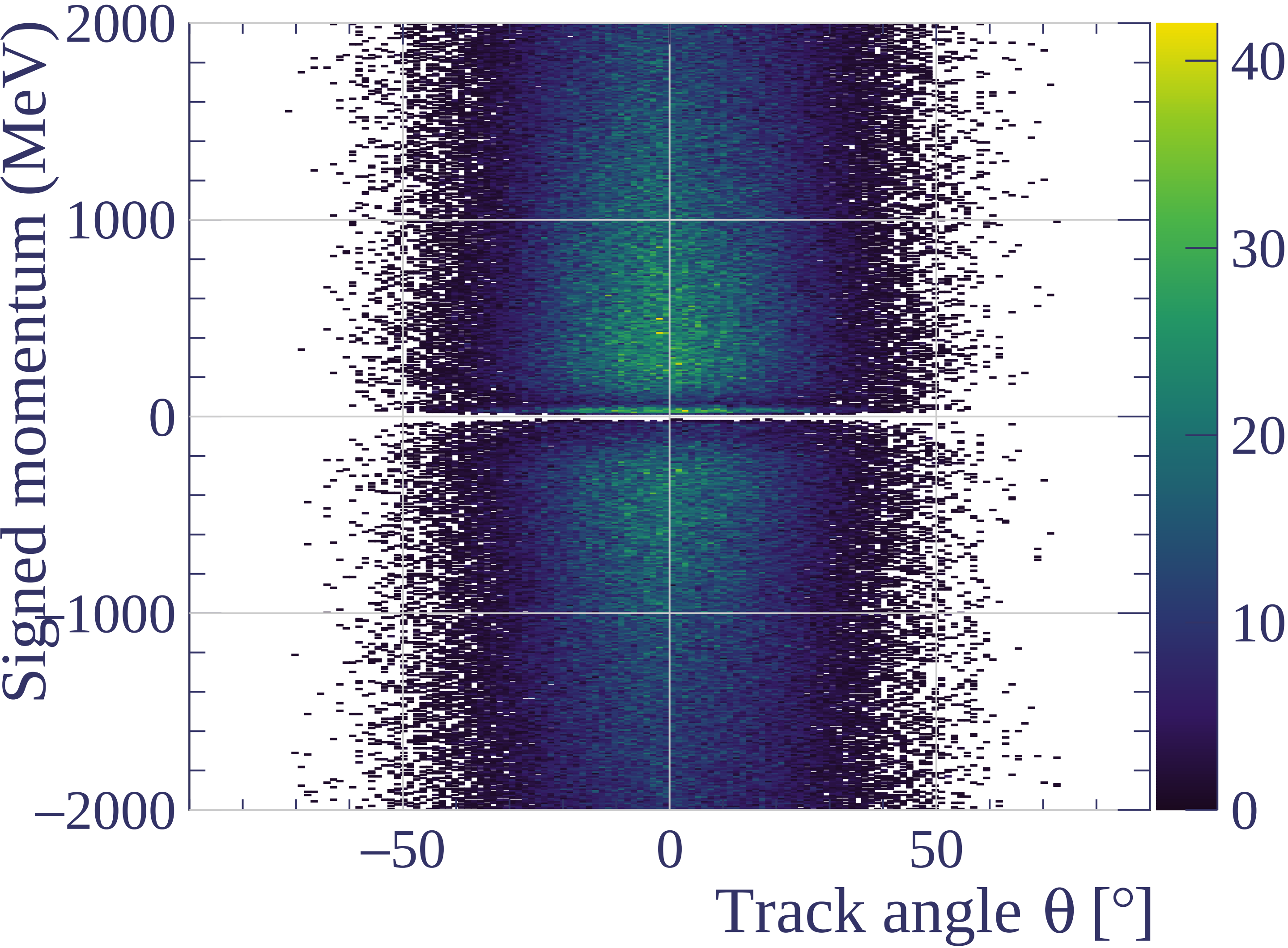
600
500
400
300
−50
0
50
Mean [ADC counts/cm]
Track angle θ [°]
WF
XP

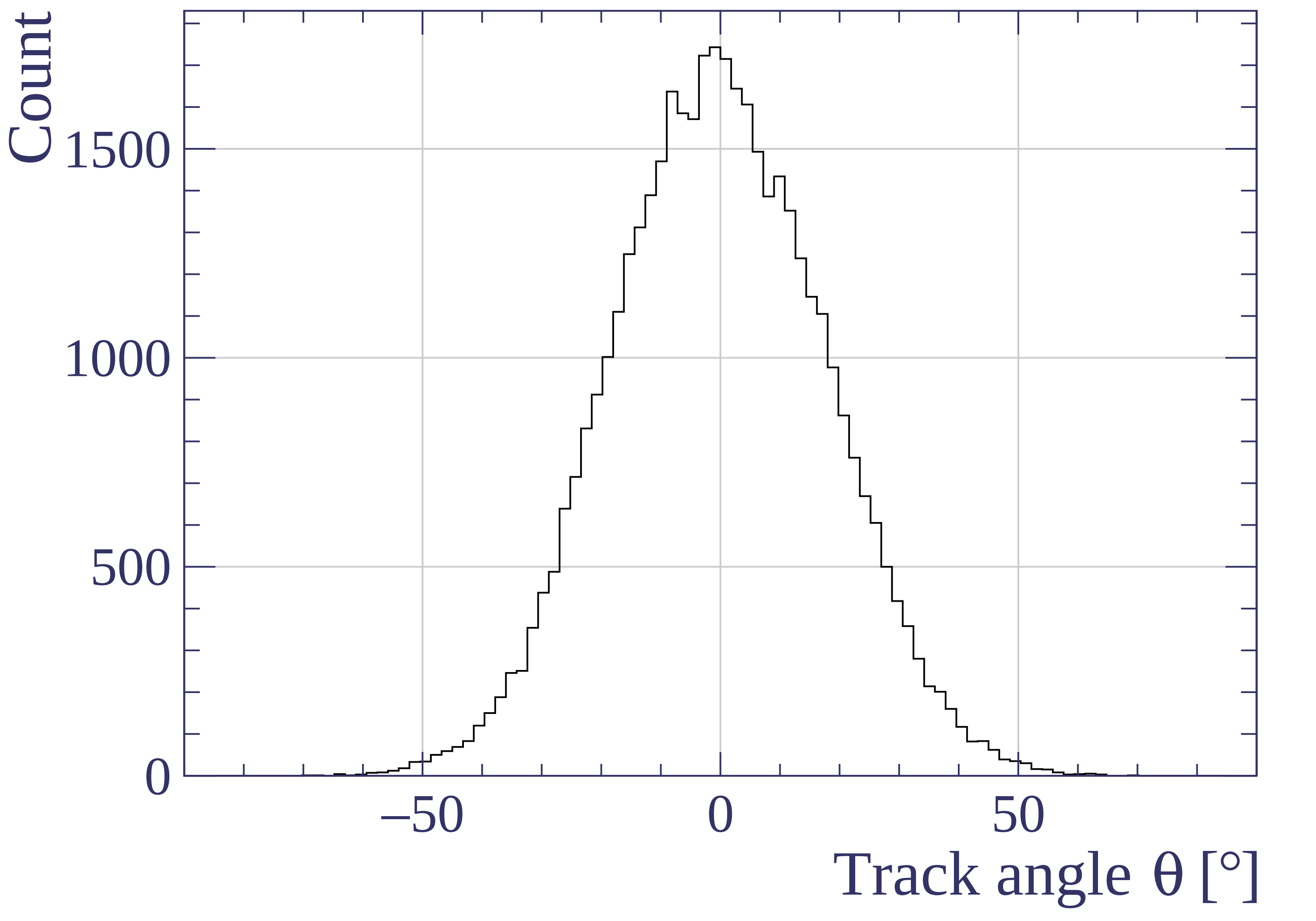


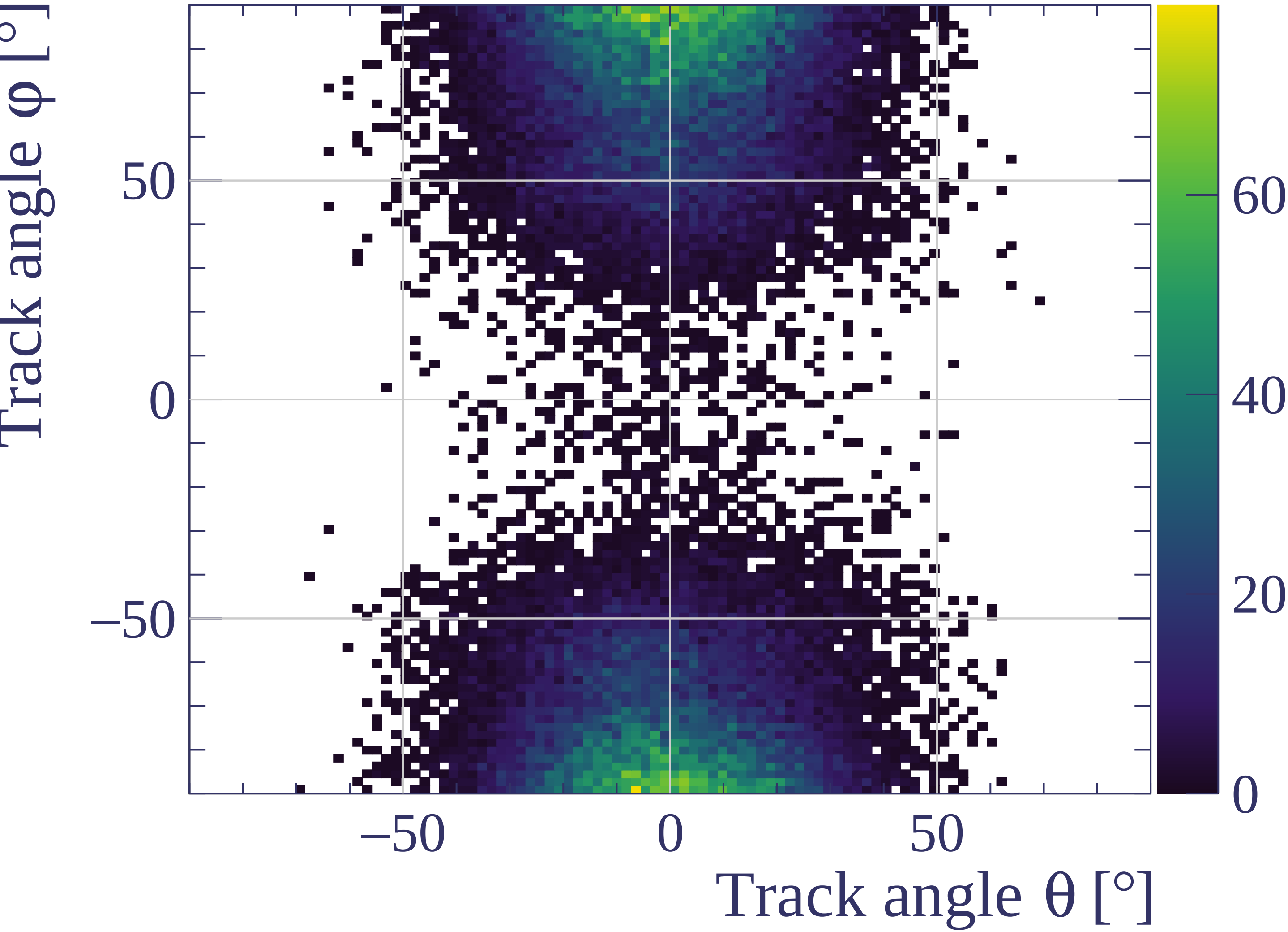
dE/dx with XP [ADC counts/cm]
Track angle θ [°]
0
200
400
600
800
1000
−50
0
50
0
10
20
30
40



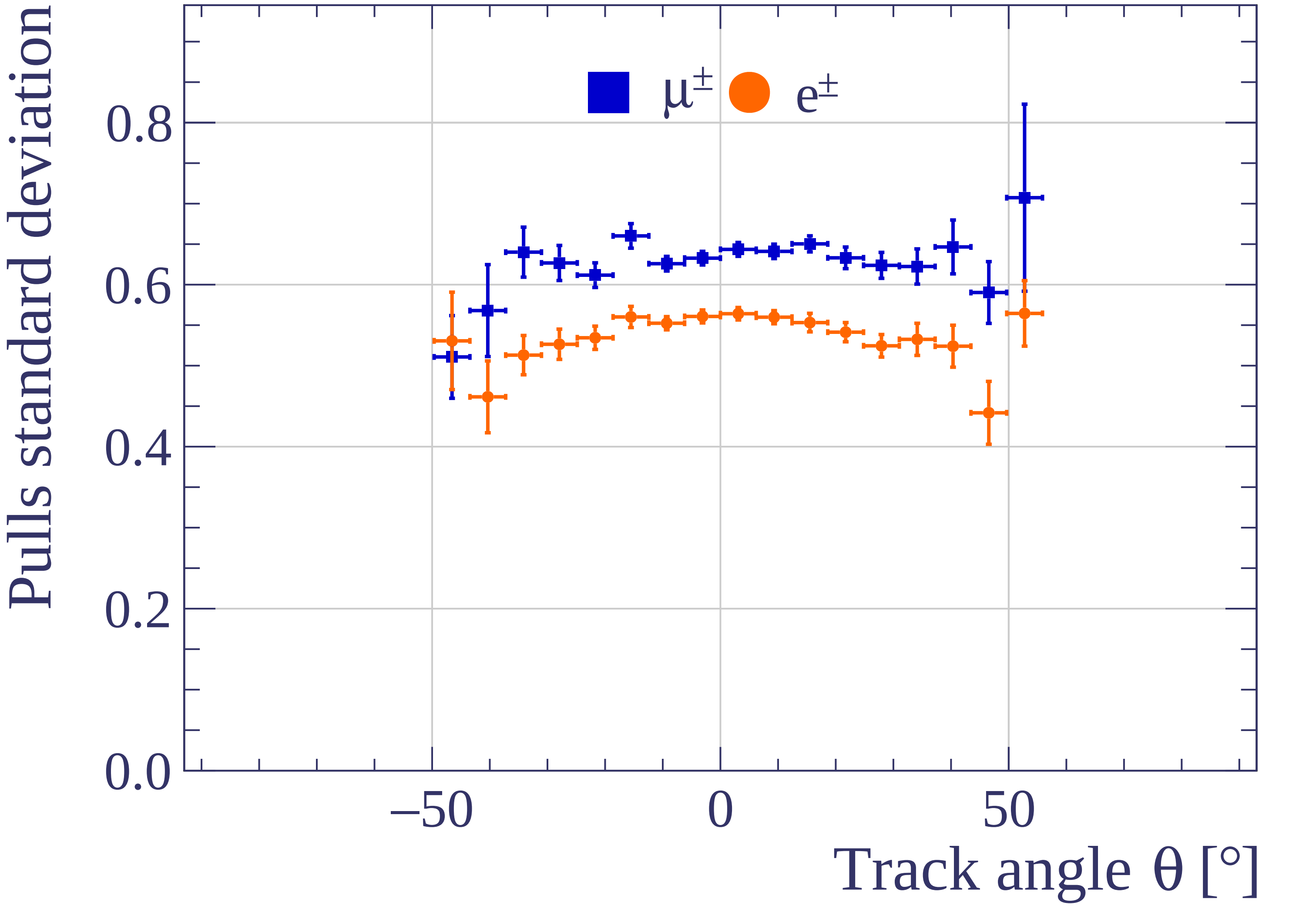
Track length [cm]
150
100
50
0
−50
0
50
Track angle θ [°]
150
100
50
0


Signed momentum (MeV)
2000
1000
0
−1000
−2000
−50
0
50
Track angle θ [°]
40
30
20
10
0

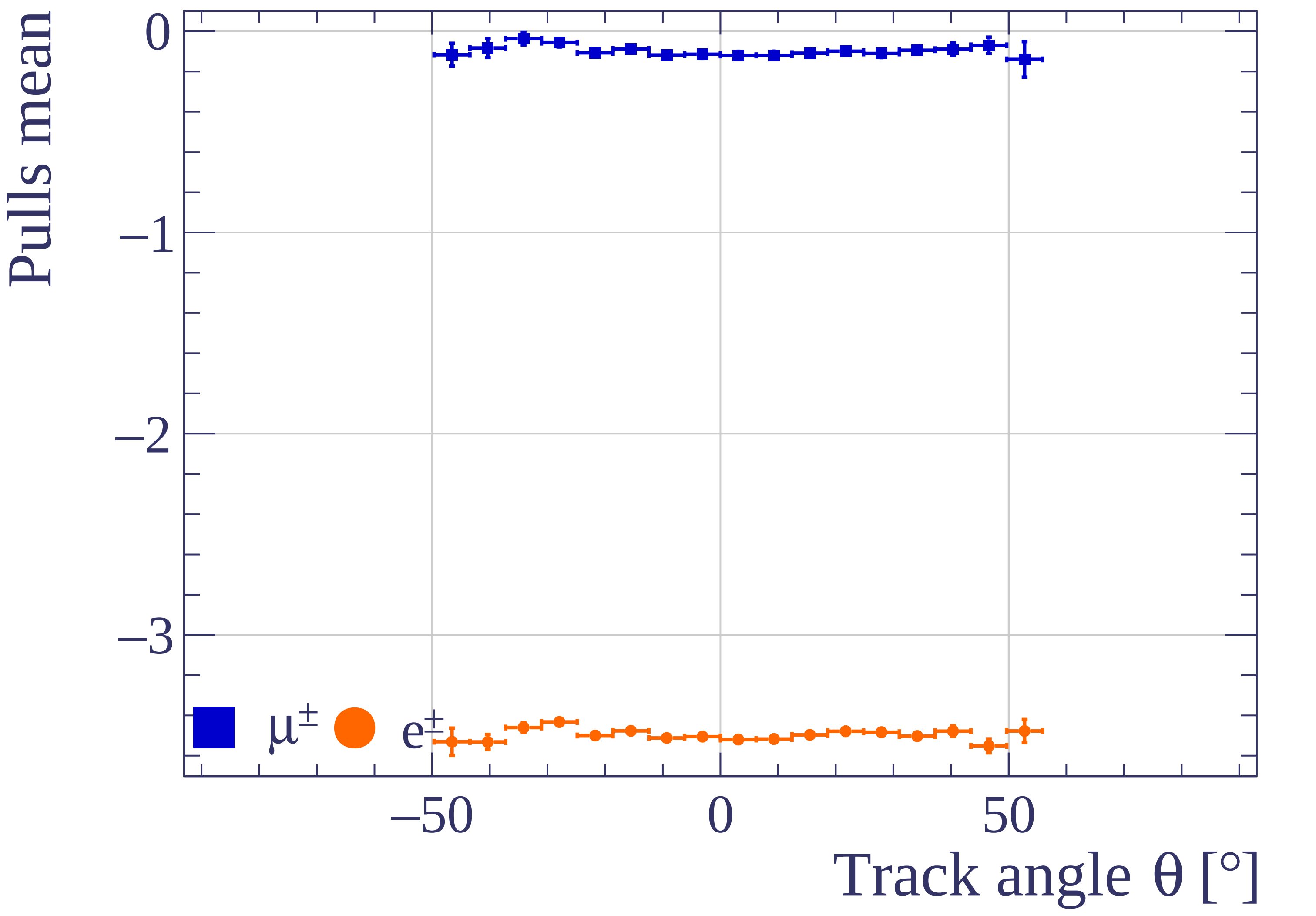

Count
1500
1000
500
0
−50
0
50
Track angle θ [°]



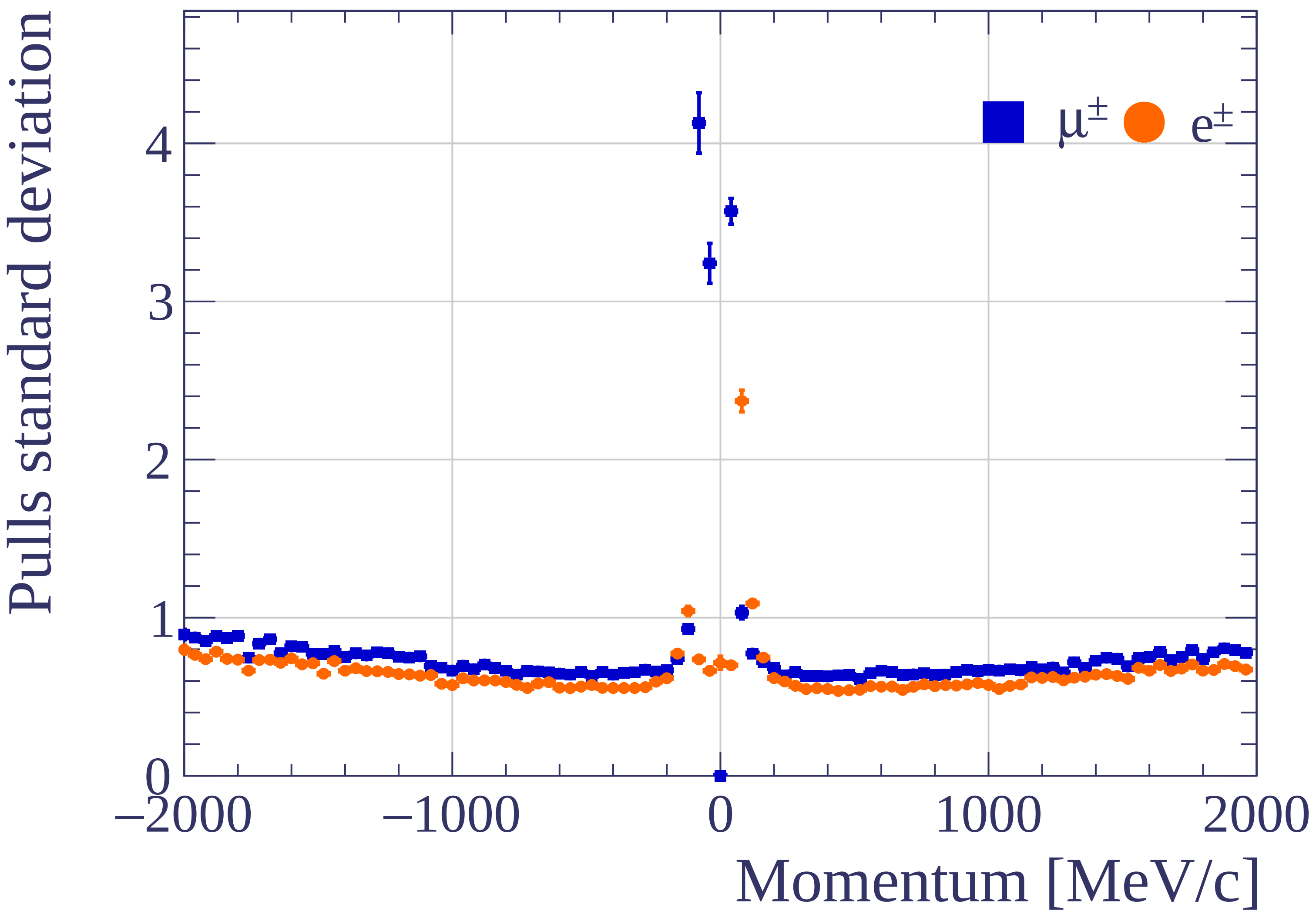
Track angle φ [°]
50
0
−50
−50
0
50
Track angle θ [°]
60
40
20
0



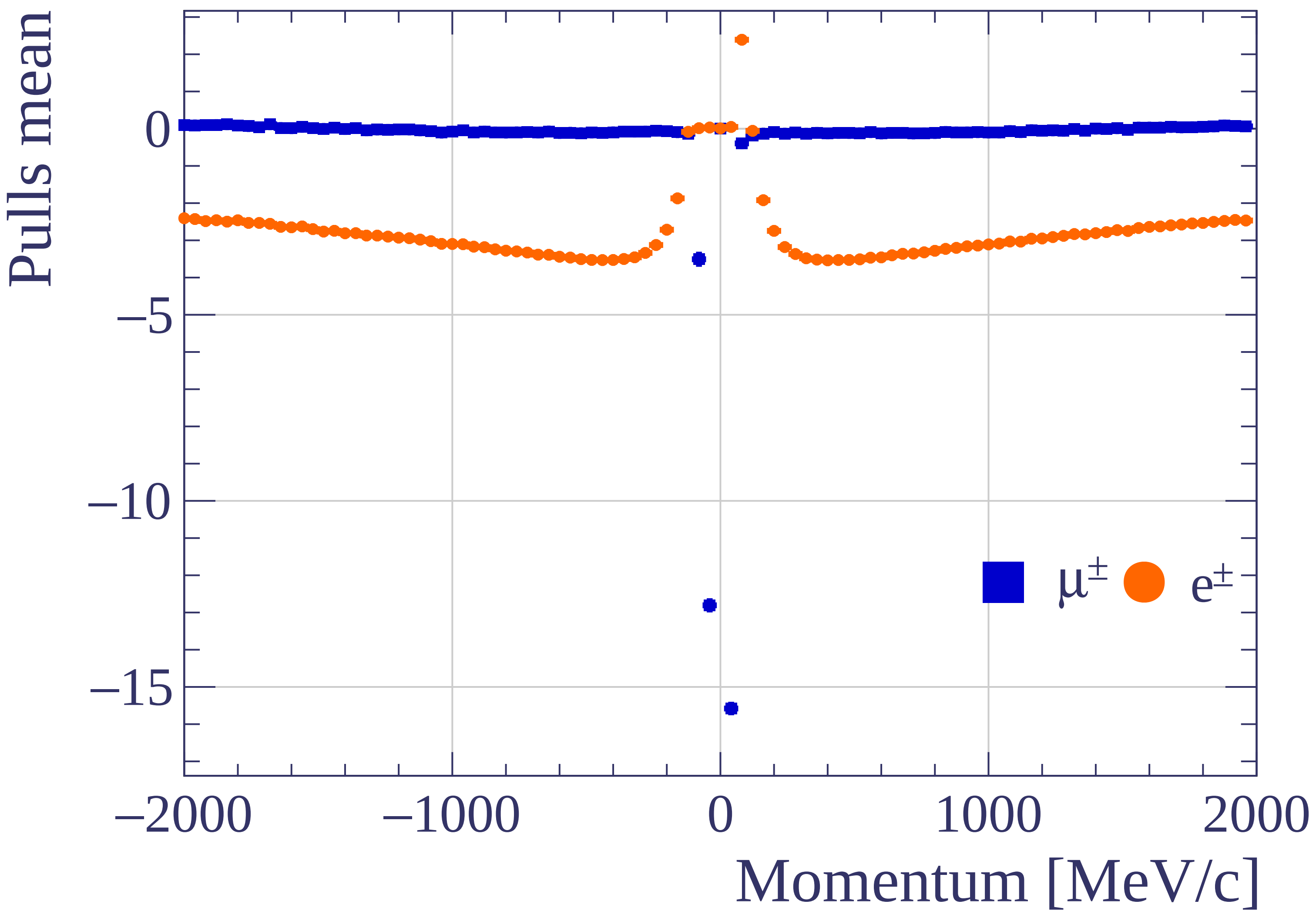
Pulls standard deviation
0.0
0.2
0.4
0.6
0.8
−50
0
50
Track angle θ [°]
μ±
e±



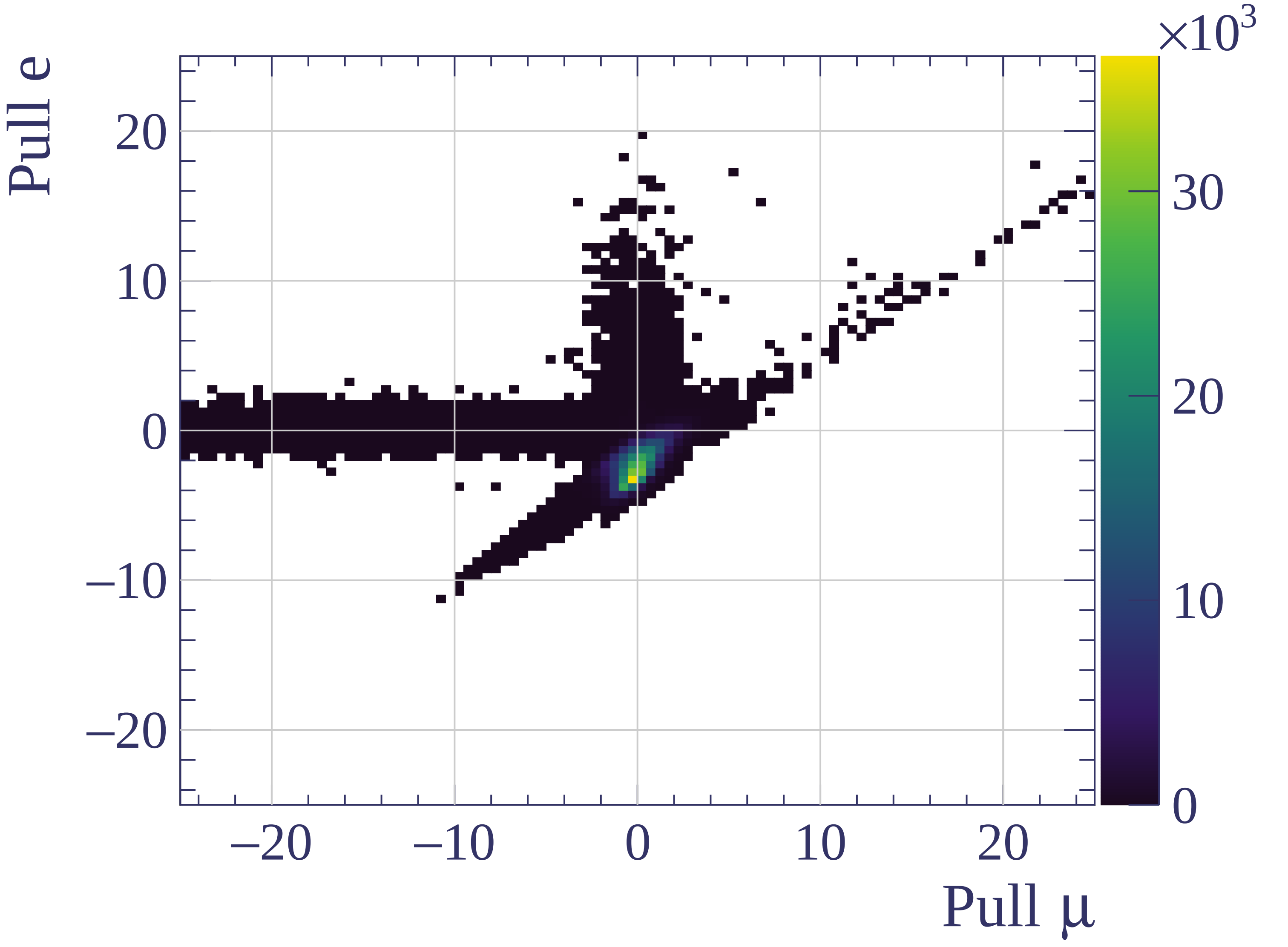
Pulls mean
0
−1
−2
−3
−50
0
50
Track angle θ [°]
μ±
e±



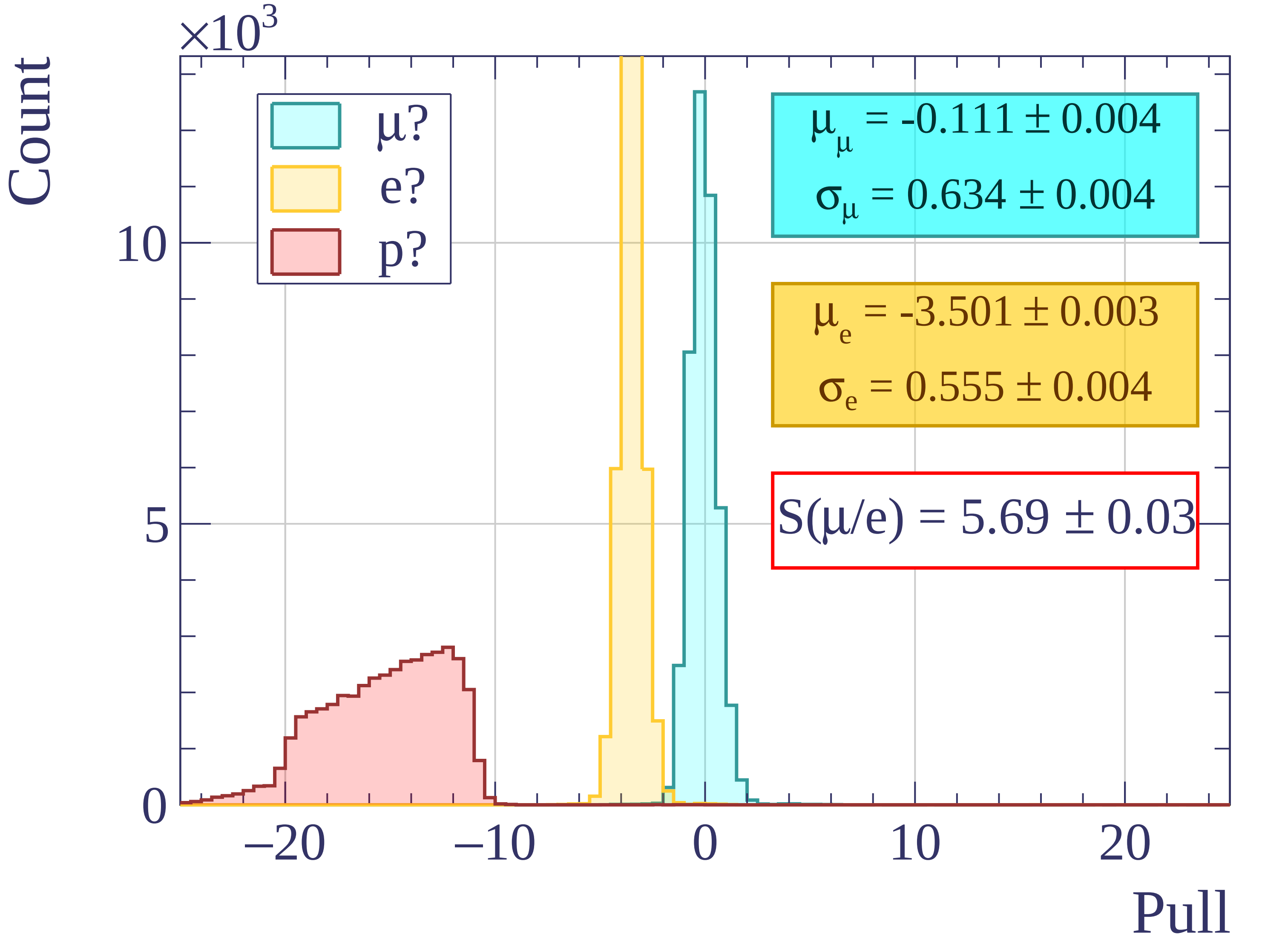
Pulls standard deviation
0
1
2
3
4
−2000
−1000
0
1000
2000
Momentum [MeV/c]
$\mu^{\pm}$
$e^{\pm}$



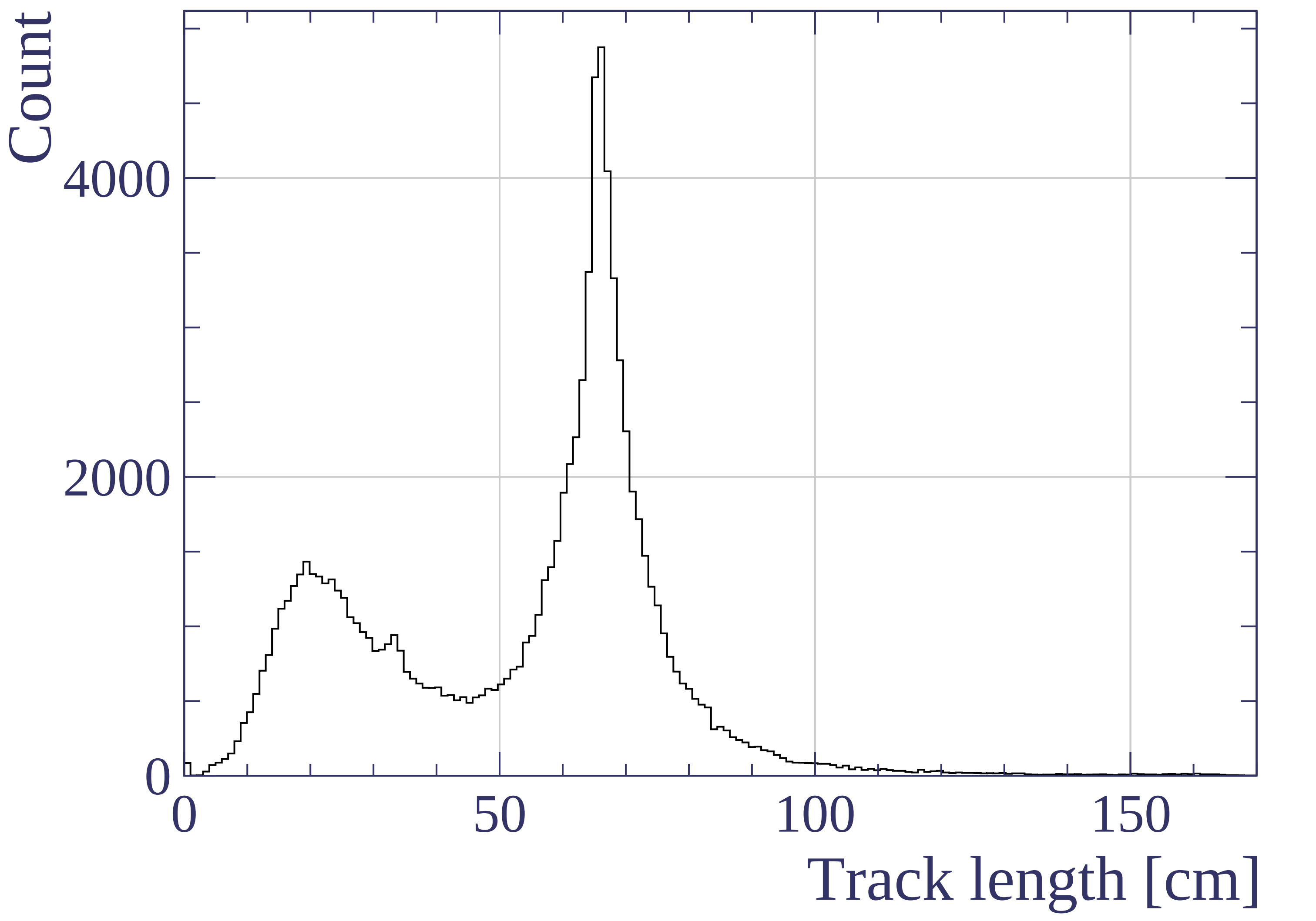
Pulls mean
0
−5
−10
−15
−2000
−1000
0
1000
2000
Momentum [MeV/c]
μ±
e±

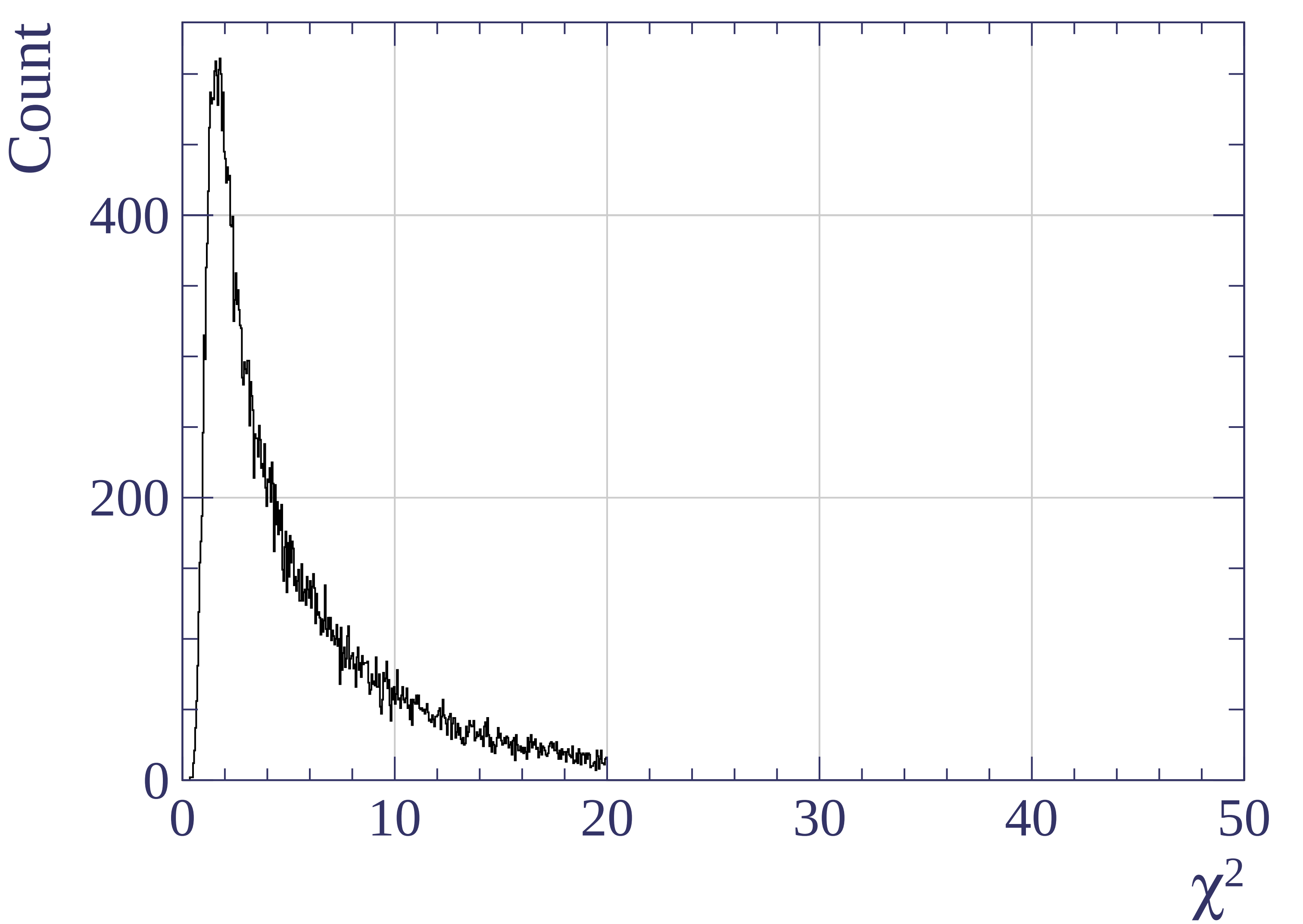


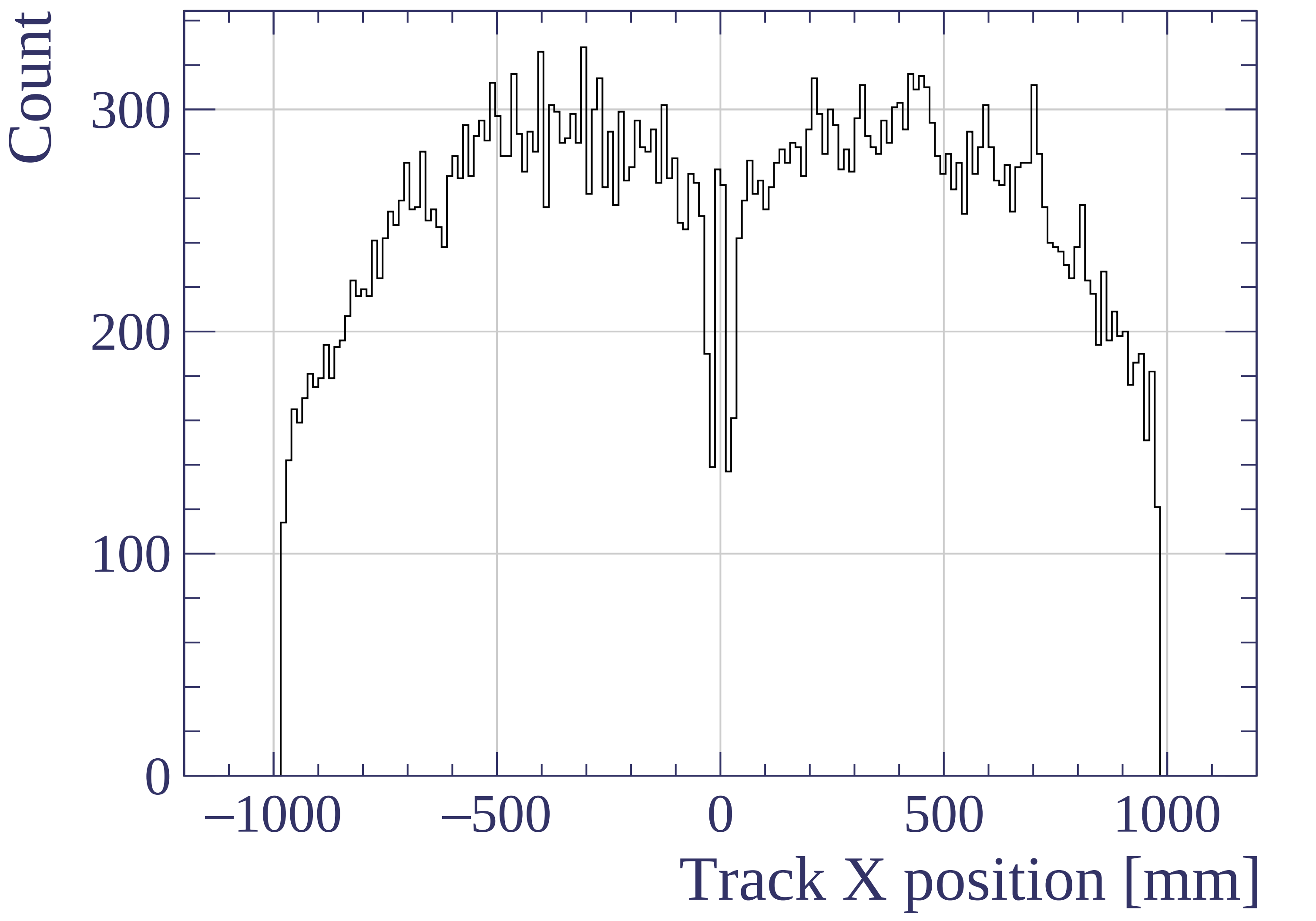
Pull e
Pull μ
×10^3
20
10
0
−10
−20
−20
−10
0
10
20
30
20
10
0



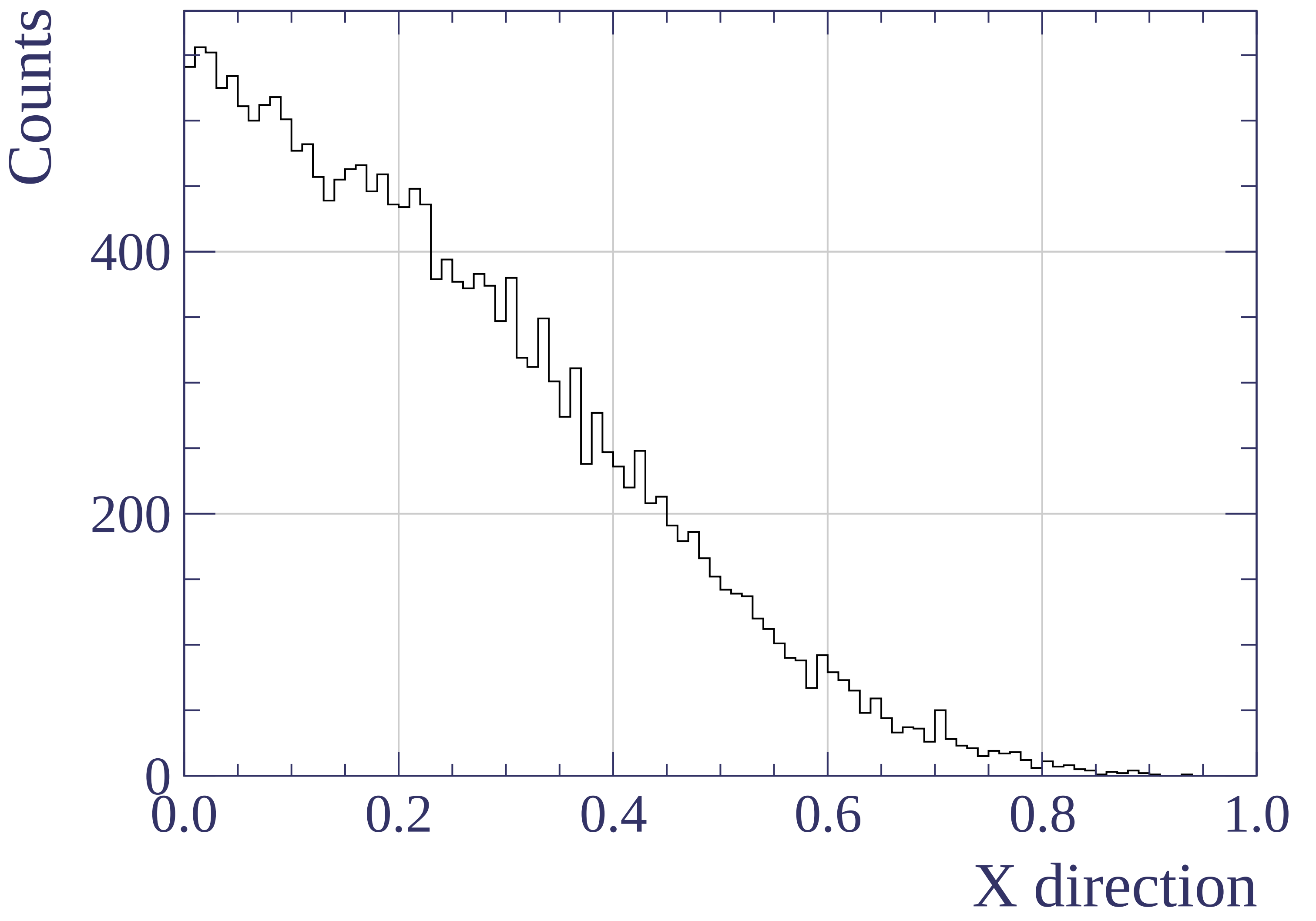
×10³
Count
μ?
e?
p?
μ_μ = -0.111 ± 0.004
σ_μ = 0.634 ± 0.004
μ_e = -3.501 ± 0.003
σ_e = 0.555 ± 0.004
S(μ/e) = 5.69 ± 0.03
0
5
10
−20
−10
0
10
20
Pull

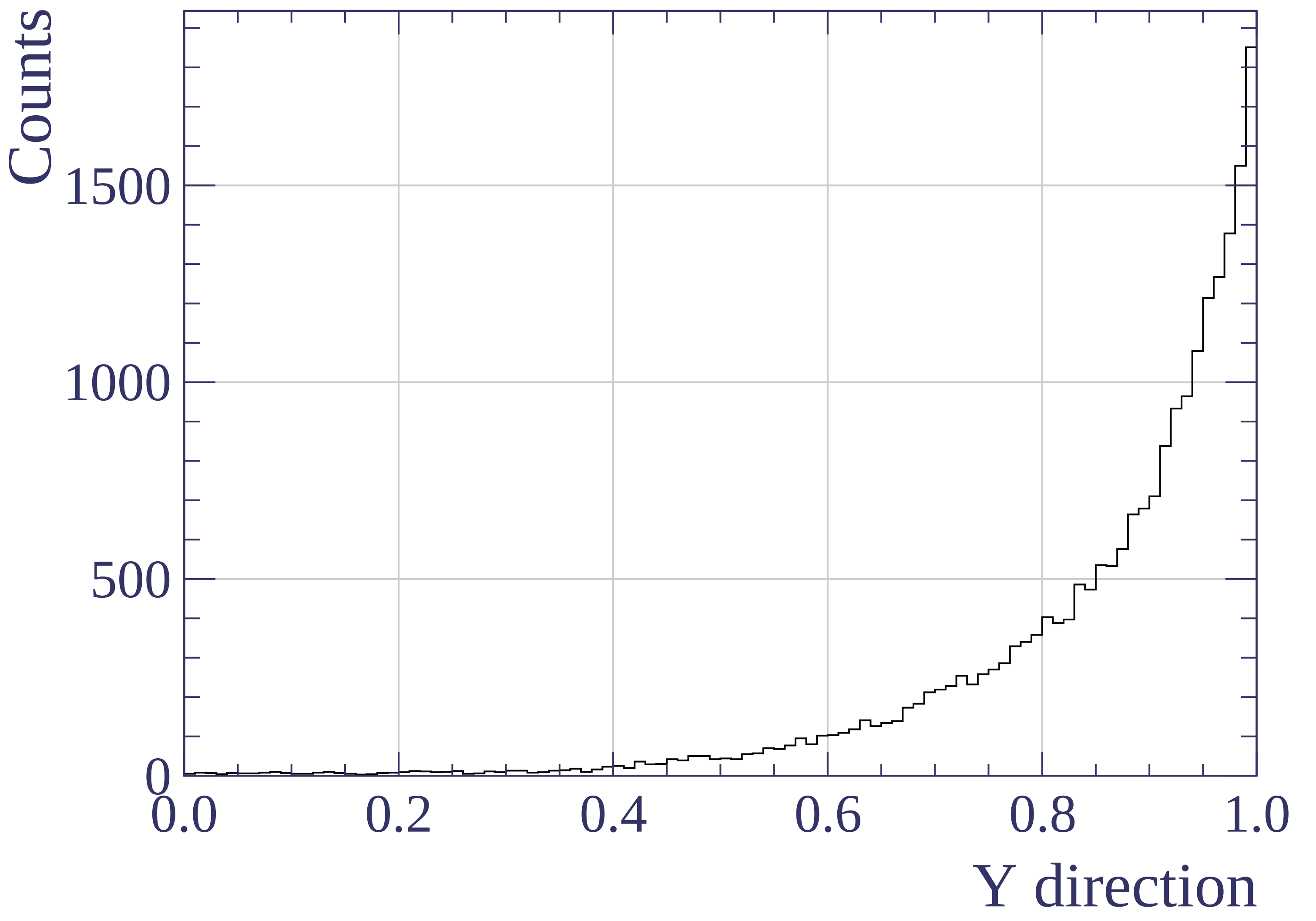

Count
4000
2000
0
0
50
100
150
Track length [cm]


Count
400
200
0
0
10
20
30
40
50
$\chi^2$

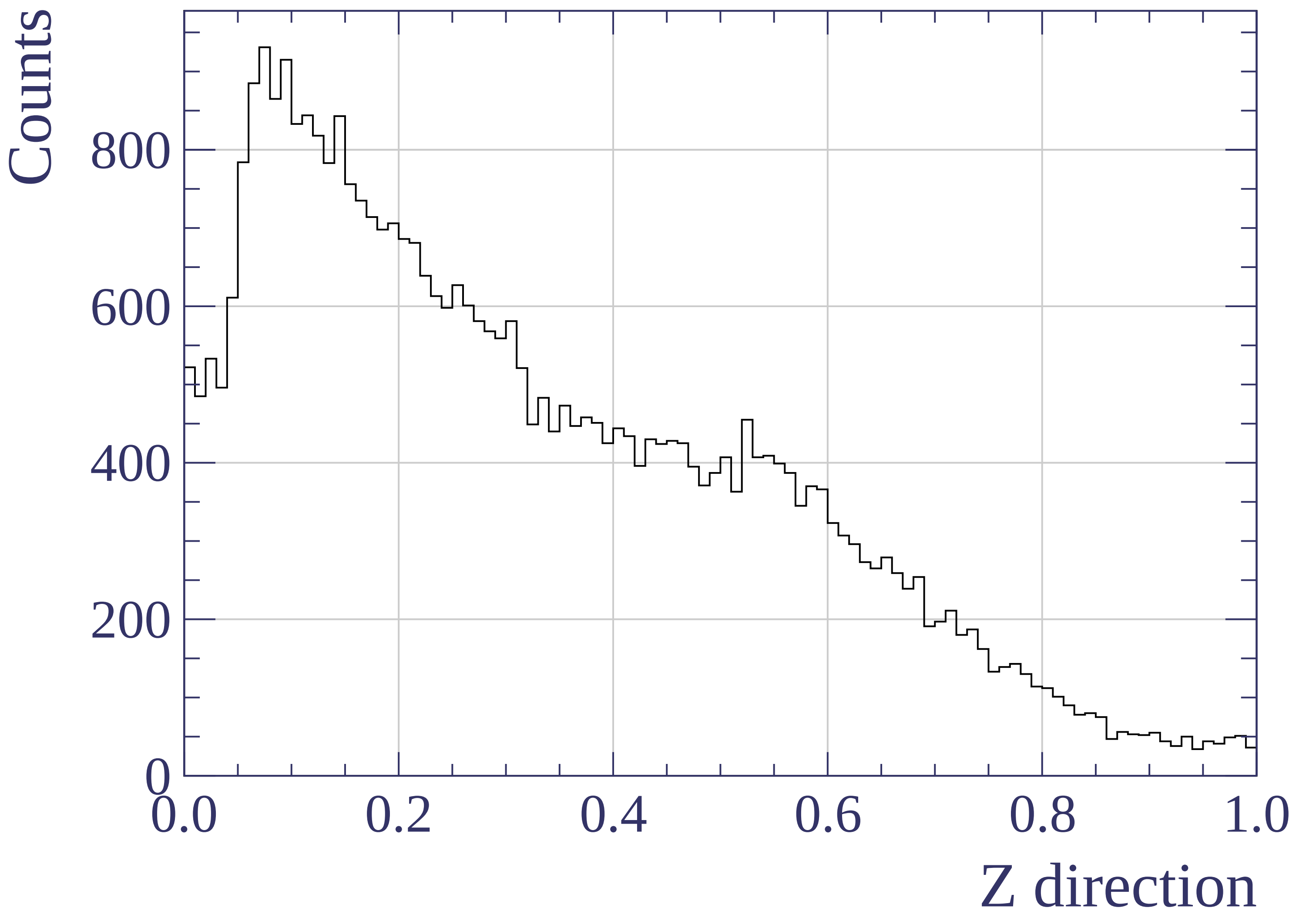


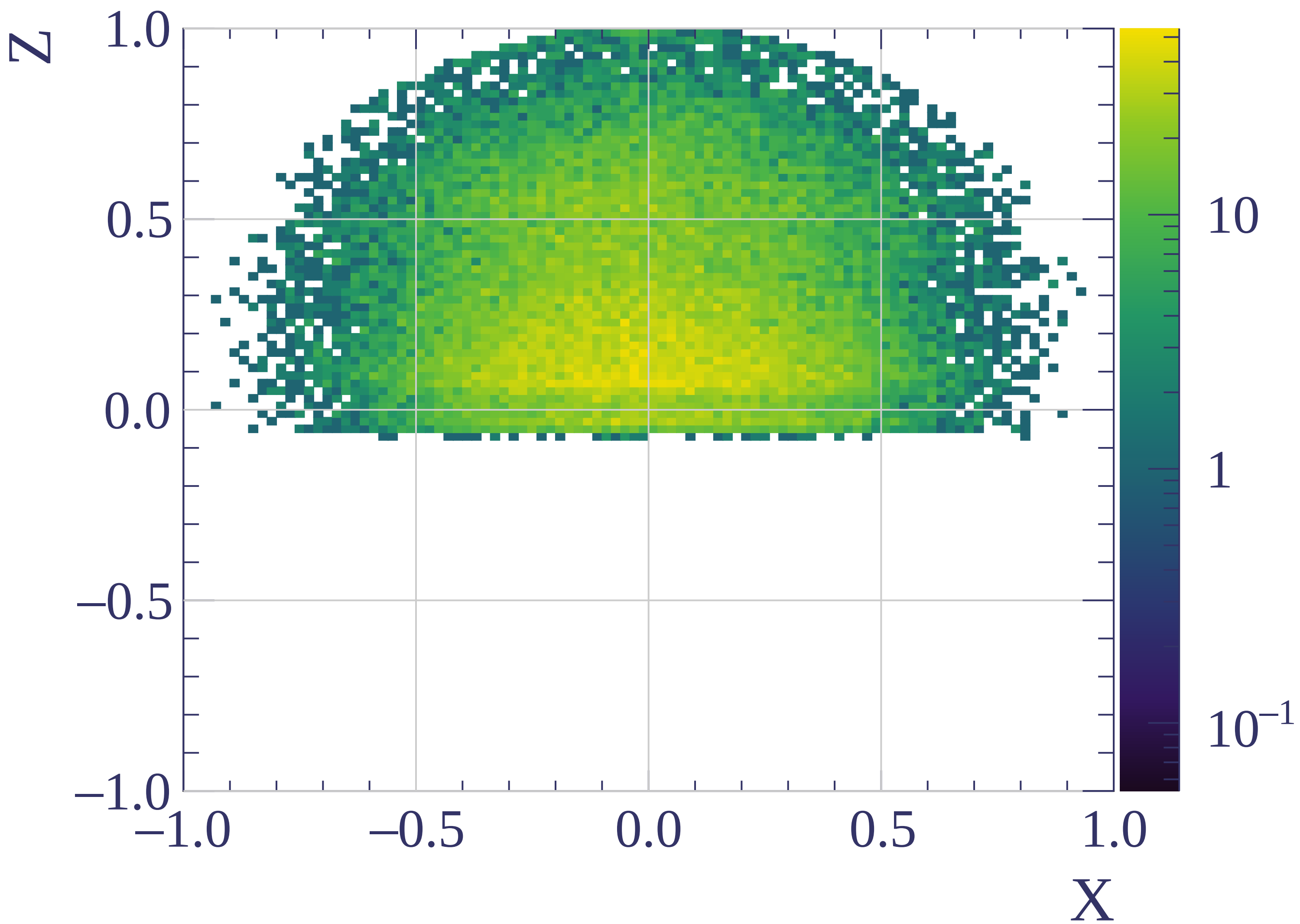
Count
300
200
100
0
−1000
−500
0
500
1000
Track X position [mm]


Counts
400
200
0
0.0
0.2
0.4
0.6
0.8
1.0
X direction

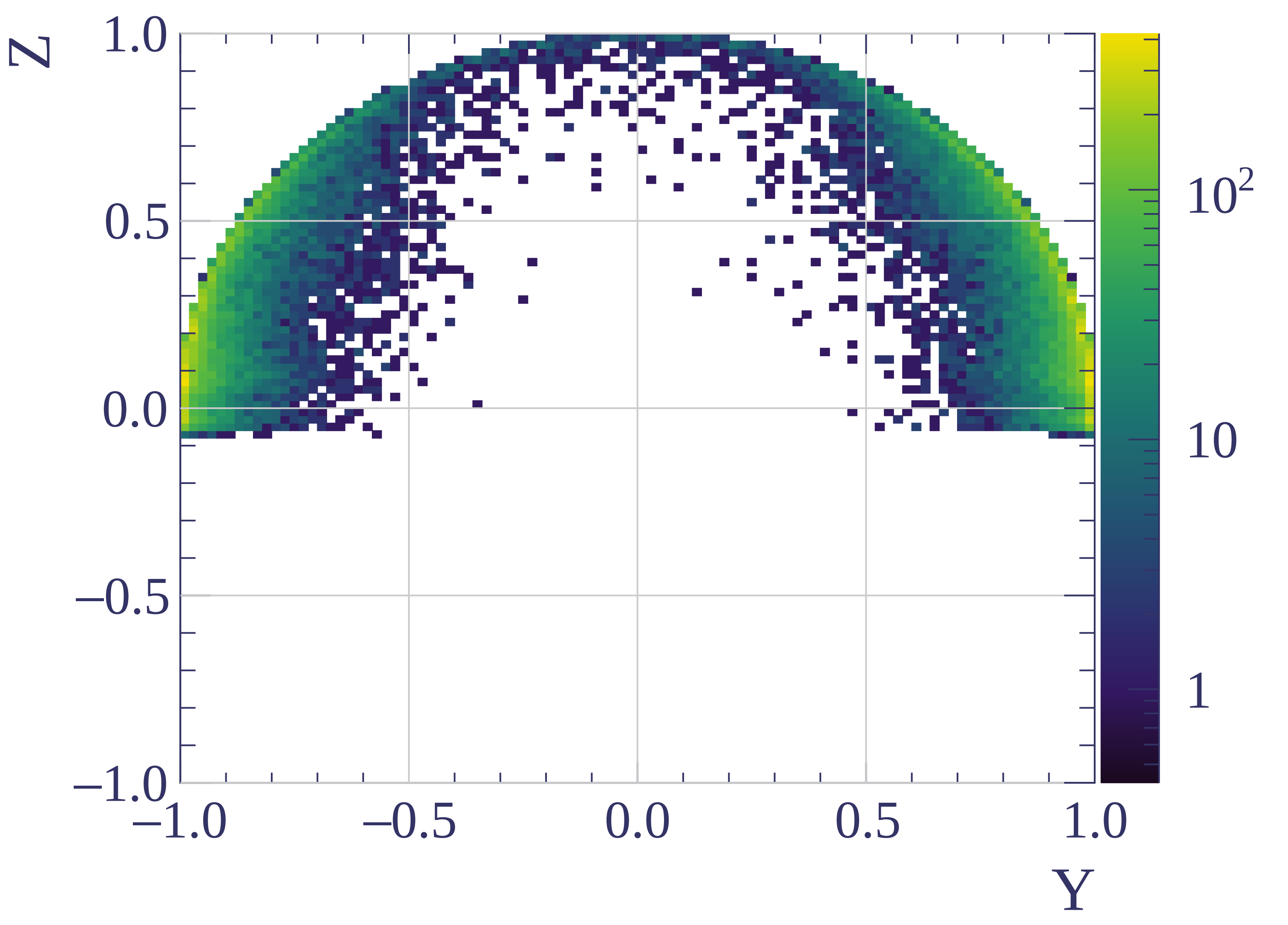


Counts
1500
1000
500
0
0.0
0.2
0.4
0.6
0.8
1.0
Y direction

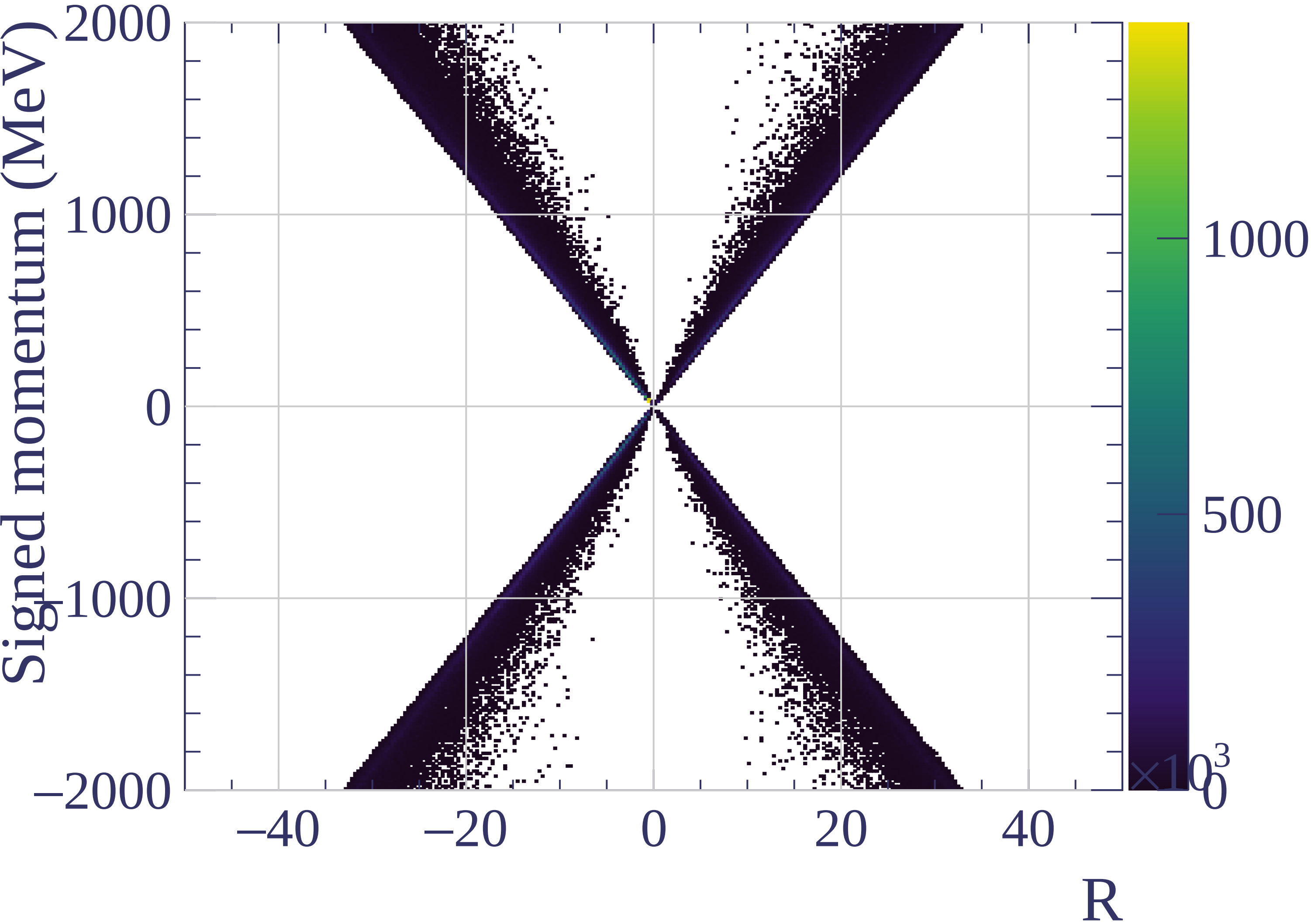

Counts
800
600
400
200
0
0.0
0.2
0.4
0.6
0.8
1.0
Z direction



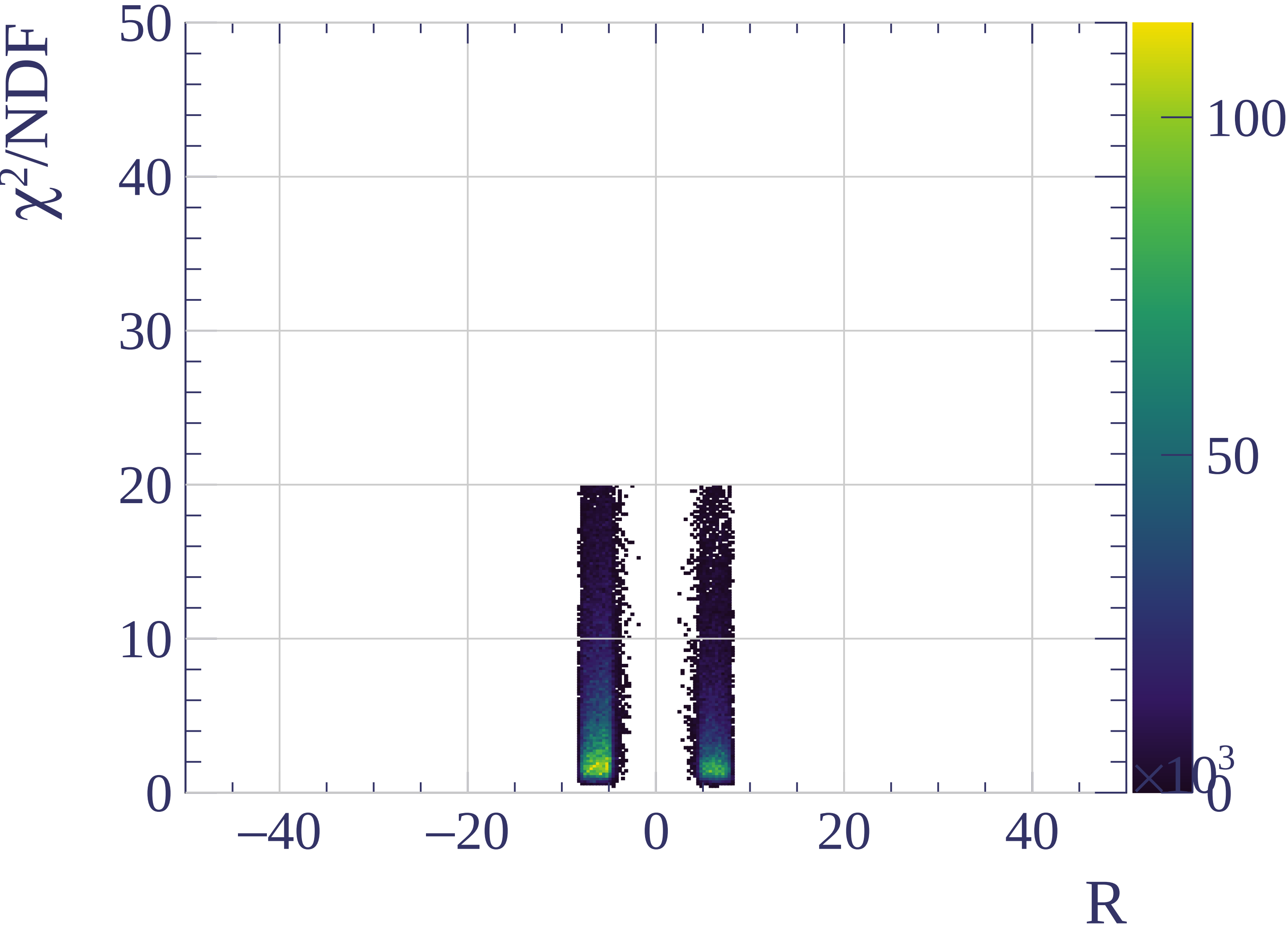
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
X
10
1
10^{−1}



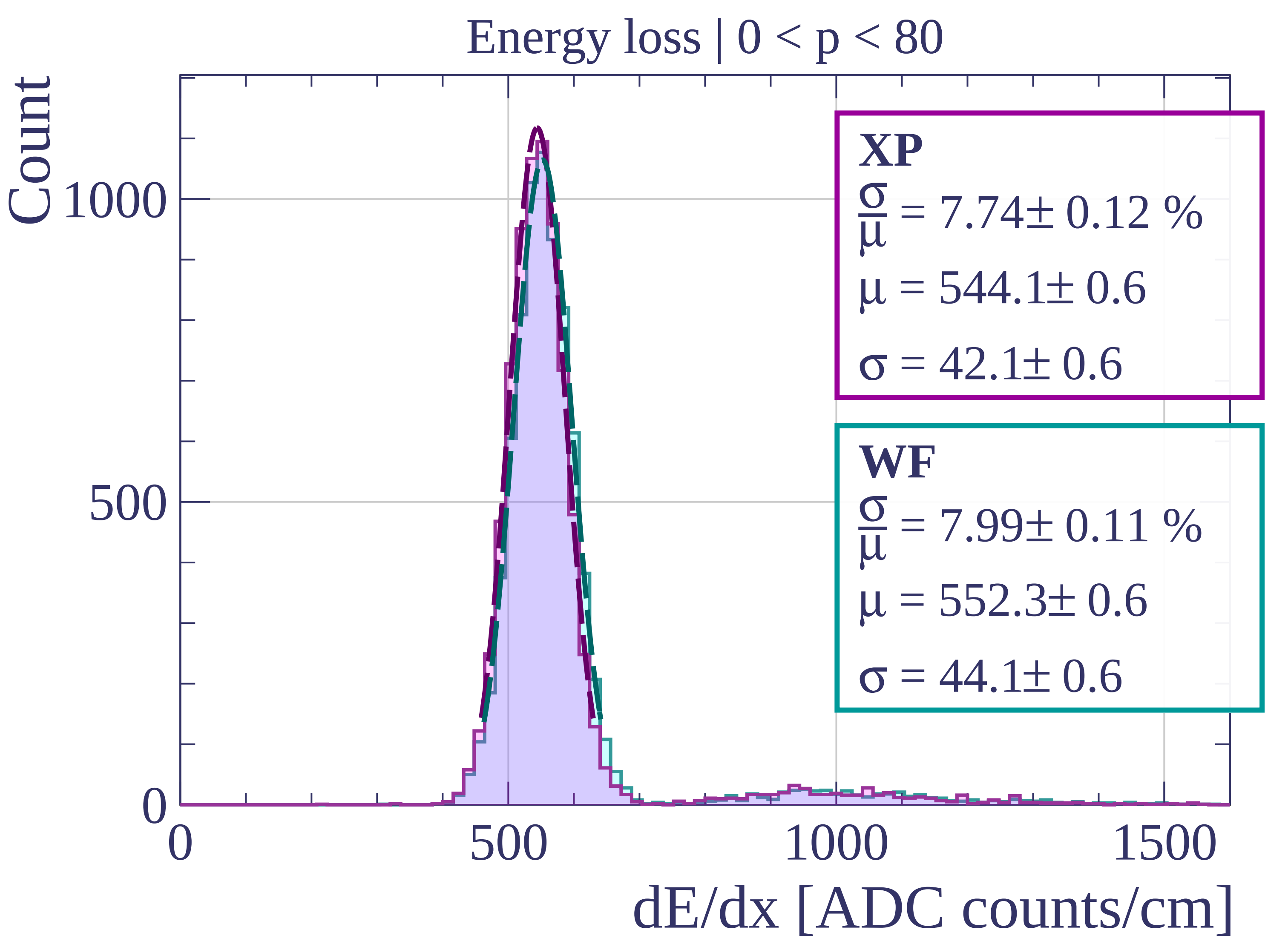
Z
1.0
0.5
0.0
−0.5
−1.0
−1.0
−0.5
0.0
0.5
1.0
Y
$10^2$
10
1

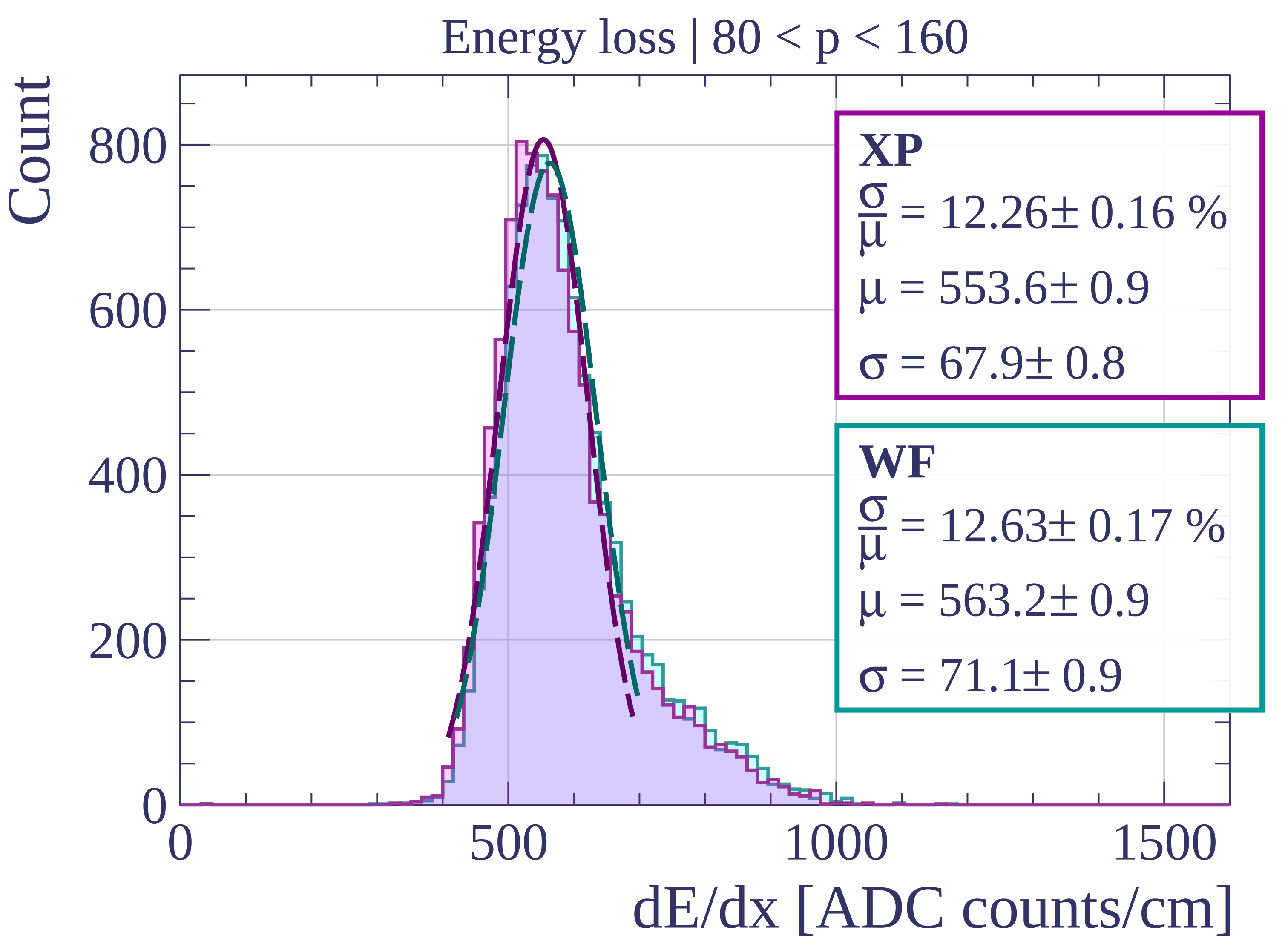


Signed momentum (MeV)
2000
1000
0
−1000
−2000
−40
−20
0
20
40
R
1000
500
0
×10^3

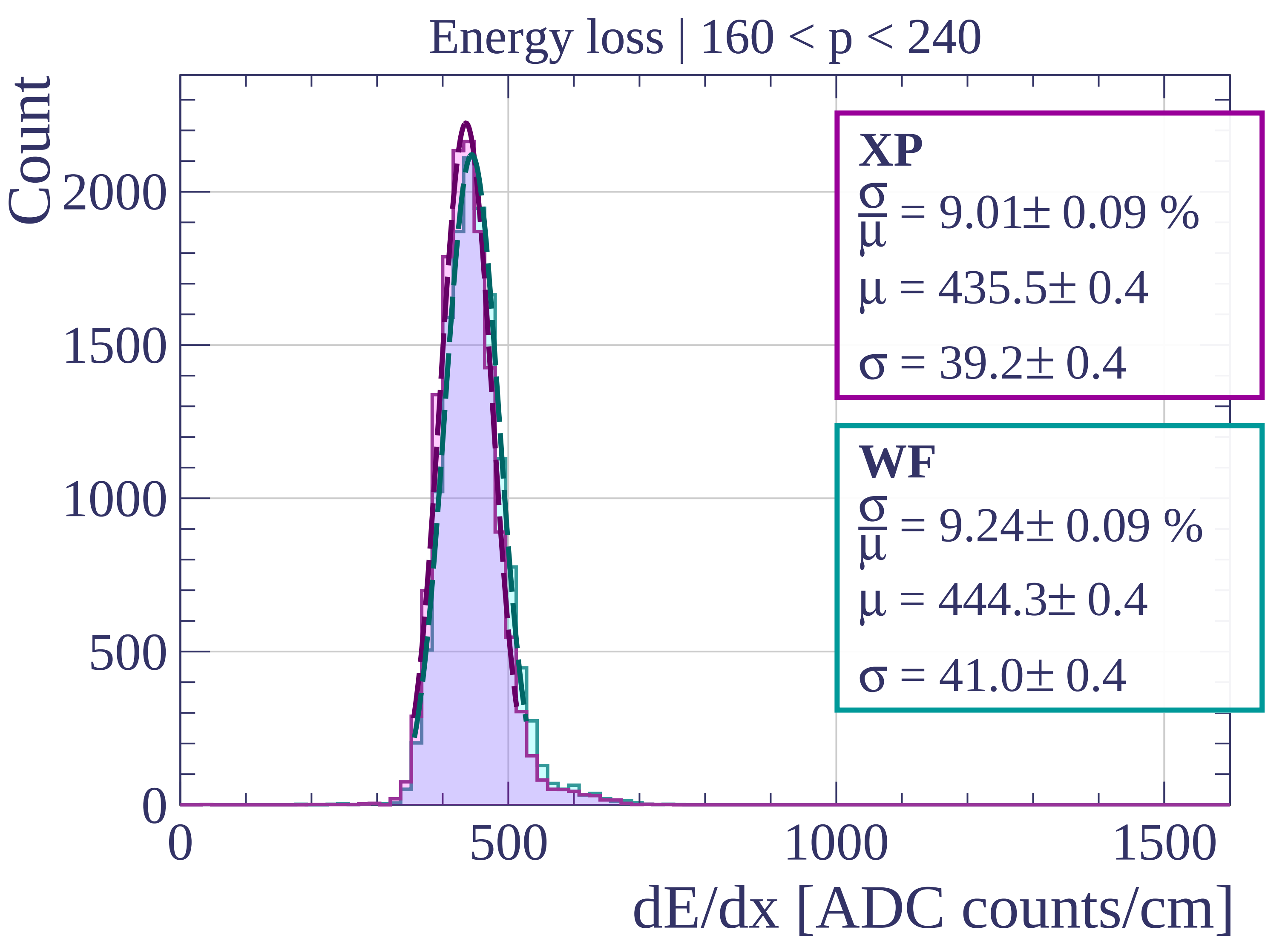


$\chi^2/NDF$
50
40
30
20
10
0
−40
−20
0
20
40
R
100
50
0
$\times 10^3$

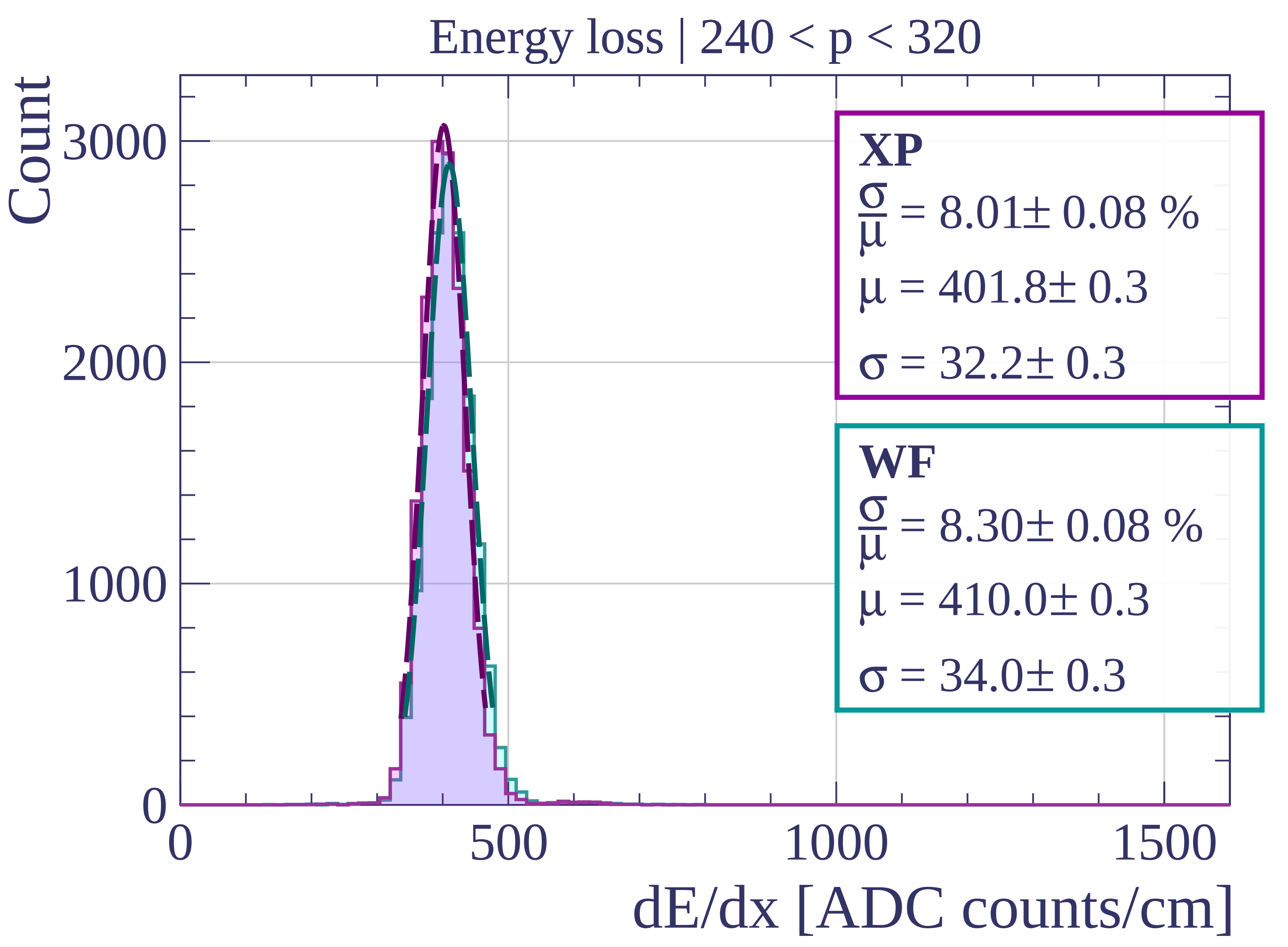

Energy loss | 0 < p < 80
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu} = 7.74 \pm 0.12$ %
$\mu = 544.1 \pm 0.6$
$\sigma = 42.1 \pm 0.6$
WF
$\frac{\sigma}{\mu} = 7.99 \pm 0.11$ %
$\mu = 552.3 \pm 0.6$
$\sigma = 44.1 \pm 0.6$



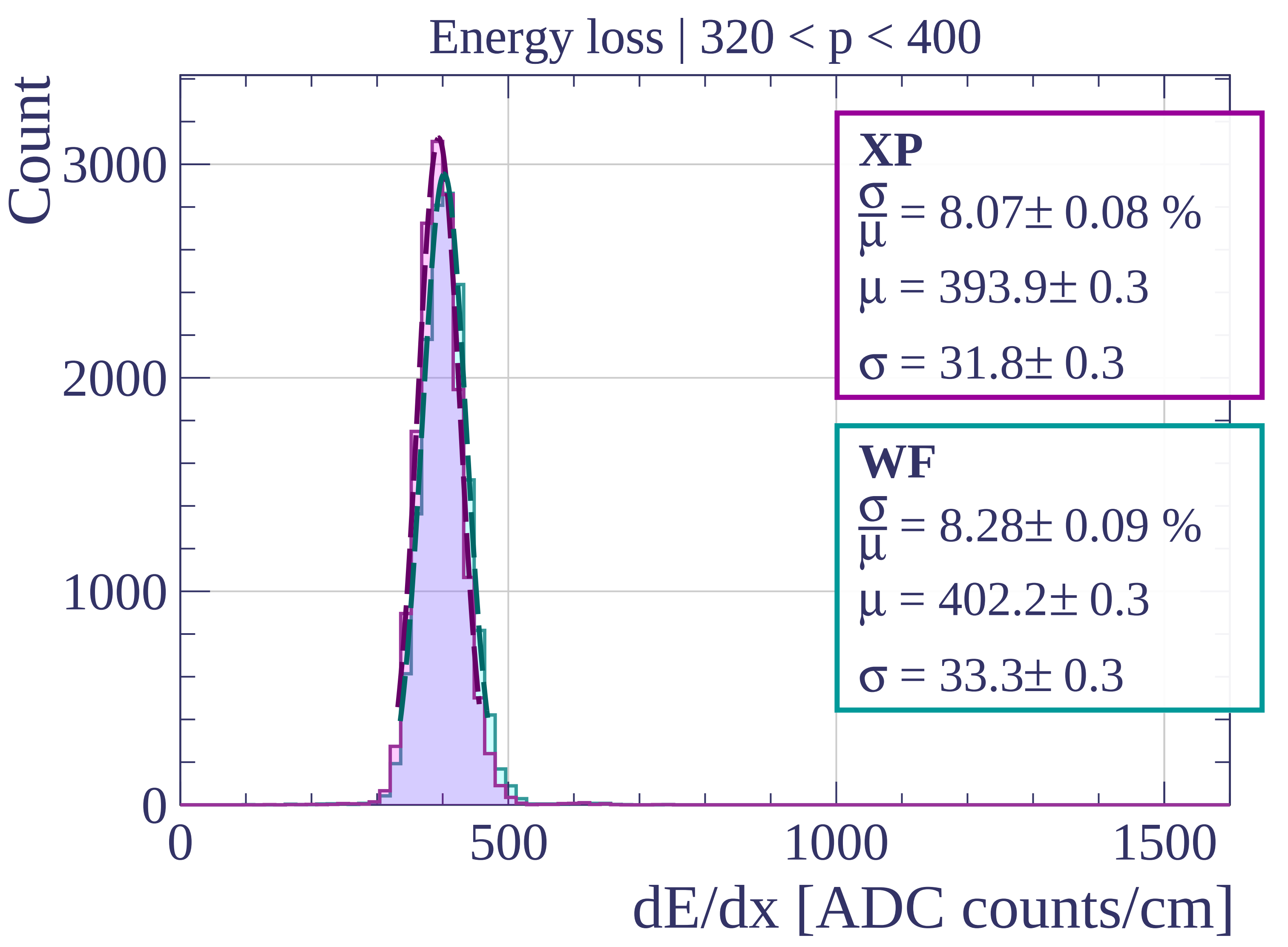
Energy loss | 80 < p < 160
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 12.26± 0.16 %
μ = 553.6± 0.9
σ = 67.9± 0.8
WF
$\frac{\sigma}{\mu}$ = 12.63± 0.17 %
μ = 563.2± 0.9
σ = 71.1± 0.9



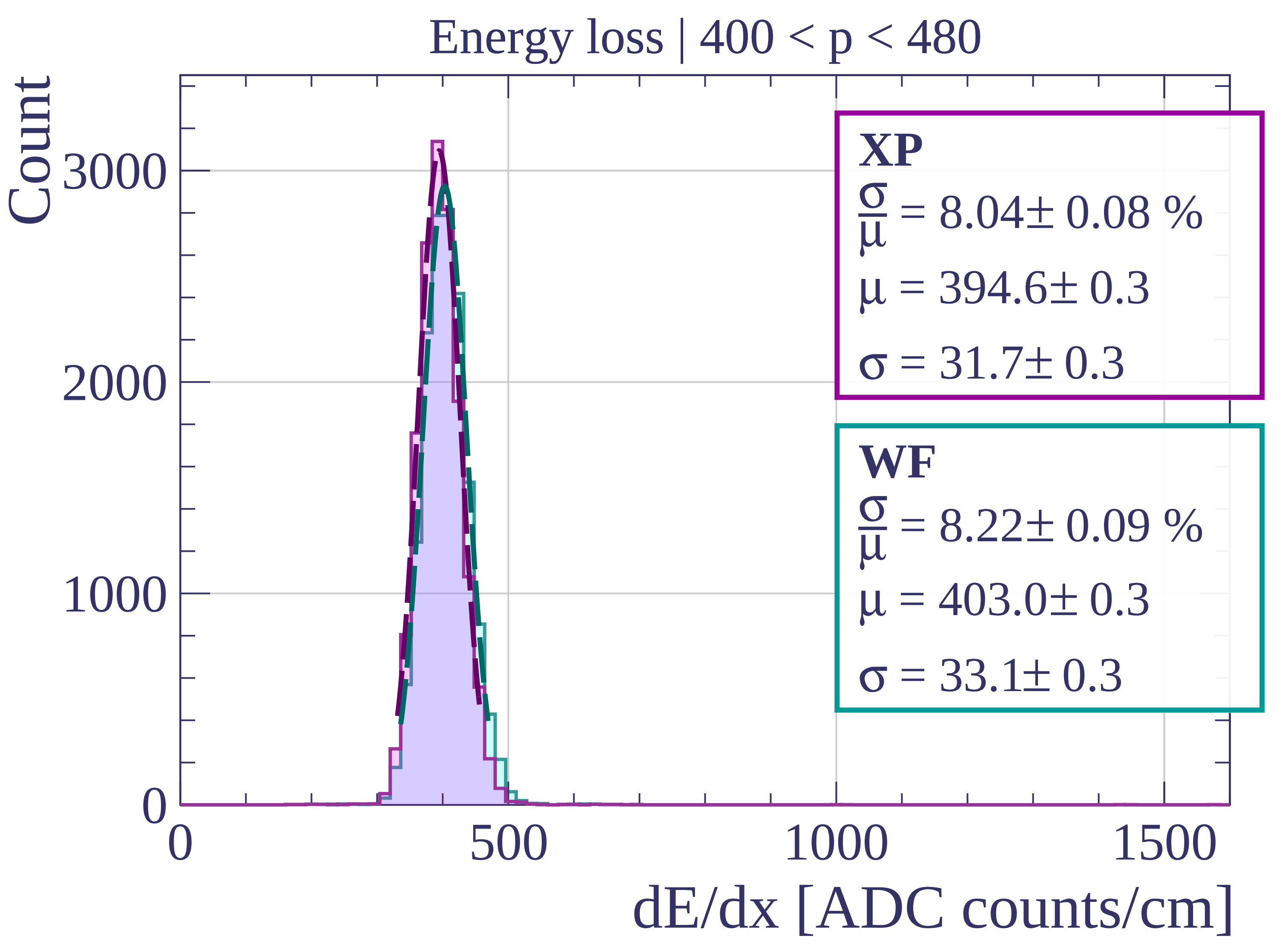
Energy loss | 160 < p < 240
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
σ/μ = 9.01± 0.09 %
μ = 435.5± 0.4
σ = 39.2± 0.4
WF
σ/μ = 9.24± 0.09 %
μ = 444.3± 0.4
σ = 41.0± 0.4



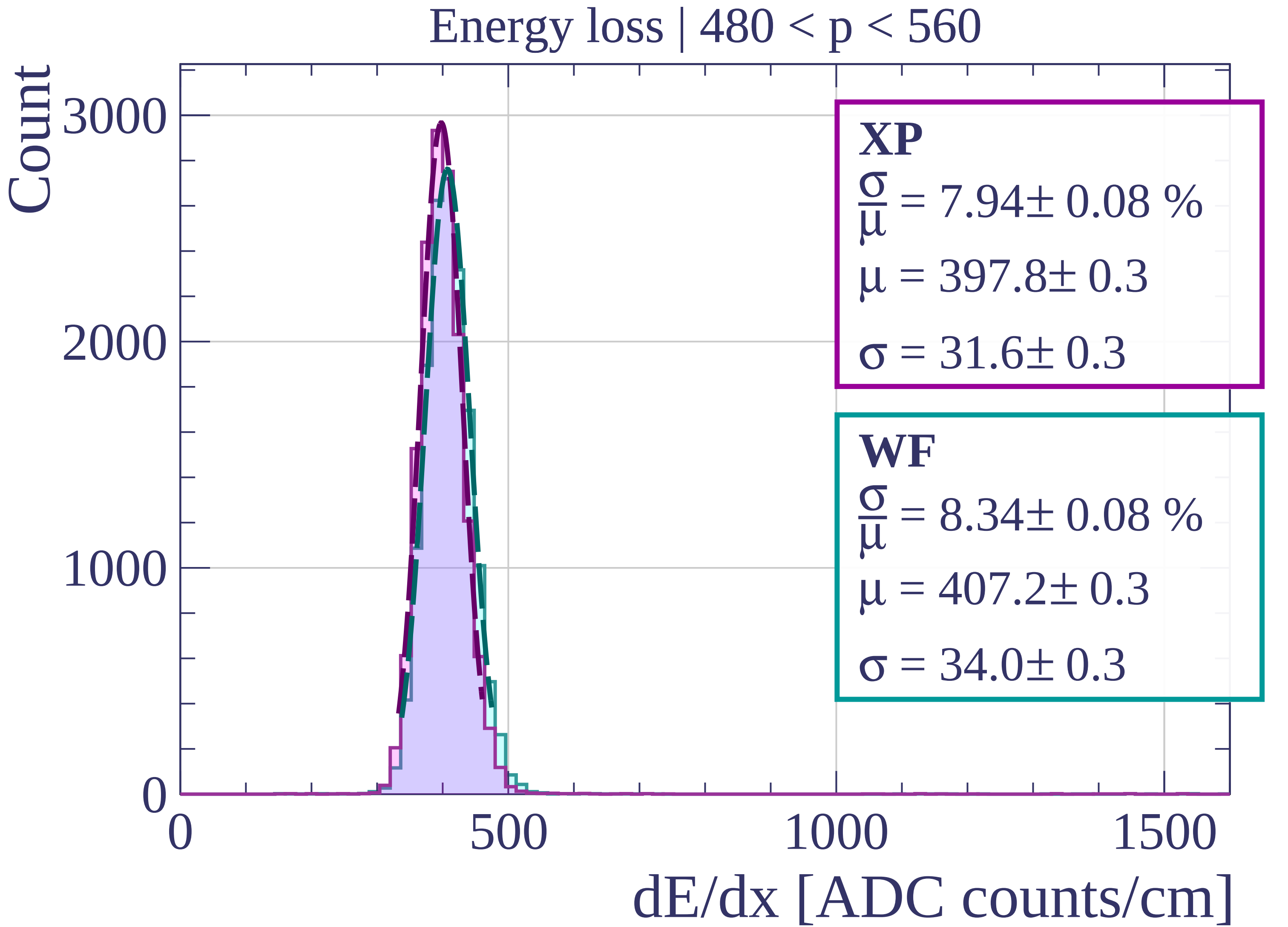
Energy loss | 240 < p < 320
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.01± 0.08 %
μ = 401.8± 0.3
σ = 32.2± 0.3
WF
σ/μ = 8.30± 0.08 %
μ = 410.0± 0.3
σ = 34.0± 0.3



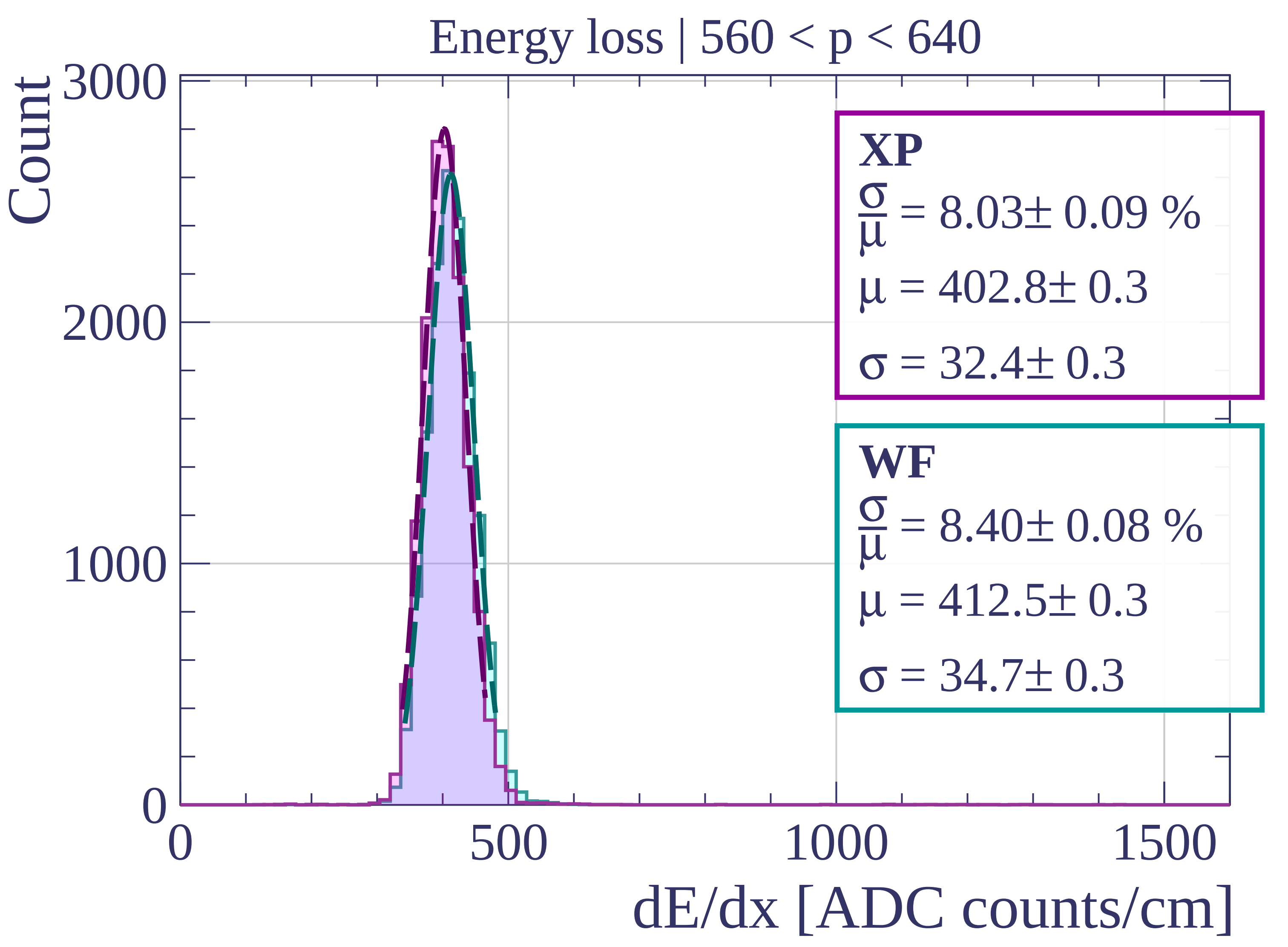
Energy loss | 320 < p < 400
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.07± 0.08 %
μ = 393.9± 0.3
σ = 31.8± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.28± 0.09 %
μ = 402.2± 0.3
σ = 33.3± 0.3



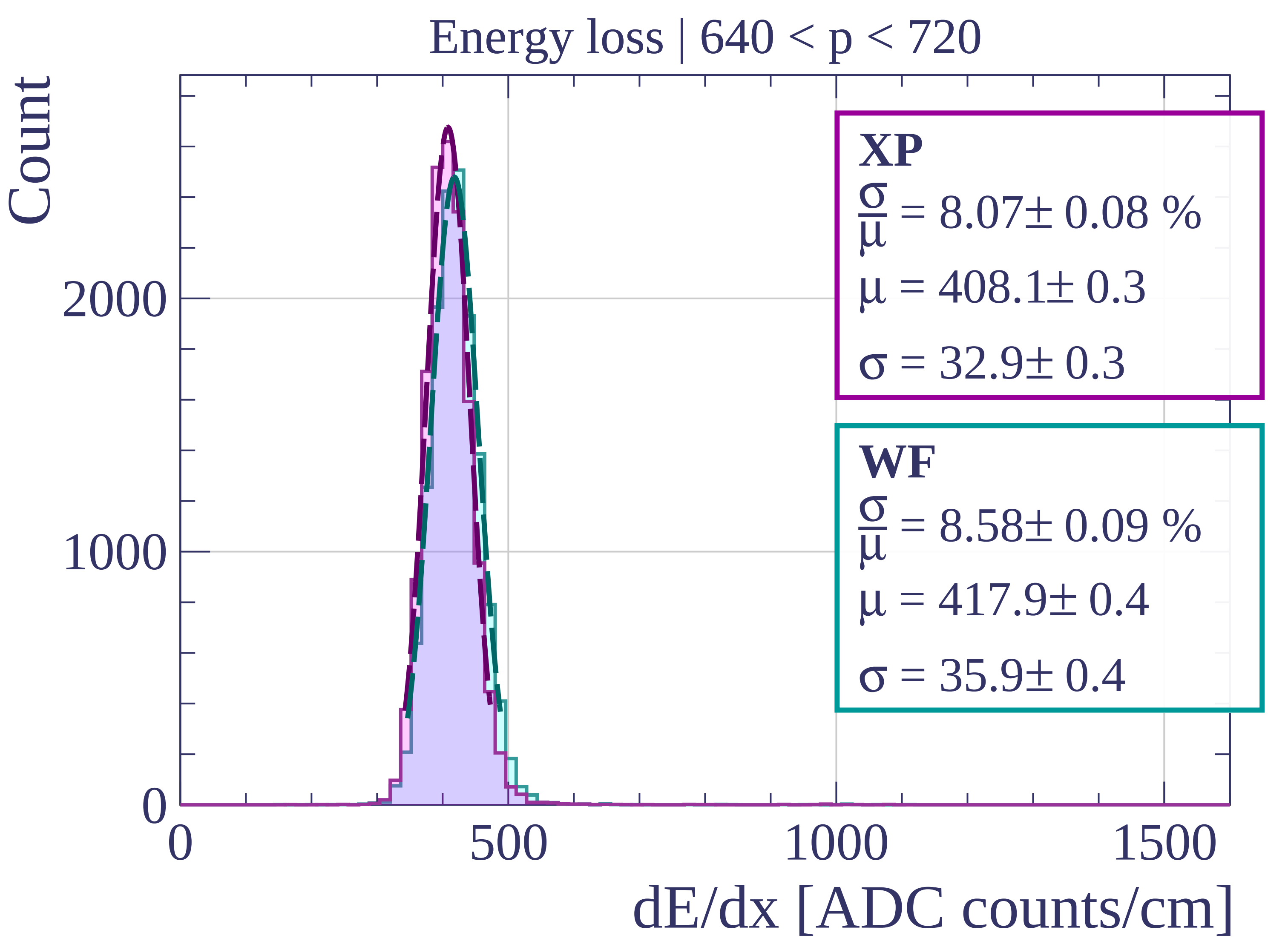
Energy loss | 400 < p < 480
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.04± 0.08 %
μ = 394.6± 0.3
σ = 31.7± 0.3
WF
σ/μ = 8.22± 0.09 %
μ = 403.0± 0.3
σ = 33.1± 0.3



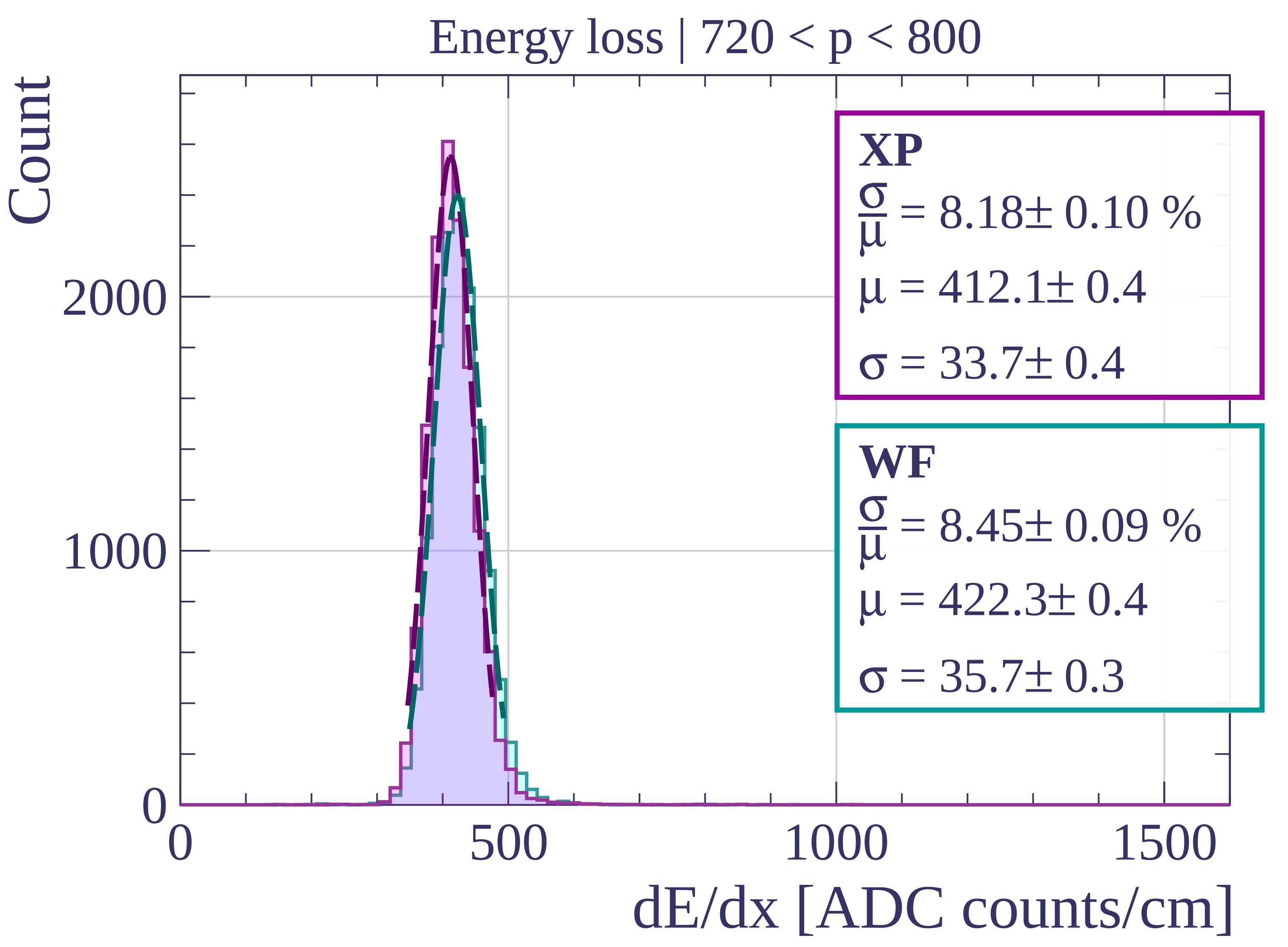
Energy loss | 480 < p < 560
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.94± 0.08 %
μ = 397.8± 0.3
σ = 31.6± 0.3
WF
σ/μ = 8.34± 0.08 %
μ = 407.2± 0.3
σ = 34.0± 0.3

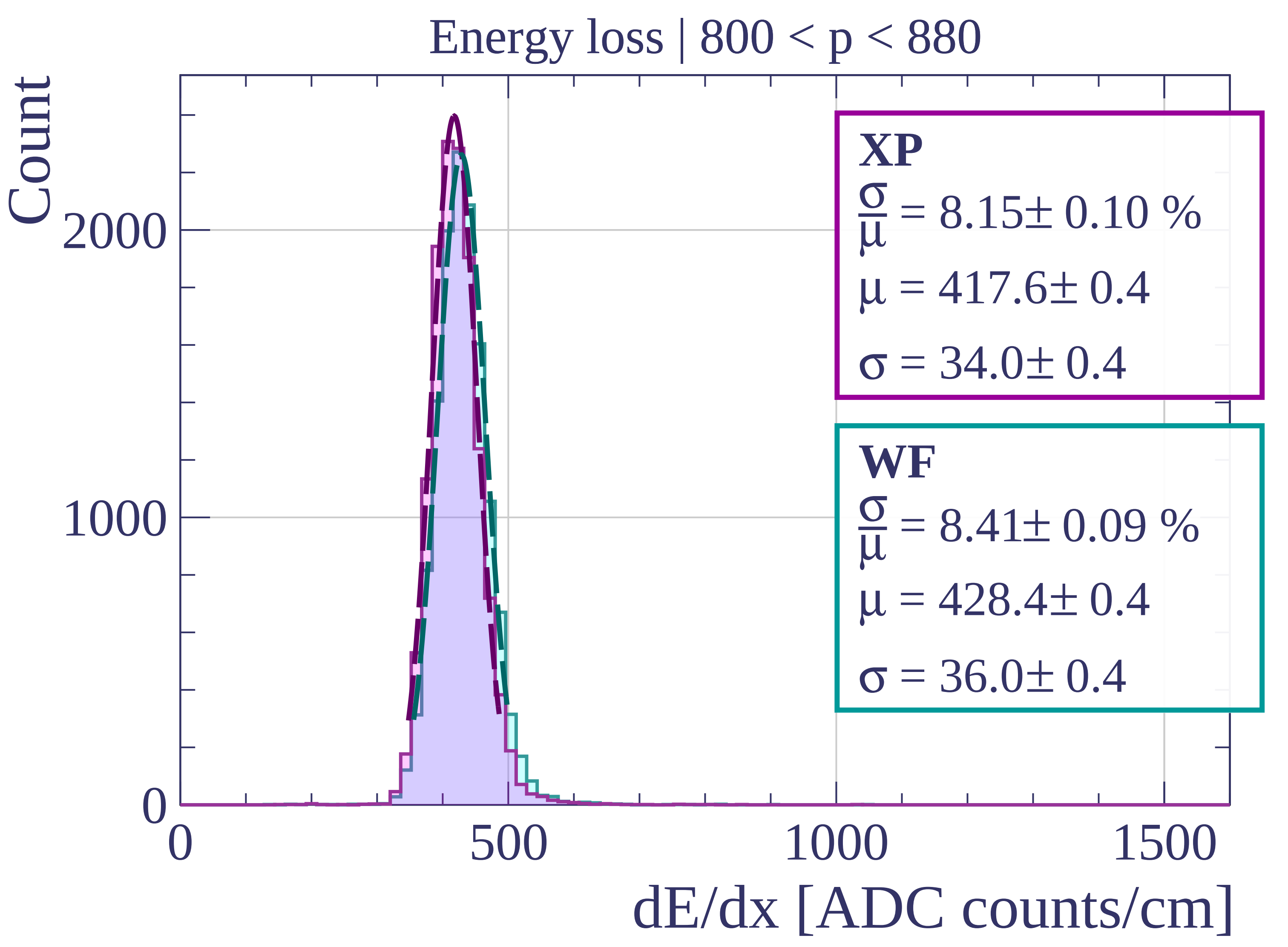


Energy loss | 560 < p < 640
Count
3000
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.03± 0.09 %
μ = 402.8± 0.3
σ = 32.4± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.40± 0.08 %
μ = 412.5± 0.3
σ = 34.7± 0.3

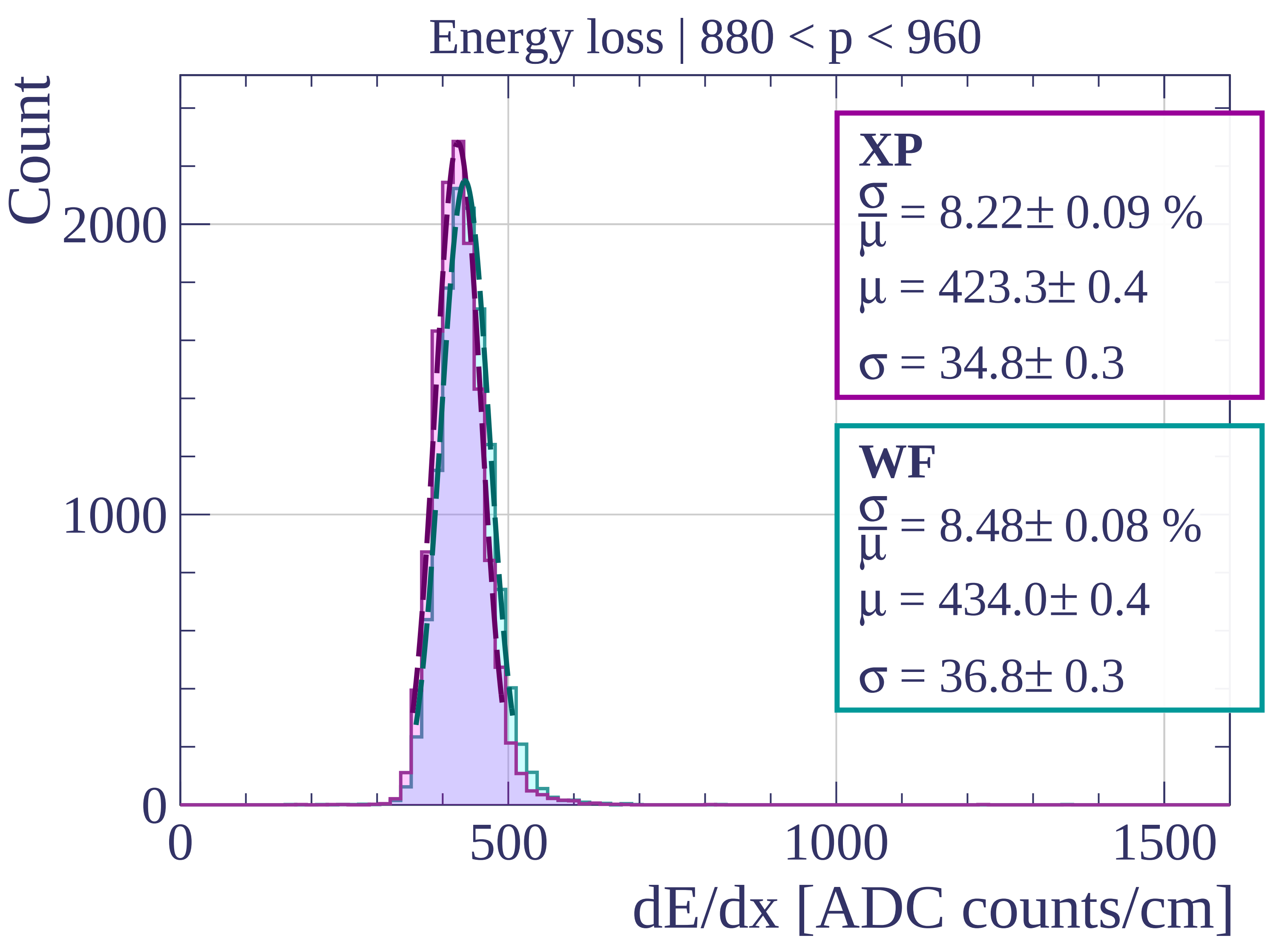


Energy loss | 640 < p < 720
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.07± 0.08 %
μ = 408.1± 0.3
σ = 32.9± 0.3
WF
σ/μ = 8.58± 0.09 %
μ = 417.9± 0.4
σ = 35.9± 0.4

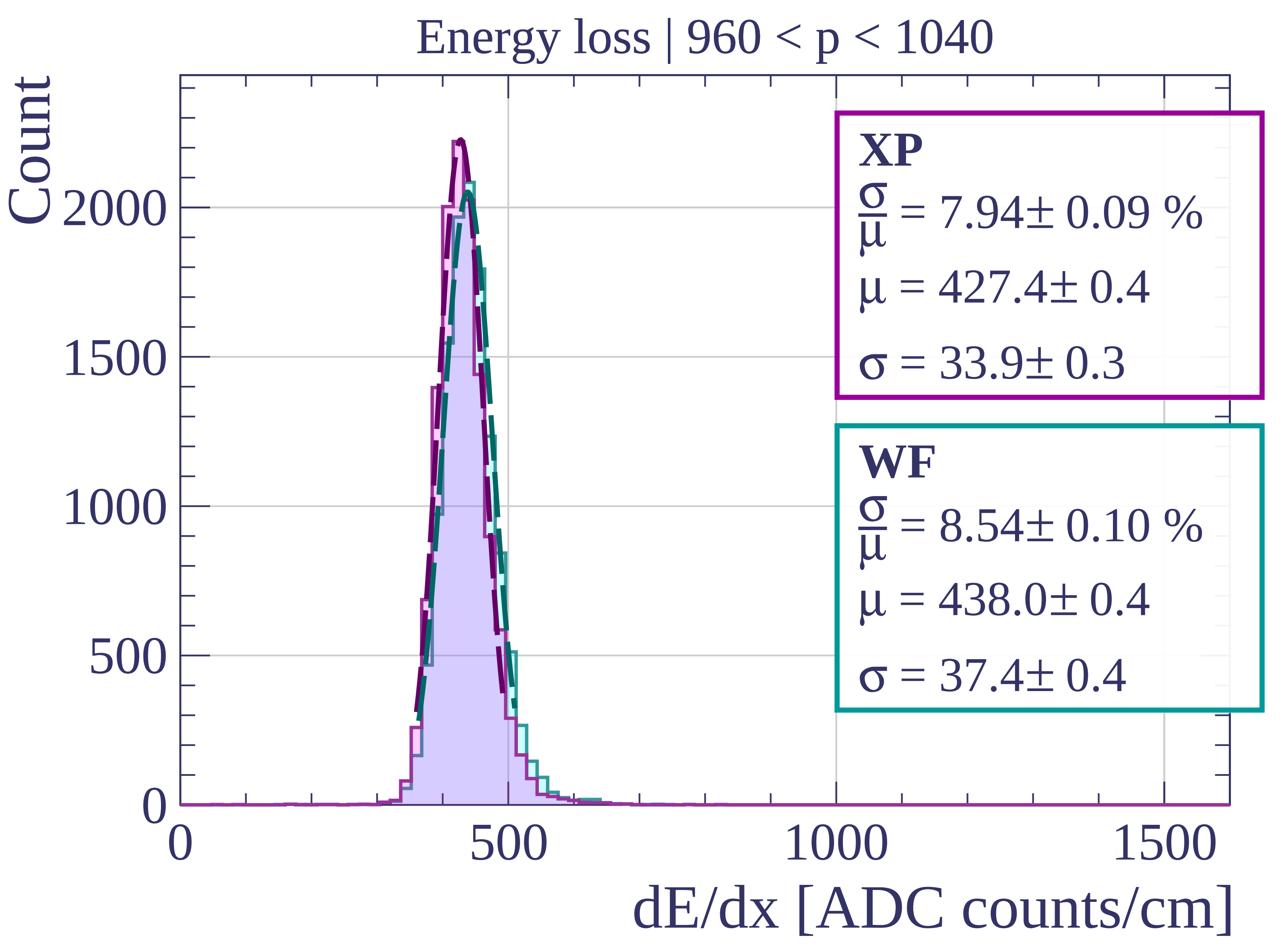

Energy loss | 720 < p < 800
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.18± 0.10 %
μ = 412.1± 0.4
σ = 33.7± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.45± 0.09 %
μ = 422.3± 0.4
σ = 35.7± 0.3



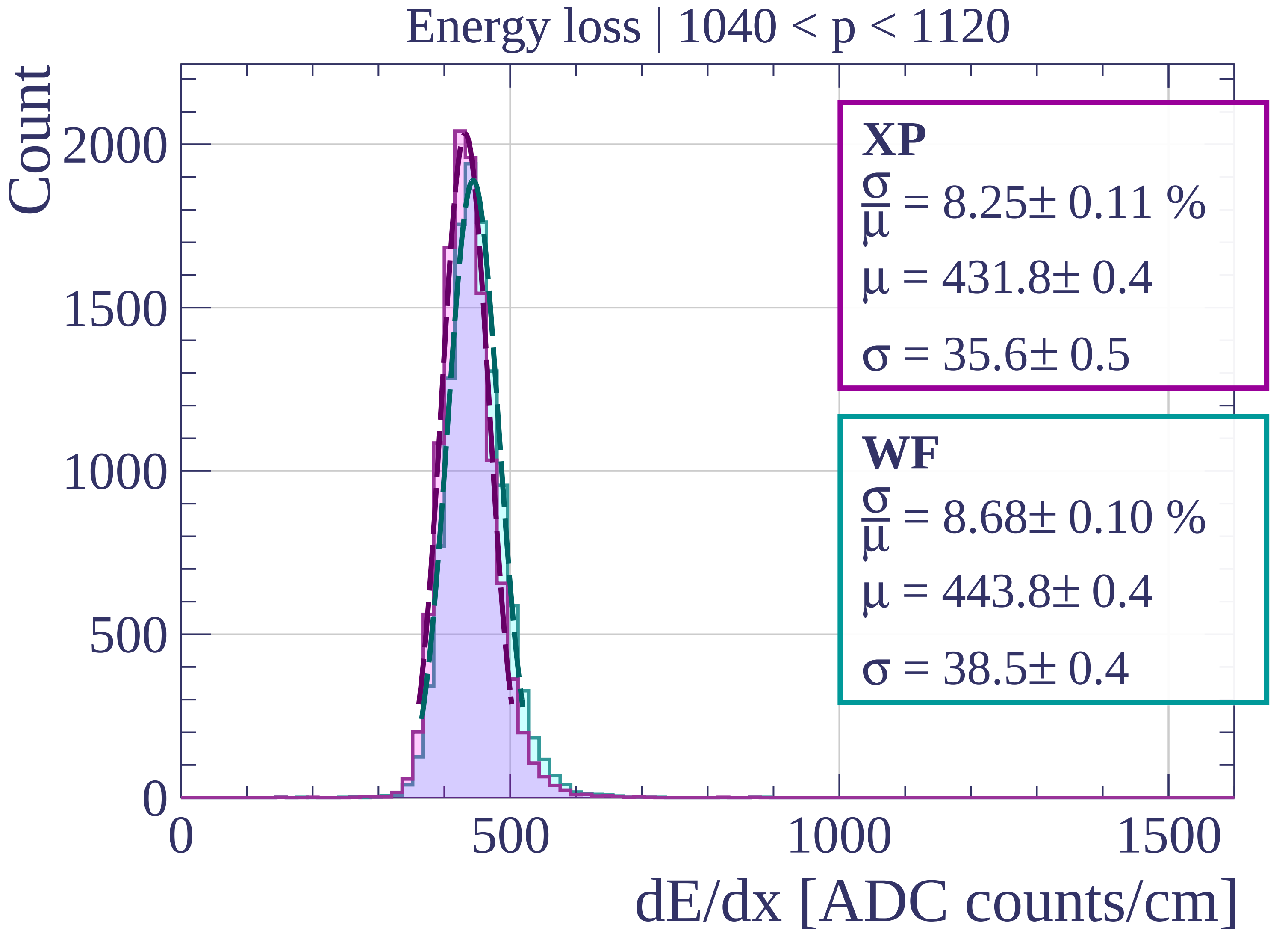
Energy loss | 800 < p < 880
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
1000
2000
XP
$\frac{\sigma}{\mu}$ = 8.15± 0.10 %
μ = 417.6± 0.4
σ = 34.0± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.41± 0.09 %
μ = 428.4± 0.4
σ = 36.0± 0.4



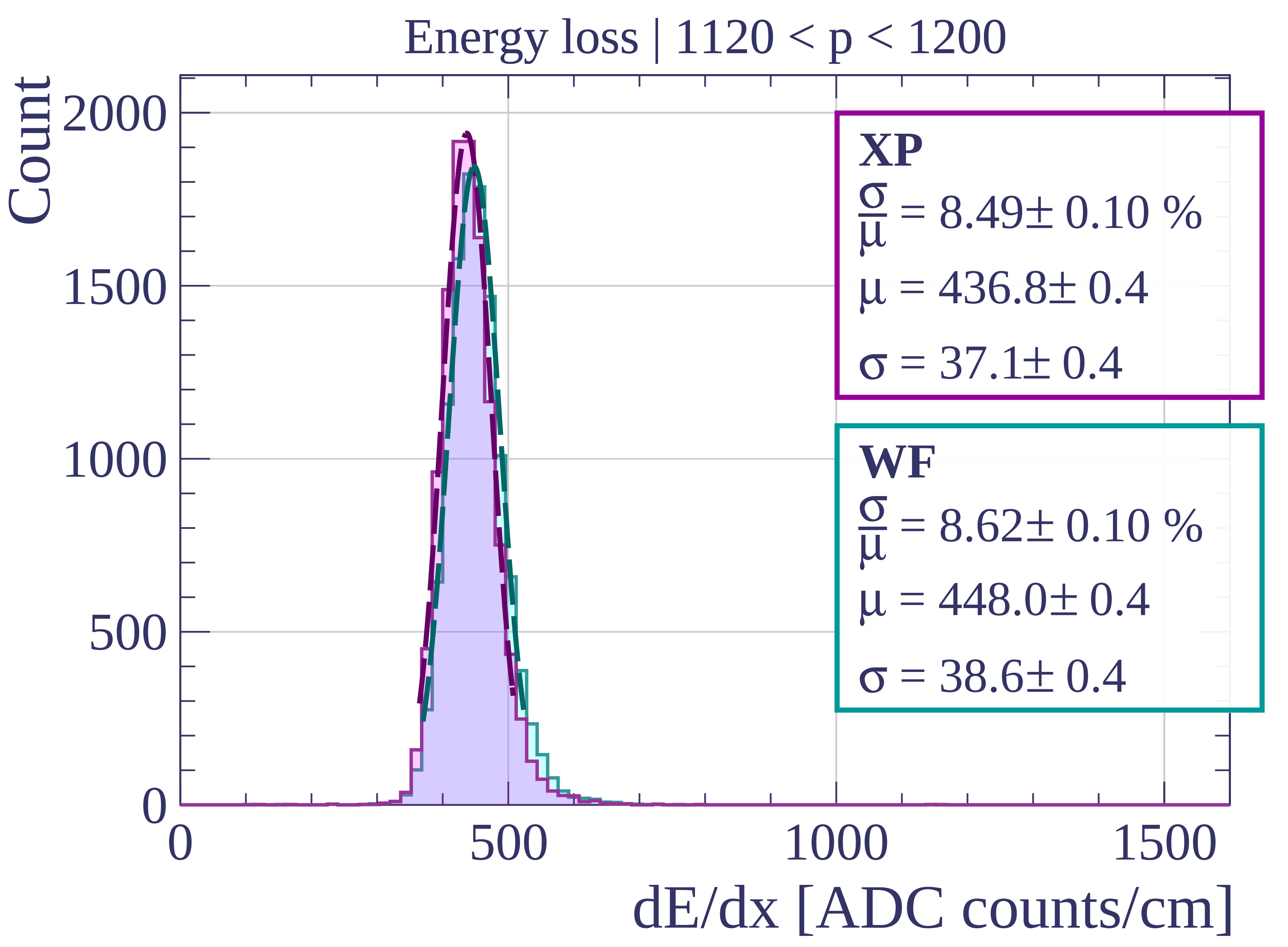
Energy loss | 880 < p < 960
Count
2000
1000
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.22± 0.09 %
μ = 423.3± 0.4
σ = 34.8± 0.3
WF
σ/μ = 8.48± 0.08 %
μ = 434.0± 0.4
σ = 36.8± 0.3



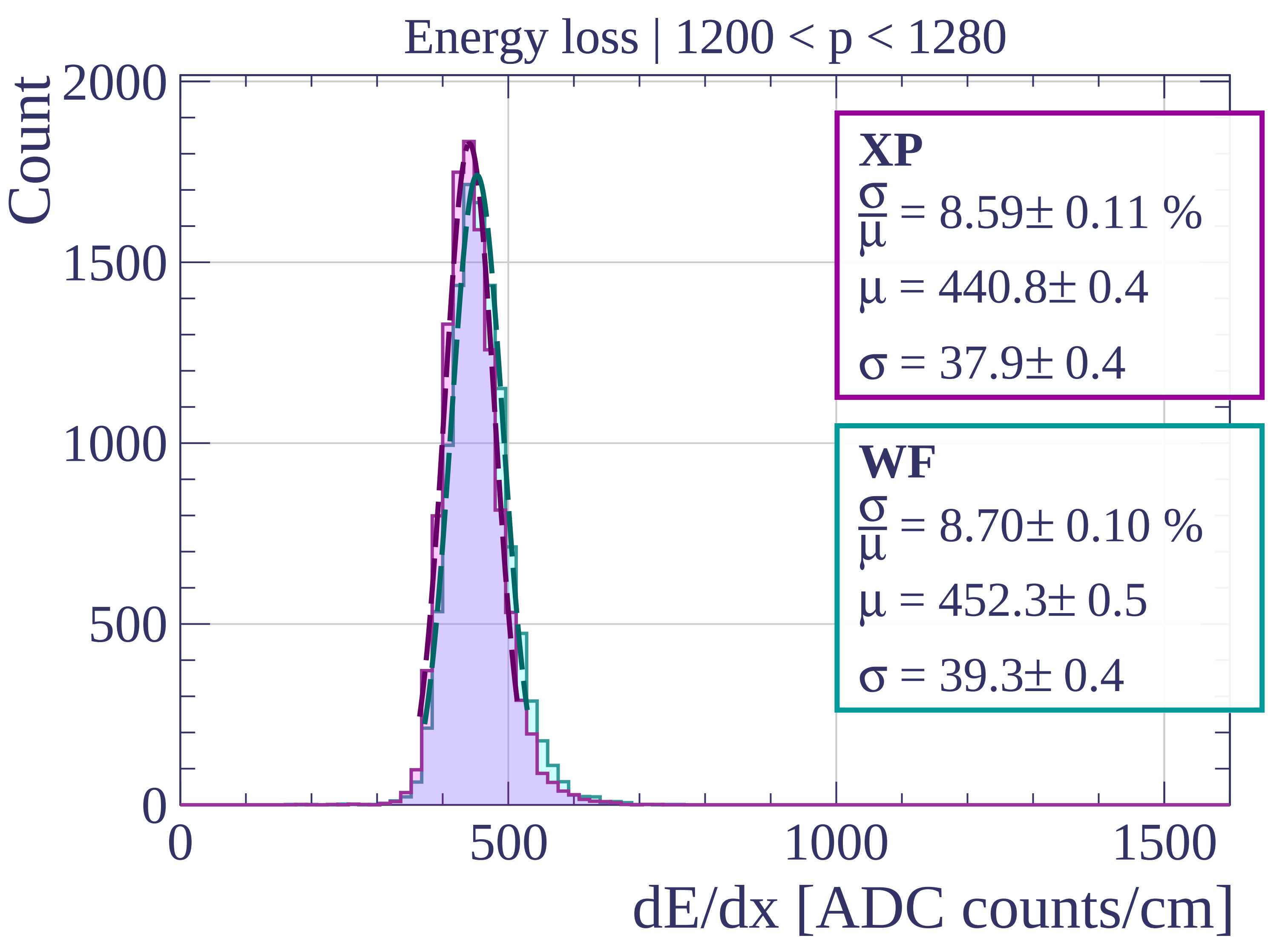
Energy loss | 960 < p < 1040
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
$\frac{\sigma}{\mu}$ = 7.94± 0.09 %
μ = 427.4± 0.4
σ = 33.9± 0.3
WF
$\frac{\sigma}{\mu}$ = 8.54± 0.10 %
μ = 438.0± 0.4
σ = 37.4± 0.4



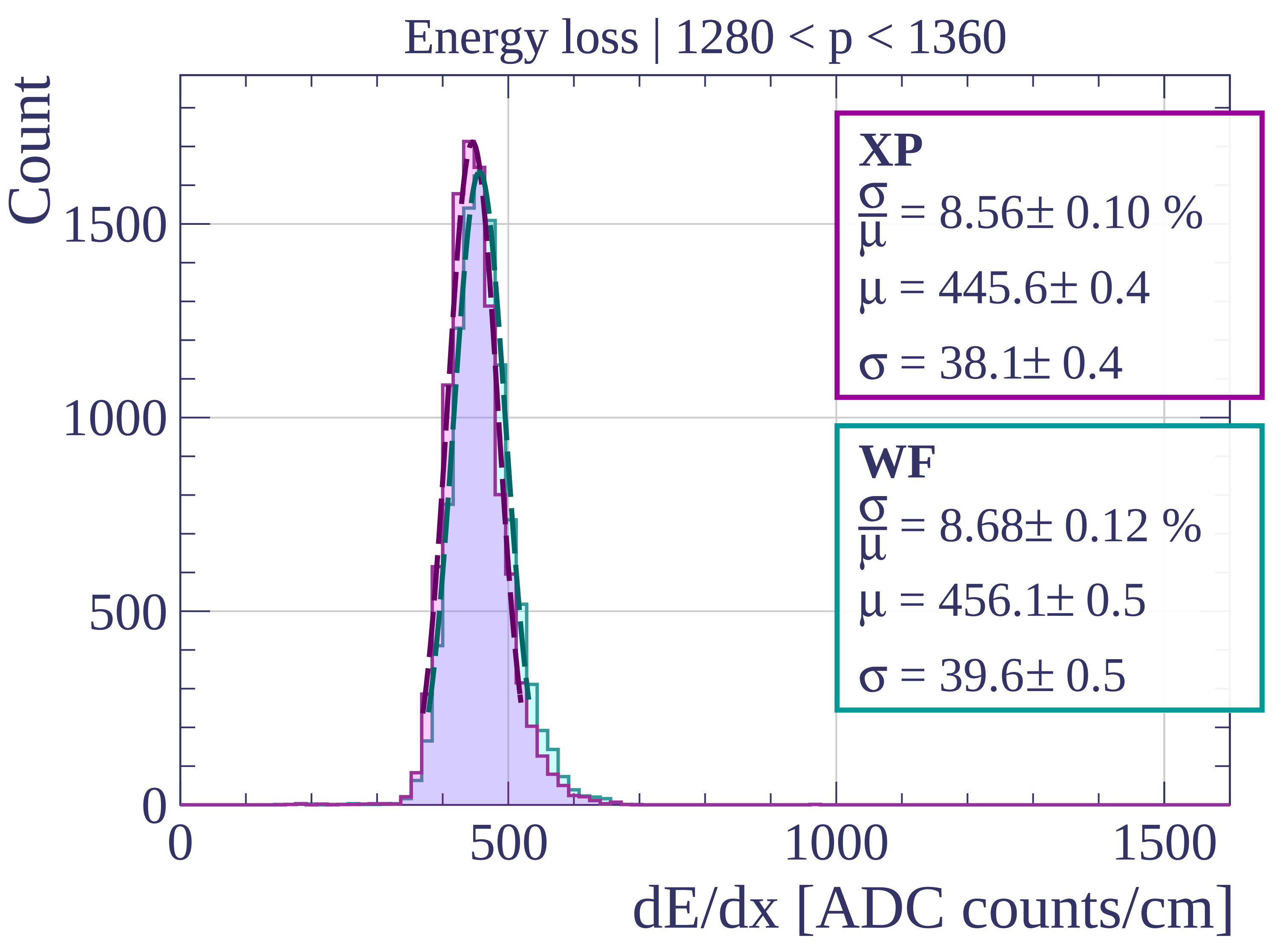
Energy loss | 1040 < p < 1120
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
2000
XP
$\frac{\sigma}{\mu}$ = 8.25± 0.11 %
μ = 431.8± 0.4
σ = 35.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 8.68± 0.10 %
μ = 443.8± 0.4
σ = 38.5± 0.4



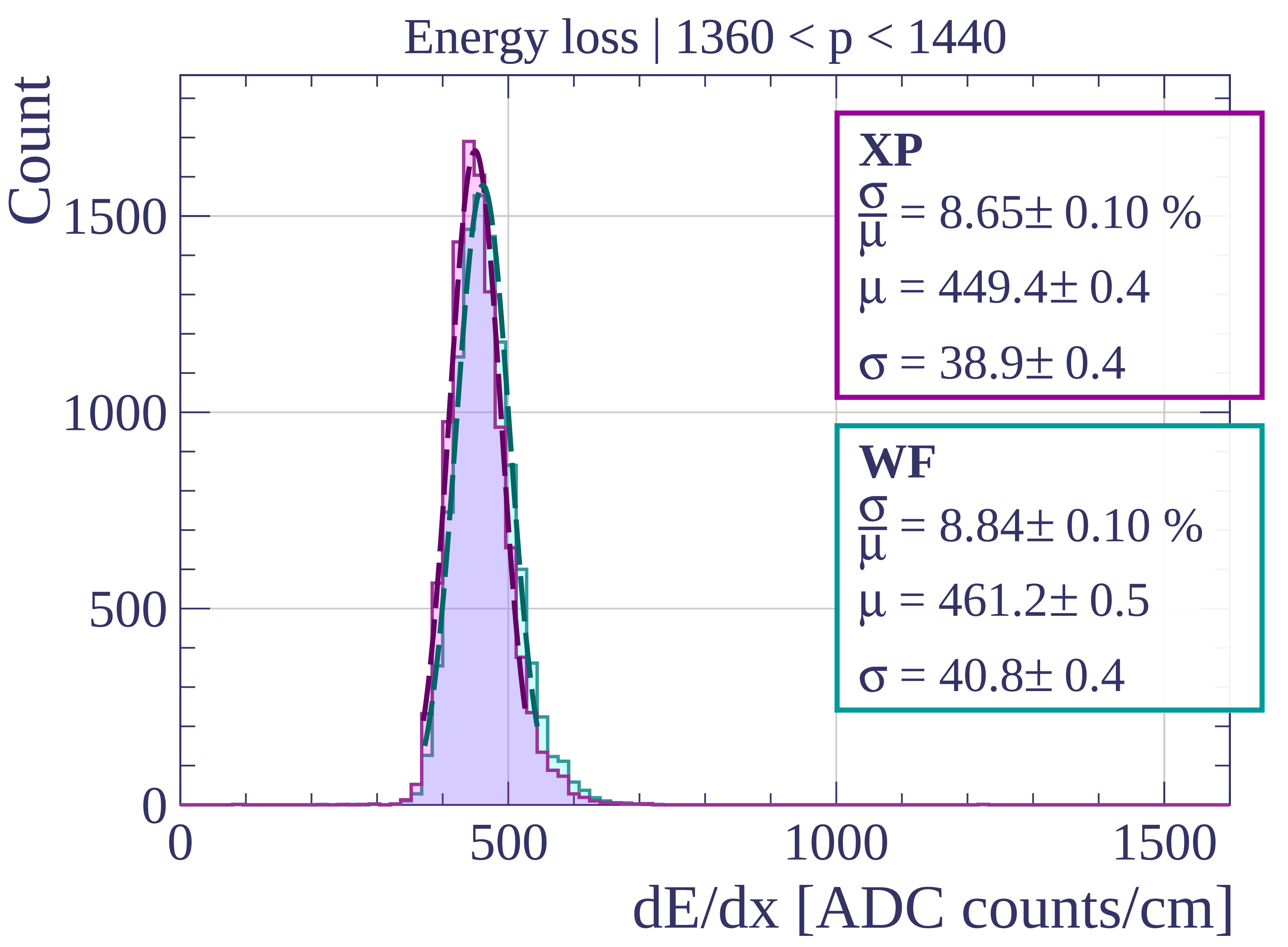
Energy loss | 1120 < p < 1200
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.49± 0.10 %
μ = 436.8± 0.4
σ = 37.1± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.62± 0.10 %
μ = 448.0± 0.4
σ = 38.6± 0.4

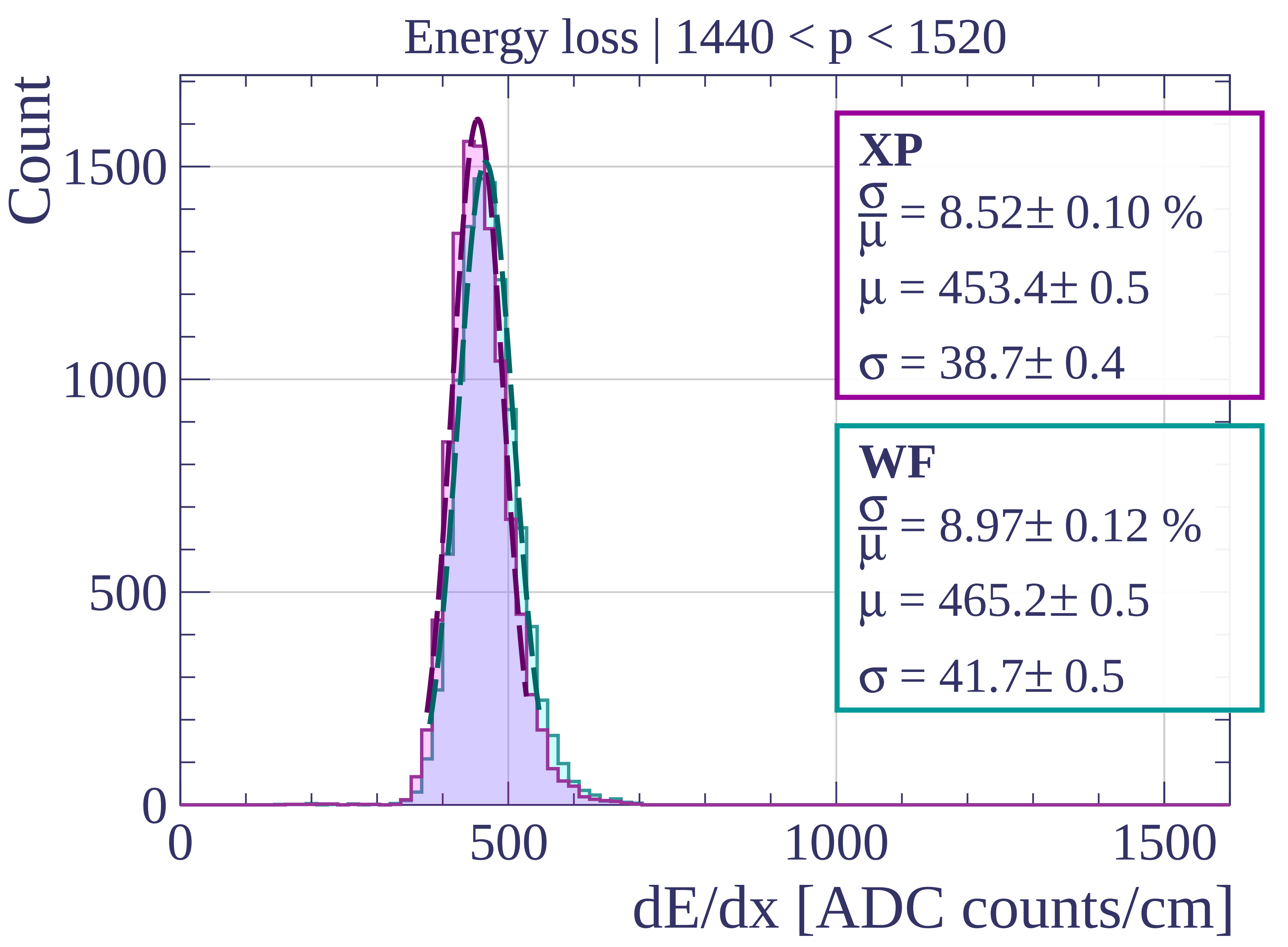


Energy loss | 1200 < p < 1280
Count
2000
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.59± 0.11 %
μ = 440.8± 0.4
σ = 37.9± 0.4
WF
σ/μ = 8.70± 0.10 %
μ = 452.3± 0.5
σ = 39.3± 0.4

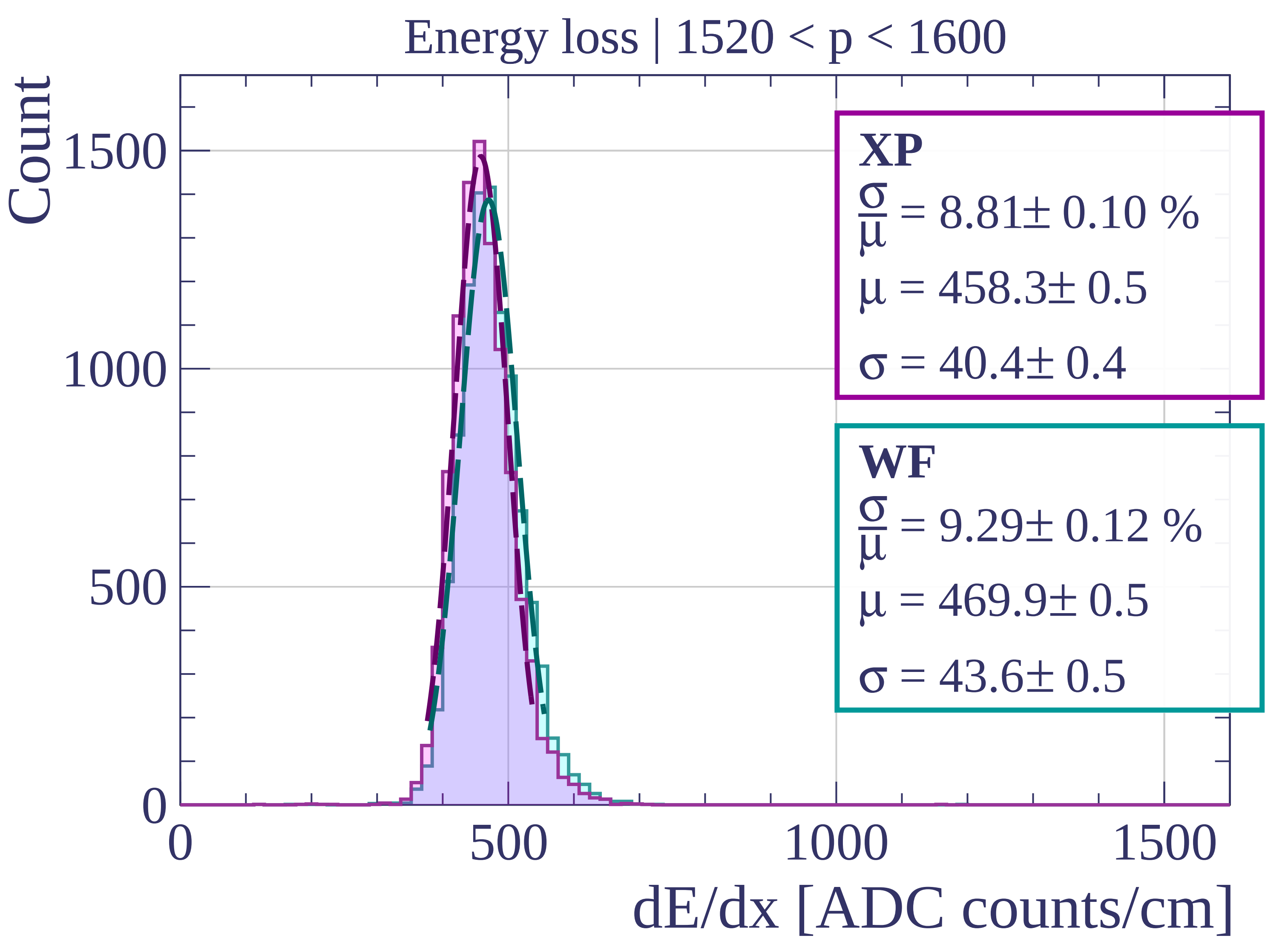


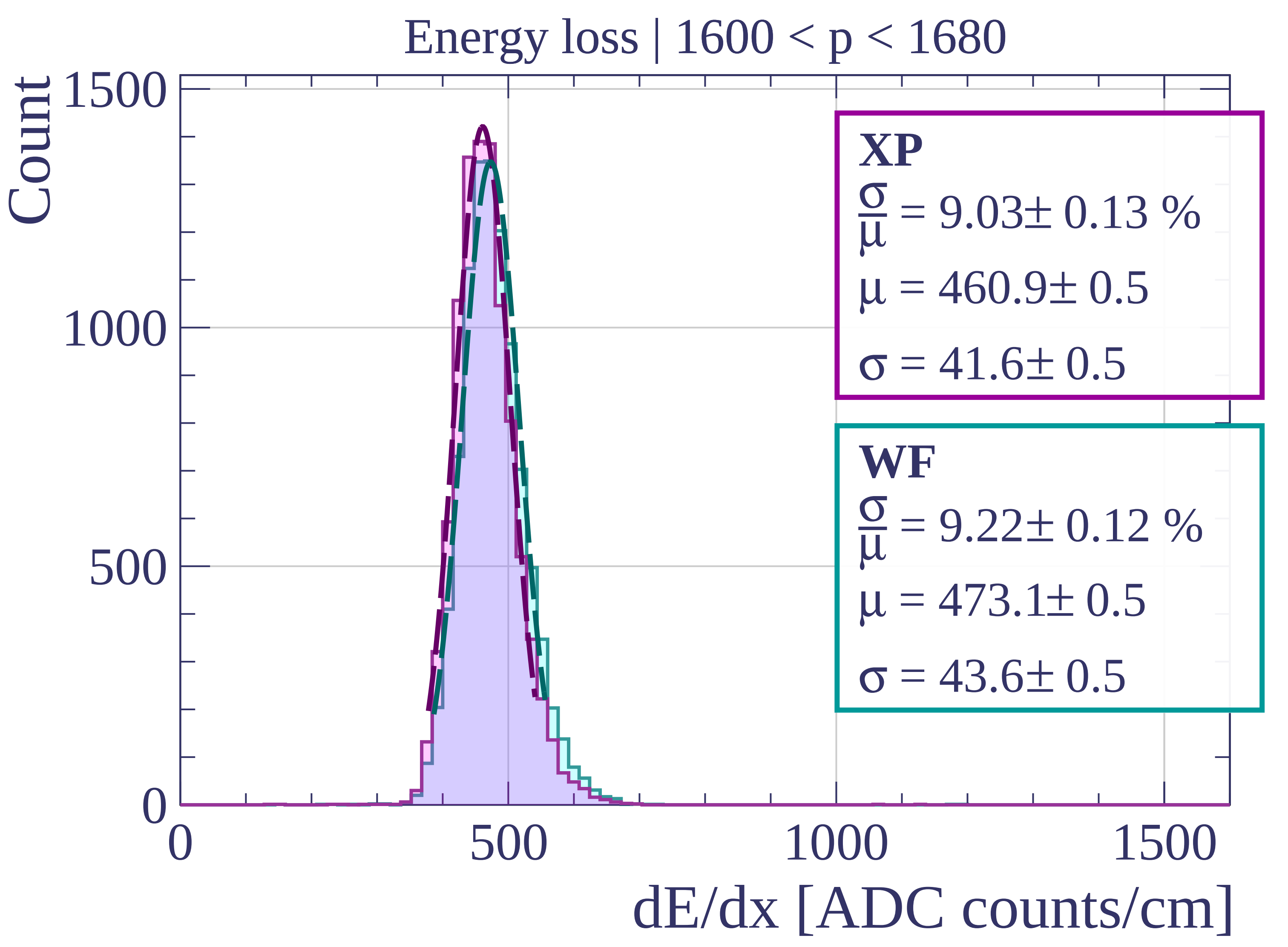
Energy loss | 1280 < p < 1360
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.56± 0.10 %
μ = 445.6± 0.4
σ = 38.1± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.68± 0.12 %
μ = 456.1± 0.5
σ = 39.6± 0.5

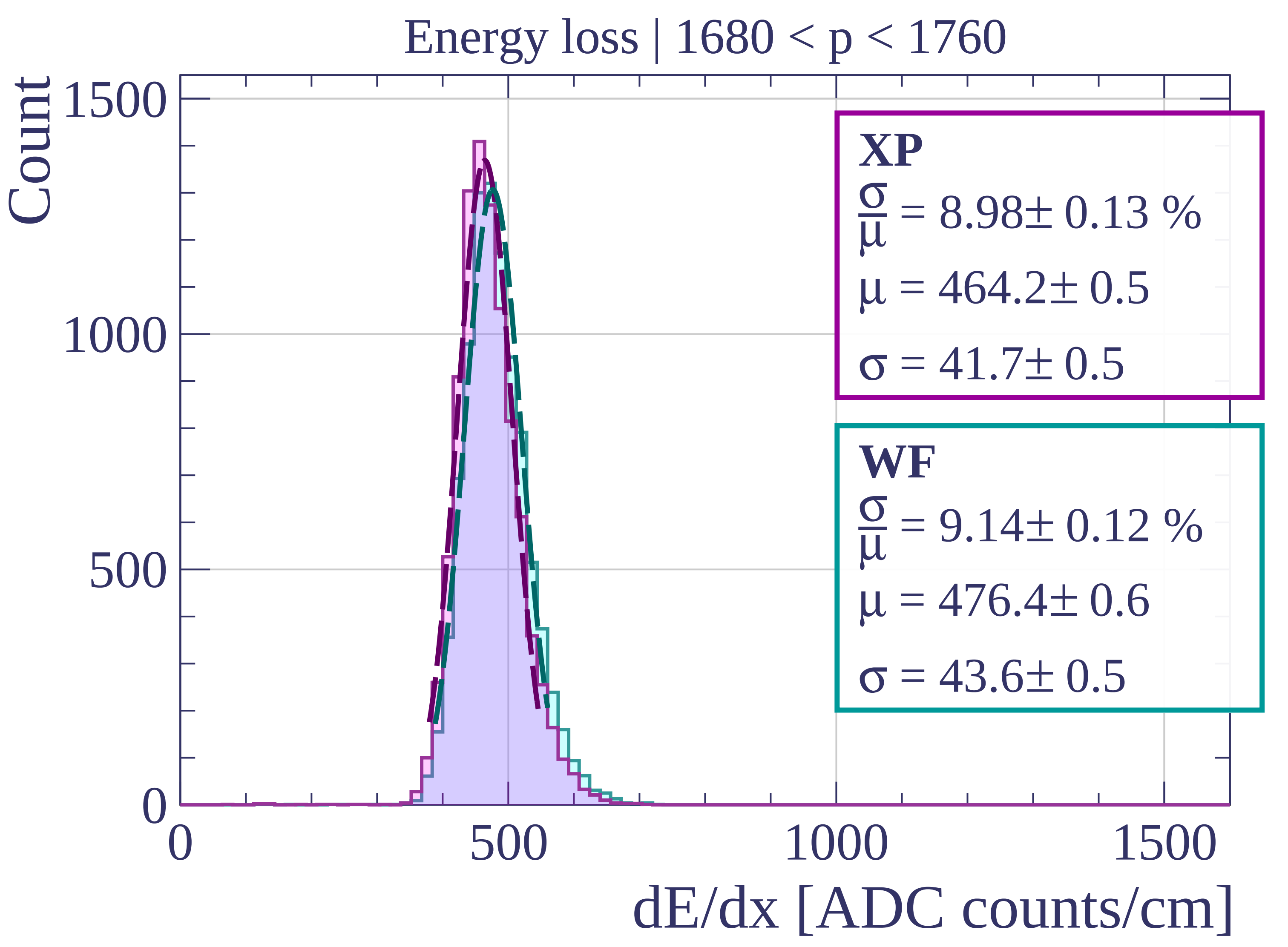

Energy loss | 1360 < p < 1440
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 8.65± 0.10 %
μ = 449.4± 0.4
σ = 38.9± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.84± 0.10 %
μ = 461.2± 0.5
σ = 40.8± 0.4



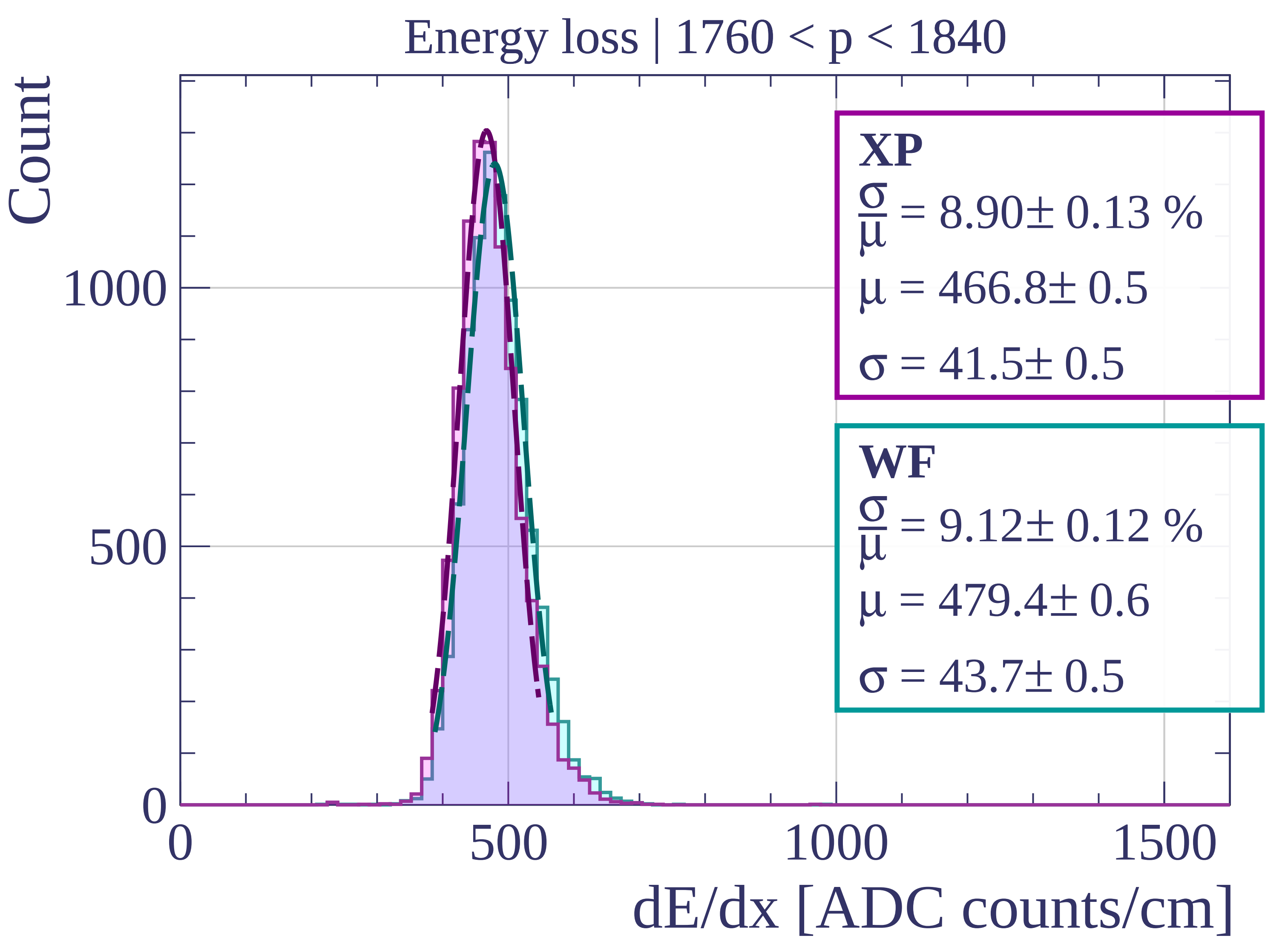
Energy loss | 1440 < p < 1520
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 8.52± 0.10 %
μ = 453.4± 0.5
σ = 38.7± 0.4
WF
$\frac{\sigma}{\mu}$ = 8.97± 0.12 %
μ = 465.2± 0.5
σ = 41.7± 0.5


Energy loss | 1520 < p < 1600
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
σ/μ = 8.81± 0.10 %
μ = 458.3± 0.5
σ = 40.4± 0.4
WF
σ/μ = 9.29± 0.12 %
μ = 469.9± 0.5
σ = 43.6± 0.5

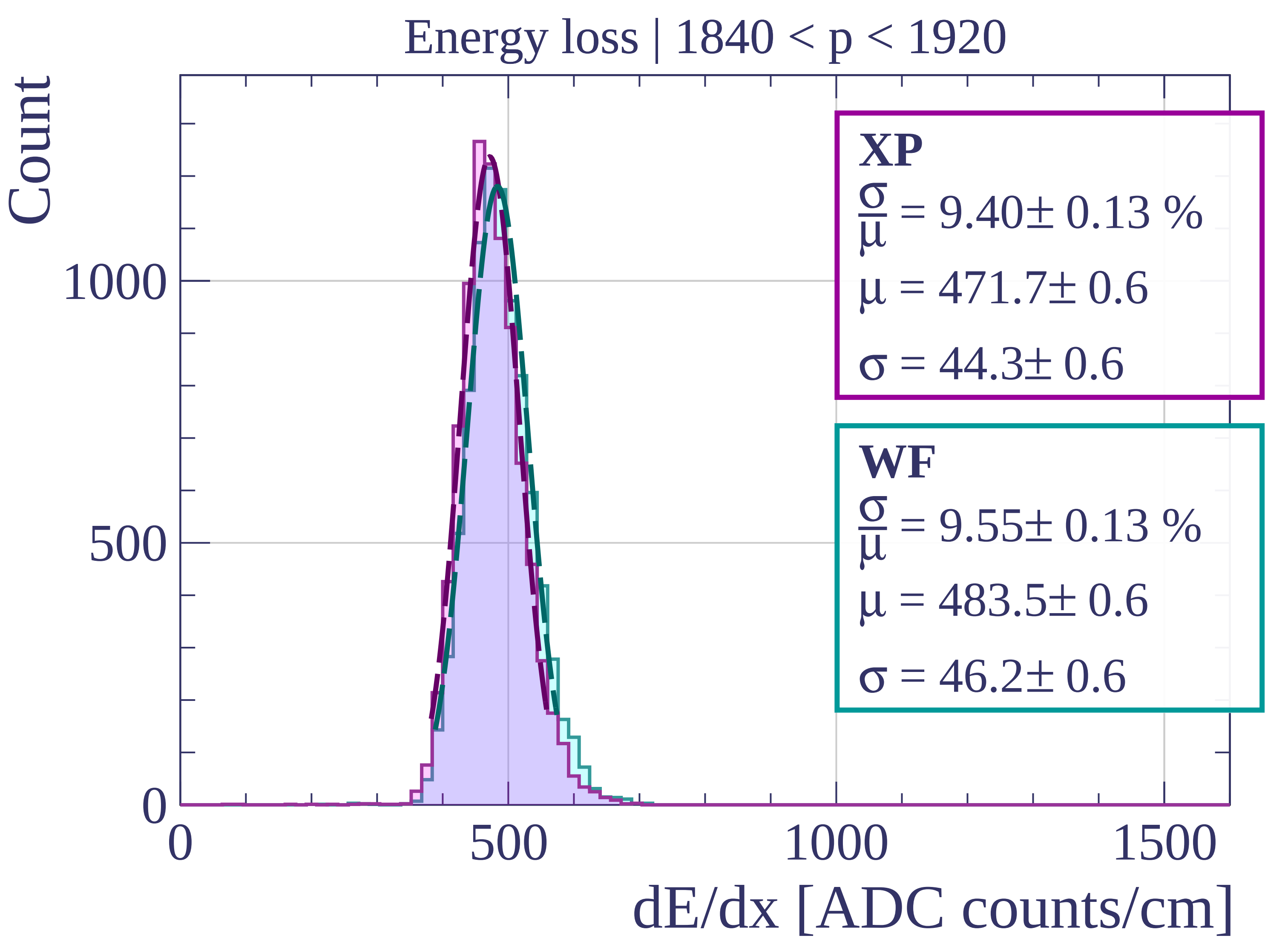

Energy loss | 1600 < p < 1680
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 9.03± 0.13 %
μ = 460.9± 0.5
σ = 41.6± 0.5
WF
σ/μ = 9.22± 0.12 %
μ = 473.1± 0.5
σ = 43.6± 0.5

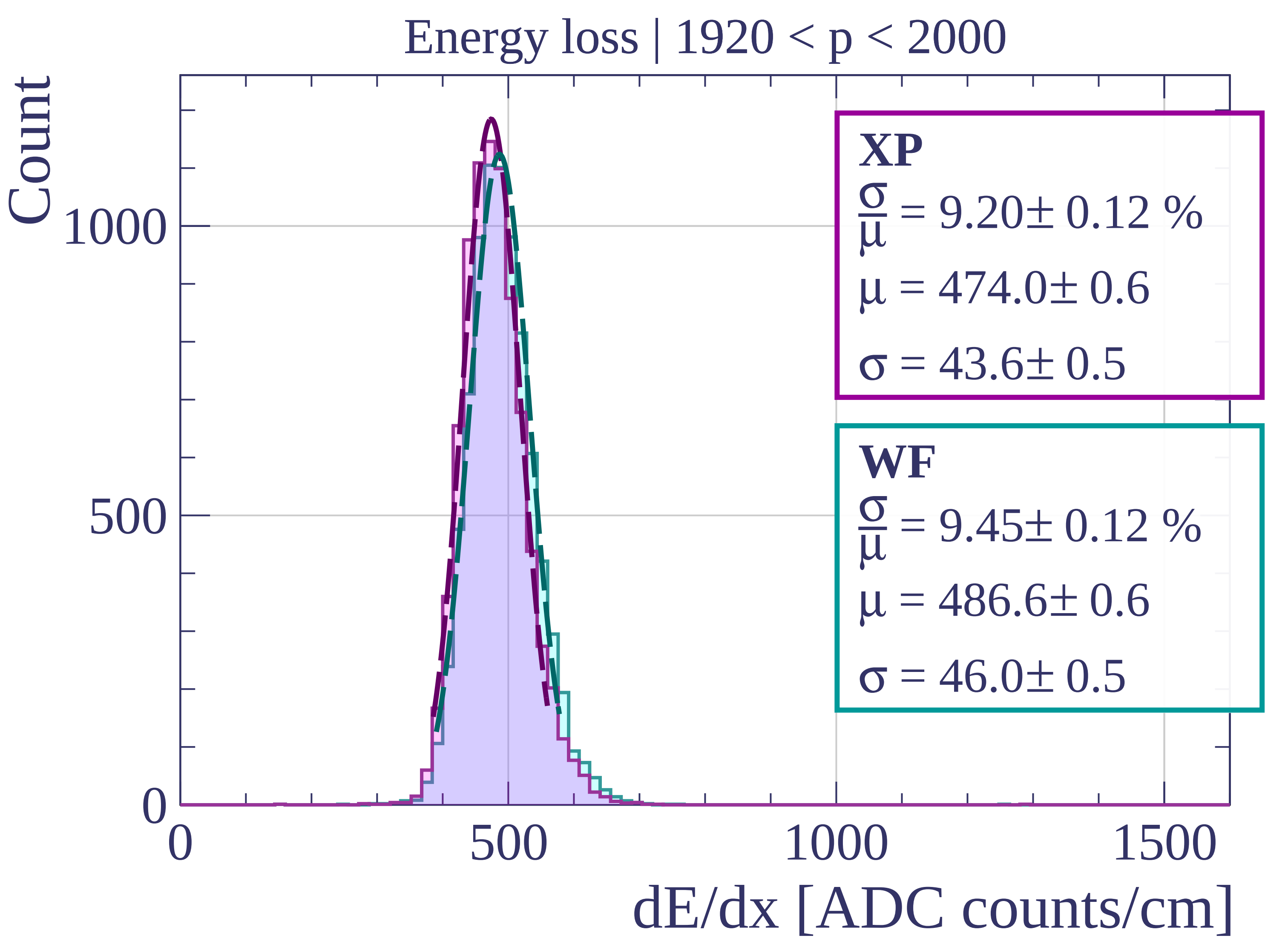


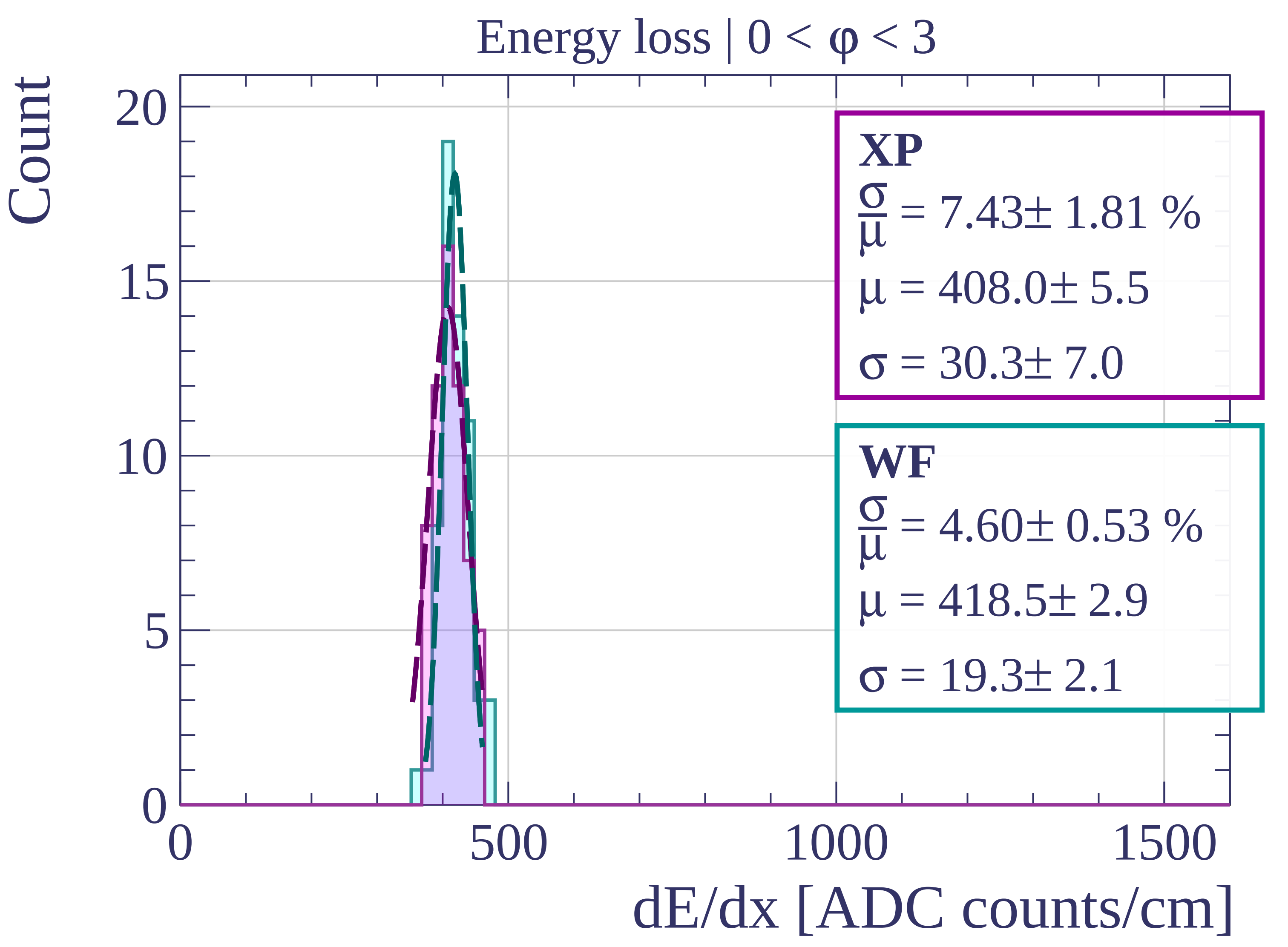
Energy loss | 1680 < p < 1760
Count
1500
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.98± 0.13 %
μ = 464.2± 0.5
σ = 41.7± 0.5
WF
σ/μ = 9.14± 0.12 %
μ = 476.4± 0.6
σ = 43.6± 0.5

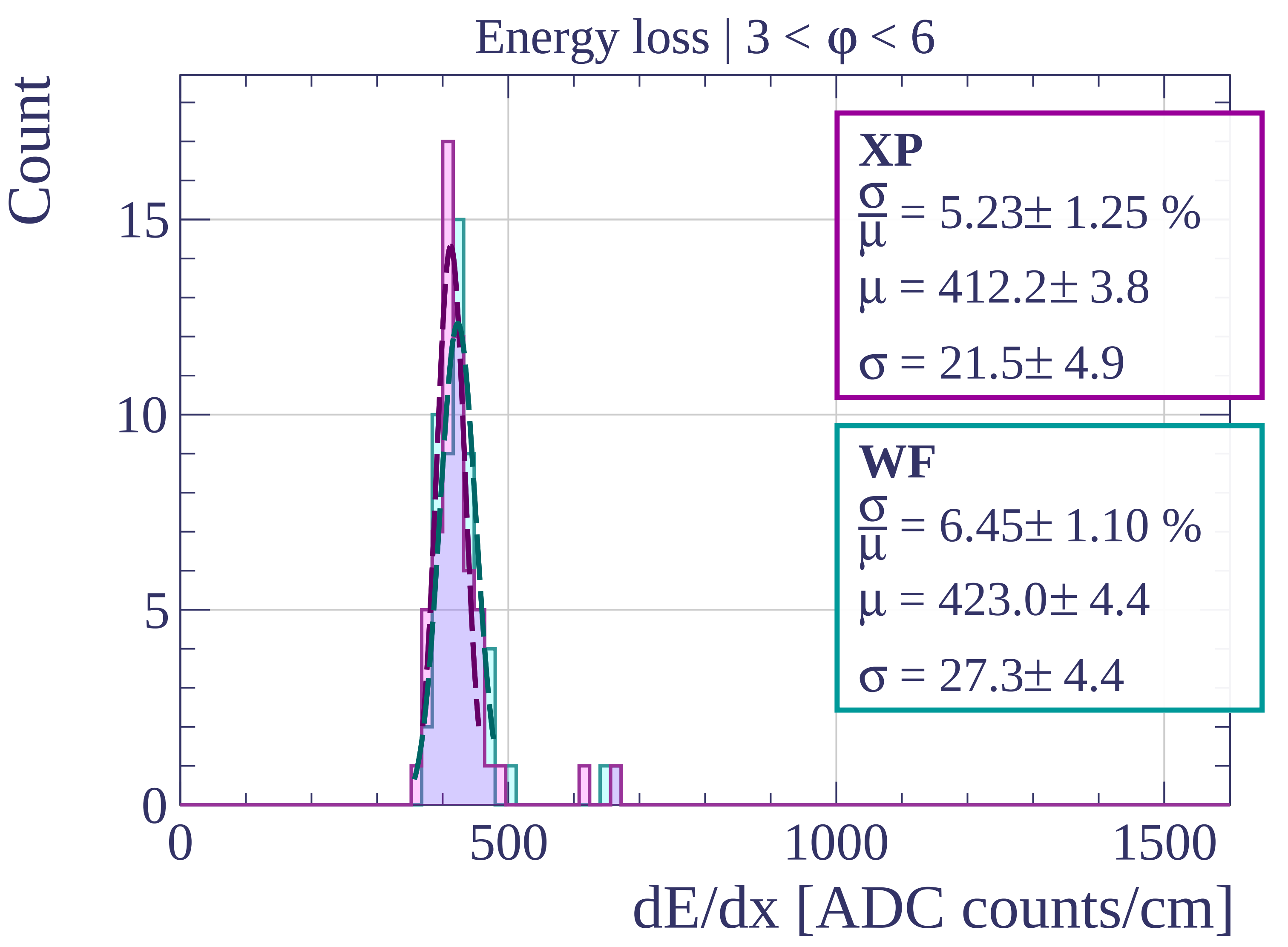


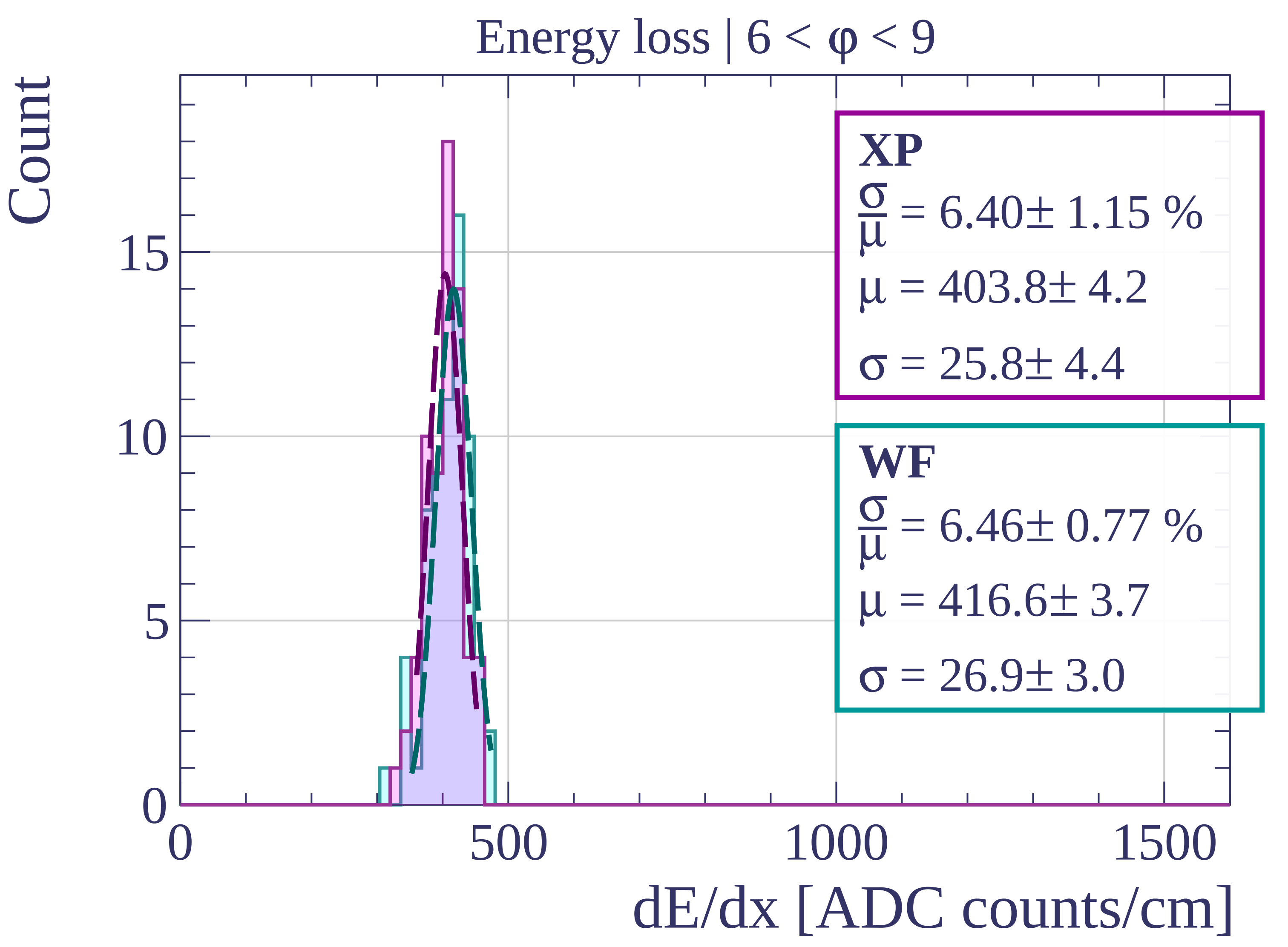
Energy loss | 1760 < p < 1840
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 8.90± 0.13 %
μ = 466.8± 0.5
σ = 41.5± 0.5
WF
$\frac{\sigma}{\mu}$ = 9.12± 0.12 %
μ = 479.4± 0.6
σ = 43.7± 0.5


Energy loss | 1840 < p < 1920
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
σ/μ = 9.40± 0.13 %
μ = 471.7± 0.6
σ = 44.3± 0.6
WF
σ/μ = 9.55± 0.13 %
μ = 483.5± 0.6
σ = 46.2± 0.6

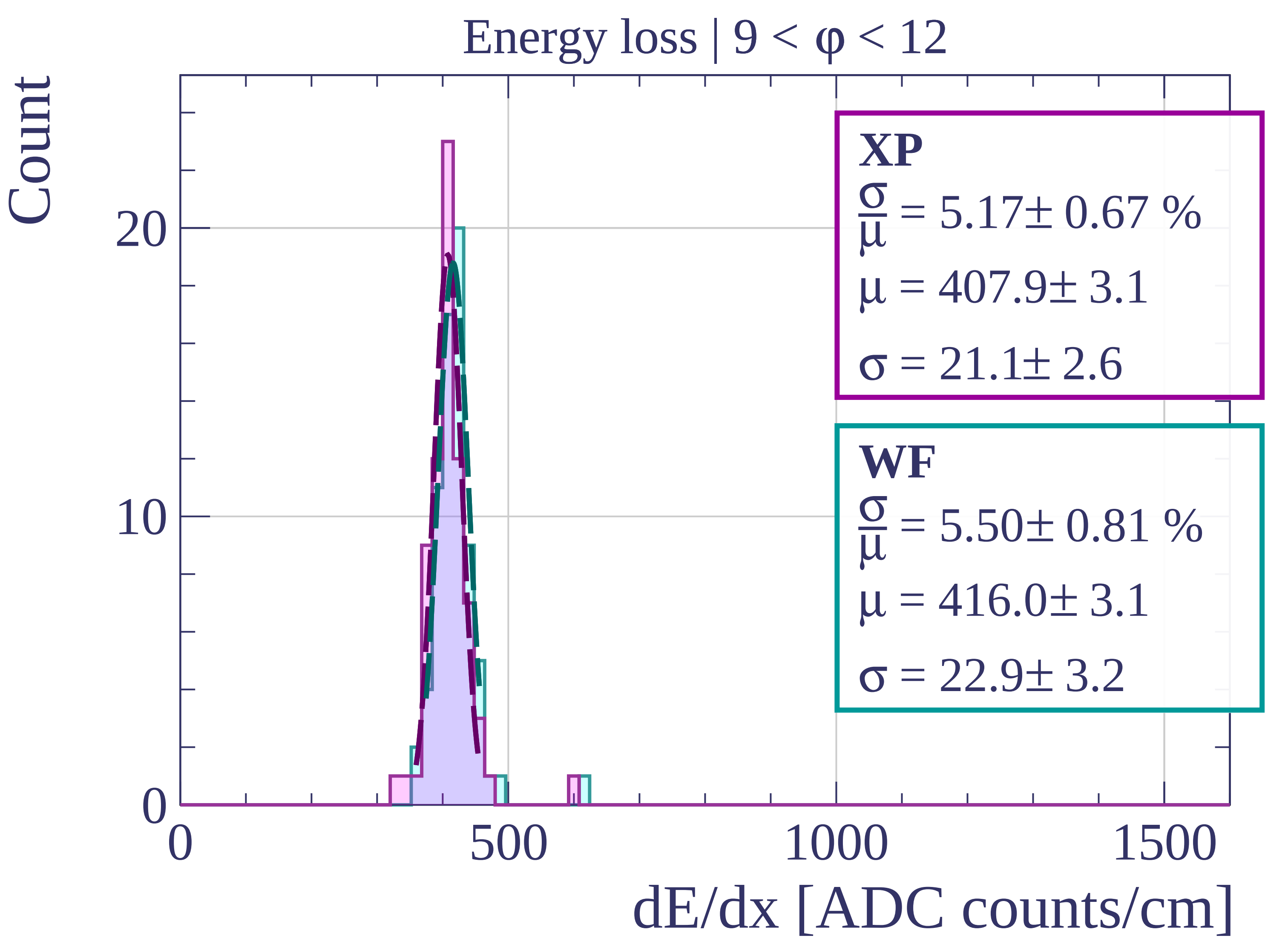

Energy loss | 1920 < p < 2000
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
0
500
1000
XP
$\frac{\sigma}{\mu}$ = 9.20± 0.12 %
μ = 474.0± 0.6
σ = 43.6± 0.5
WF
$\frac{\sigma}{\mu}$ = 9.45± 0.12 %
μ = 486.6± 0.6
σ = 46.0± 0.5


Energy loss | 0 < φ < 3
Count
dE/dx [ADC counts/cm]
0
5
10
15
20
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 7.43± 1.81 %
μ = 408.0± 5.5
σ = 30.3± 7.0
WF
$\frac{\sigma}{\mu}$ = 4.60± 0.53 %
μ = 418.5± 2.9
σ = 19.3± 2.1

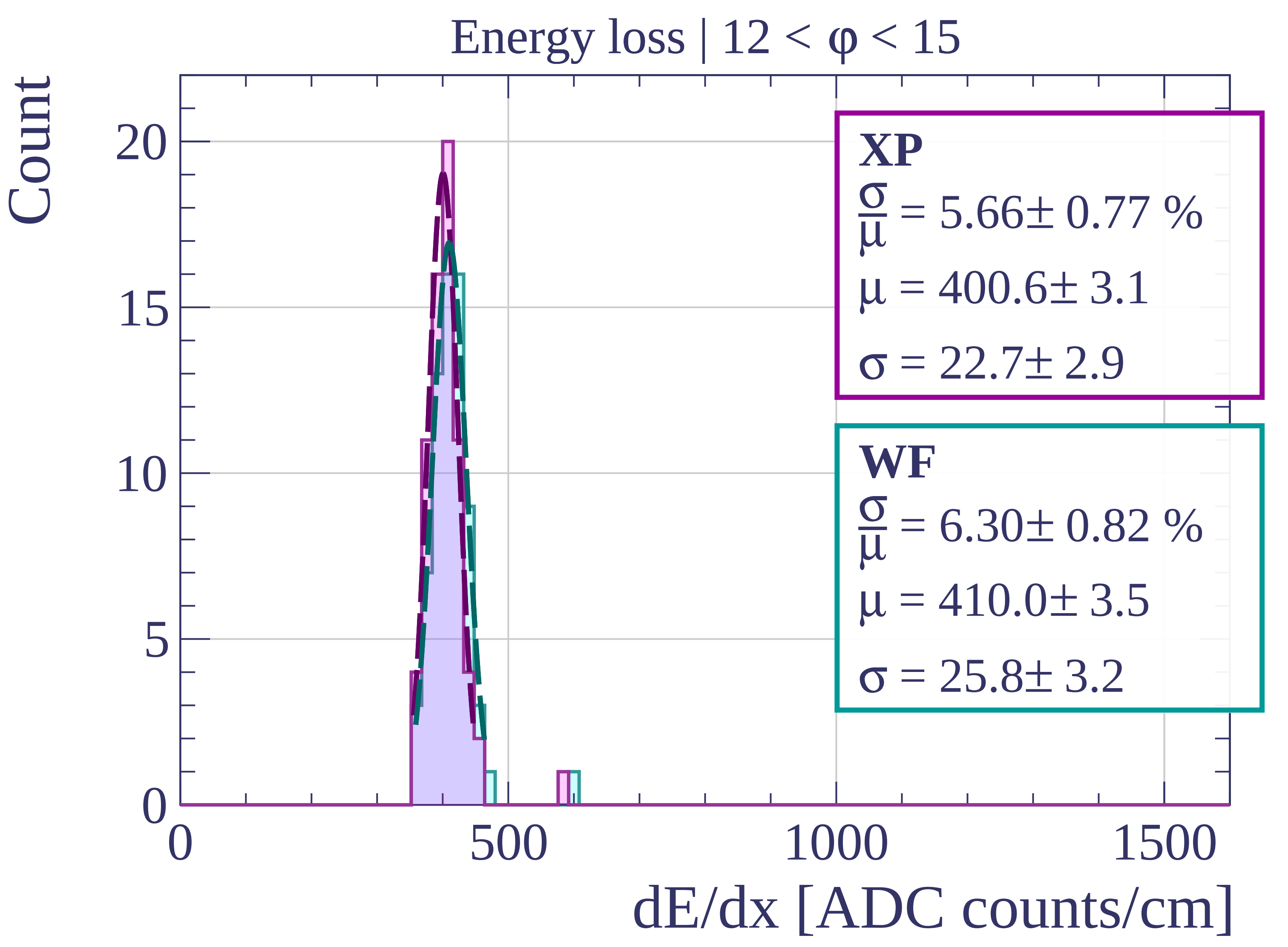

Energy loss | 3 < φ < 6
Count
15
10
5
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 5.23± 1.25 %
μ = 412.2± 3.8
σ = 21.5± 4.9
WF
σ/μ = 6.45± 1.10 %
μ = 423.0± 4.4
σ = 27.3± 4.4



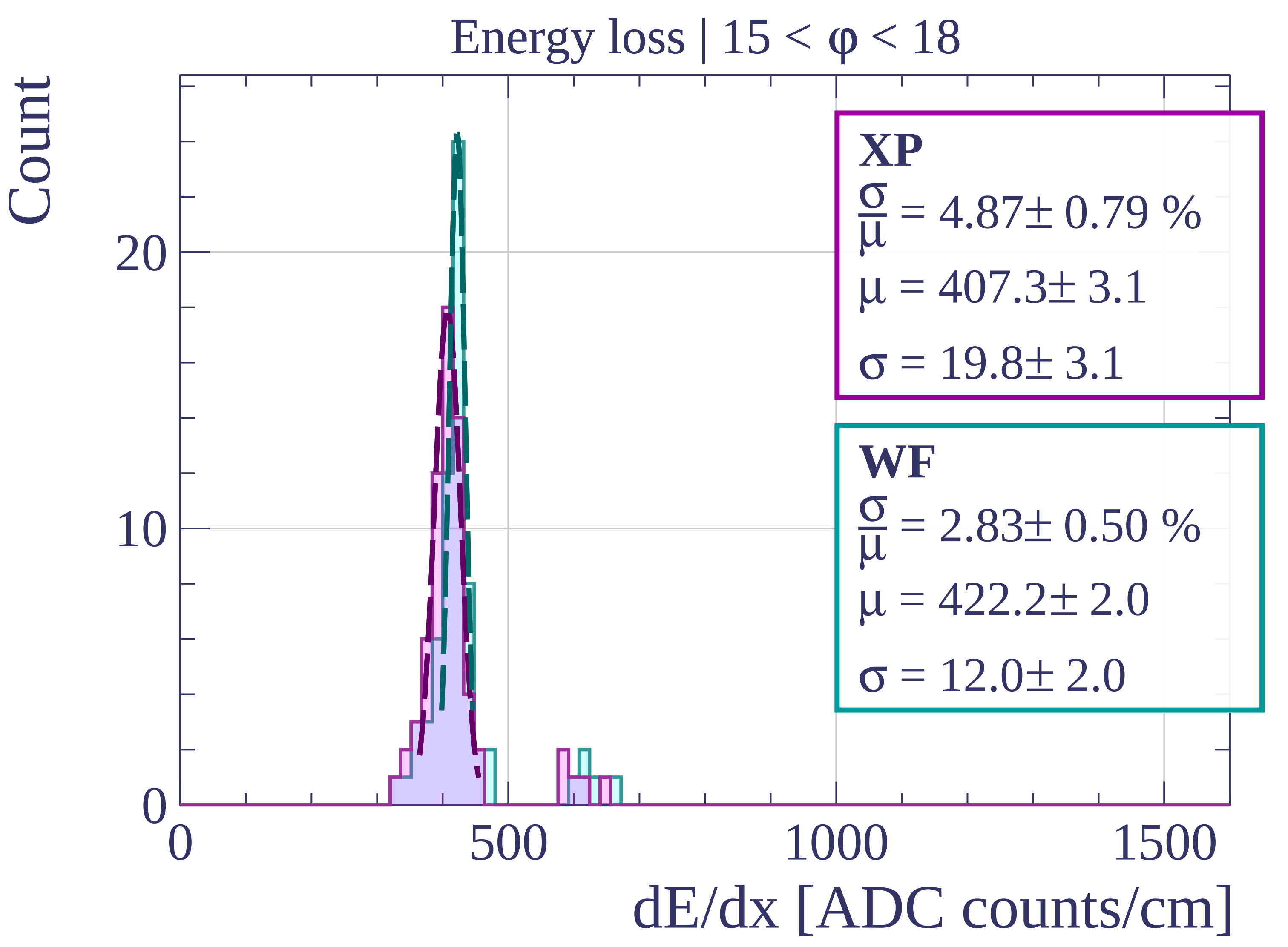
Energy loss | 6 < φ < 9
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
5
10
15
XP
σ/μ = 6.40± 1.15 %
μ = 403.8± 4.2
σ = 25.8± 4.4
WF
σ/μ = 6.46± 0.77 %
μ = 416.6± 3.7
σ = 26.9± 3.0

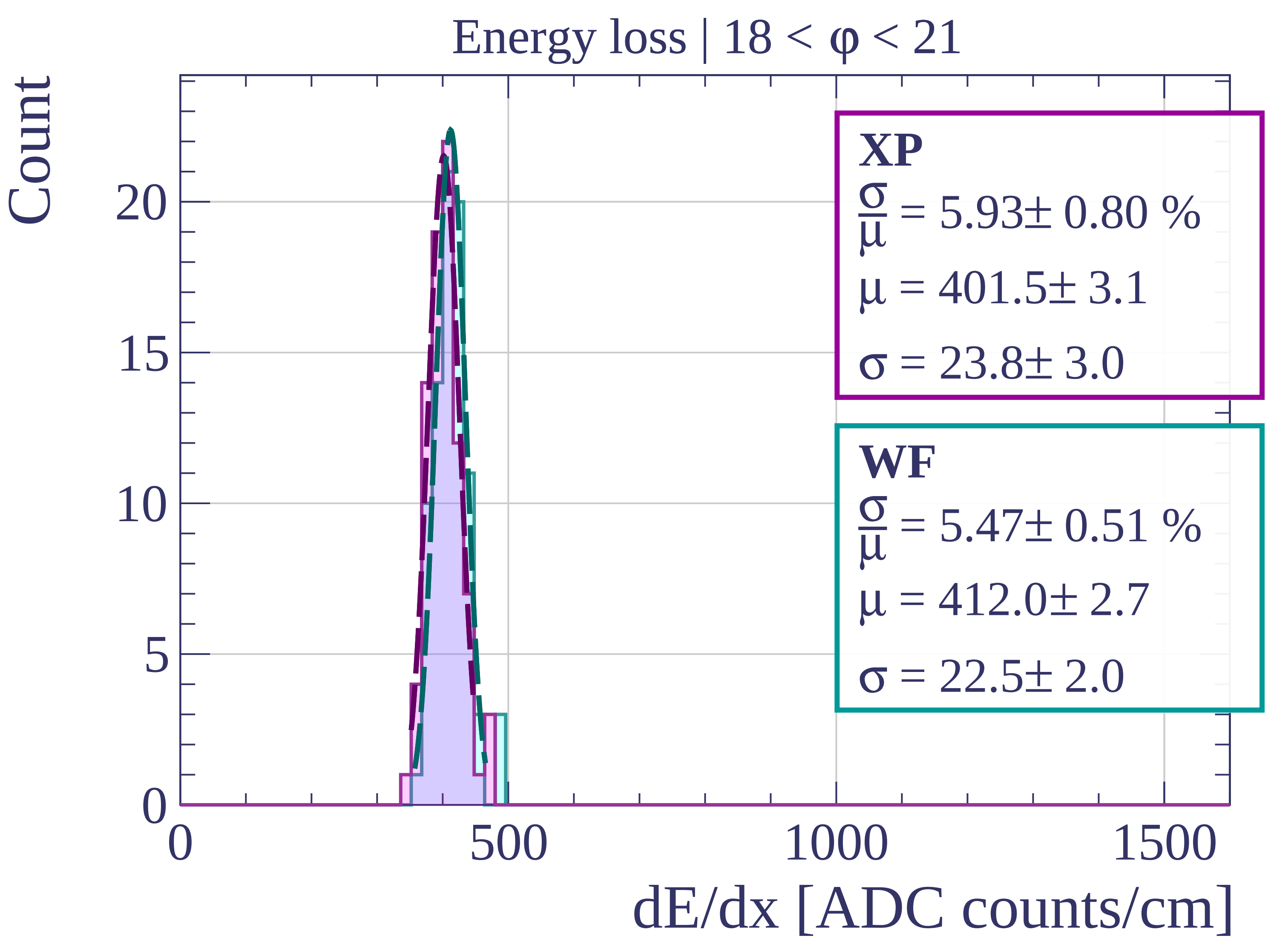


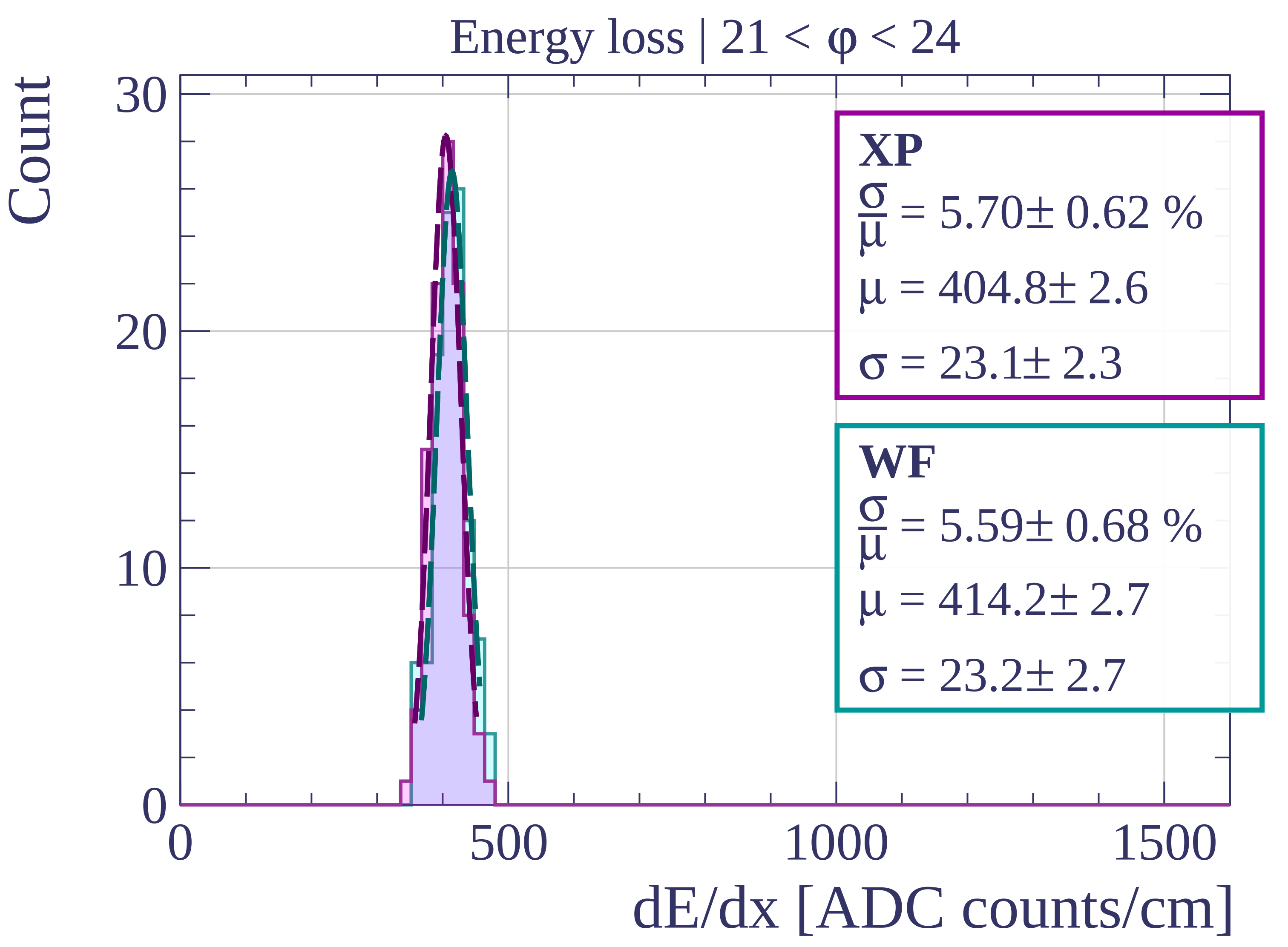
Energy loss | 9 < φ < 12
Count
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 5.17± 0.67 %
μ = 407.9± 3.1
σ = 21.1± 2.6
WF
$\frac{\sigma}{\mu}$ = 5.50± 0.81 %
μ = 416.0± 3.1
σ = 22.9± 3.2



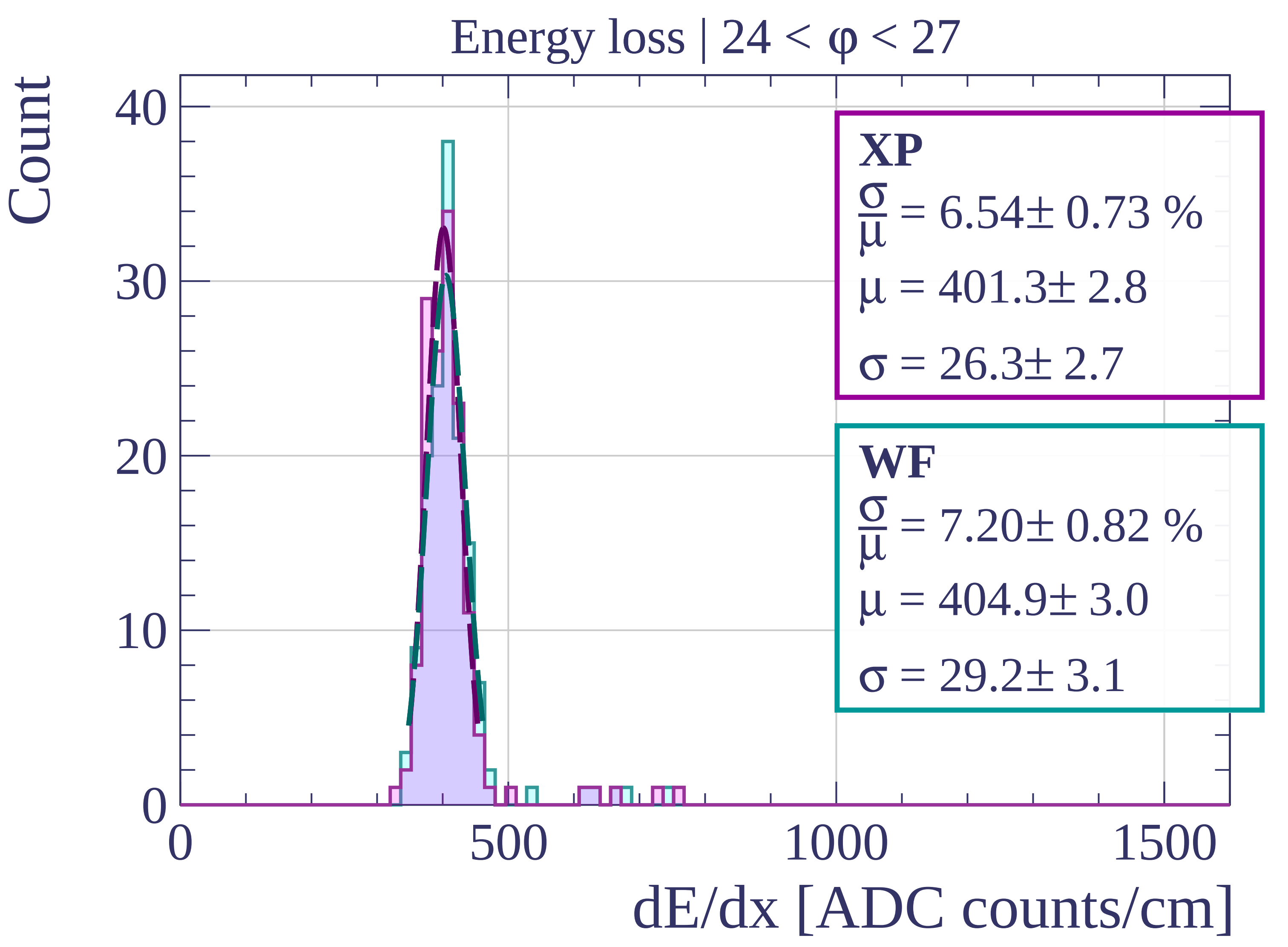
Energy loss | 12 < φ < 15
Count
dE/dx [ADC counts/cm]
0
5
10
15
20
0
500
1000
1500
XP
σ/μ = 5.66± 0.77 %
μ = 400.6± 3.1
σ = 22.7± 2.9
WF
σ/μ = 6.30± 0.82 %
μ = 410.0± 3.5
σ = 25.8± 3.2

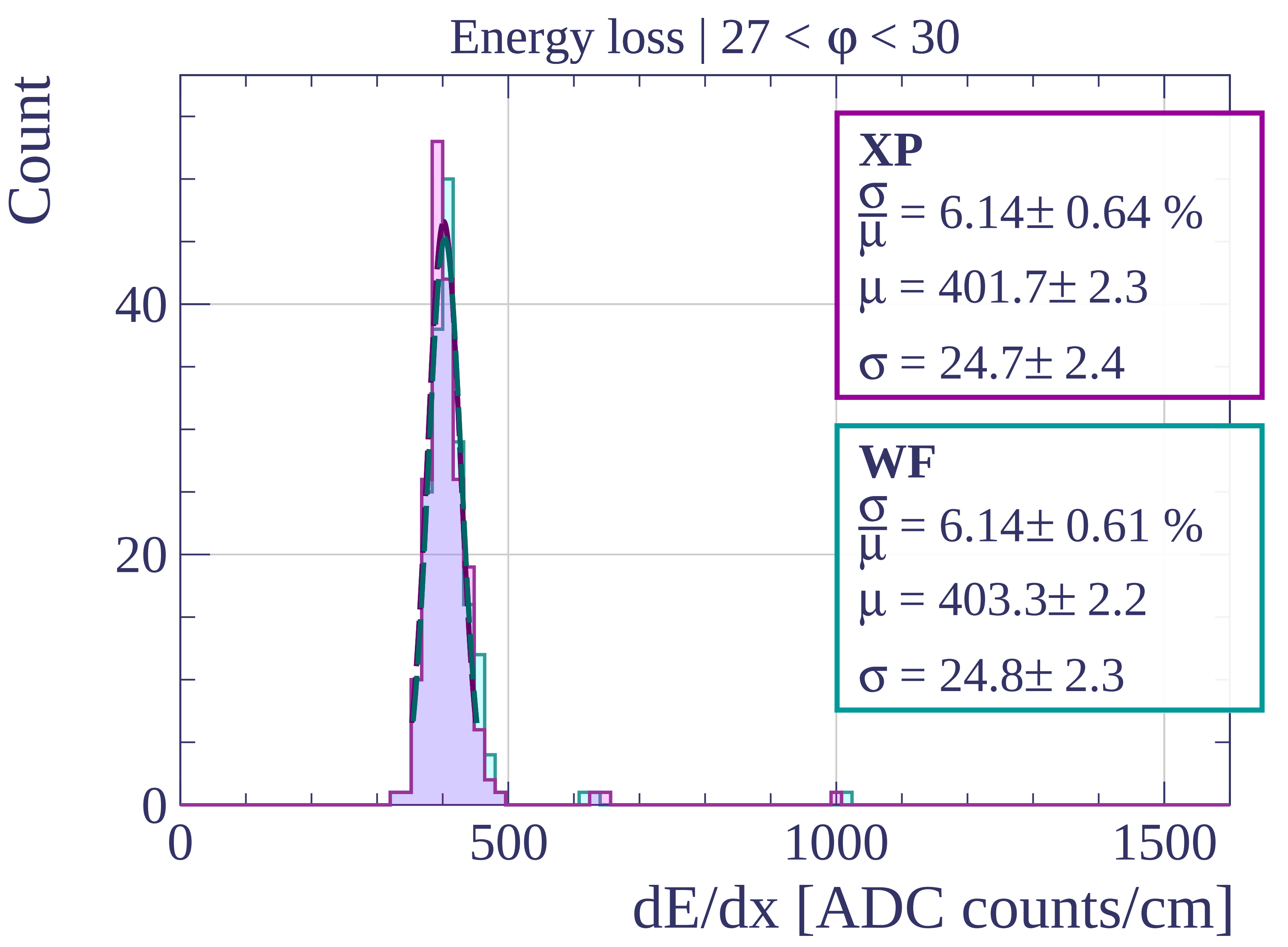

Energy loss | 15 < φ < 18
Count
dE/dx [ADC counts/cm]
0
10
20
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 4.87± 0.79 %
μ = 407.3± 3.1
σ = 19.8± 3.1
WF
$\frac{\sigma}{\mu}$ = 2.83± 0.50 %
μ = 422.2± 2.0
σ = 12.0± 2.0

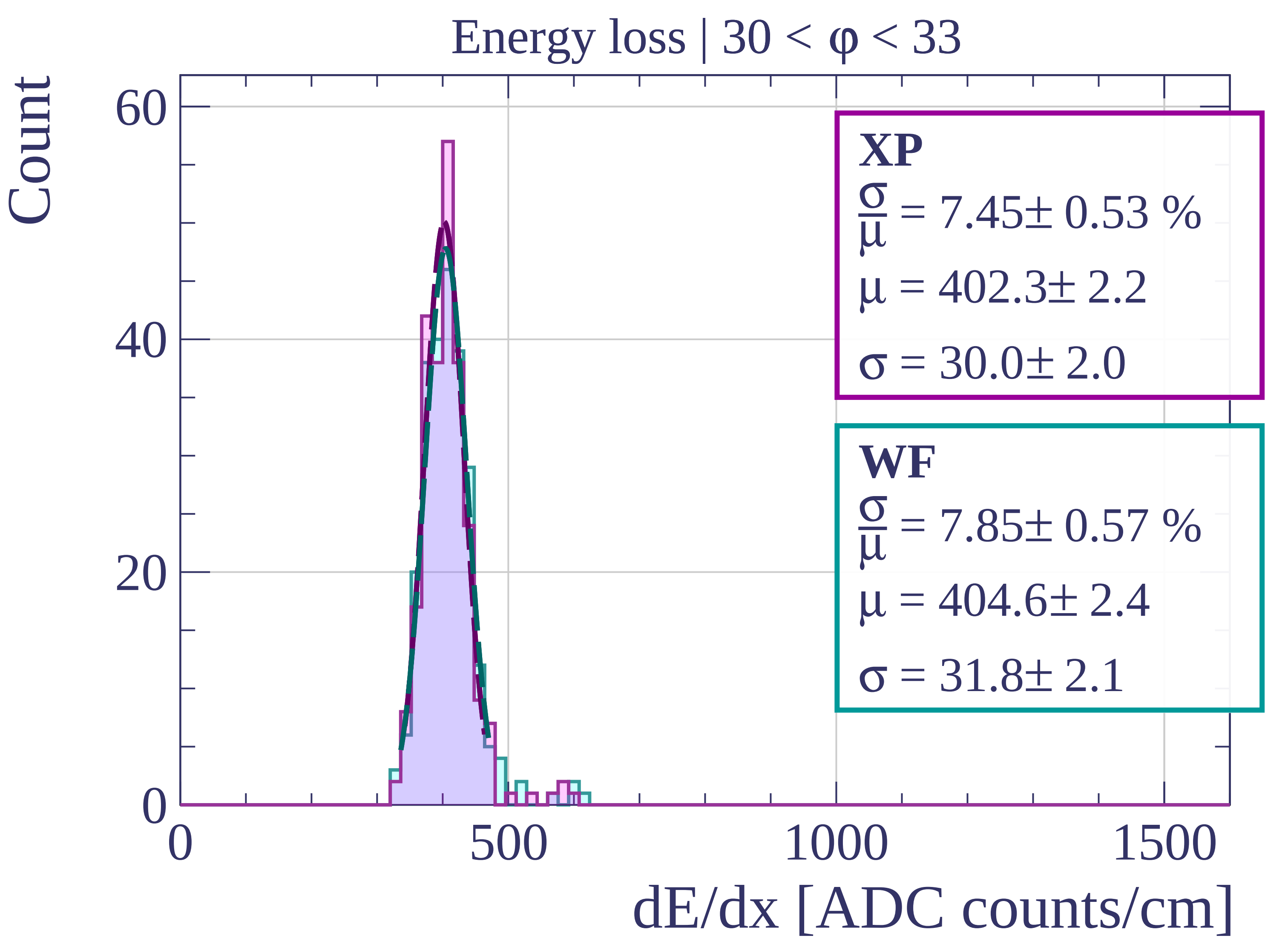

Energy loss | 18 < φ < 21
Count
dE/dx [ADC counts/cm]
0
5
10
15
20
0
500
1000
1500
XP
σ/μ = 5.93± 0.80 %
μ = 401.5± 3.1
σ = 23.8± 3.0
WF
σ/μ = 5.47± 0.51 %
μ = 412.0± 2.7
σ = 22.5± 2.0



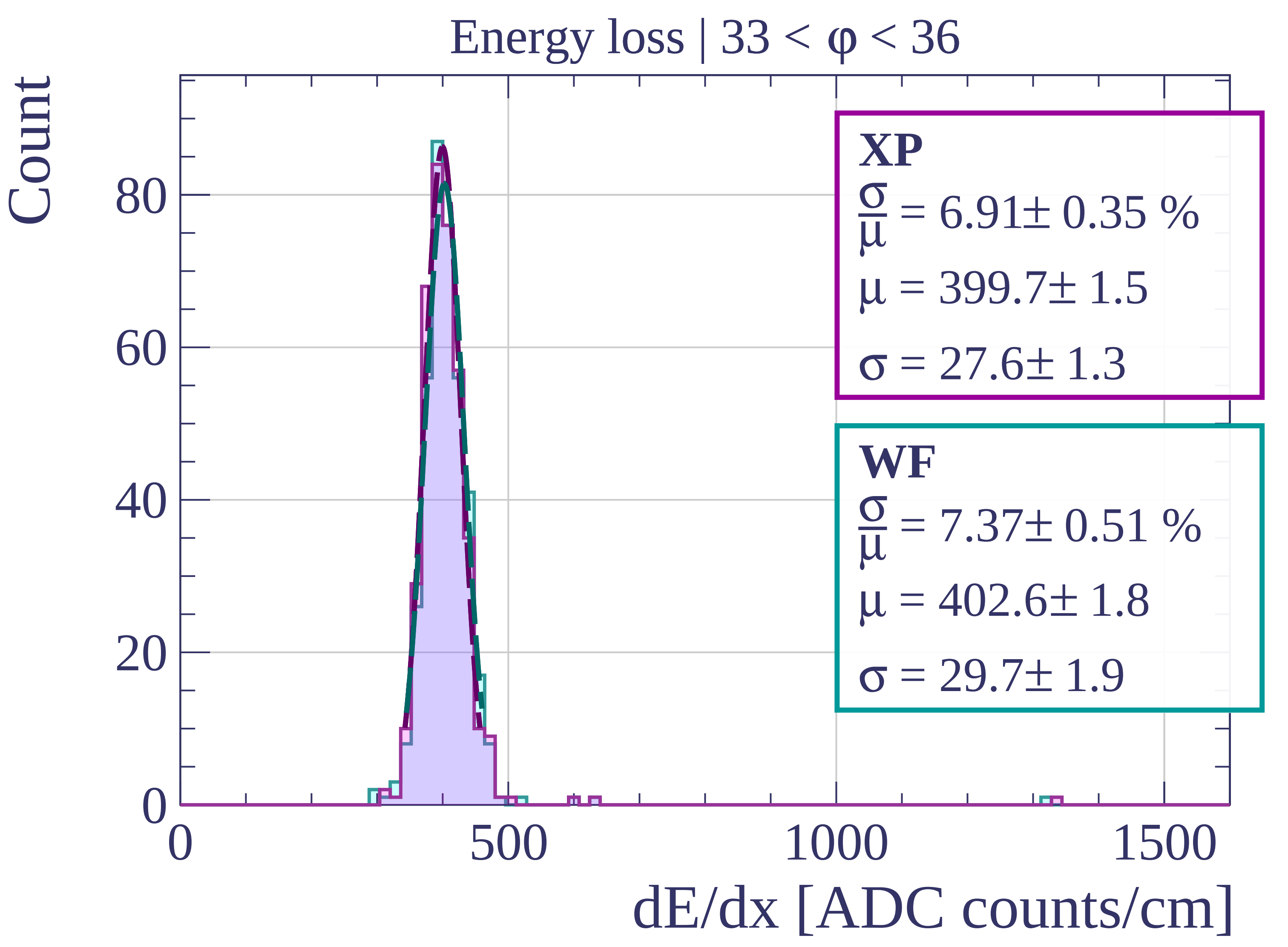
Energy loss | 21 < φ < 24
Count
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 5.70± 0.62 %
μ = 404.8± 2.6
σ = 23.1± 2.3
WF
σ/μ = 5.59± 0.68 %
μ = 414.2± 2.7
σ = 23.2± 2.7


Energy loss | 24 < φ < 27
Count
dE/dx [ADC counts/cm]
0
10
20
30
40
0
500
1000
1500
XP
σ/μ = 6.54± 0.73 %
μ = 401.3± 2.8
σ = 26.3± 2.7
WF
σ/μ = 7.20± 0.82 %
μ = 404.9± 3.0
σ = 29.2± 3.1

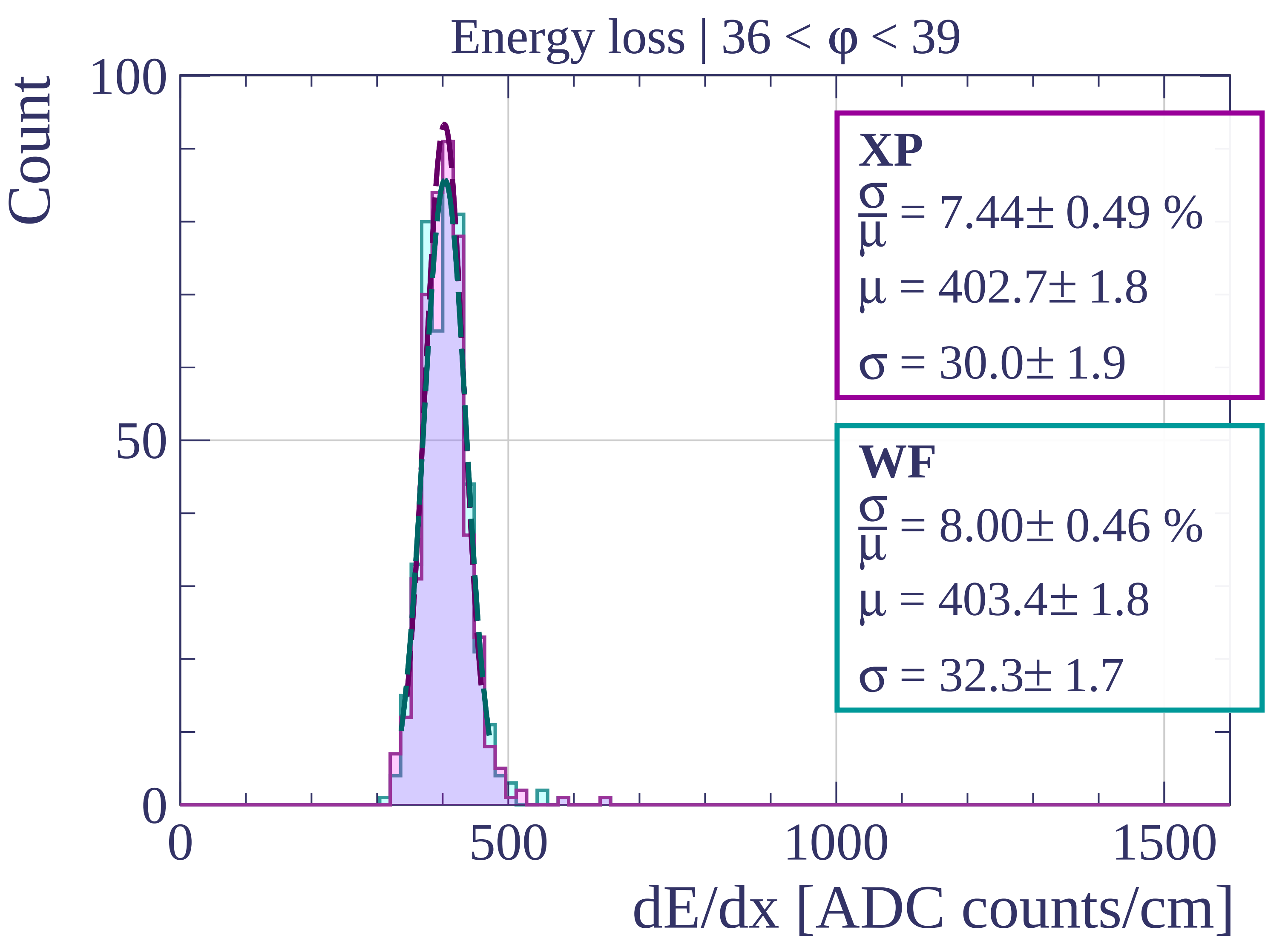

Energy loss | 27 < φ < 30
Count
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 6.14± 0.64 %
μ = 401.7± 2.3
σ = 24.7± 2.4
WF
$\frac{\sigma}{\mu}$ = 6.14± 0.61 %
μ = 403.3± 2.2
σ = 24.8± 2.3



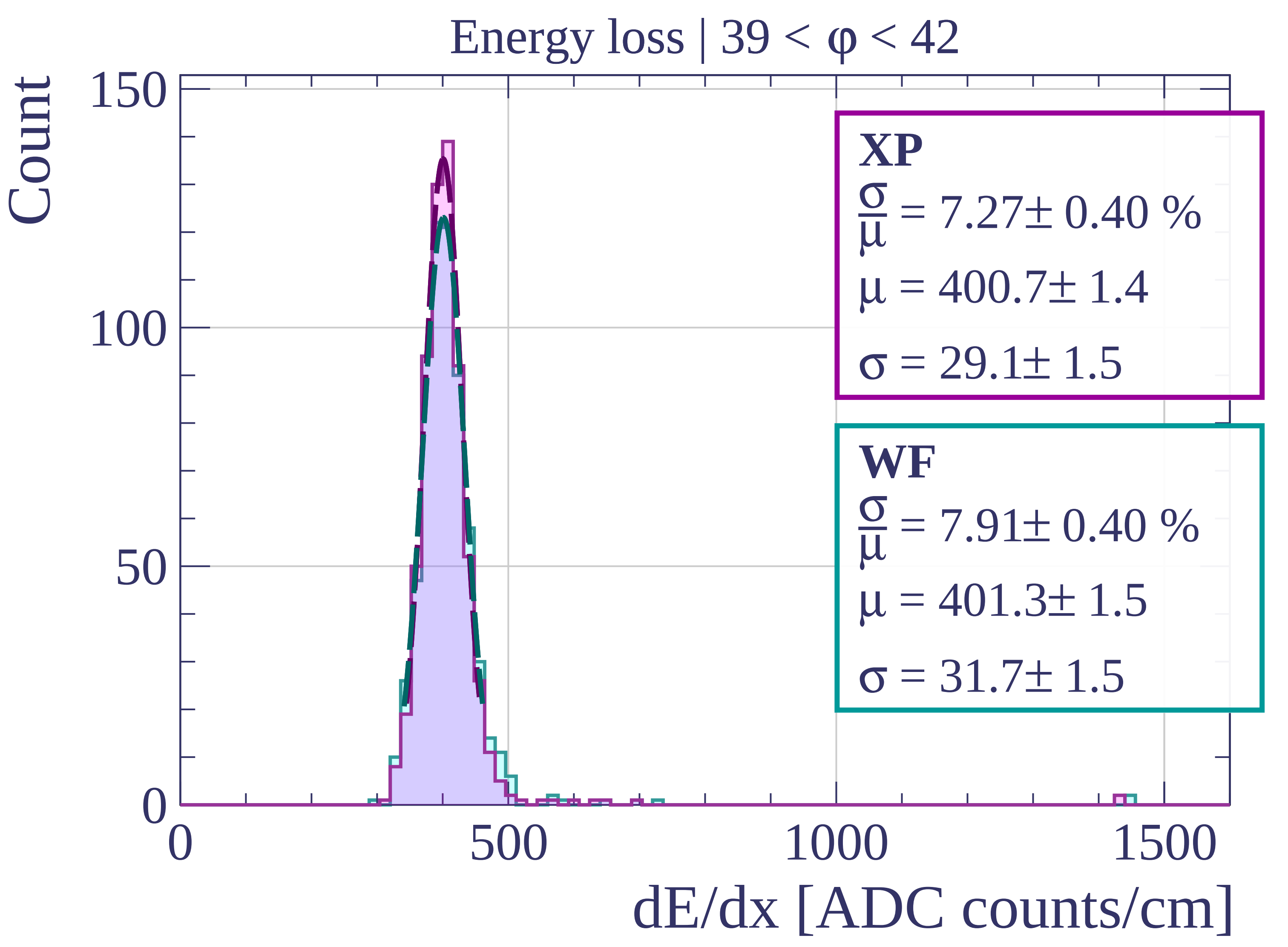
Energy loss | 30 < φ < 33
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 7.45± 0.53 %
μ = 402.3± 2.2
σ = 30.0± 2.0
WF
$\frac{\sigma}{\mu}$ = 7.85± 0.57 %
μ = 404.6± 2.4
σ = 31.8± 2.1



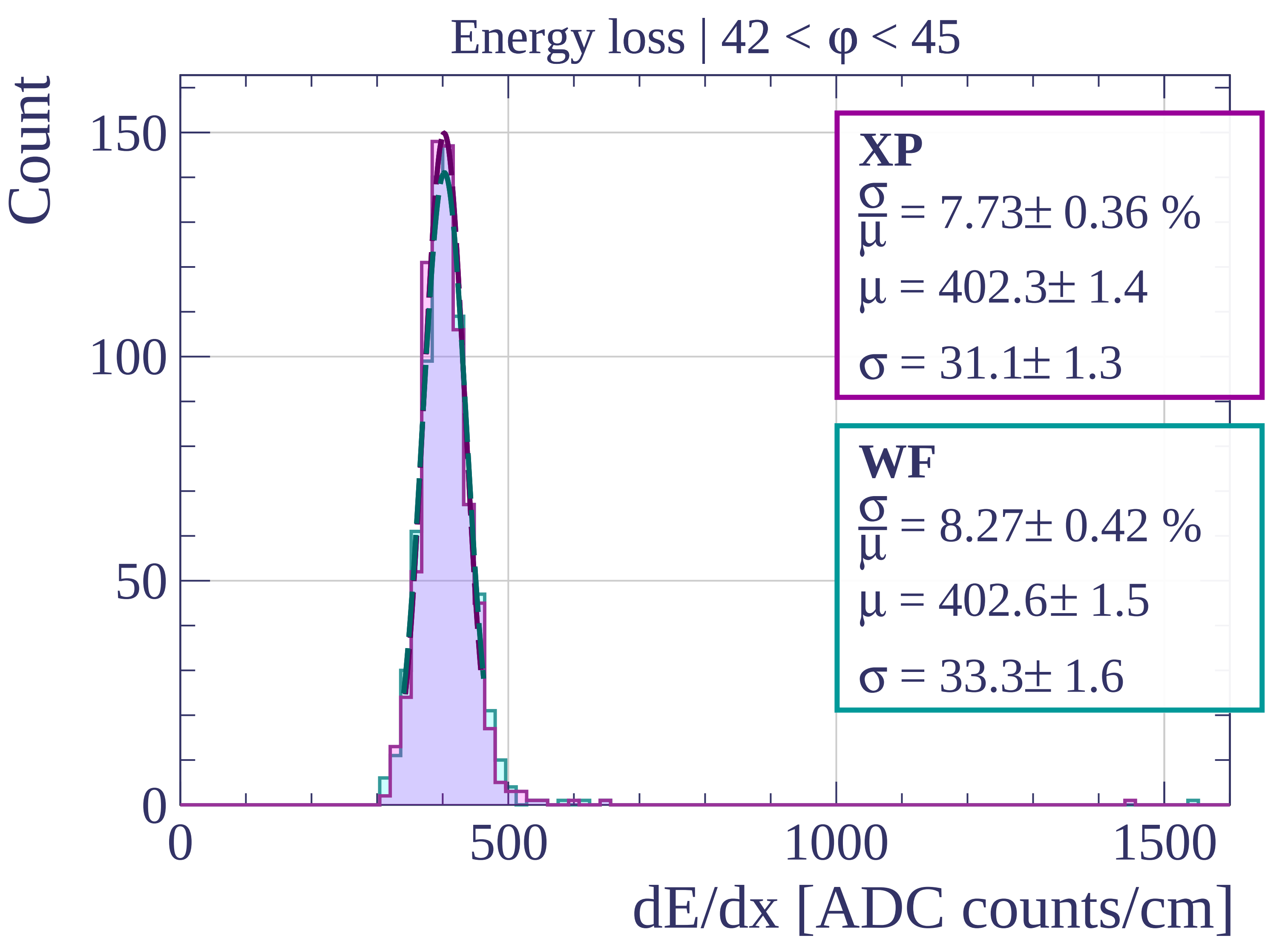
Energy loss | 33 < φ < 36
Count
0
20
40
60
80
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 6.91± 0.35 %
μ = 399.7± 1.5
σ = 27.6± 1.3
WF
σ/μ = 7.37± 0.51 %
μ = 402.6± 1.8
σ = 29.7± 1.9

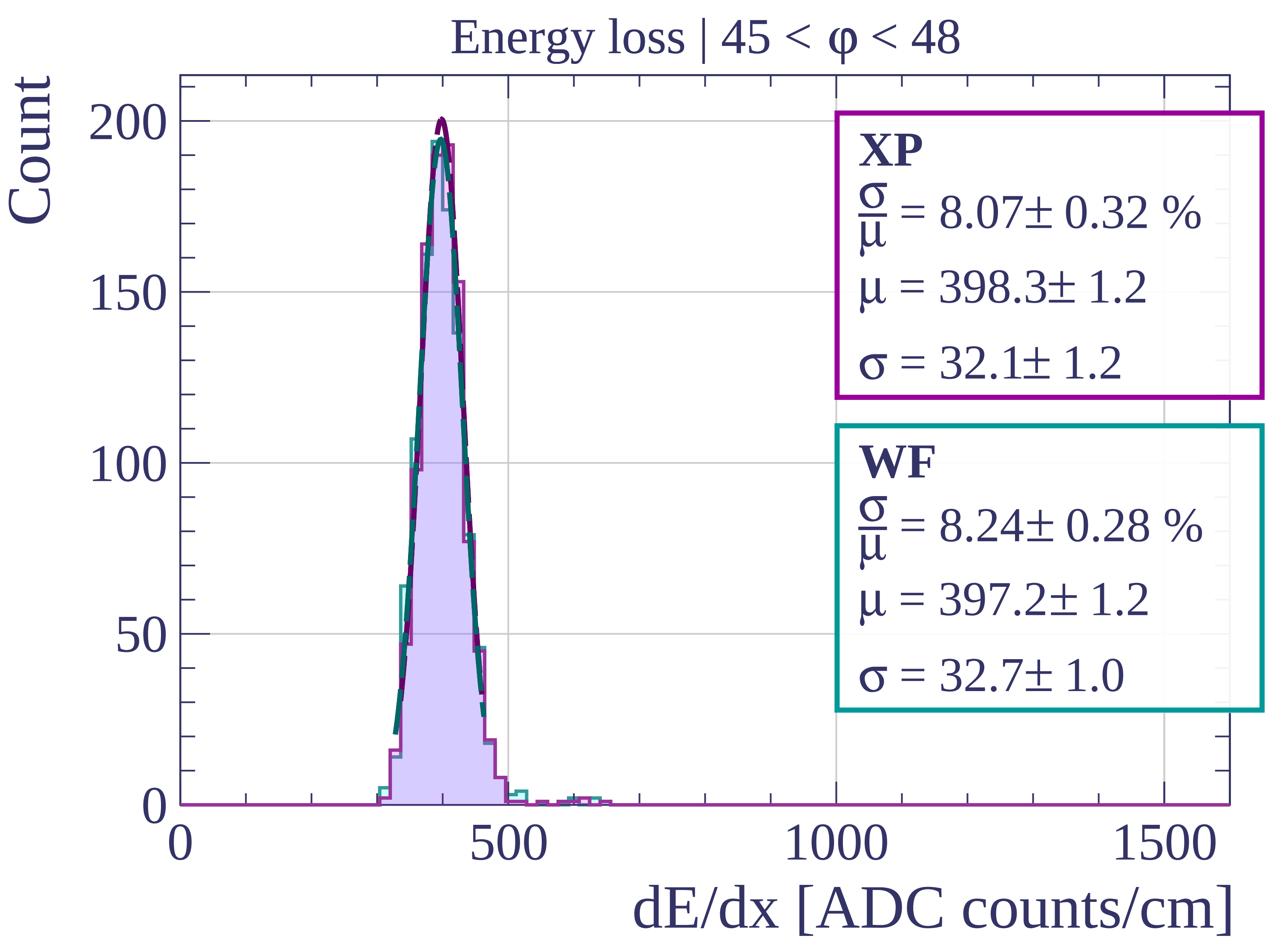


Energy loss | 36 < φ < 39
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.44± 0.49 %
μ = 402.7± 1.8
σ = 30.0± 1.9
WF
σ/μ = 8.00± 0.46 %
μ = 403.4± 1.8
σ = 32.3± 1.7

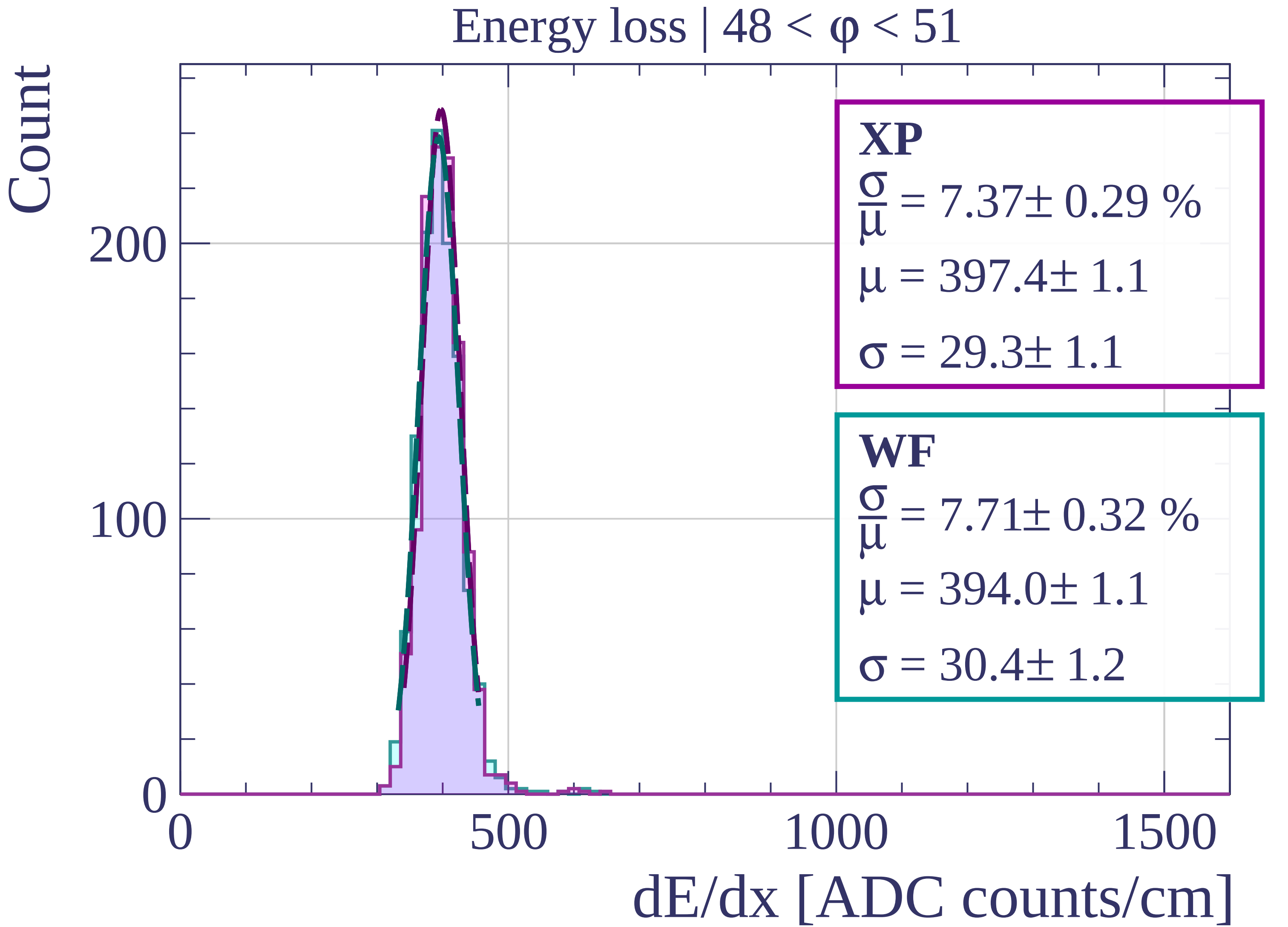


Energy loss | 39 < φ < 42
Count
dE/dx [ADC counts/cm]
0
50
100
150
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 7.27± 0.40 %
μ = 400.7± 1.4
σ = 29.1± 1.5
WF
$\frac{\sigma}{\mu}$ = 7.91± 0.40 %
μ = 401.3± 1.5
σ = 31.7± 1.5

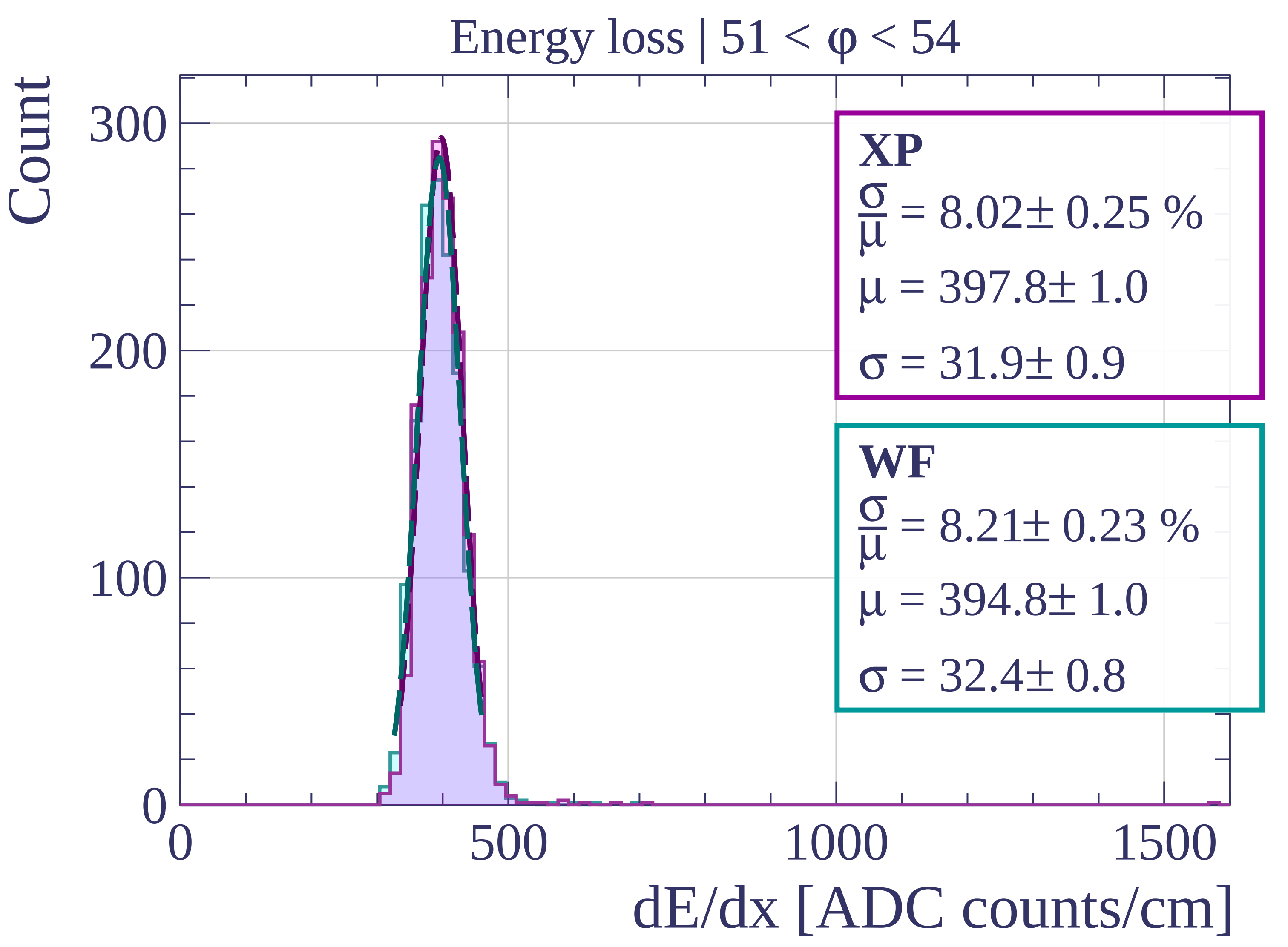

Energy loss | 42 < φ < 45
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 7.73± 0.36 %
μ = 402.3± 1.4
σ = 31.1± 1.3
WF
$\frac{\sigma}{\mu}$ = 8.27± 0.42 %
μ = 402.6± 1.5
σ = 33.3± 1.6

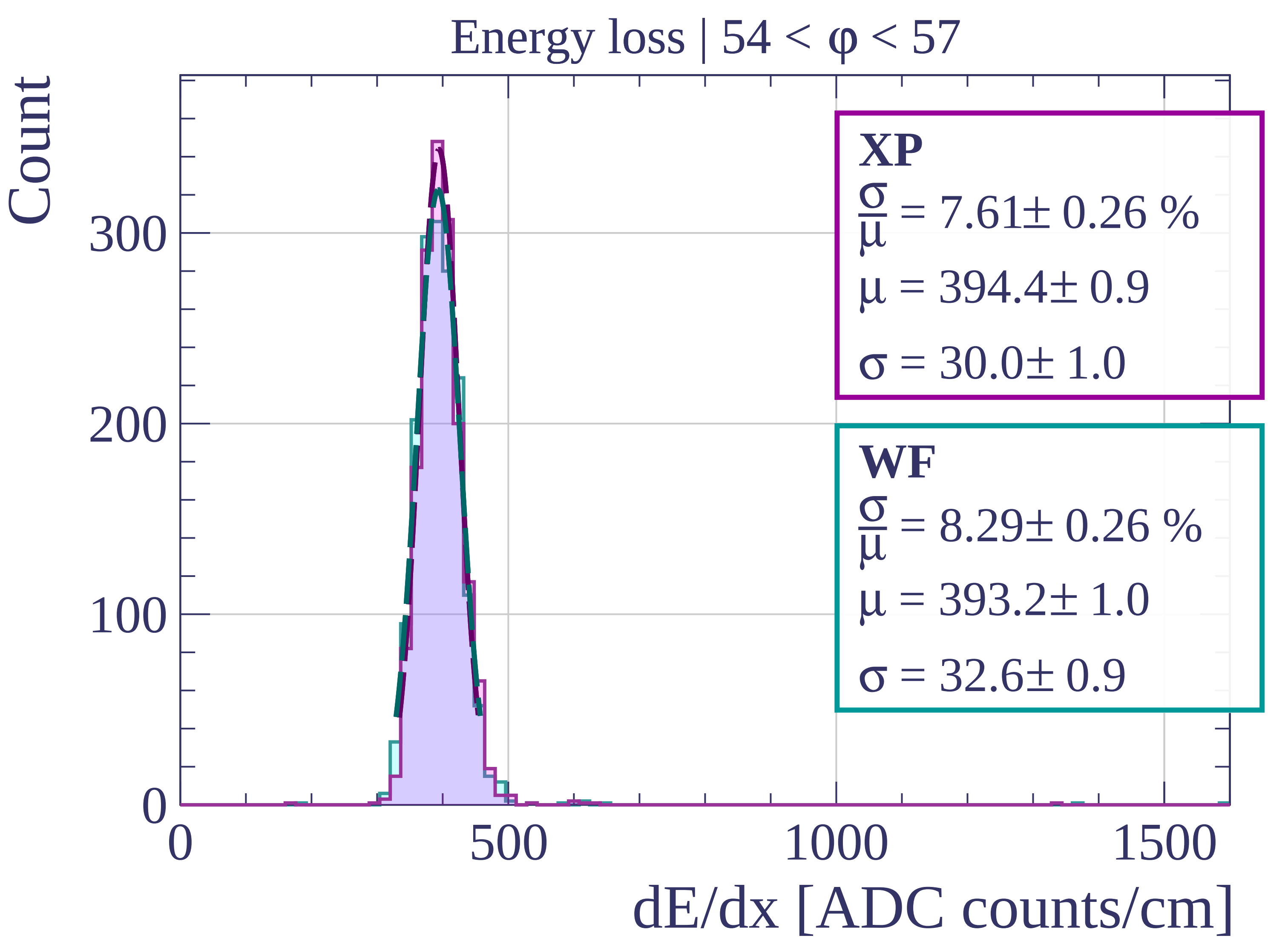


Energy loss | 45 < φ < 48
Count
dE/dx [ADC counts/cm]
0
50
100
150
200
0
500
1000
1500
XP
σ/μ = 8.07± 0.32 %
μ = 398.3± 1.2
σ = 32.1± 1.2
WF
σ/μ = 8.24± 0.28 %
μ = 397.2± 1.2
σ = 32.7± 1.0

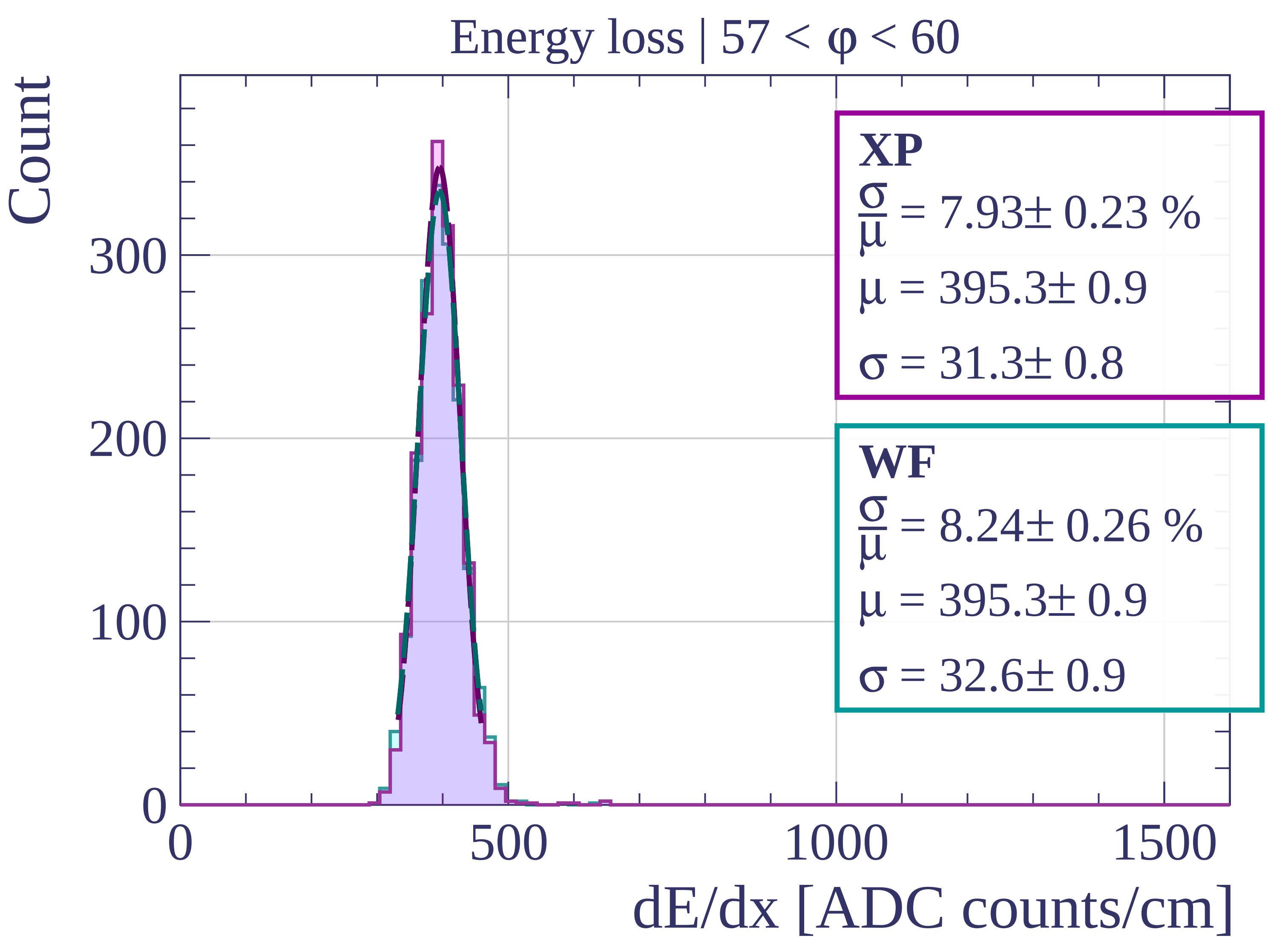


Energy loss | 48 < φ < 51
Count
dE/dx [ADC counts/cm]
0
100
200
0
500
1000
1500
XP
σ/μ = 7.37± 0.29 %
μ = 397.4± 1.1
σ = 29.3± 1.1
WF
σ/μ = 7.71± 0.32 %
μ = 394.0± 1.1
σ = 30.4± 1.2

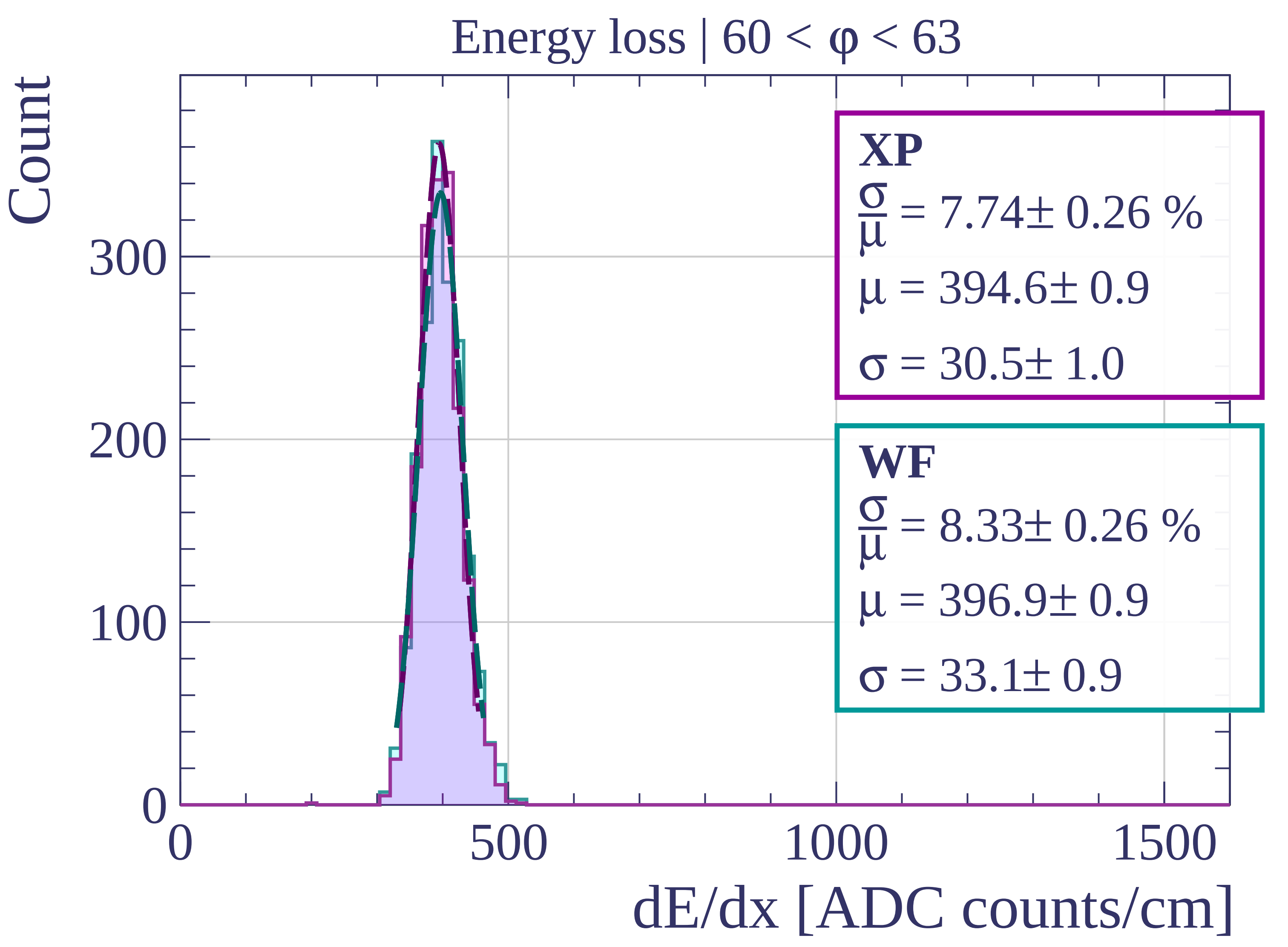

Energy loss | 51 < φ < 54
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.02± 0.25 %
μ = 397.8± 1.0
σ = 31.9± 0.9
WF
σ/μ = 8.21± 0.23 %
μ = 394.8± 1.0
σ = 32.4± 0.8



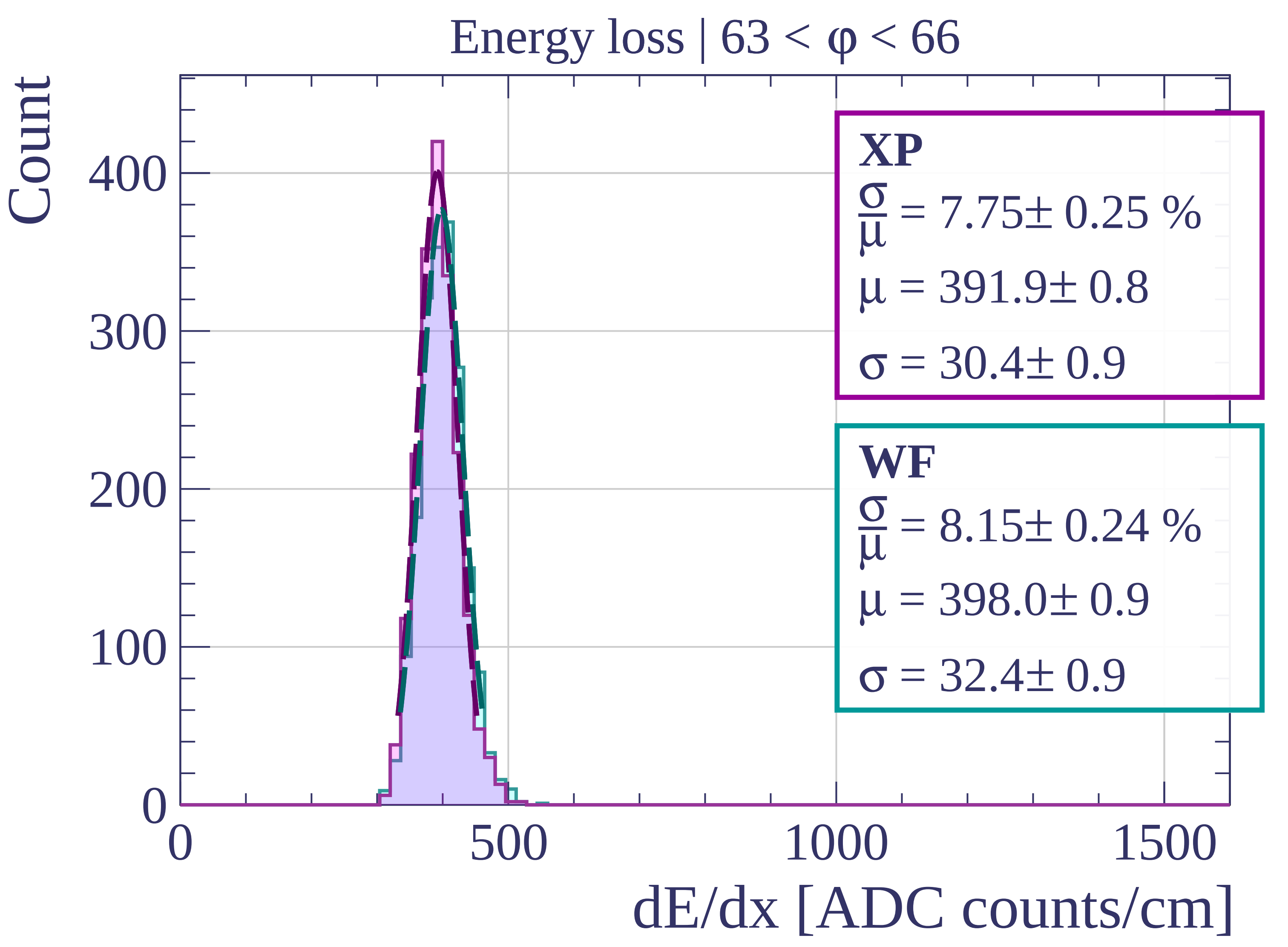
Energy loss | 54 < φ < 57
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 7.61± 0.26 %
μ = 394.4± 0.9
σ = 30.0± 1.0
WF
$\frac{\sigma}{\mu}$ = 8.29± 0.26 %
μ = 393.2± 1.0
σ = 32.6± 0.9



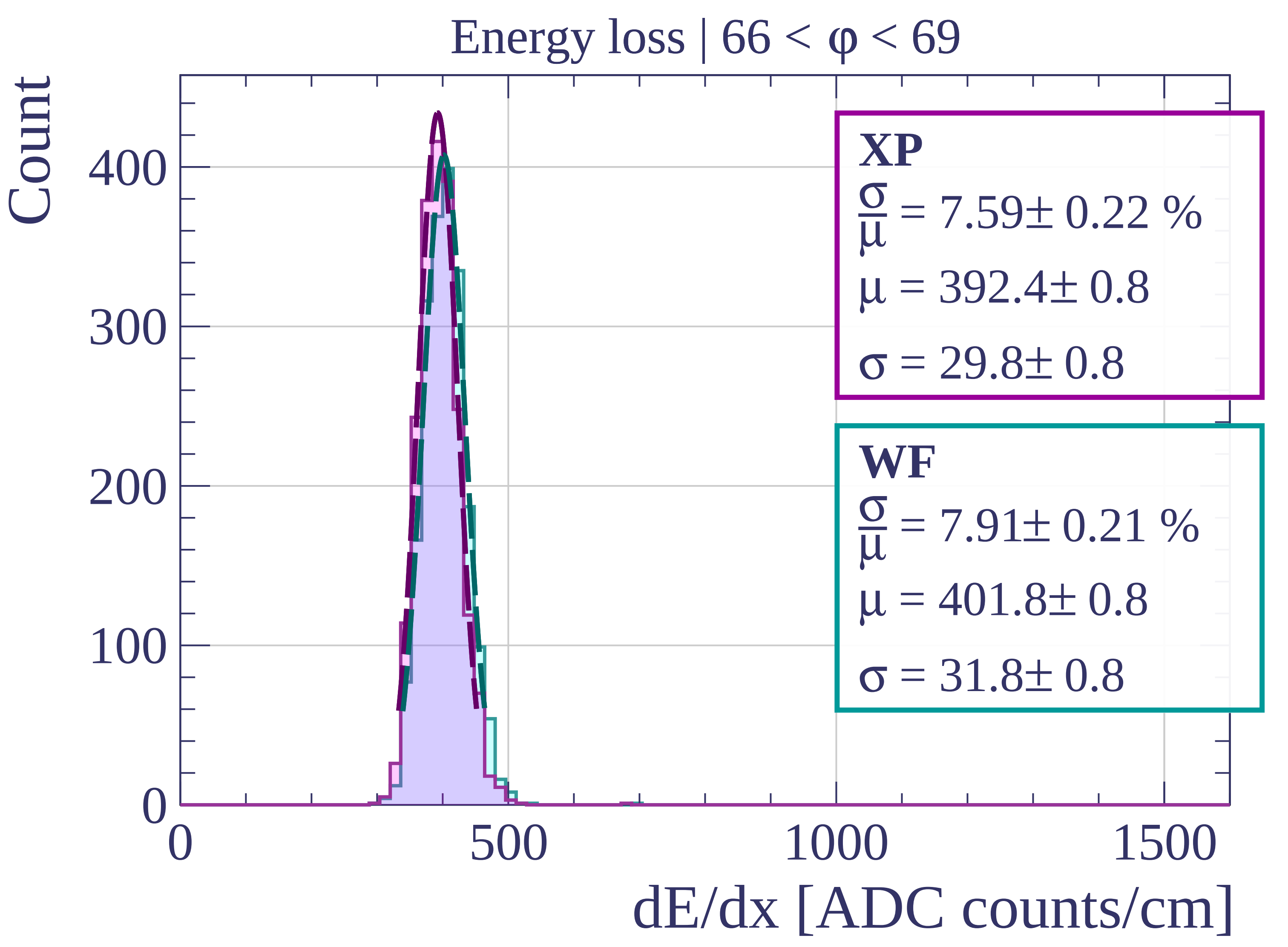
Energy loss | 57 < φ < 60
Count
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.93± 0.23 %
μ = 395.3± 0.9
σ = 31.3± 0.8
WF
σ/μ = 8.24± 0.26 %
μ = 395.3± 0.9
σ = 32.6± 0.9



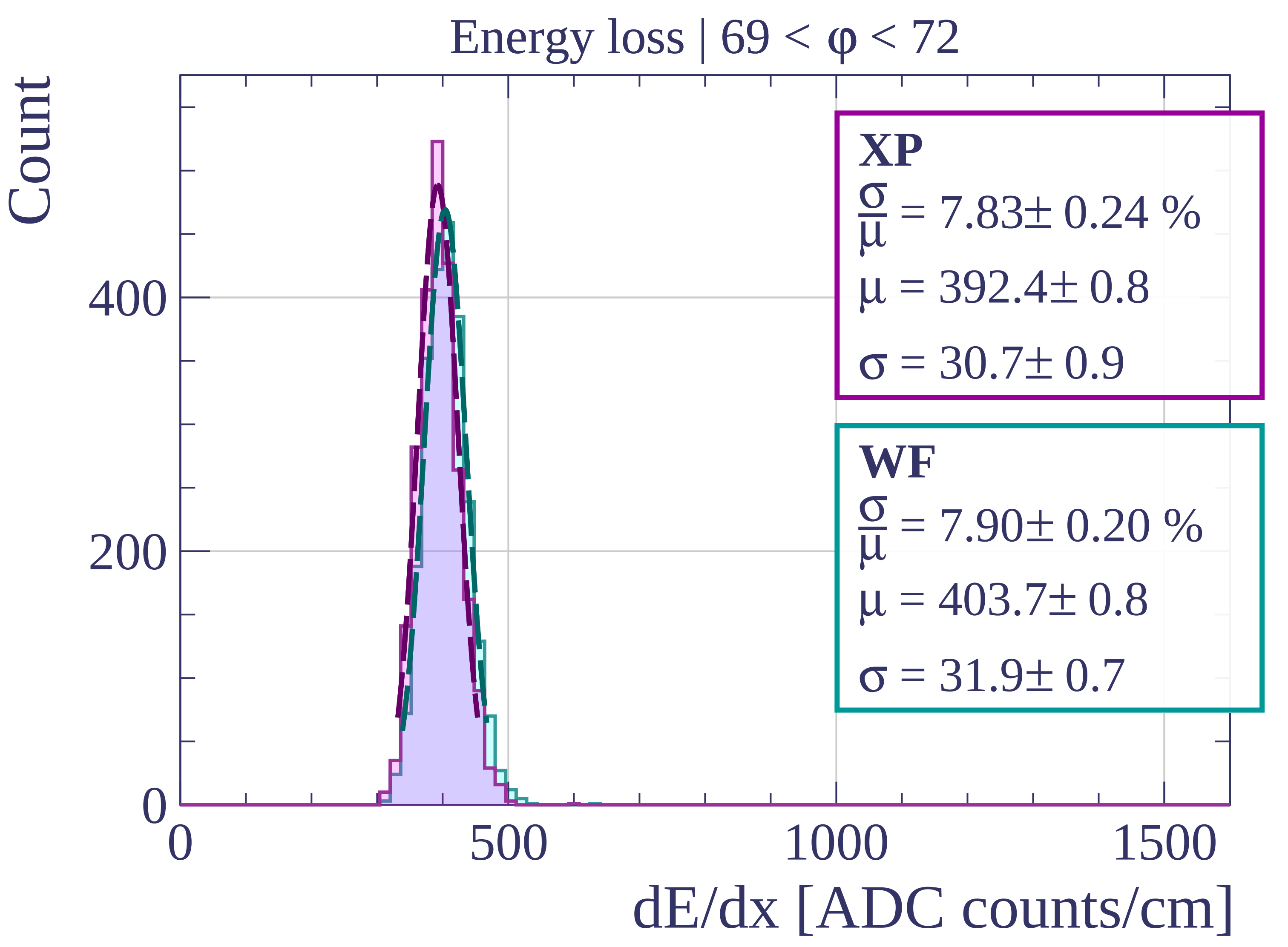
Energy loss | 60 < φ < 63
Count
0
100
200
300
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.74± 0.26 %
μ = 394.6± 0.9
σ = 30.5± 1.0
WF
σ/μ = 8.33± 0.26 %
μ = 396.9± 0.9
σ = 33.1± 0.9



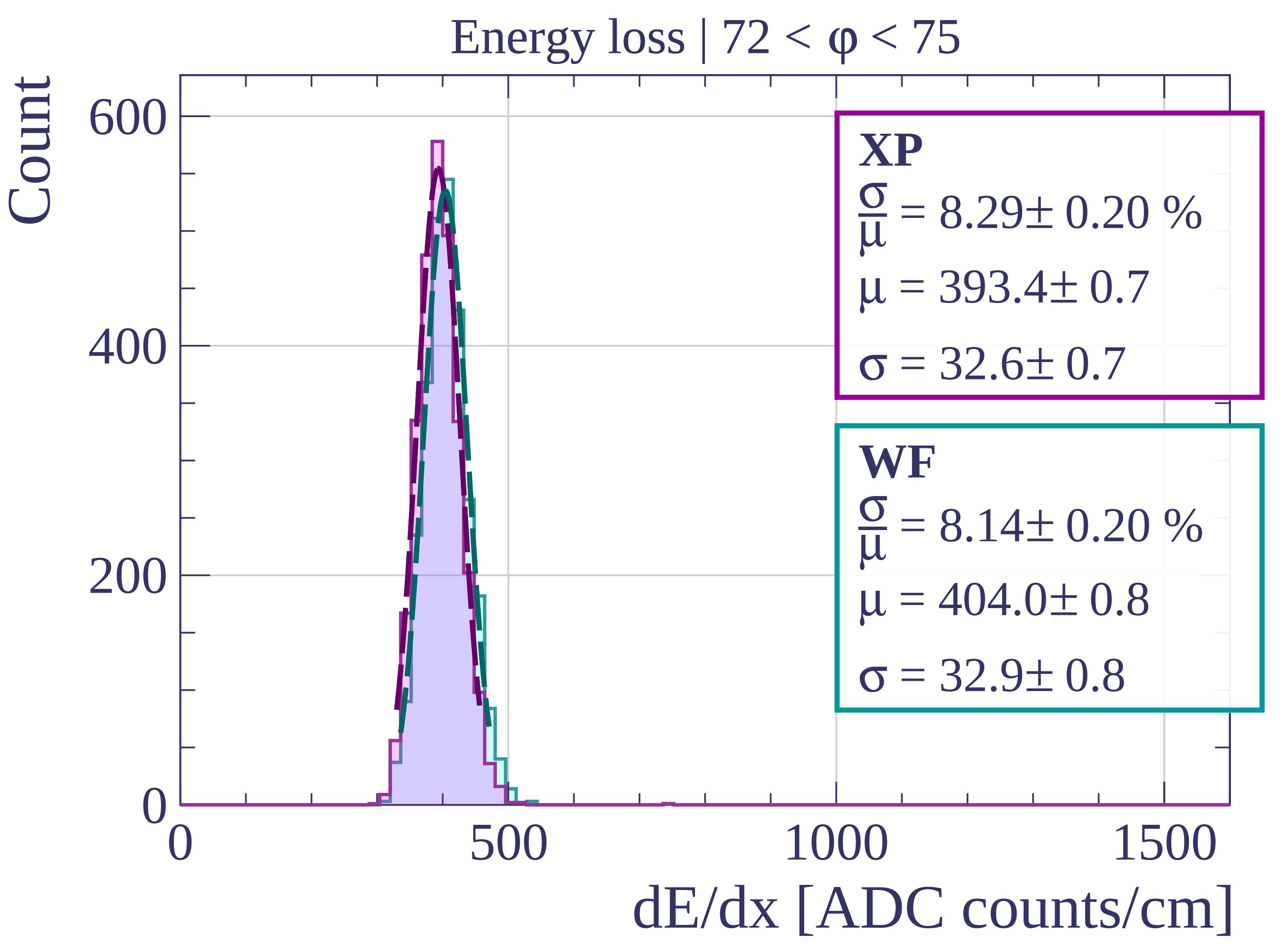
Energy loss | 63 < φ < 66
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
500
1000
1500
XP
σ/μ = 7.75± 0.25 %
μ = 391.9± 0.8
σ = 30.4± 0.9
WF
σ/μ = 8.15± 0.24 %
μ = 398.0± 0.9
σ = 32.4± 0.9



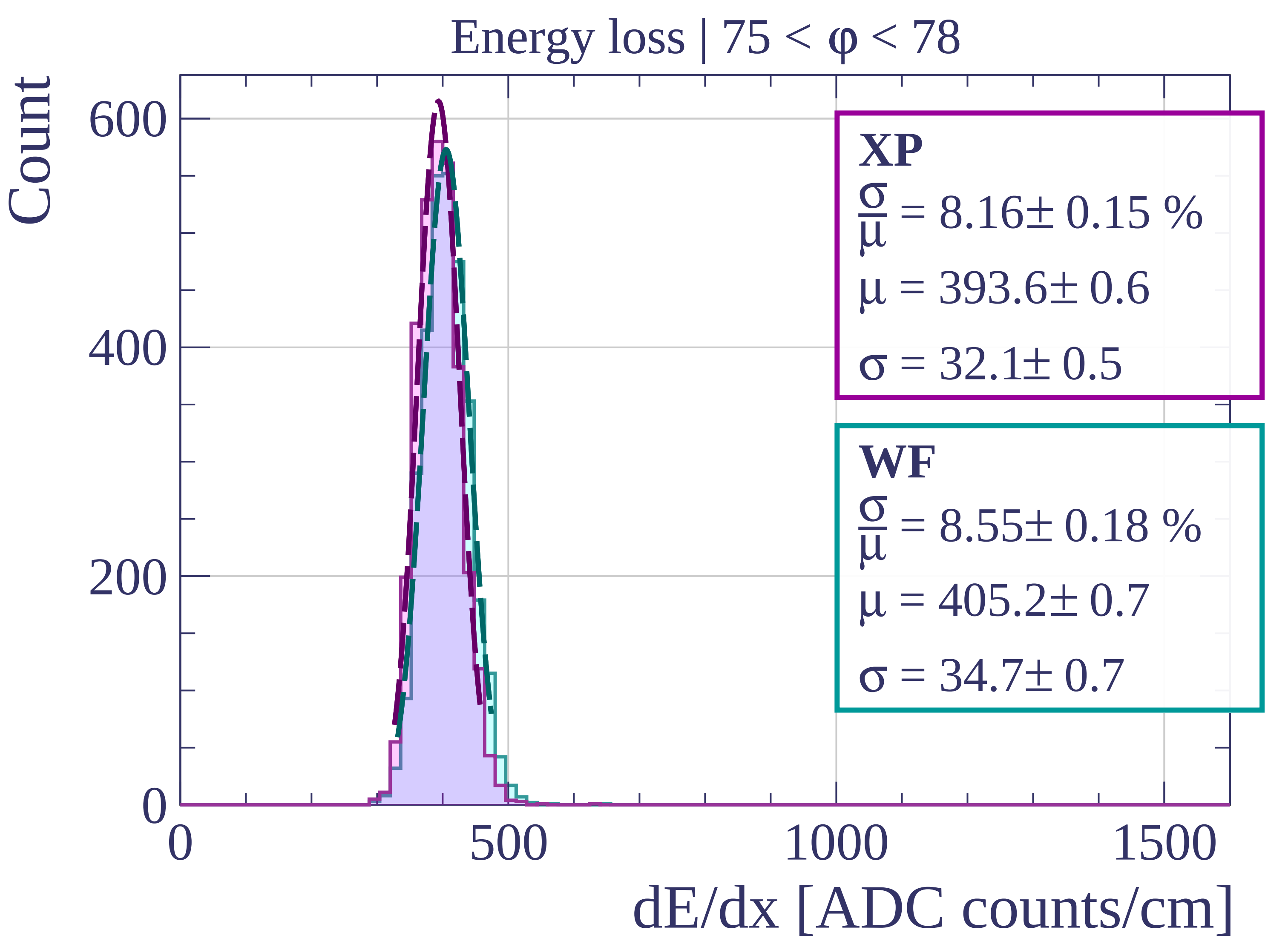
Energy loss | 66 < φ < 69
Count
400
300
200
100
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.59± 0.22 %
μ = 392.4± 0.8
σ = 29.8± 0.8
WF
σ/μ = 7.91± 0.21 %
μ = 401.8± 0.8
σ = 31.8± 0.8



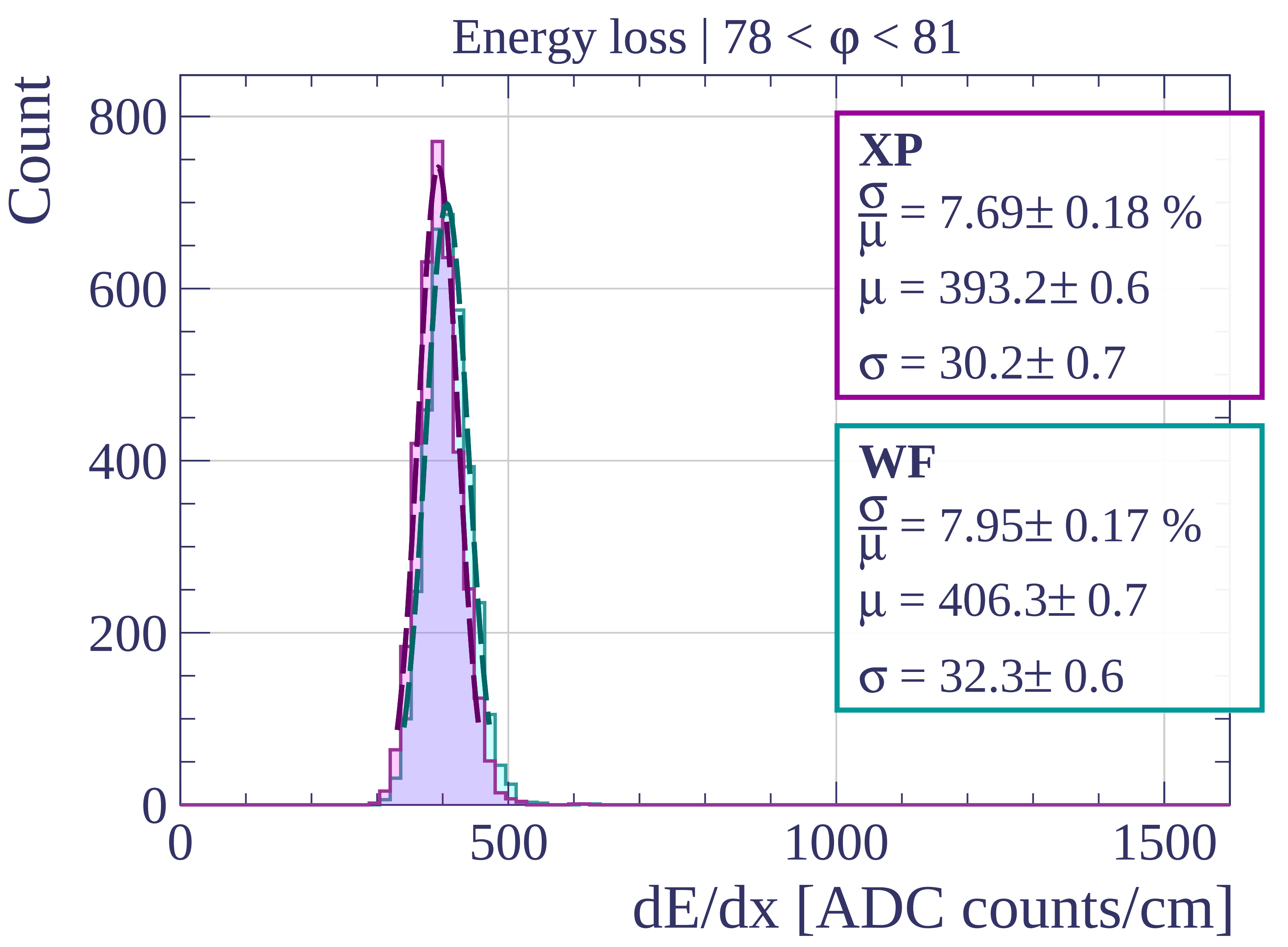
Energy loss | 69 < φ < 72
Count
0
200
400
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 7.83± 0.24 %
μ = 392.4± 0.8
σ = 30.7± 0.9
WF
$\frac{\sigma}{\mu}$ = 7.90± 0.20 %
μ = 403.7± 0.8
σ = 31.9± 0.7

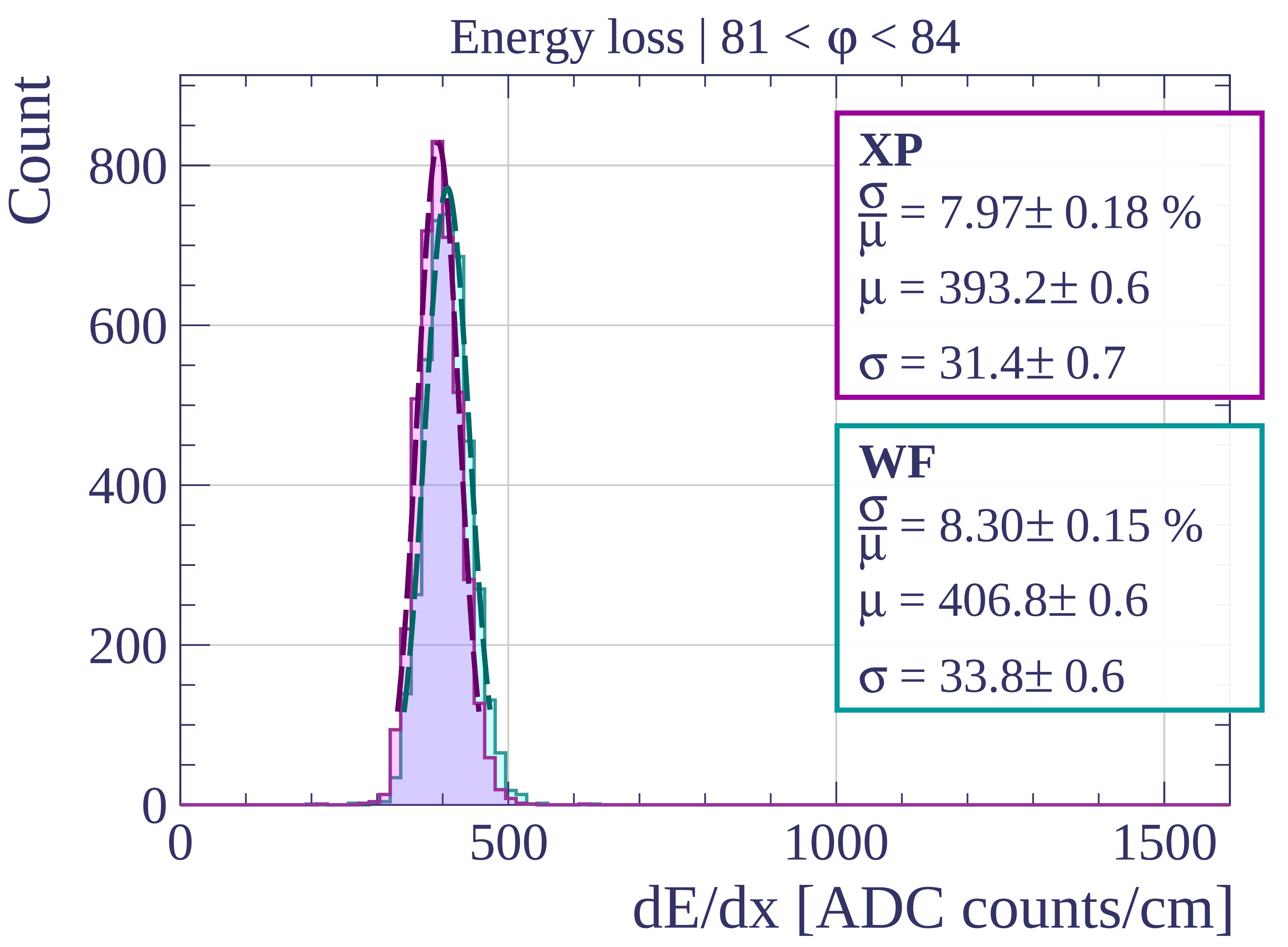


Energy loss | 72 < φ < 75
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.29± 0.20 %
μ = 393.4± 0.7
σ = 32.6± 0.7
WF
σ/μ = 8.14± 0.20 %
μ = 404.0± 0.8
σ = 32.9± 0.8

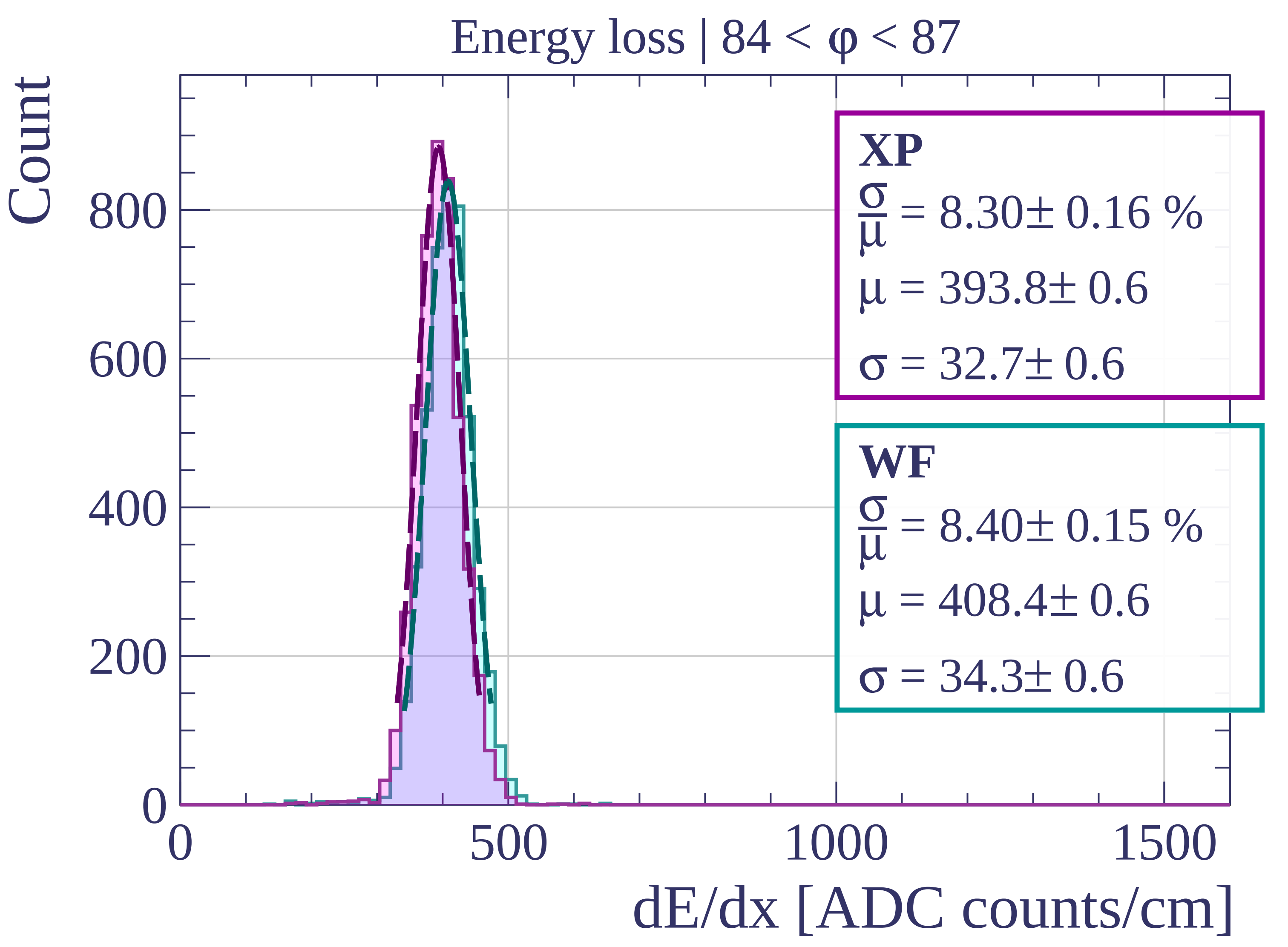


Energy loss | 75 < φ < 78
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.16± 0.15 %
μ = 393.6± 0.6
σ = 32.1± 0.5
WF
σ/μ = 8.55± 0.18 %
μ = 405.2± 0.7
σ = 34.7± 0.7

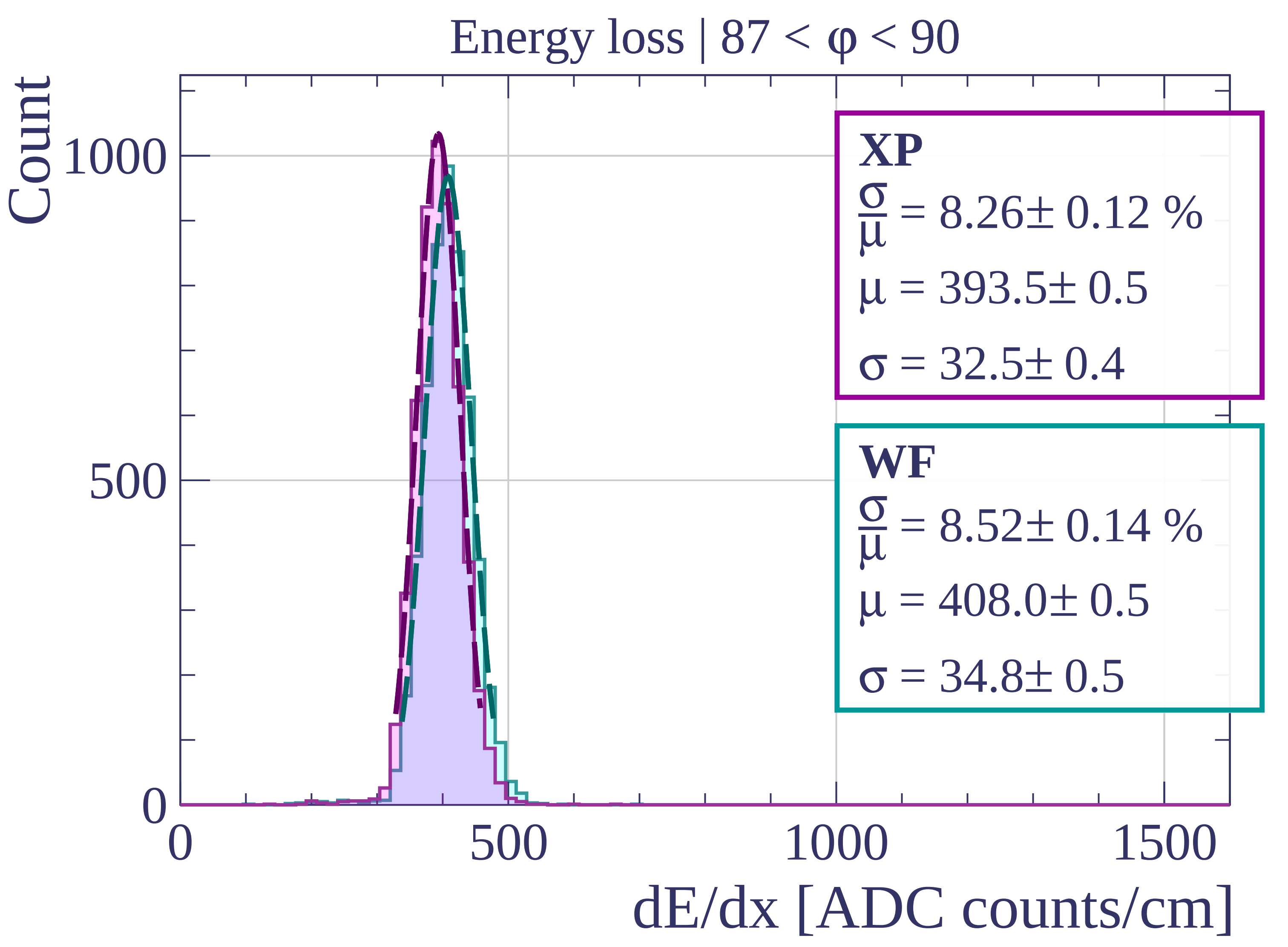

Energy loss | 78 < φ < 81
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 7.69± 0.18 %
μ = 393.2± 0.6
σ = 30.2± 0.7
WF
$\frac{\sigma}{\mu}$ = 7.95± 0.17 %
μ = 406.3± 0.7
σ = 32.3± 0.6

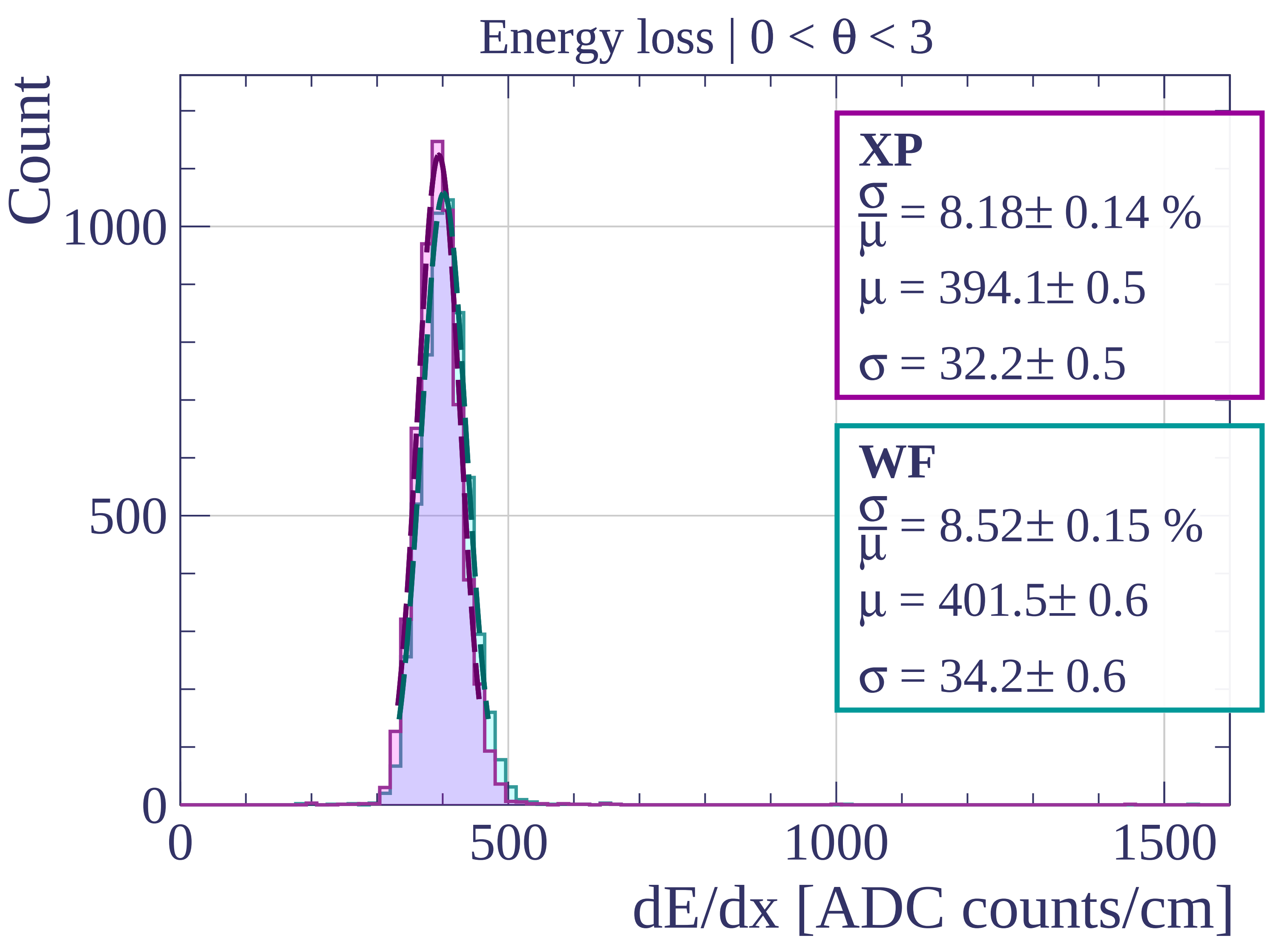


Energy loss | 81 < φ < 84
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
σ/μ = 7.97± 0.18 %
μ = 393.2± 0.6
σ = 31.4± 0.7
WF
σ/μ = 8.30± 0.15 %
μ = 406.8± 0.6
σ = 33.8± 0.6

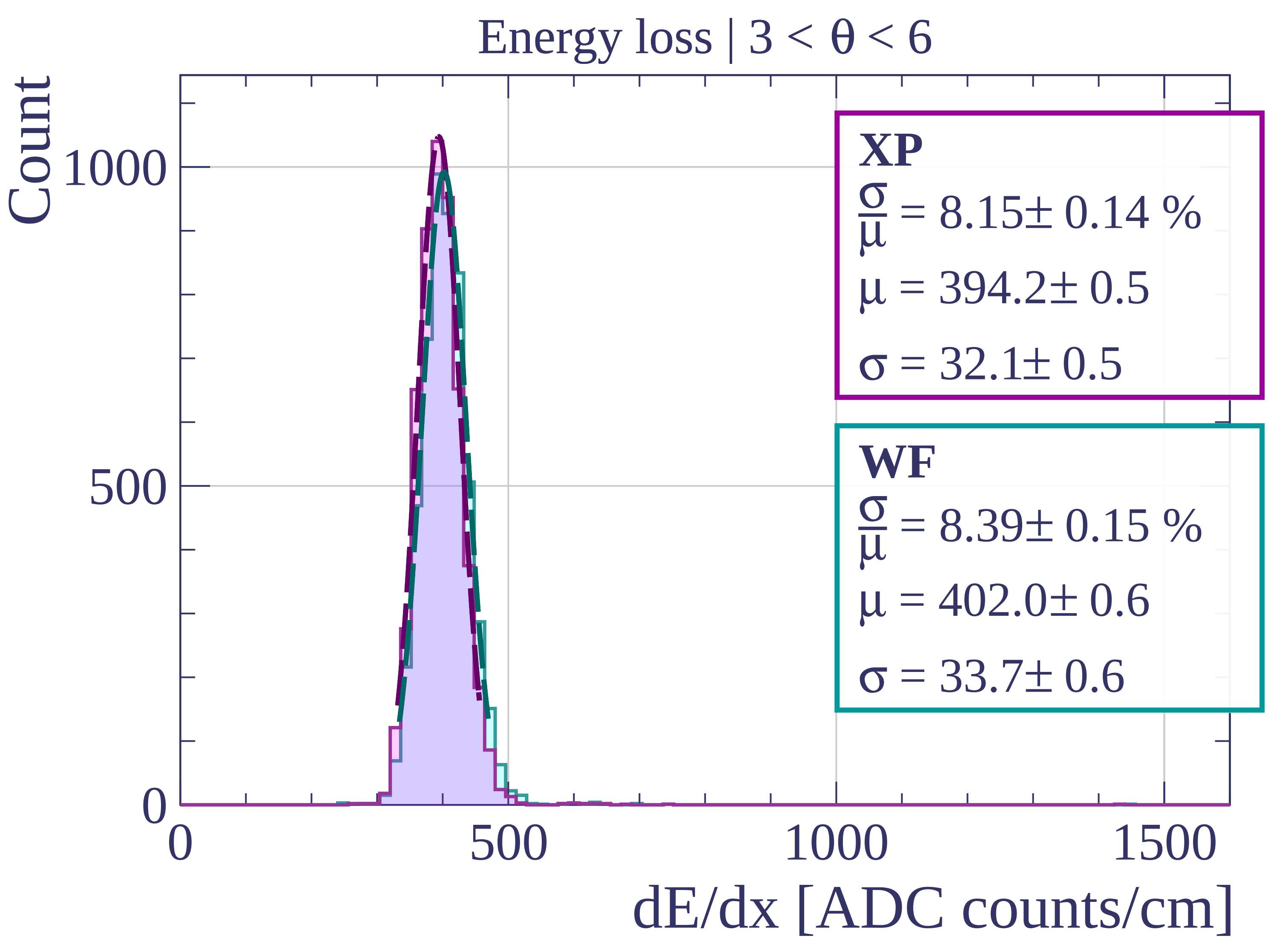


Energy loss | 84 < φ < 87
Count
dE/dx [ADC counts/cm]
0
200
400
600
800
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 8.30± 0.16 %
μ = 393.8± 0.6
σ = 32.7± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.40± 0.15 %
μ = 408.4± 0.6
σ = 34.3± 0.6

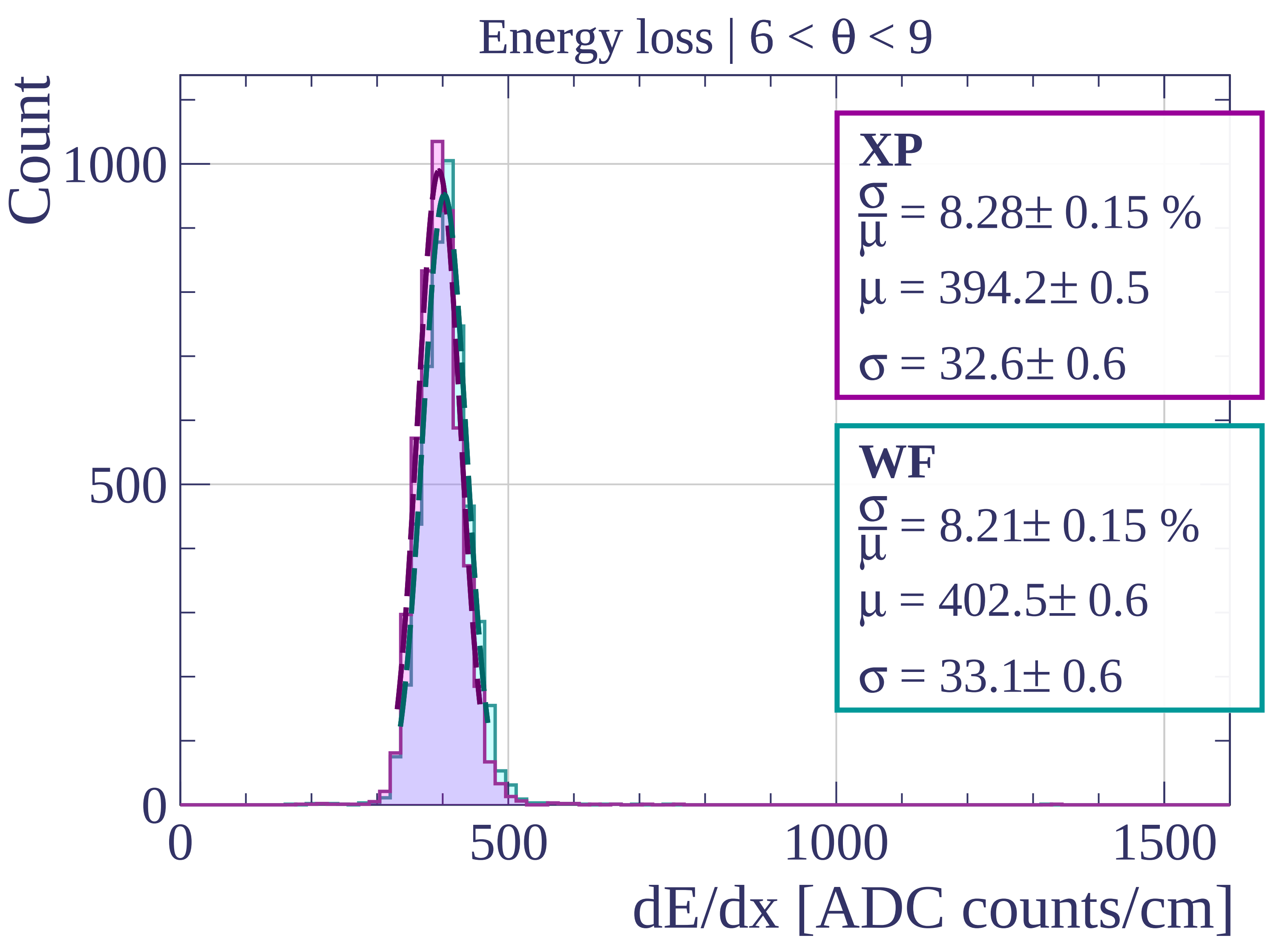

Energy loss | 87 < φ < 90
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.26± 0.12 %
μ = 393.5± 0.5
σ = 32.5± 0.4
WF
σ/μ = 8.52± 0.14 %
μ = 408.0± 0.5
σ = 34.8± 0.5



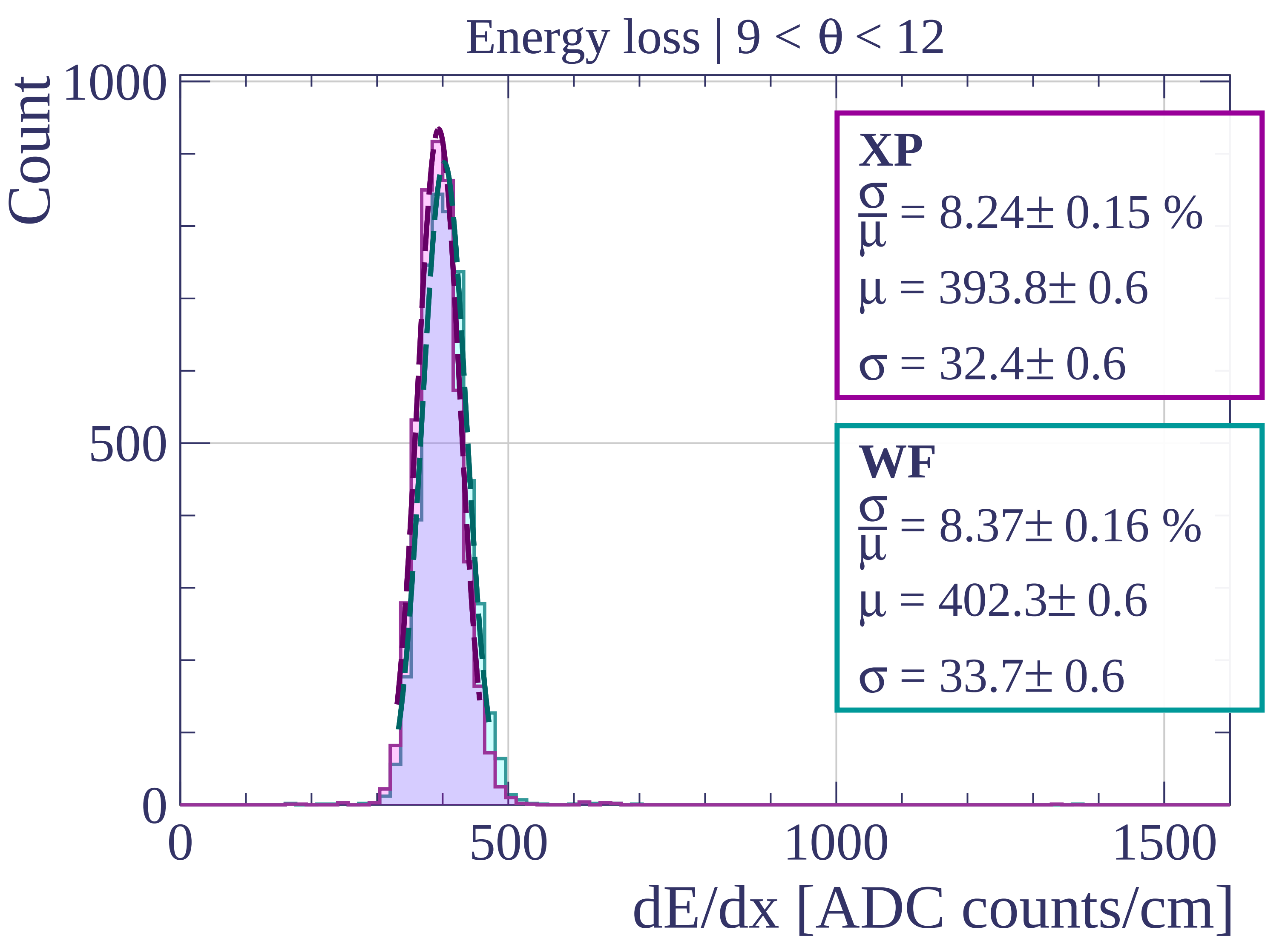
Energy loss | 0 < θ < 3
Count
dE/dx [ADC counts/cm]
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 8.18± 0.14 %
μ = 394.1± 0.5
σ = 32.2± 0.5
WF
$\frac{\sigma}{\mu}$ = 8.52± 0.15 %
μ = 401.5± 0.6
σ = 34.2± 0.6



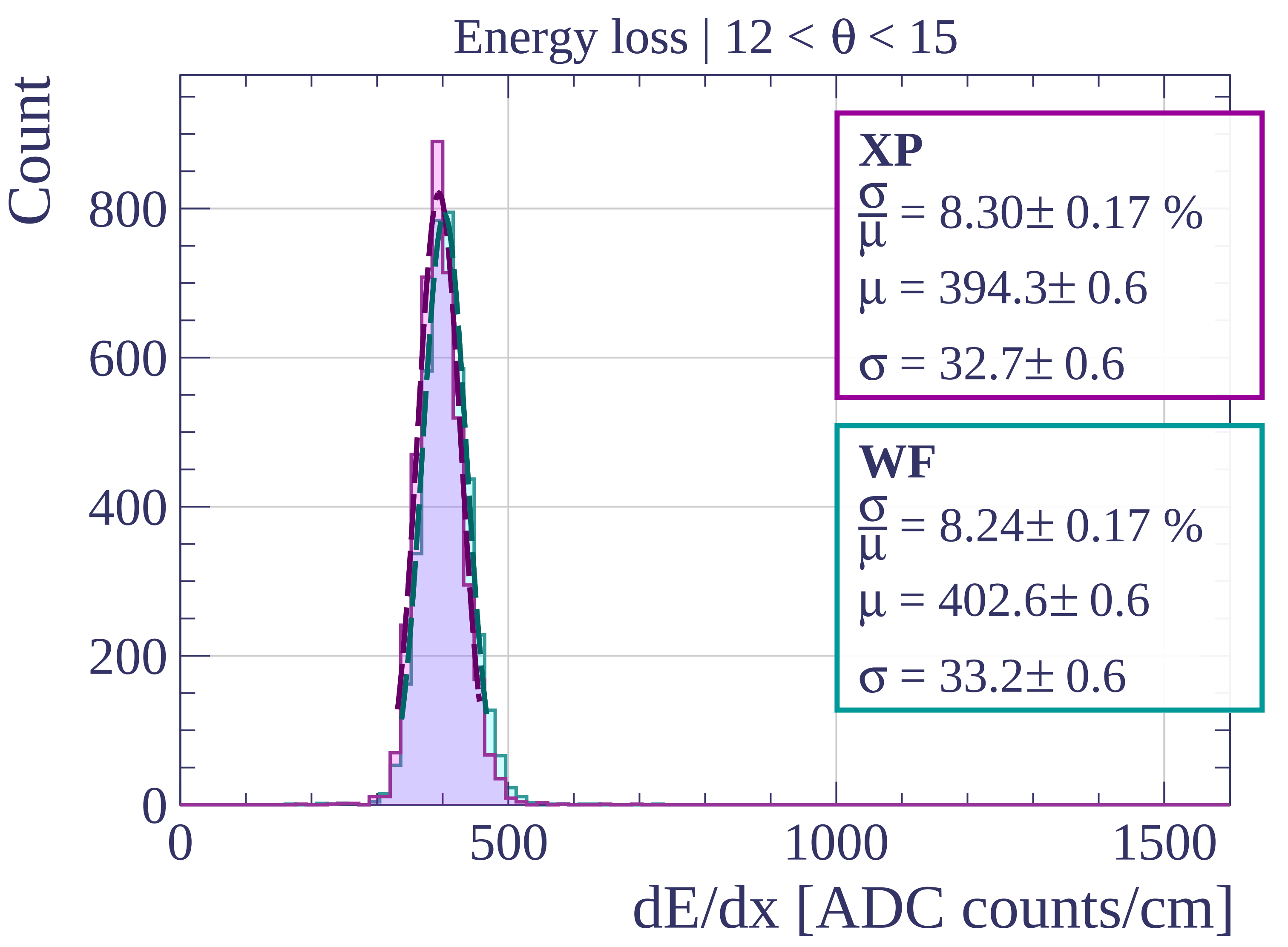
Energy loss | 3 < θ < 6
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.15± 0.14 %
μ = 394.2± 0.5
σ = 32.1± 0.5
WF
σ/μ = 8.39± 0.15 %
μ = 402.0± 0.6
σ = 33.7± 0.6



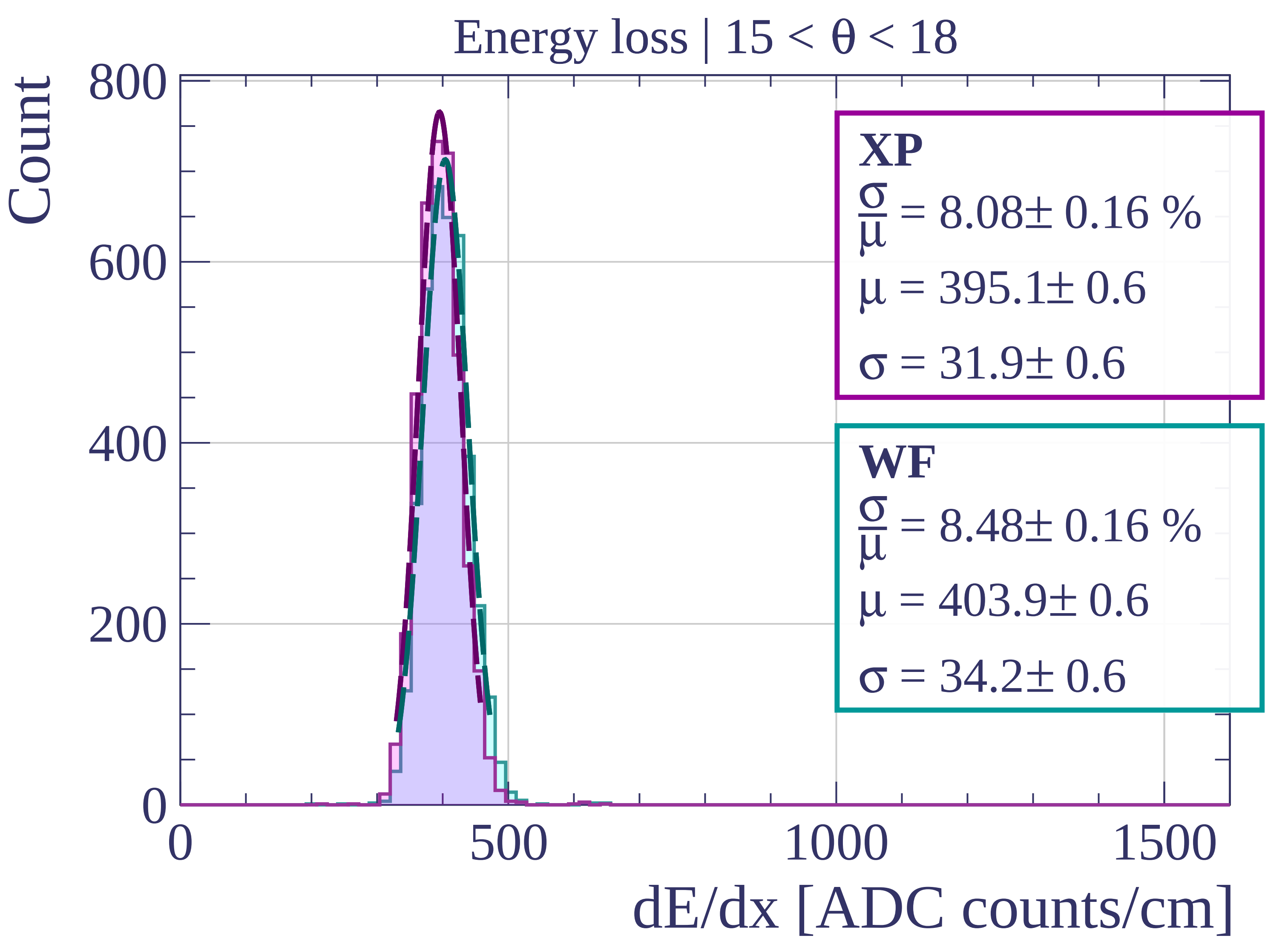
Energy loss | 6 < θ < 9
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.28± 0.15 %
μ = 394.2± 0.5
σ = 32.6± 0.6
WF
σ/μ = 8.21± 0.15 %
μ = 402.5± 0.6
σ = 33.1± 0.6



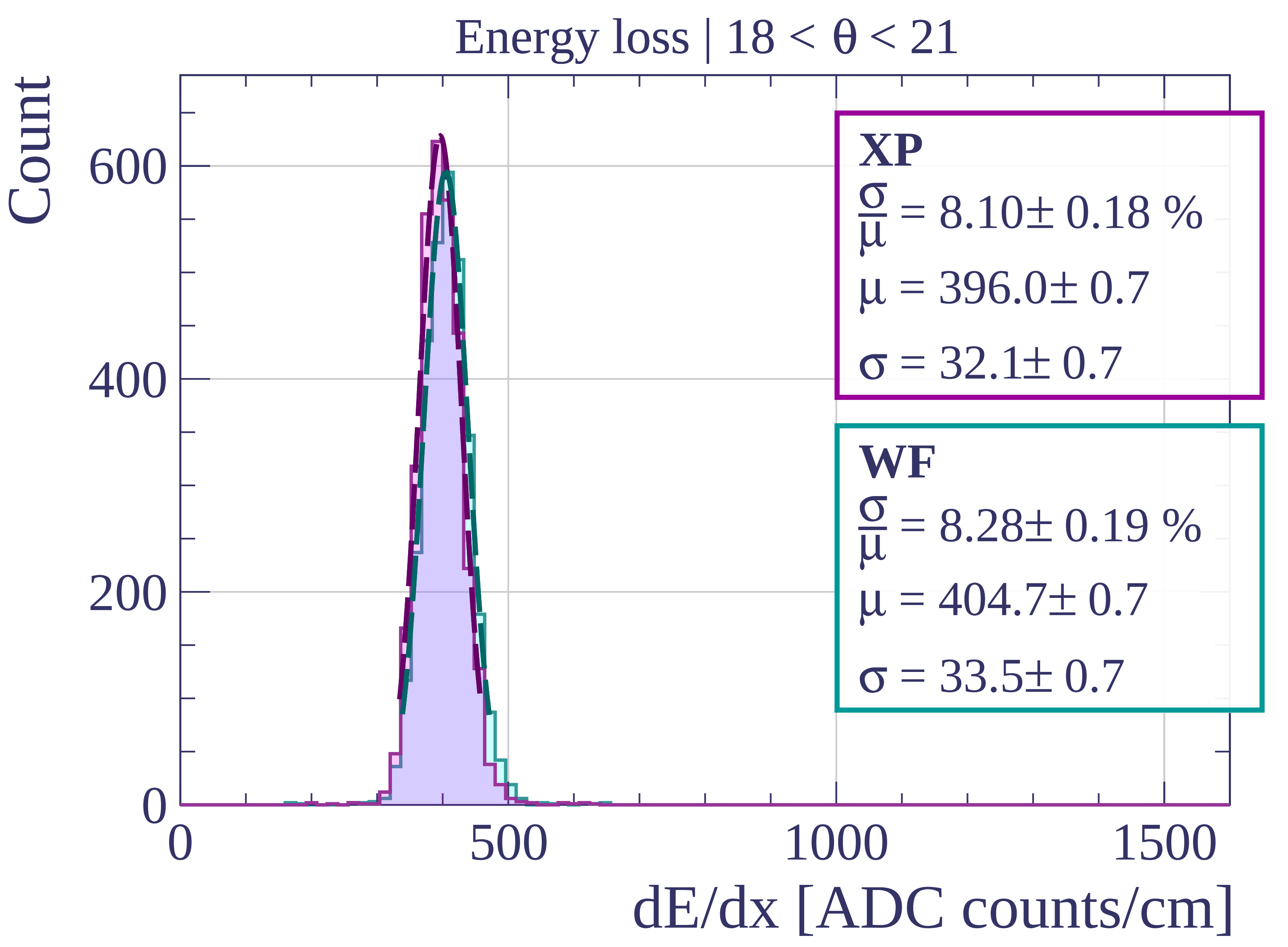
Energy loss | 9 < θ < 12
Count
1000
500
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.24± 0.15 %
μ = 393.8± 0.6
σ = 32.4± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.37± 0.16 %
μ = 402.3± 0.6
σ = 33.7± 0.6



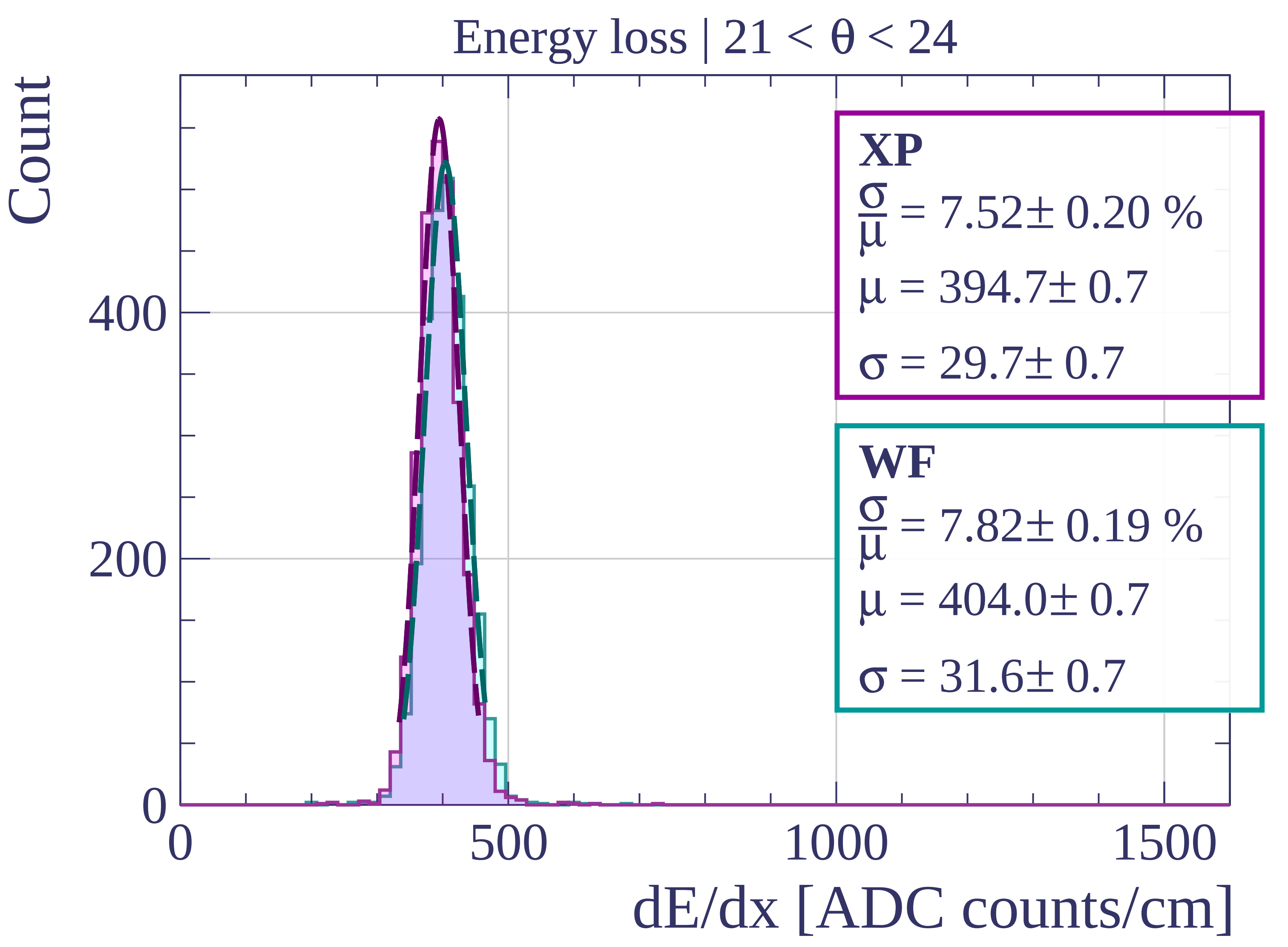
Energy loss | 12 < θ < 15
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 8.30± 0.17 %
μ = 394.3± 0.6
σ = 32.7± 0.6
WF
σ/μ = 8.24± 0.17 %
μ = 402.6± 0.6
σ = 33.2± 0.6



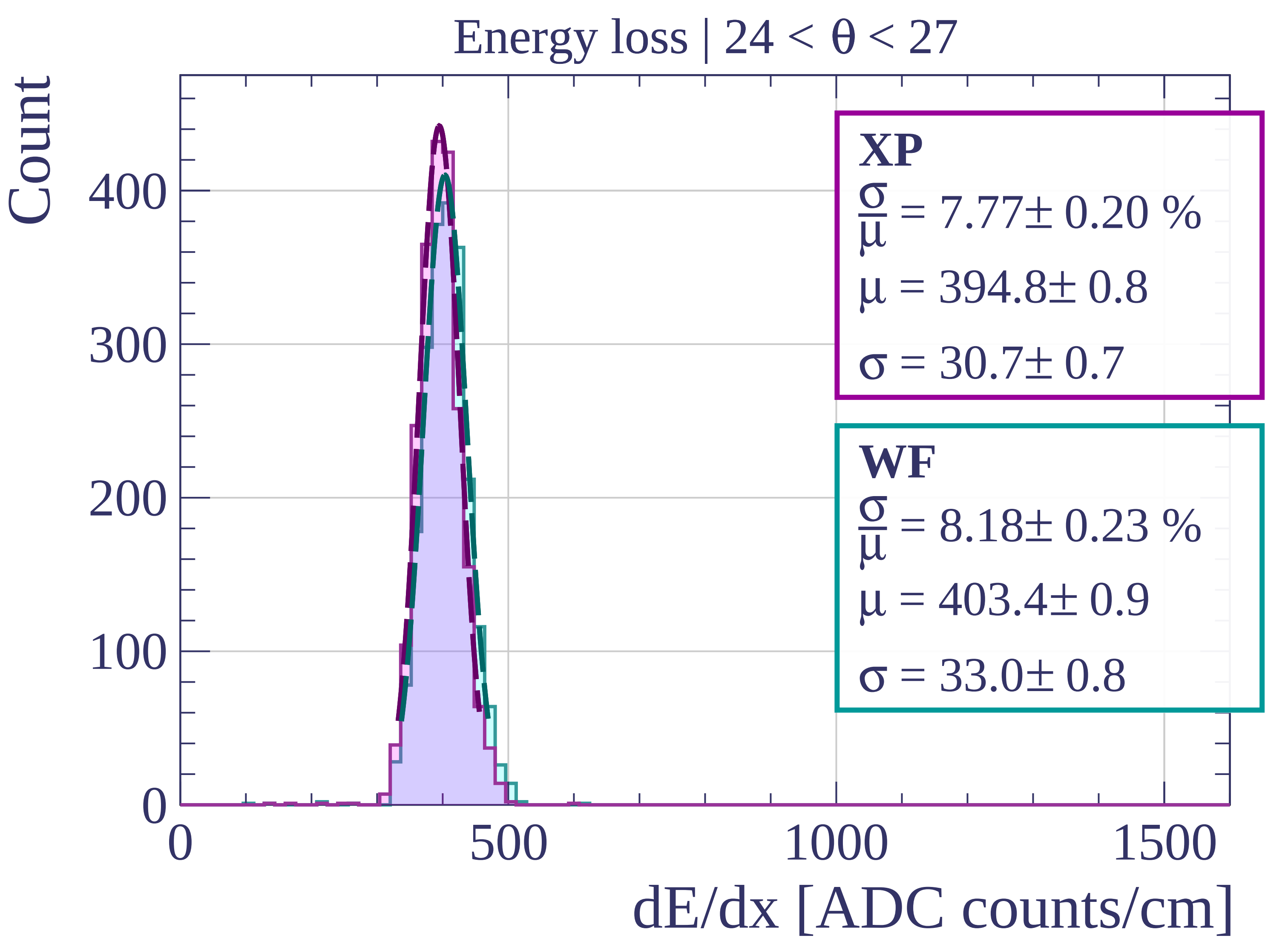
Energy loss | 15 < θ < 18
Count
800
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.08± 0.16 %
μ = 395.1± 0.6
σ = 31.9± 0.6
WF
$\frac{\sigma}{\mu}$ = 8.48± 0.16 %
μ = 403.9± 0.6
σ = 34.2± 0.6

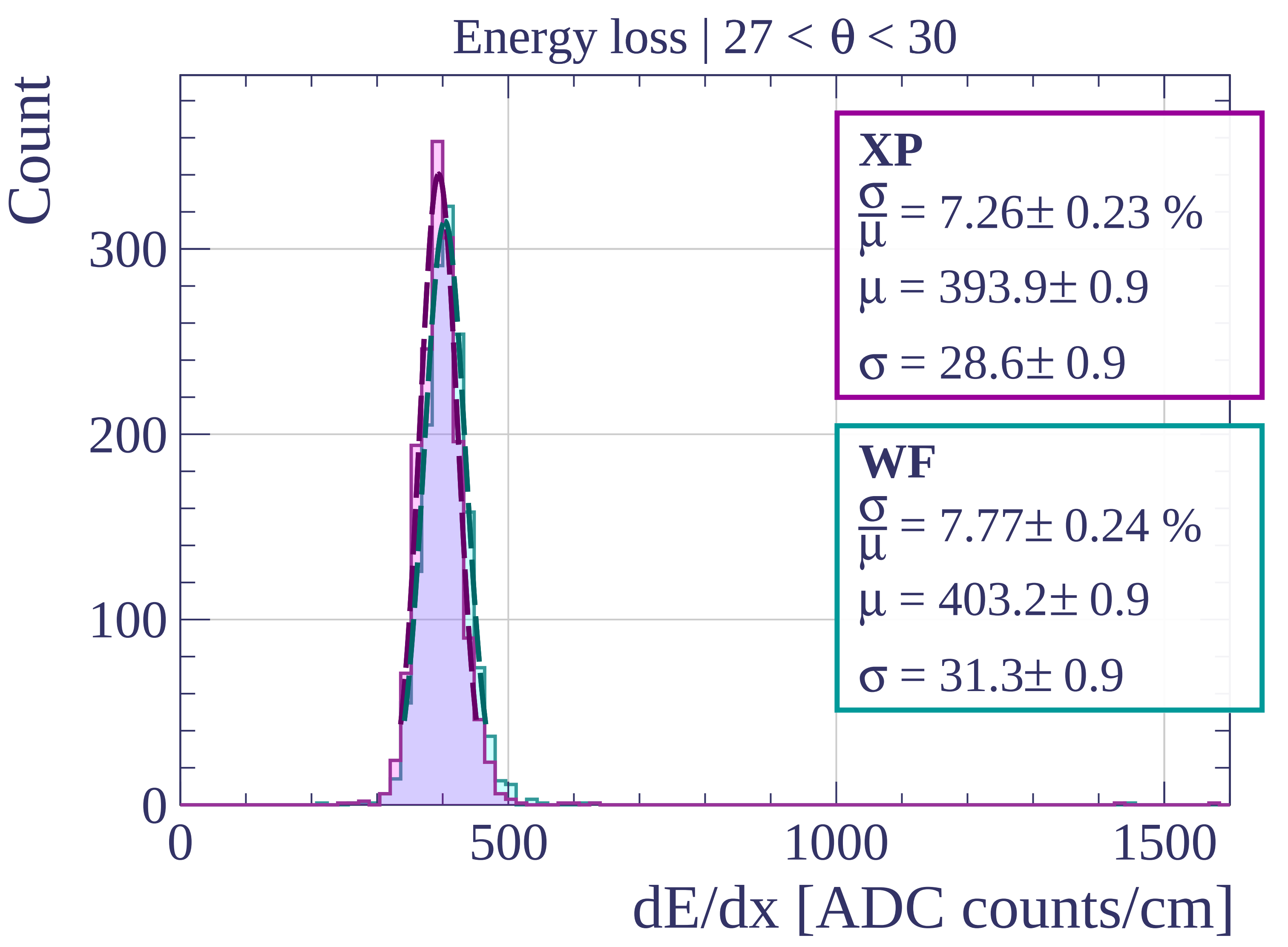


Energy loss | 18 < θ < 21
Count
600
400
200
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 8.10± 0.18 %
μ = 396.0± 0.7
σ = 32.1± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.28± 0.19 %
μ = 404.7± 0.7
σ = 33.5± 0.7

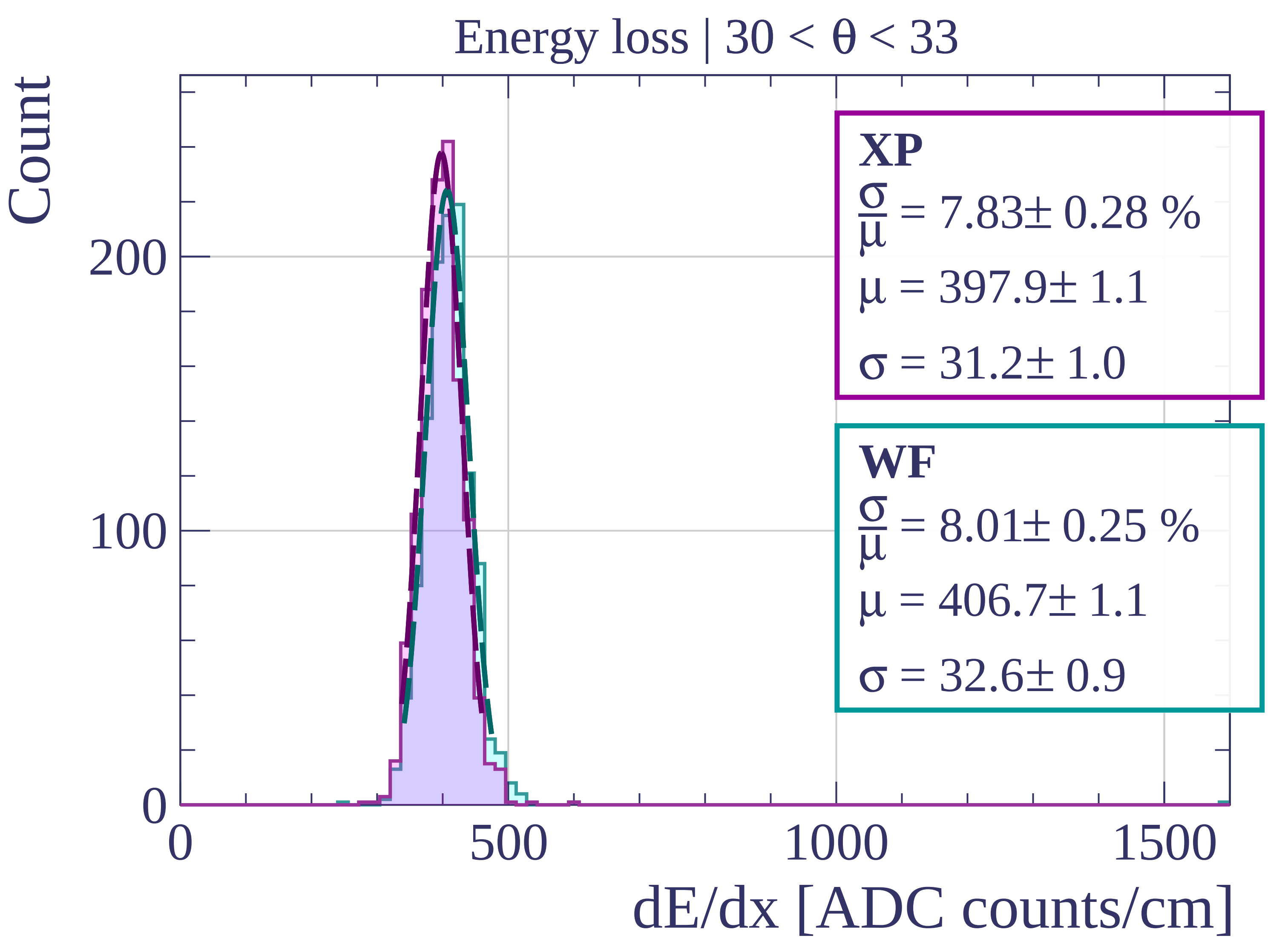


Energy loss | 21 < θ < 24
Count
dE/dx [ADC counts/cm]
0
200
400
0
500
1000
1500
XP
σ/μ = 7.52± 0.20 %
μ = 394.7± 0.7
σ = 29.7± 0.7
WF
σ/μ = 7.82± 0.19 %
μ = 404.0± 0.7
σ = 31.6± 0.7

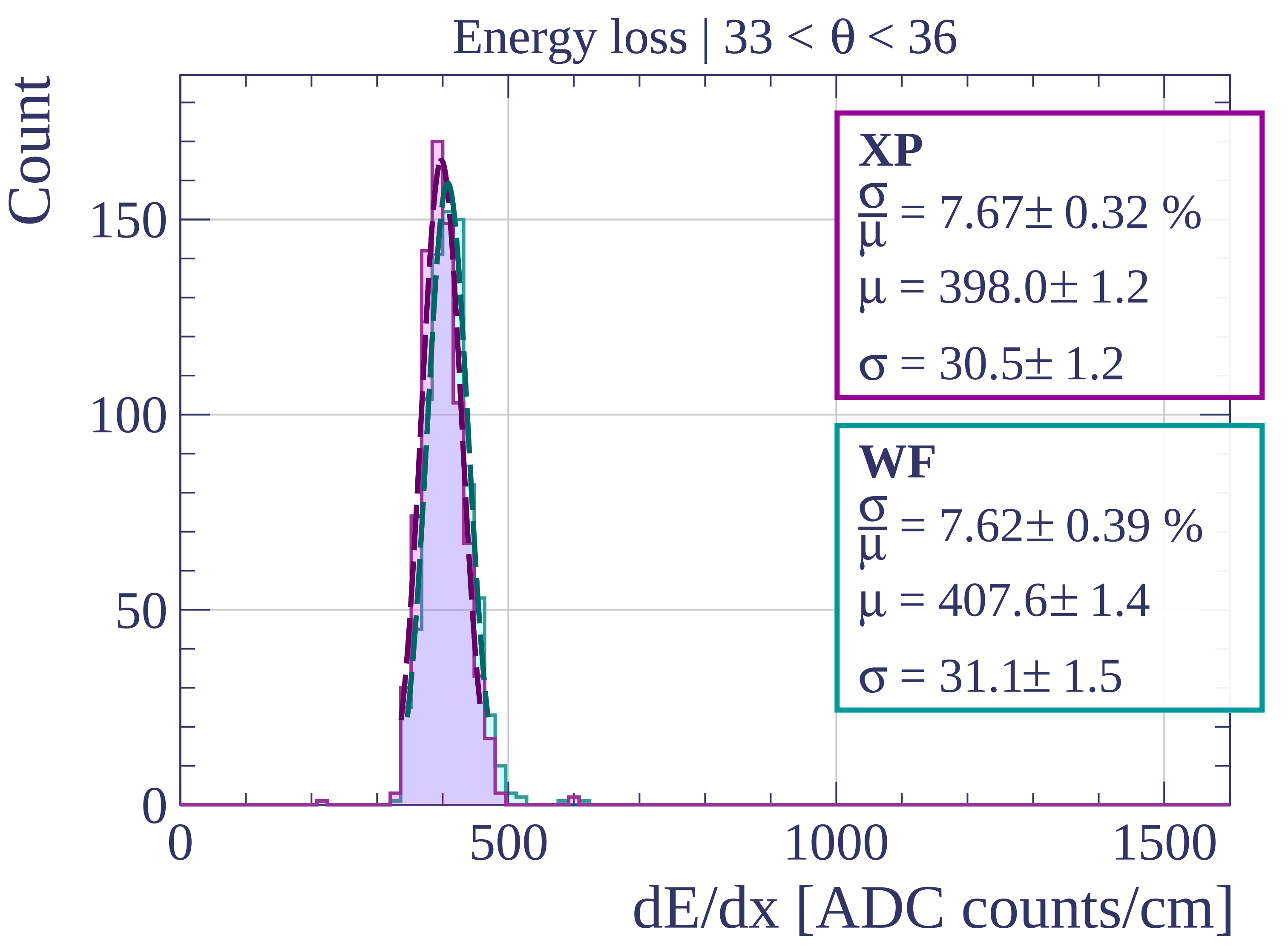

Energy loss | 24 < θ < 27
Count
dE/dx [ADC counts/cm]
0
100
200
300
400
0
500
1000
1500
XP
$\frac{\sigma}{\mu}$ = 7.77± 0.20 %
μ = 394.8± 0.8
σ = 30.7± 0.7
WF
$\frac{\sigma}{\mu}$ = 8.18± 0.23 %
μ = 403.4± 0.9
σ = 33.0± 0.8

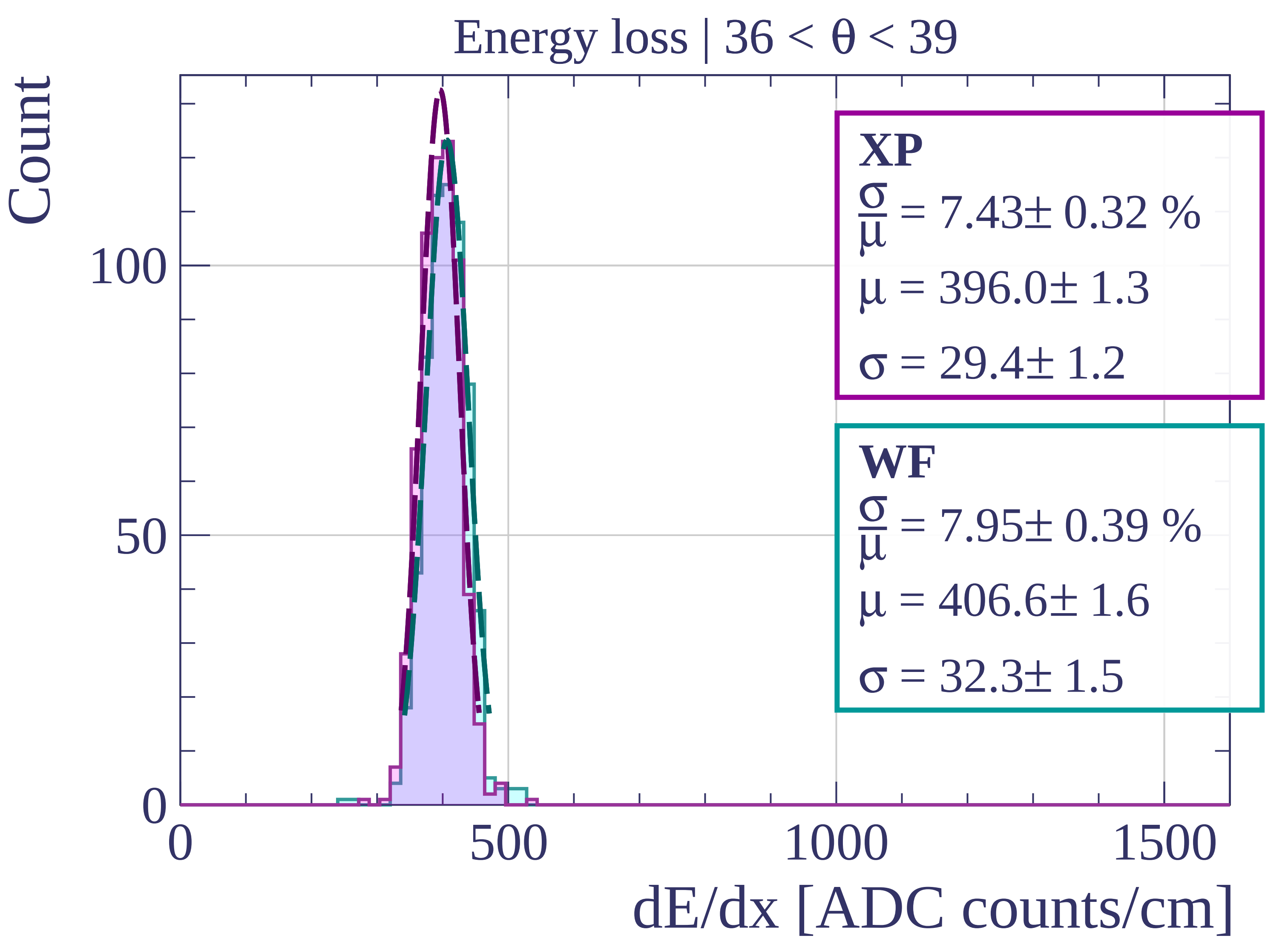


Energy loss | 27 < θ < 30
Count
dE/dx [ADC counts/cm]
0
100
200
300
0
500
1000
1500
XP
σ/μ = 7.26± 0.23 %
μ = 393.9± 0.9
σ = 28.6± 0.9
WF
σ/μ = 7.77± 0.24 %
μ = 403.2± 0.9
σ = 31.3± 0.9

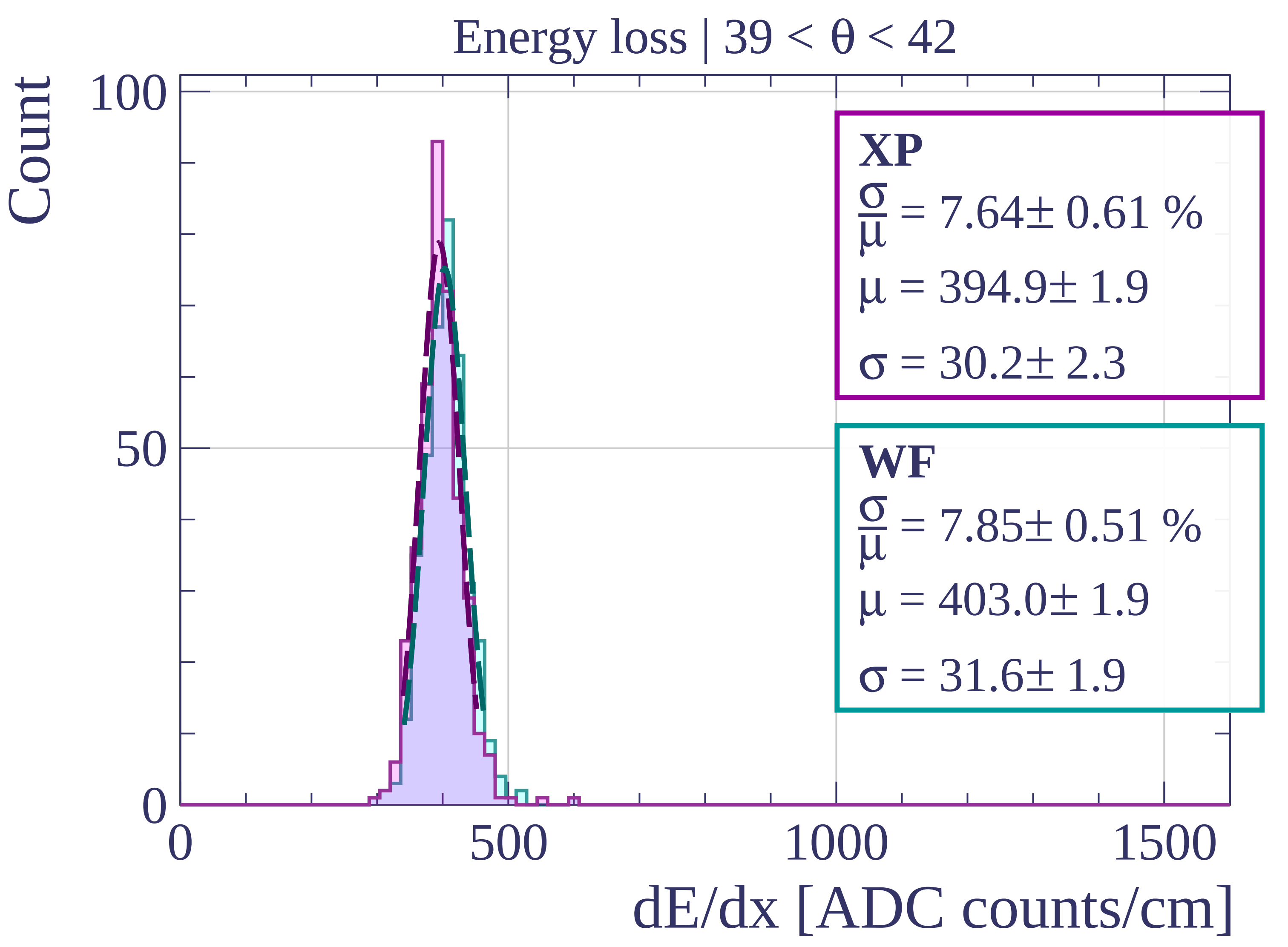


Energy loss | 30 < θ < 33
Count
0
100
200
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.83± 0.28 %
μ = 397.9± 1.1
σ = 31.2± 1.0
WF
σ/μ = 8.01± 0.25 %
μ = 406.7± 1.1
σ = 32.6± 0.9

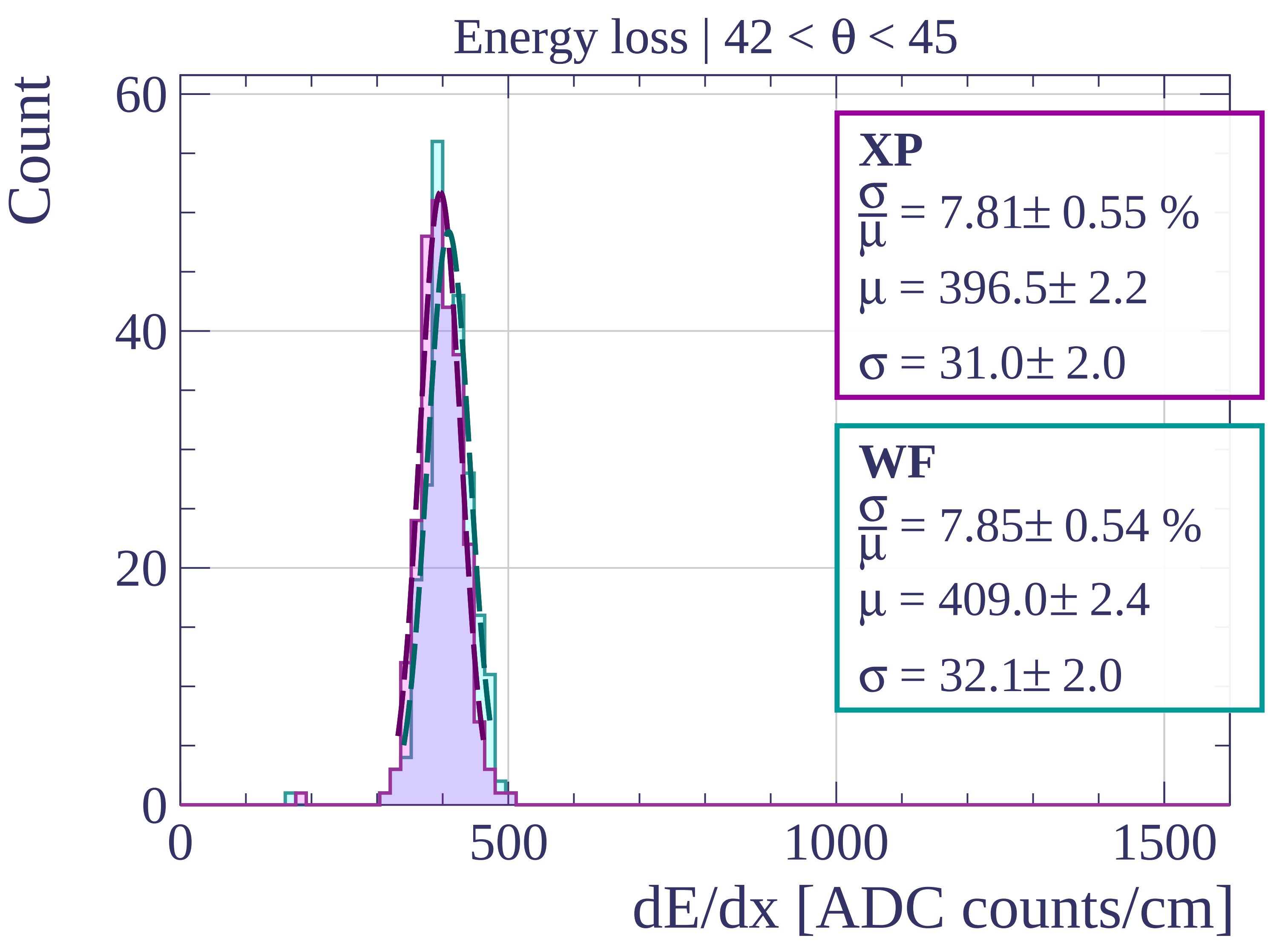

Energy loss | 33 < θ < 36
Count
150
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.67± 0.32 %
μ = 398.0± 1.2
σ = 30.5± 1.2
WF
σ/μ = 7.62± 0.39 %
μ = 407.6± 1.4
σ = 31.1± 1.5



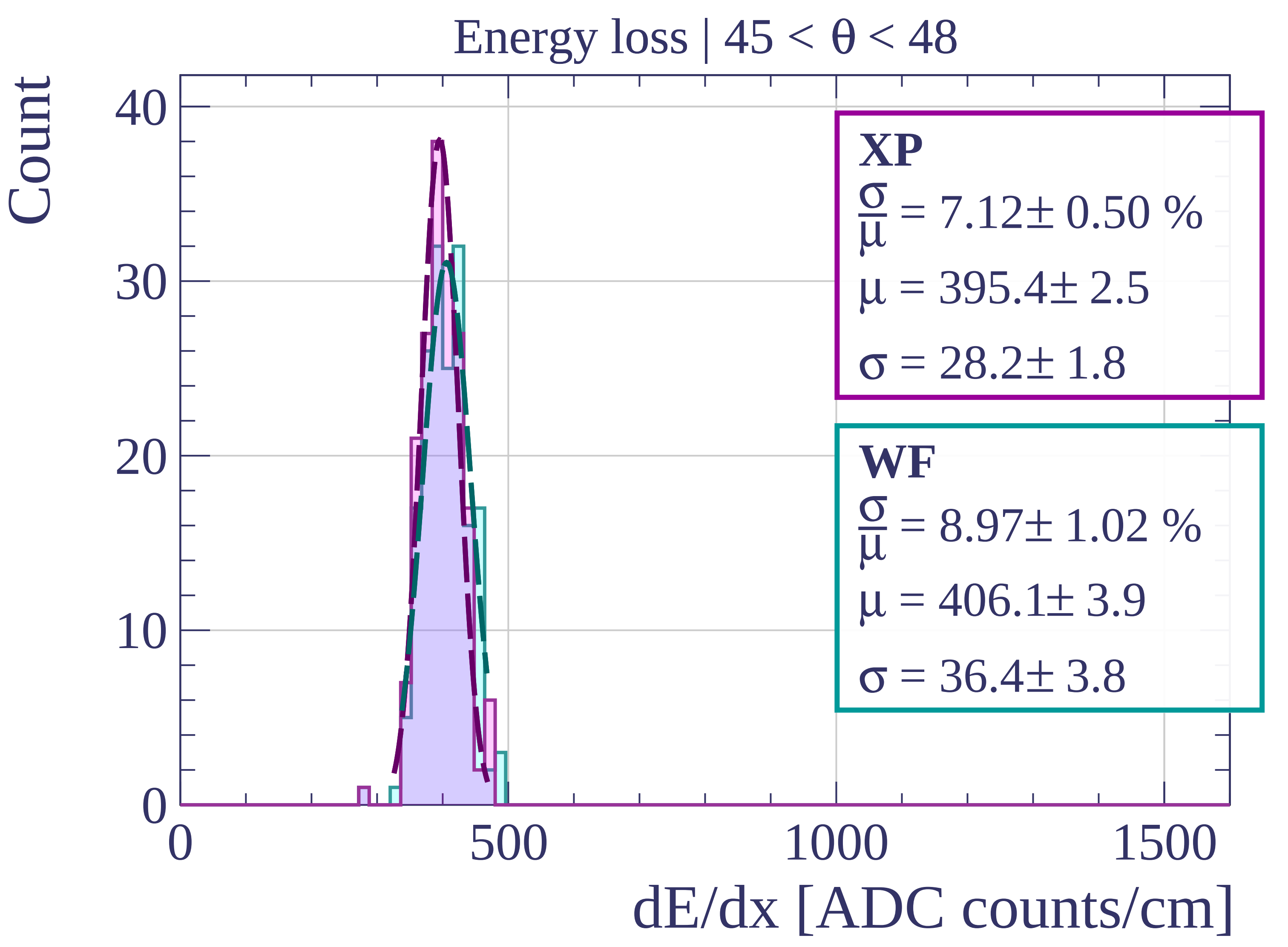
Energy loss | 36 < θ < 39
Count
dE/dx [ADC counts/cm]
0
50
100
0
500
1000
1500
XP
σ/μ = 7.43± 0.32 %
μ = 396.0± 1.3
σ = 29.4± 1.2
WF
σ/μ = 7.95± 0.39 %
μ = 406.6± 1.6
σ = 32.3± 1.5



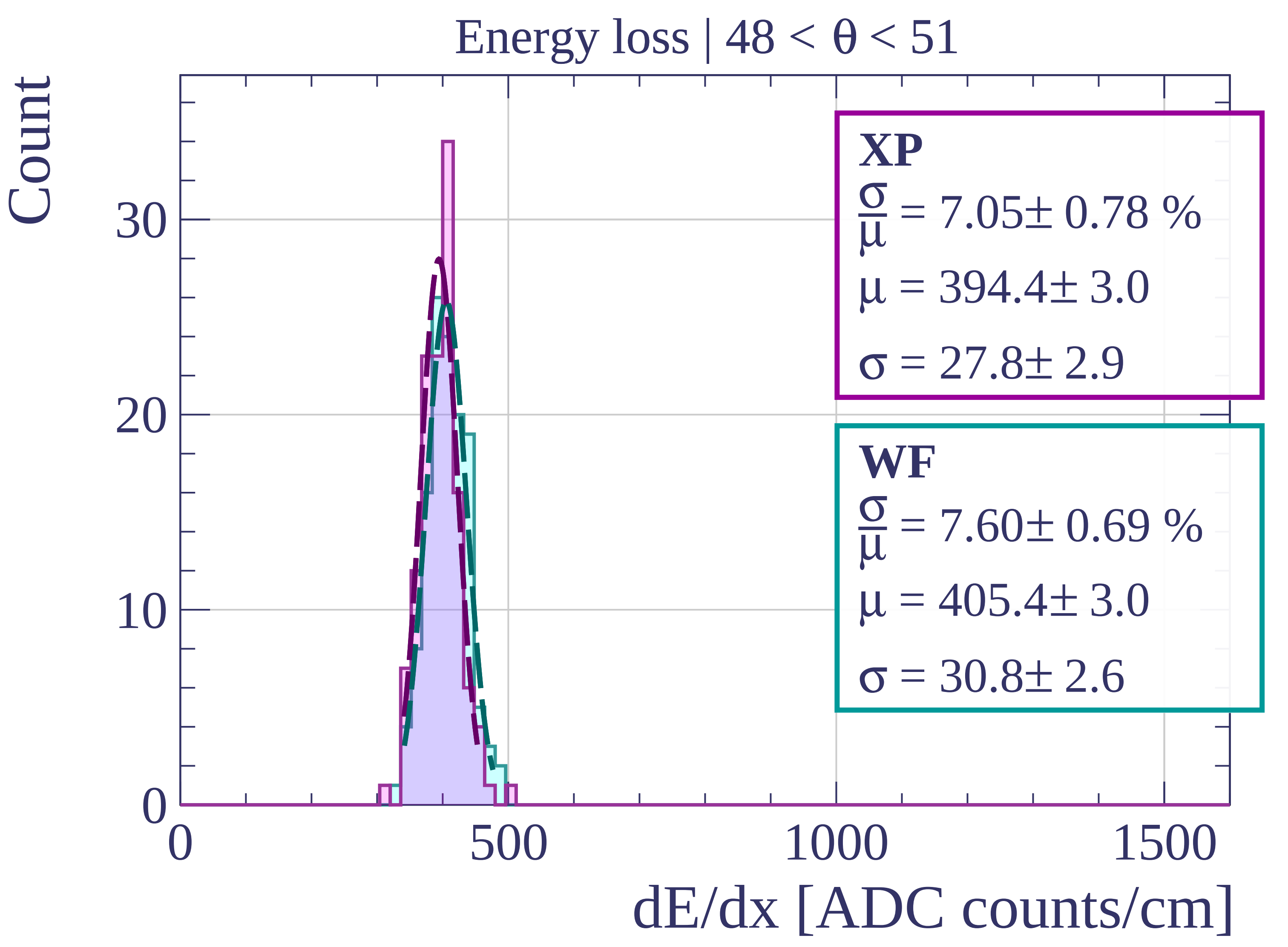
Energy loss | 39 < θ < 42
Count
100
50
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.64± 0.61 %
μ = 394.9± 1.9
σ = 30.2± 2.3
WF
σ/μ = 7.85± 0.51 %
μ = 403.0± 1.9
σ = 31.6± 1.9

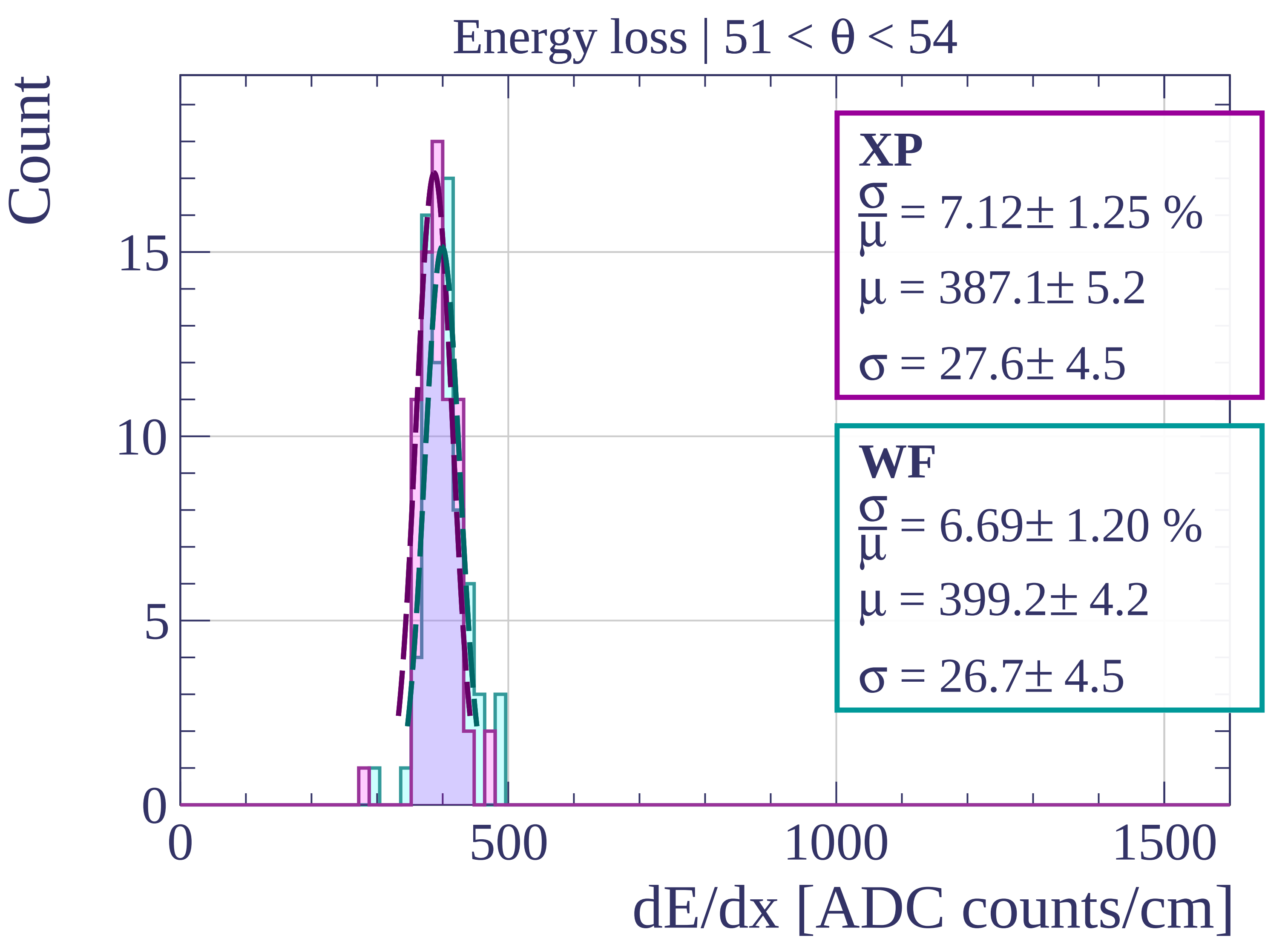


Energy loss | 42 < θ < 45
Count
60
40
20
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.81± 0.55 %
μ = 396.5± 2.2
σ = 31.0± 2.0
WF
σ/μ = 7.85± 0.54 %
μ = 409.0± 2.4
σ = 32.1± 2.0

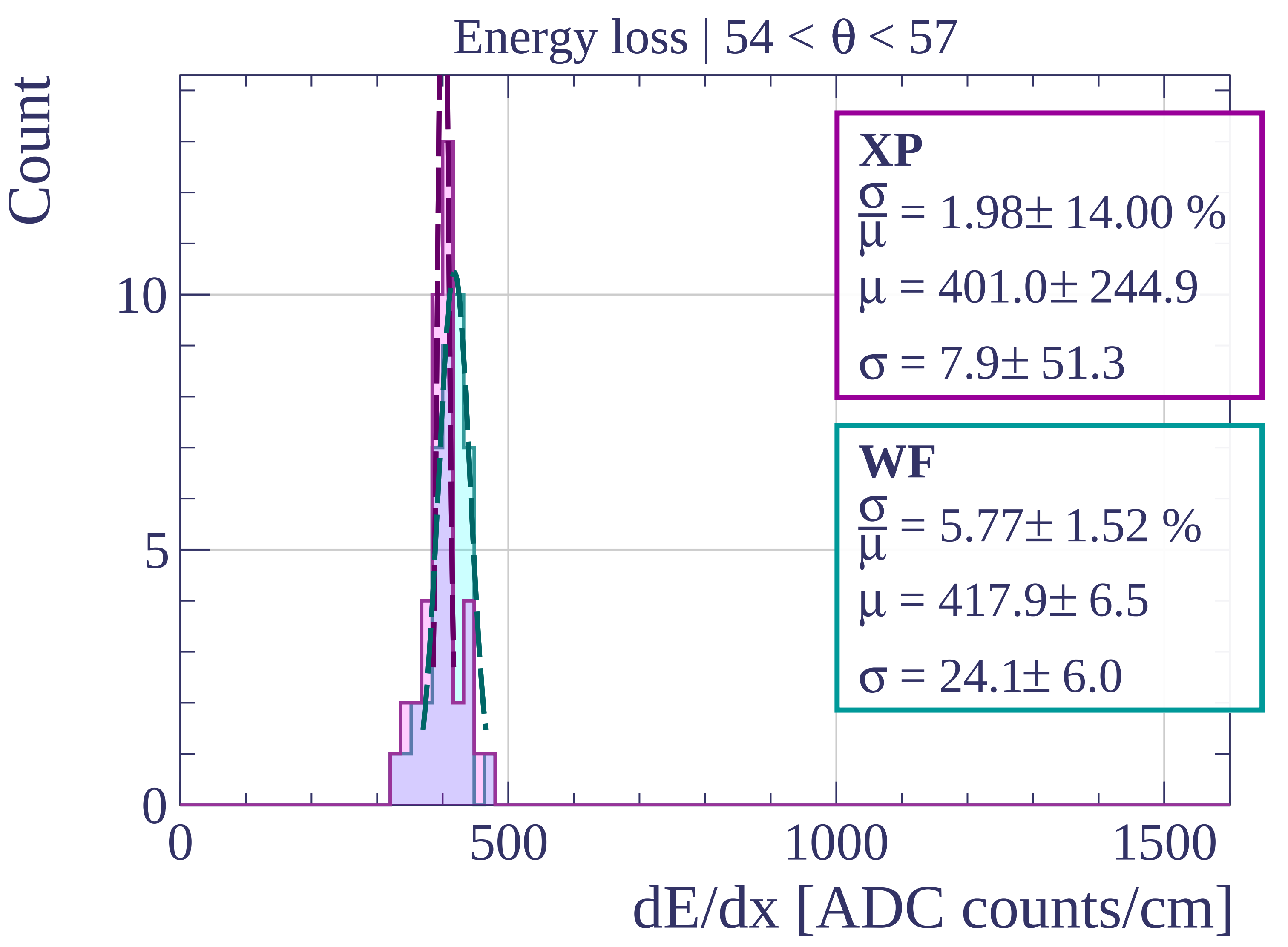


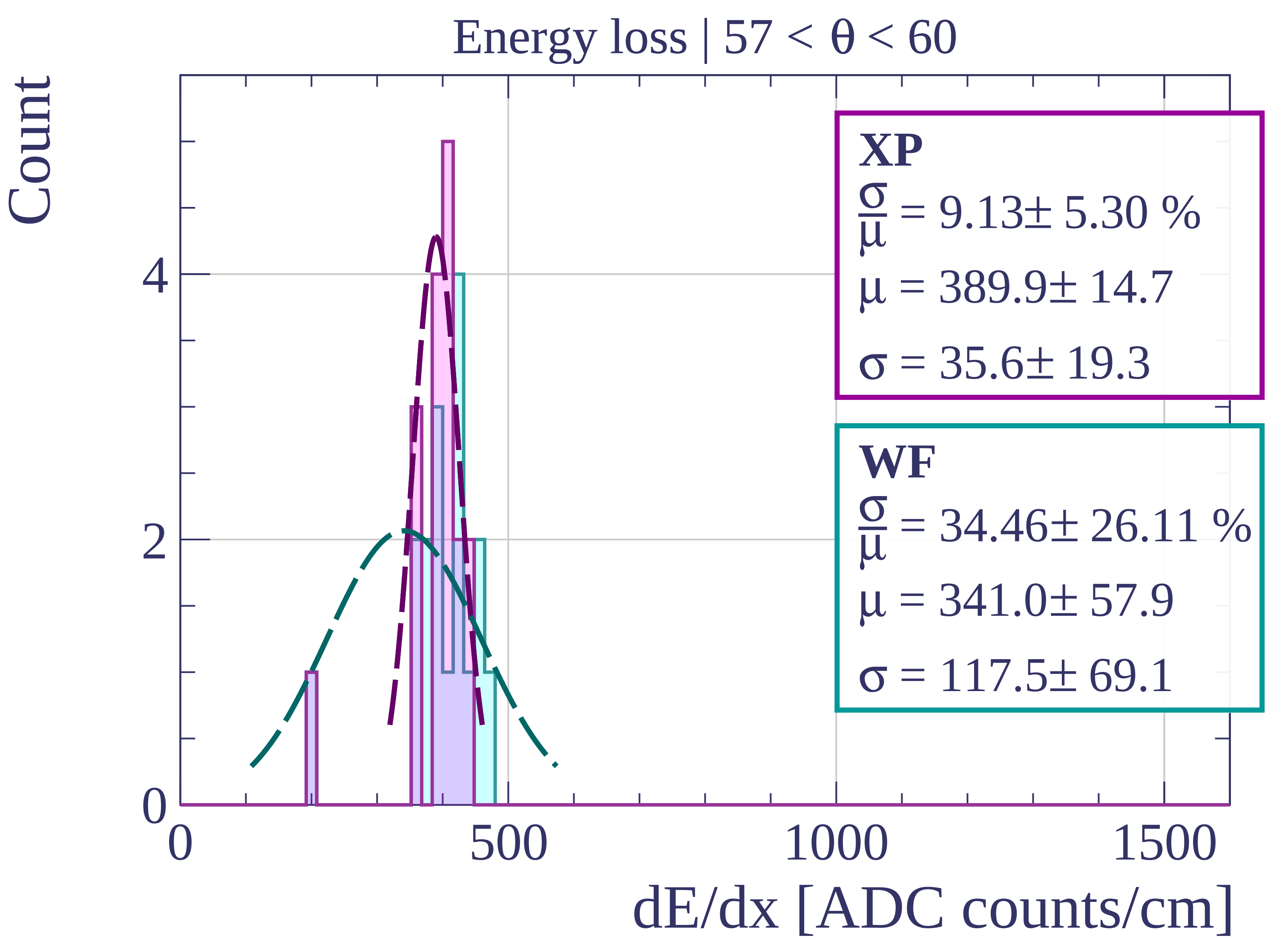
Energy loss | 45 < θ < 48
Count
40
30
20
10
0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 7.12± 0.50 %
μ = 395.4± 2.5
σ = 28.2± 1.8
WF
σ/μ = 8.97± 1.02 %
μ = 406.1± 3.9
σ = 36.4± 3.8

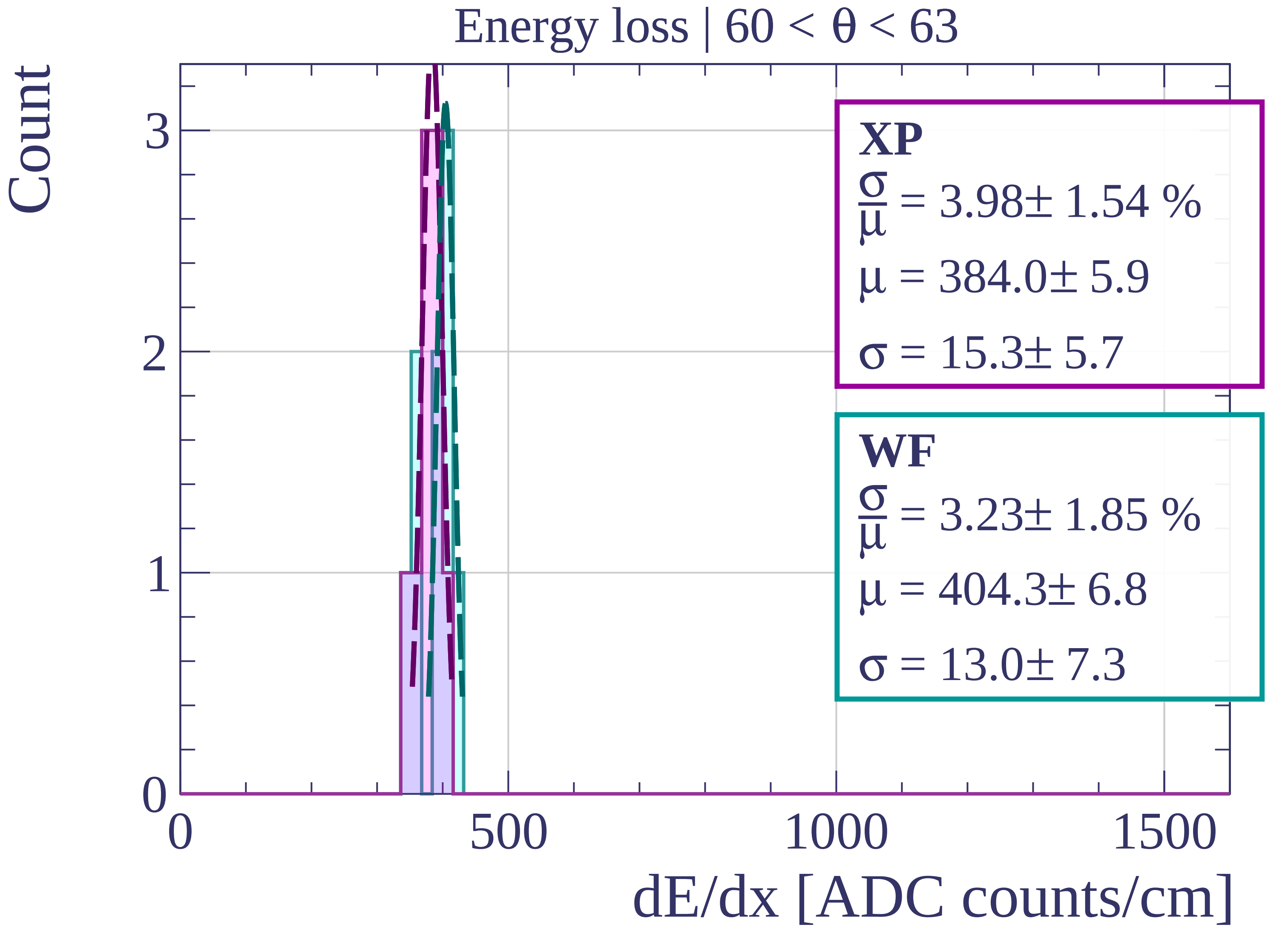

Energy loss | 48 < θ < 51
Count
dE/dx [ADC counts/cm]
0
10
20
30
0
500
1000
1500
XP
σ/μ = 7.05± 0.78 %
μ = 394.4± 3.0
σ = 27.8± 2.9
WF
σ/μ = 7.60± 0.69 %
μ = 405.4± 3.0
σ = 30.8± 2.6



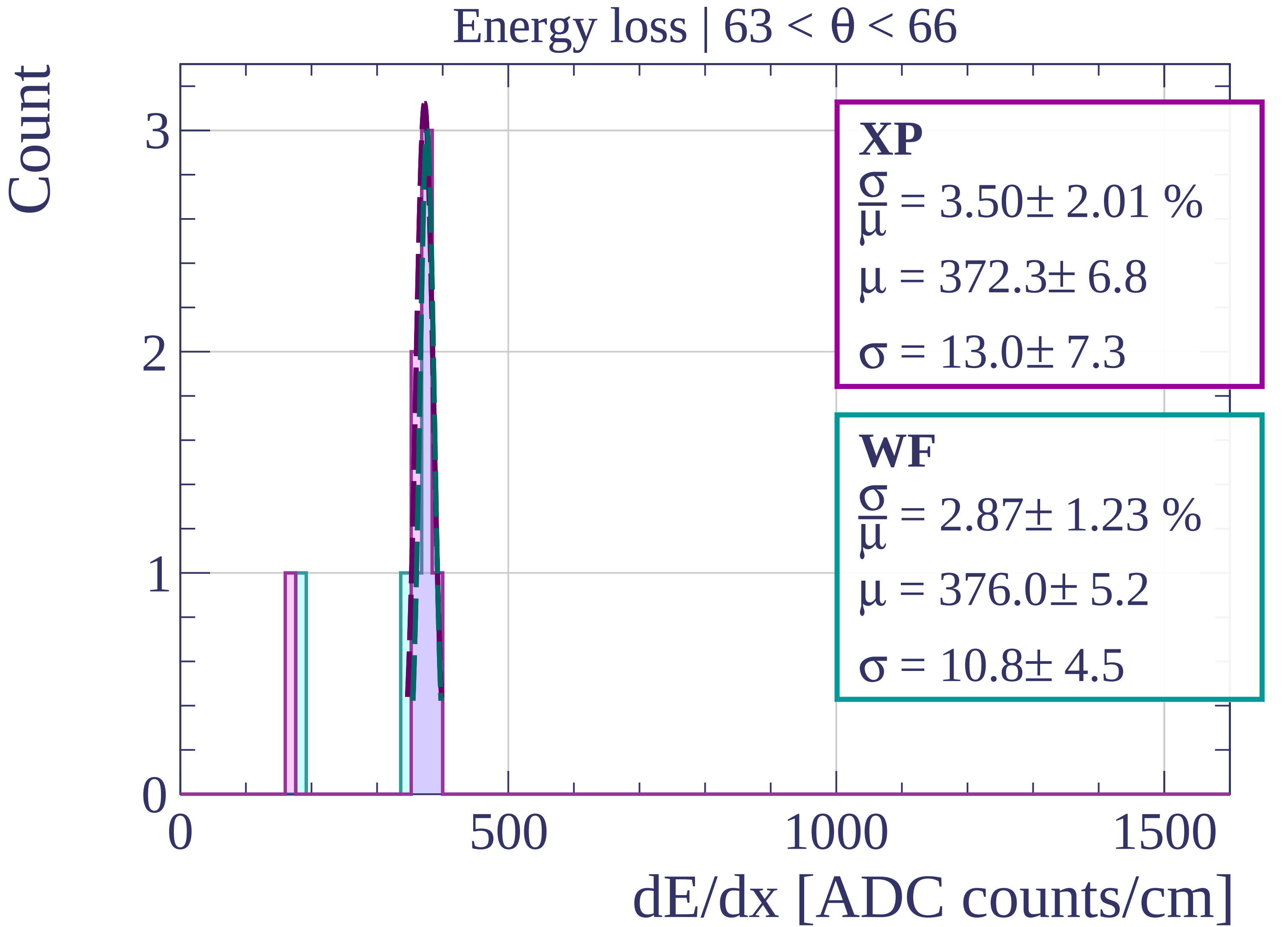
Energy loss | 51 < θ < 54
Count
dE/dx [ADC counts/cm]
0
5
10
15
0
500
1000
1500
XP
σ/μ = 7.12± 1.25 %
μ = 387.1± 5.2
σ = 27.6± 4.5
WF
σ/μ = 6.69± 1.20 %
μ = 399.2± 4.2
σ = 26.7± 4.5


Energy loss | 54 < θ < 57
Count
dE/dx [ADC counts/cm]
0
5
10
0
500
1000
1500
XP
σ/μ = 1.98± 14.00 %
μ = 401.0± 244.9
σ = 7.9± 51.3
WF
σ/μ = 5.77± 1.52 %
μ = 417.9± 6.5
σ = 24.1± 6.0

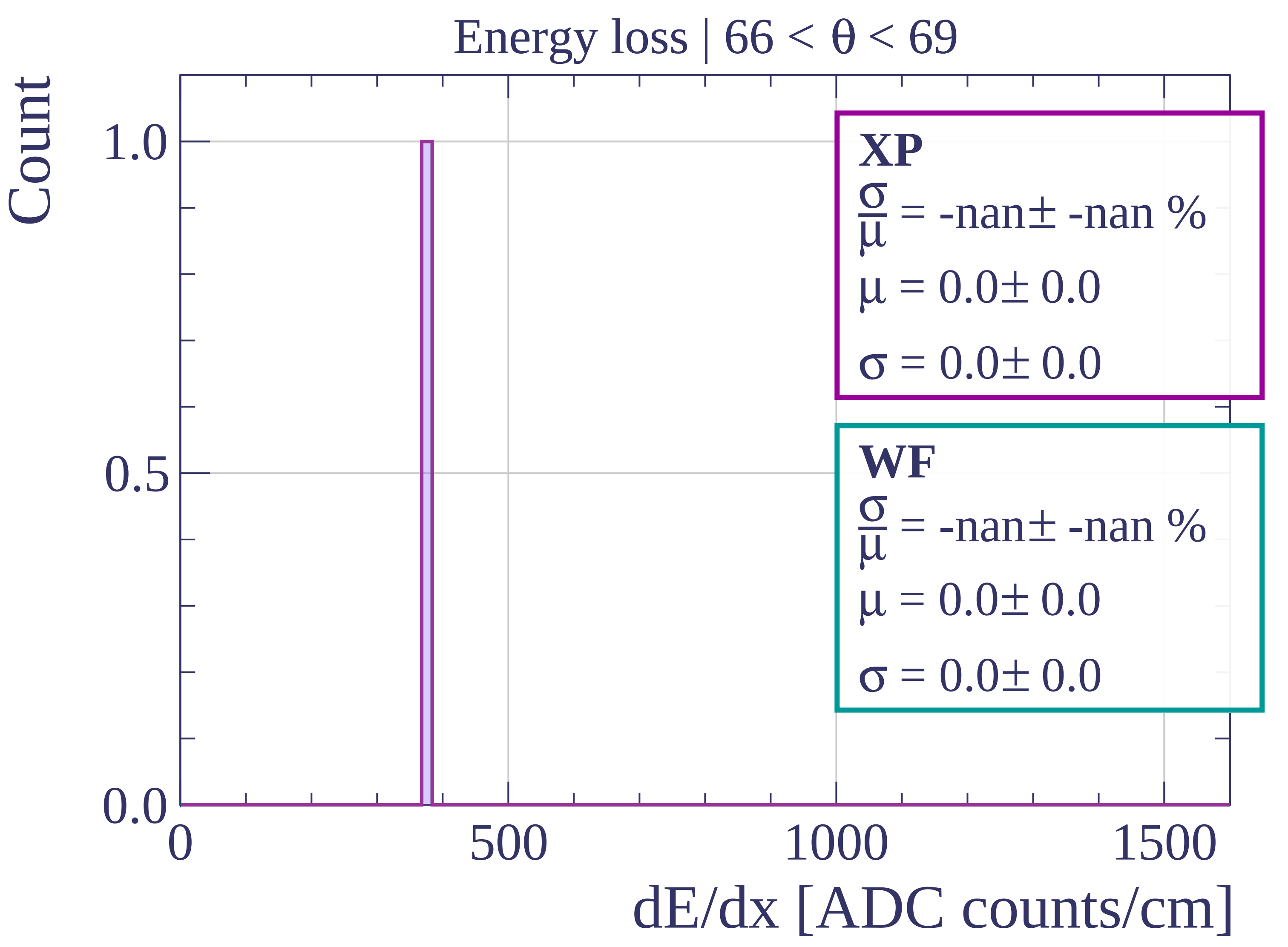

Energy loss | 57 < θ < 60
Count
dE/dx [ADC counts/cm]
0
2
4
0
500
1000
1500
XP
σ/μ = 9.13± 5.30 %
μ = 389.9± 14.7
σ = 35.6± 19.3
WF
σ/μ = 34.46± 26.11 %
μ = 341.0± 57.9
σ = 117.5± 69.1



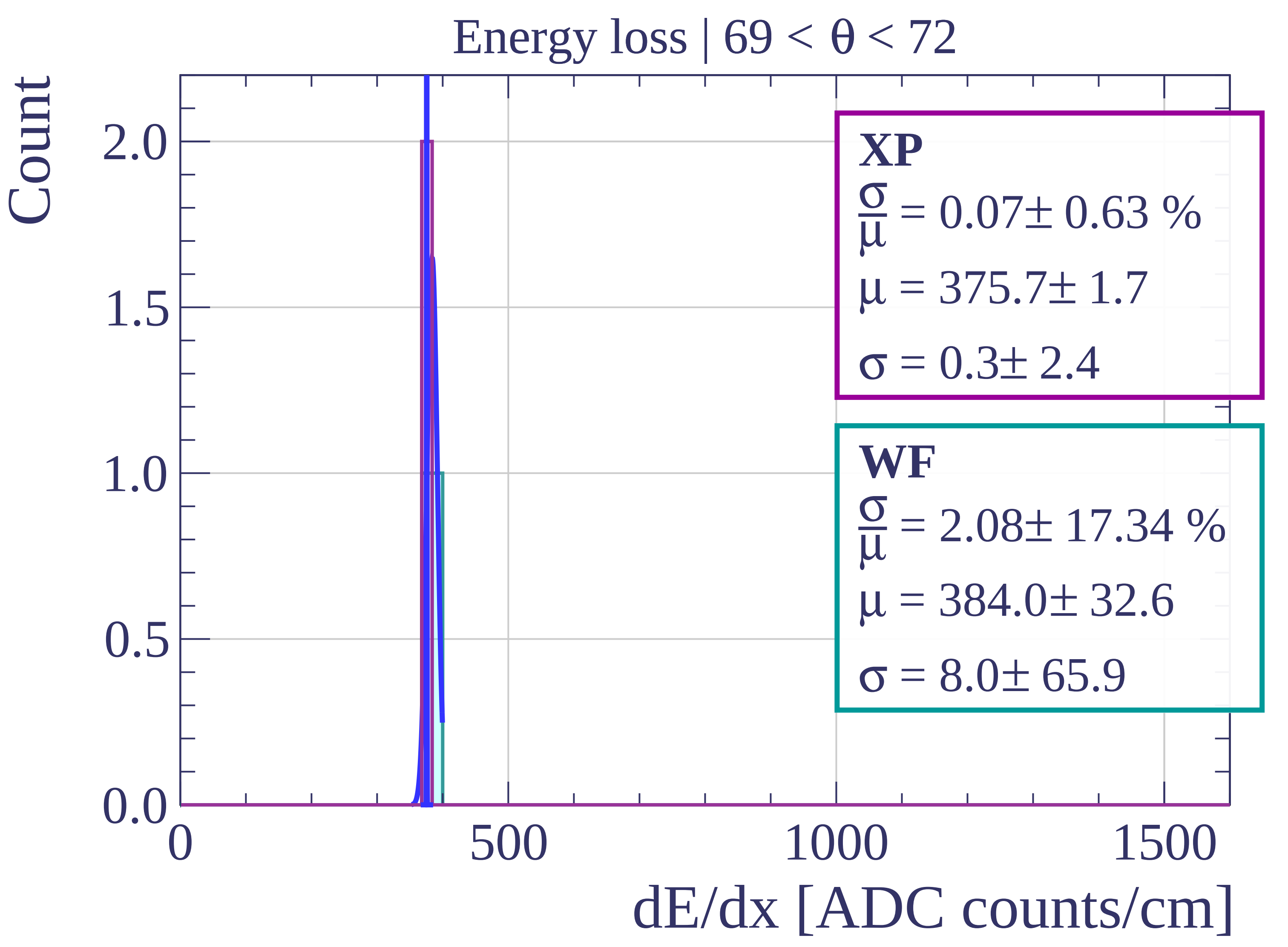
Energy loss | 60 < θ < 63
Count
0
1
2
3
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = 3.98± 1.54 %
μ = 384.0± 5.9
σ = 15.3± 5.7
WF
$\frac{\sigma}{\mu}$ = 3.23± 1.85 %
μ = 404.3± 6.8
σ = 13.0± 7.3

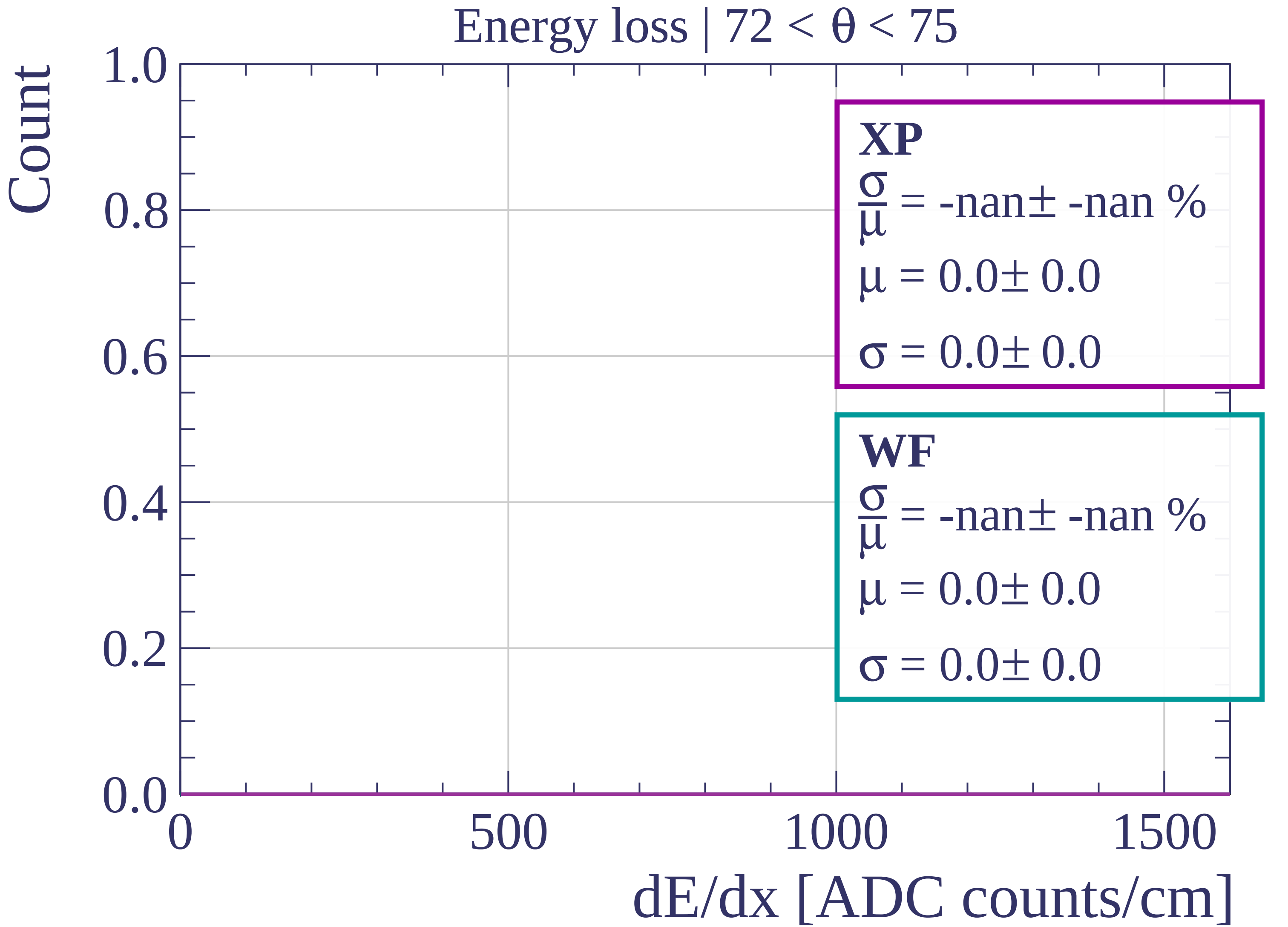


Energy loss | 63 < θ < 66
Count
dE/dx [ADC counts/cm]
0
1
2
3
0
500
1000
1500
XP
σ/μ = 3.50± 2.01 %
μ = 372.3± 6.8
σ = 13.0± 7.3
WF
σ/μ = 2.87± 1.23 %
μ = 376.0± 5.2
σ = 10.8± 4.5

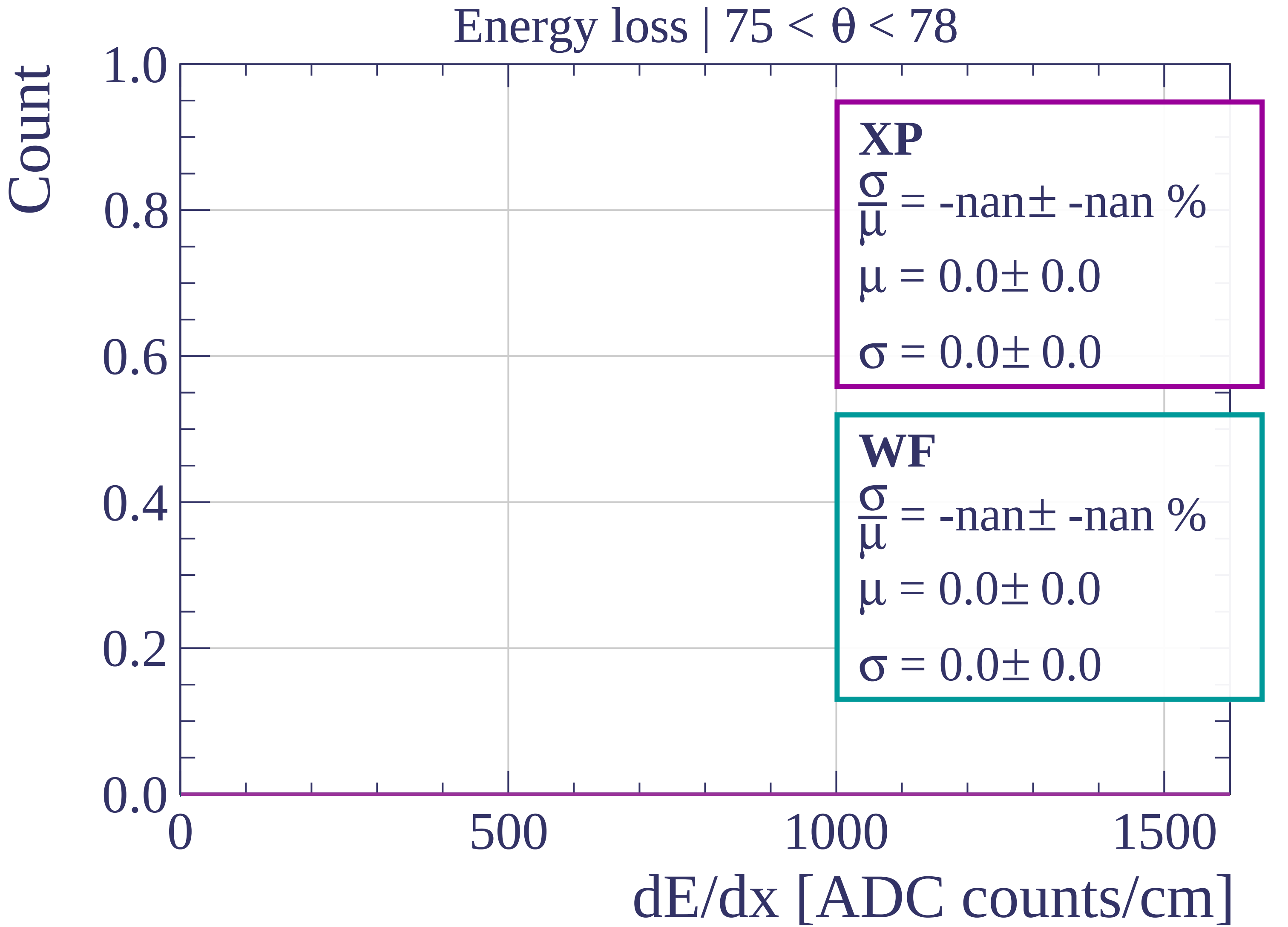


Energy loss | 66 < θ < 69
Count
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0

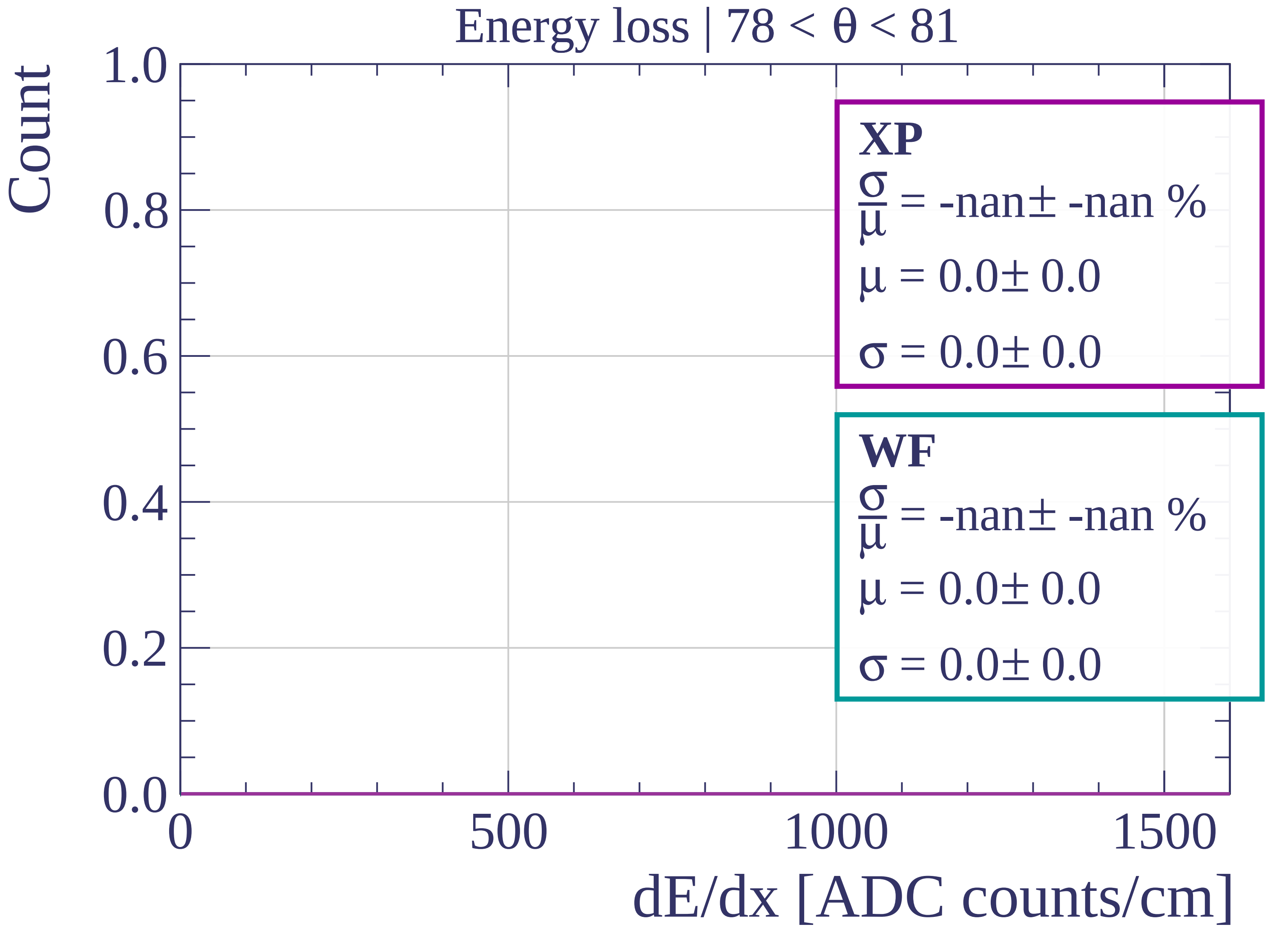

Energy loss | 69 < θ < 72
Count
2.0
1.5
1.0
0.5
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = 0.07± 0.63 %
μ = 375.7± 1.7
σ = 0.3± 2.4
WF
σ/μ = 2.08± 17.34 %
μ = 384.0± 32.6
σ = 8.0± 65.9



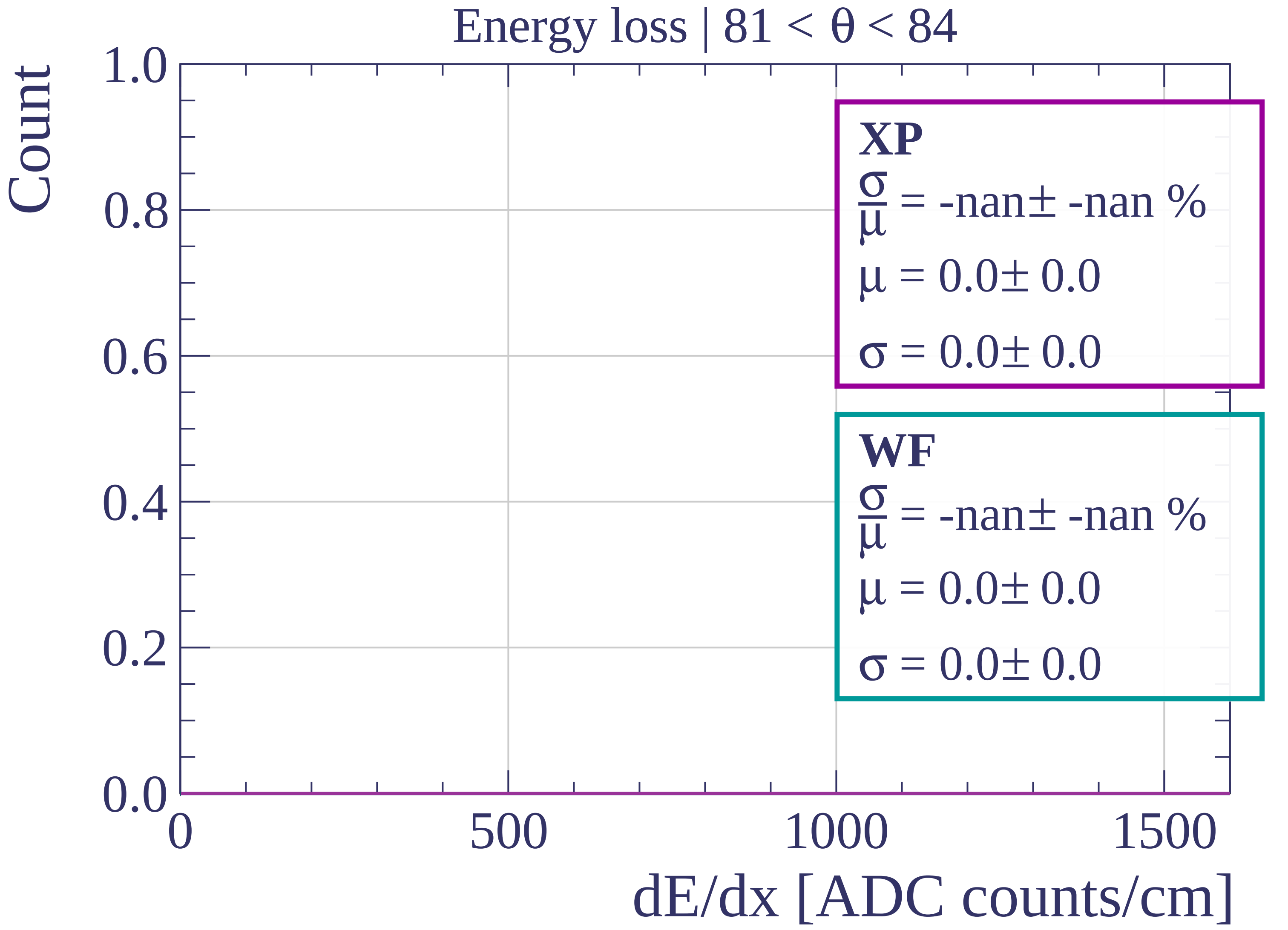
Energy loss | 72 < θ < 75
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



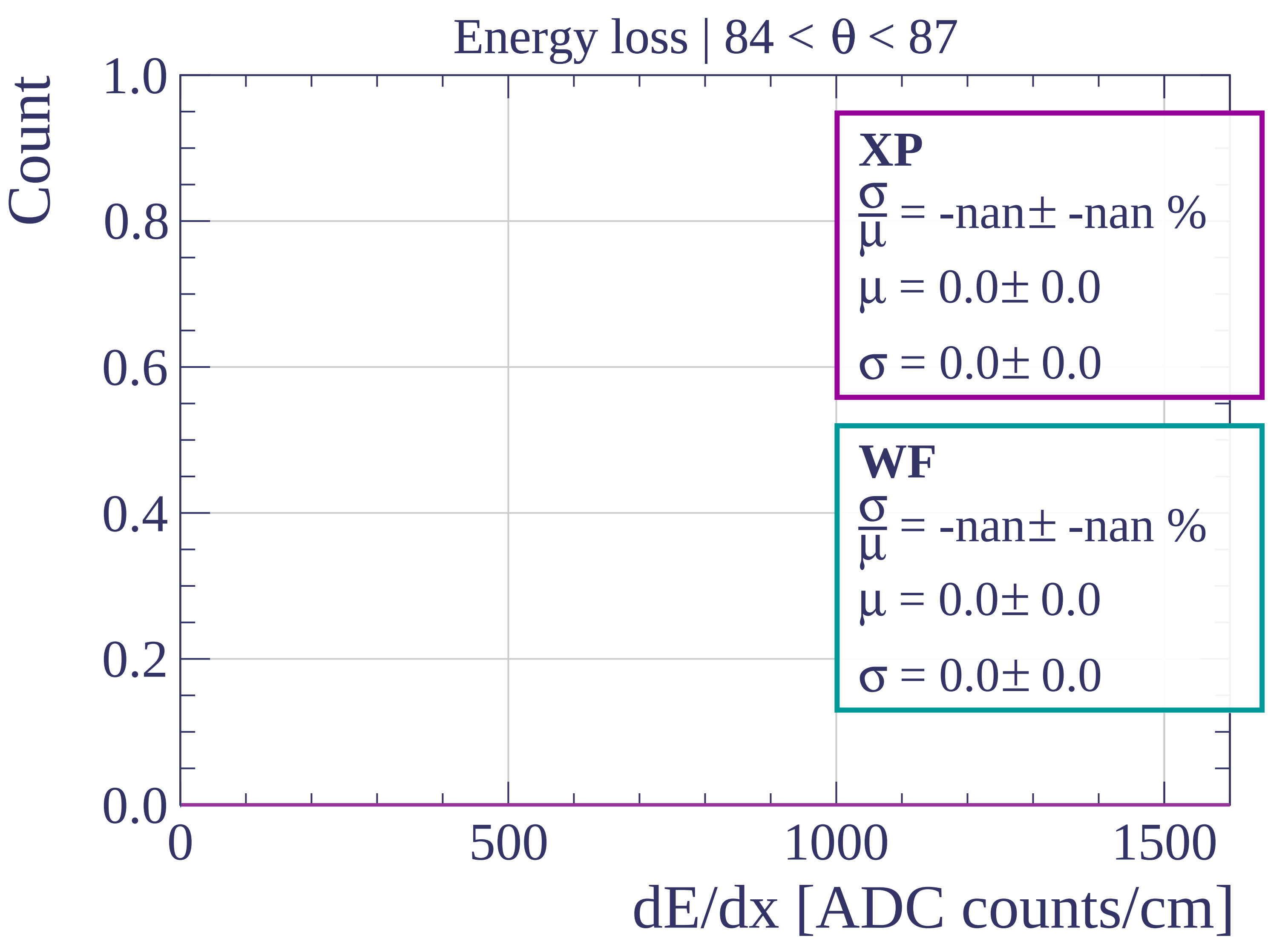
Energy loss | 75 < θ < 78
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



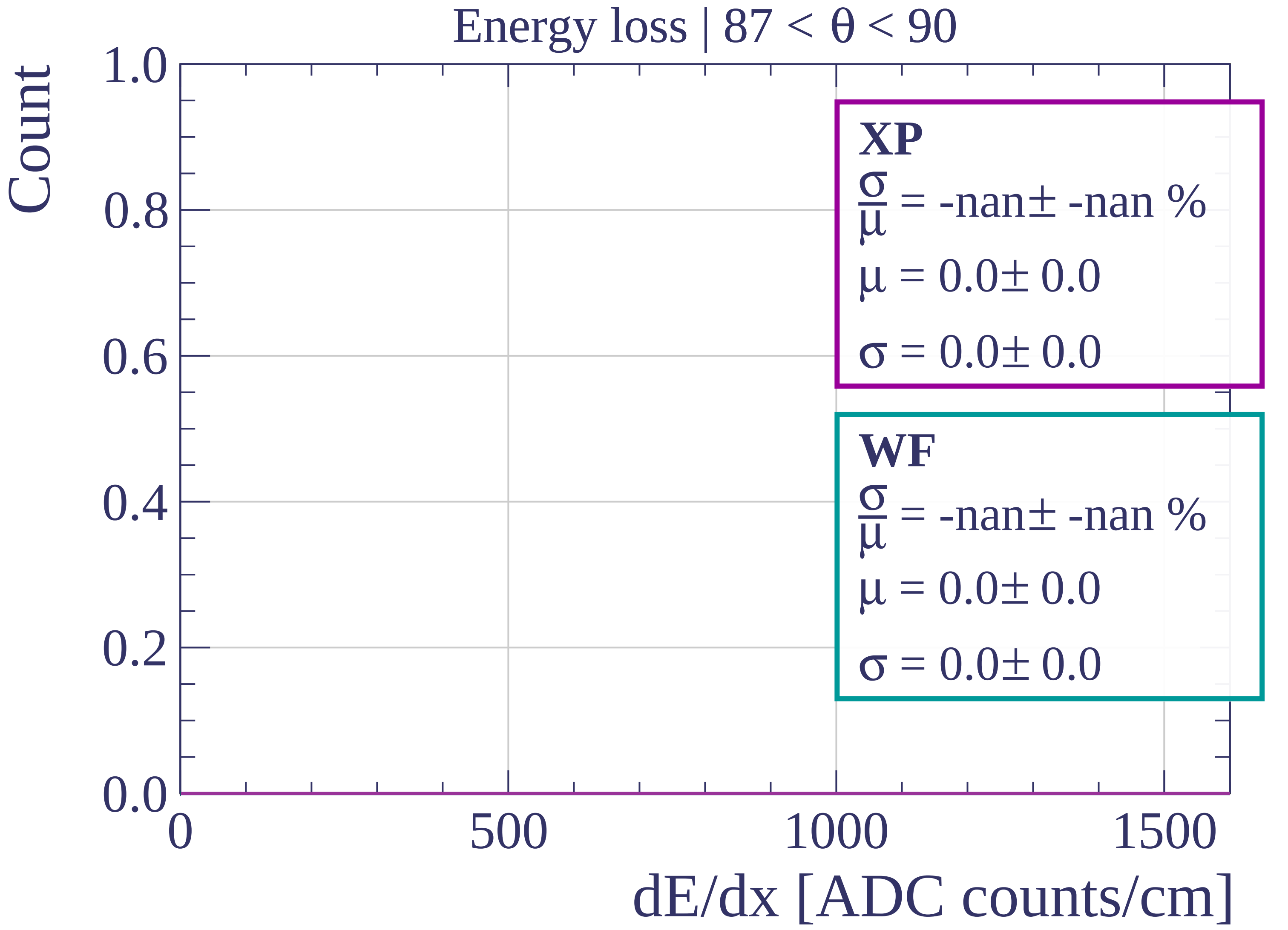
Energy loss | 78 < θ < 81
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
$\frac{\sigma}{\mu}$ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



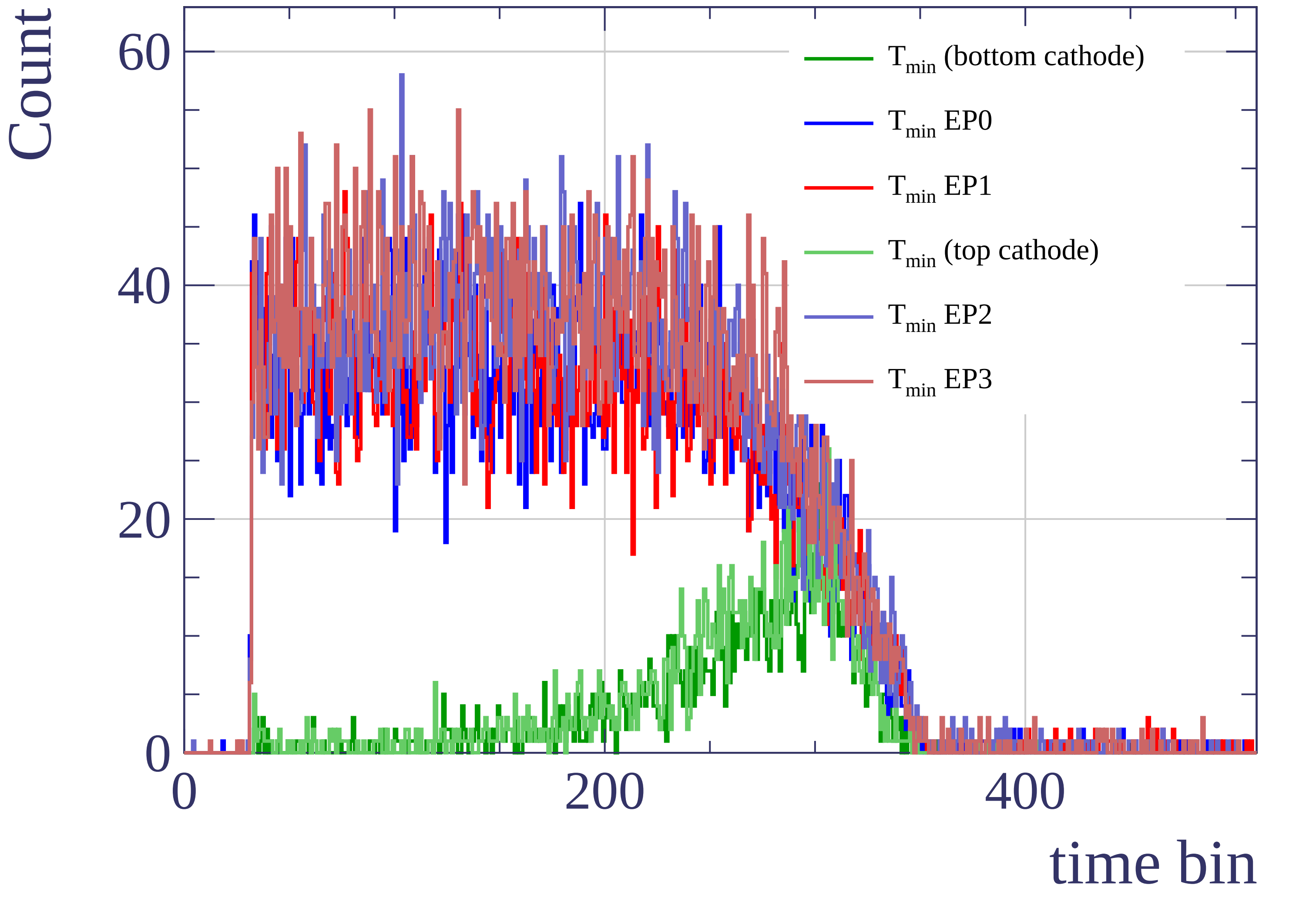
Energy loss | 81 < θ < 84
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



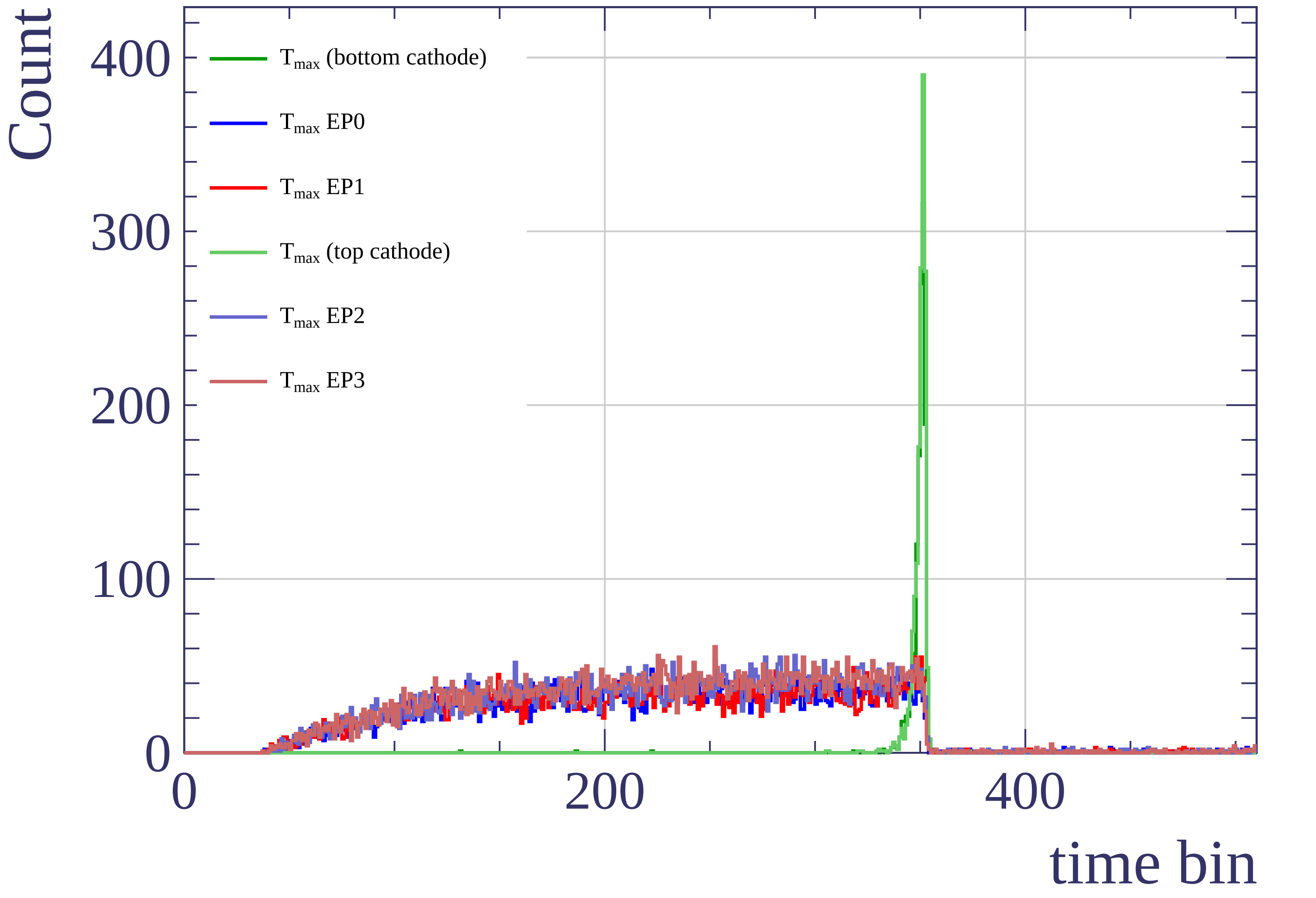
Energy loss | 84 < θ < 87
Count
dE/dx [ADC counts/cm]
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Energy loss | 87 < θ < 90
Count
1.0
0.8
0.6
0.4
0.2
0.0
0
500
1000
1500
dE/dx [ADC counts/cm]
XP
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0
WF
σ/μ = -nan± -nan %
μ = 0.0± 0.0
σ = 0.0± 0.0



Count
time bin
0
20
40
60
0
200
400
$T_{min}$ (bottom cathode)
$T_{min}$ EP0
$T_{min}$ EP1
$T_{min}$ (top cathode)
$T_{min}$ EP2
$T_{min}$ EP3



Count
time bin
$T_{max}$ (bottom cathode)
$T_{max}$ EP0
$T_{max}$ EP1
$T_{max}$ (top cathode)
$T_{max}$ EP2
$T_{max}$ EP3
0
100
200
300
400
0
200
400